

# MATH-4 FINAL

## Matrix Decomposition Theorem

Matrix **decomposition** is a factorization of a matrix into some canonical form (Standard form/Normal form). There are many different matrix decompositions; each finds use among a particular class of problems.

### Decompositions related to solving systems of linear equations

- LU decomposition
- Cholesky decomposition
- QR decomposition

### Decompositions based on eigenvalues and related concepts

- Eigendecomposition (also called spectral decomposition)
- Singular value decomposition

### A. Eigen Decomposition Theorem:

Let A be the given Matrix, D is a diagonal matrix with the corresponding eigen values of Matrix A and P be a matrix of eigenvectors of a given Matrix A, Then A can be written as an eigen decomposition  $A = PDP^{-1}$

**Example 42:** Let  $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

$$\therefore A - \lambda I = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1$$

$$\Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow (16 - 4\lambda - 4\lambda + \lambda^2) - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 5, \lambda = 3$$

$$\therefore D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now to find out the eigenvector of the matrix A,

$$\text{Let a non-zero vector } v = \begin{bmatrix} x \\ y \end{bmatrix}$$

We have,  $Av = \lambda v$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 3]$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow 4x + y = 3x$$

$$x + 4y = 3y$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular  $(2-1)=1$  free variable, which is y.

Let  $y = 1$  then  $x = -1$

$$\therefore \text{Eigenvector } v = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ for } \lambda = 3$$

Again for  $\lambda = 5$

$$\text{Let a non-zero vector } v = \begin{bmatrix} x \\ y \end{bmatrix}$$

We have,  $Av = \lambda v$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 5]$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow 4x + y = 5x$$

$$x + 4y = 5y$$

$$\Rightarrow -x + y = 0$$

$$x - y = 0$$

$$\Rightarrow x - y = 0$$

$$x - y = 0$$

$$\Rightarrow \cancel{x} - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular  $(2-1)=1$  free variable, which is  $y$ .

Let  $y=1$ , then  $x=1$

$$\therefore \text{Eigenvector } \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ for } \lambda = 5$$

$P$  be a matrix of eigenvectors of a given Matrix  $A$ .

$$\therefore P = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

Now to find out  $P^{-1}$

$$c_{11} = 1, c_{12} = -1, c_{21} = -1, c_{22} = -1$$

$$P^c = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$\text{Adj } P = (P^c)^T = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$|P| = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2$$

$$\therefore P^{-1} = \frac{\text{Adj } P}{|P|} = -\frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

Then  $A$  can be written as an eigen decomposition  $A = PDP^{-1}$

**Justification:**

$$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$\begin{aligned}
&= -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & -3 \\ -5 & -5 \end{pmatrix} \\
&= -\frac{1}{2} \begin{pmatrix} -3-5 & 3-5 \\ 3-5 & -3-5 \end{pmatrix} \\
&= -\frac{1}{2} \begin{pmatrix} -8 & -2 \\ -2 & -8 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = \mathbf{A} \quad (\text{Proved})
\end{aligned}$$

**$\mathbf{A} = \mathbf{PDP}^{-1}$  (Proved)**

Where D is a diagonal matrix, Furthermore, if A is symmetric, then the columns of P are orthogonal vectors.

If P is not a square matrix (for example, the space of eigenvectors of  $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  is one-dimensional), then P cannot have a matrix inverse and A does not have an Eigen decomposition. However, if P is  $\mathbf{m} \times \mathbf{n}$  (with  $\mathbf{m} > \mathbf{n}$ ), then A can be written using a so-called singular value decomposition.

### **What is singular value of a Matrix?**

The singular values of a  $\mathbf{m} \times \mathbf{n}$  matrix A are the square roots of the eigenvalues of the symmetric  $\mathbf{n} \times \mathbf{n}$  matrix  $\mathbf{A}^T \mathbf{A}$ . The Eigen values of  $\mathbf{A}^T \mathbf{A}$  are

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > \lambda_{r+1} = \dots = \lambda_n = 0$$

$\sigma_i = \sqrt{\lambda_i}$  are called the singular value of A where  $i = 1, 2, \dots, n$ .

It is customary to denote the singular values by  $\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_n$  and to list them in decreasing order:

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq \sigma_n$$

### **Singular Value Decomposition:**

In linear algebra, the **singular value decomposition (SVD)** is an important factorization of a rectangular real or complex matrix, with several applications **in signal processing , image processing and compression, gene expression analysis and statistics**. Applications which employ the SVD include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of eigenvectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

The singular value decomposition (SVD) is a generalization of the eigen-decomposition, which can be used to analyze rectangular matrices (the eigen-decomposition is defined only for squared matrices). By analogy with the eigen-decomposition, which decomposes a matrix into two simple matrices, the main idea of the SVD is to decompose a rectangular matrix into three simple matrices: Two orthogonal matrices and one diagonal matrix.

Matrix  $A$  is decomposed using SVD to acquire an orthogonal base in the multidimensional feature space:

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} * \\ * \end{bmatrix}}_S \underbrace{\begin{bmatrix} * & * \\ * & * \end{bmatrix}}_{V^T}$$

$$A = USV^T$$

Here,

$A \rightarrow$  Given Matrix

$U \rightarrow$  Matrix of Eigen Vectors of the Matrix  $AA^T$

$V \rightarrow$  Matrix of Eigen Vectors of the Matrix  $A^T A$

$S \rightarrow$  Roots of Eigen Values of the Matrix  $AA^T$  or  $A^T A$

$V^T \rightarrow$  Transpose of the Matrix  $V$

**Example 43: Compute Singular Value Decomposition (SVD) for the given matrix**

$$\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

**Answer:** Let,  $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

In order to find  $U$ , Now we need to find the eigenvectors for  $AA^T$

Given,  $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore A^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

So,  $AA^T = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 4+4 & 2-2 \\ 2-2 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

------(i)

Now we need to find the eigenvalues of  $\mathbf{AA}^T$

$$\begin{aligned}\mathbf{AA}^T - \lambda \mathbf{I} &= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix}\end{aligned}$$

$$|\mathbf{AA}^T - \lambda \mathbf{I}| = 0$$

$$\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)(2-\lambda) = 0$$

$$\Rightarrow (\lambda-8)(\lambda-2) = 0$$

$$\Rightarrow \lambda = 8, 2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\mathbf{D} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } \mathbf{S} = \sqrt{\mathbf{D}} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{-----(ii)}$$

Now we need to find the eigenvector of  $\mathbf{AA}^T$

$$\text{We have, } \mathbf{A}\mathbf{u}_1 = \lambda \mathbf{u}_1$$

$$[\mathbf{A}\mathbf{V} = \lambda \mathbf{V}]$$

$$\text{Let } \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore \mathbf{AA}^T \mathbf{u}_1 = \lambda \mathbf{u}_1$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For  $\lambda = 8$  & from (i) putting the value of  $\mathbf{AA}^T$ ]

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 8x = 8x$$

$$2y = 8y$$

$$\Rightarrow 8x - 8x = 0$$

$$2y - 8y = 0$$

$$\Rightarrow 8x = 8x$$

$$-6y = 0$$

$$\Rightarrow 8x = 8x$$

$$y = 0$$

$$\therefore y = 0$$

$$\text{Let } x = -1$$

$$\therefore \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \text{ It is a unit vector or normal vector. Since the length of } \mathbf{u}_1$$

$$\text{is } |\mathbf{u}_1| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$$

$$\text{Again, Let } \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

Here,

$$\therefore \mathbf{A}\mathbf{A}^T \mathbf{u}_2 = \lambda \mathbf{u}_2$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 2 \text{ \& from (i) putting the value of } \mathbf{A}\mathbf{A}^T]$$

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 8x = 2x$$

$$2y = 2y$$

$$\Rightarrow 8x - 2x = 0$$

$$2y = 2y$$

$$\Rightarrow 6x = 0$$

$$2y = 2y$$

$$\therefore x = 0$$

$$2y = 2y$$

$$\text{Let } y = 1$$

$$\therefore \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ It is a unit vector or normal vector. Since the length of } \mathbf{u}_1$$

$$\text{is } |\mathbf{u}_2| = \sqrt{(0)^2 + (1)^2} = \sqrt{1} = 1$$

$$\therefore U = (u_1 \quad u_2) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \text{-----(iii)}$$

Now we need to find the eigenvectors for  $A^T A$

$$\text{Given, } A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

$$\therefore A^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\begin{aligned} \therefore A^T A &= \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \text{-----(iv)} \end{aligned}$$

Now we need to find the eigenvalues of  $A^T A$

$$\begin{aligned} \therefore A^T A - \lambda I &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{pmatrix} \end{aligned}$$

$$\therefore |A^T A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(5-\lambda) - 9 = 0$$

$$\Rightarrow 25 - 10\lambda + \lambda^2 - 9 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

$$\Rightarrow (\lambda - 8)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 8, 2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } S = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$$

$$S = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{-----}(v)$$

Now we need find the eigenvectors of  $A^T A$  which are the columns of  $V$ .

We have,  $A\mathbf{v}_1 = \lambda\mathbf{v}_1$

$$[AV = \lambda V]$$

$$\text{Let } \mathbf{v}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore A^T A \mathbf{v}_1 = \lambda \mathbf{v}_1$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For  $\lambda = 8$  & from (iv) putting the value of  $A^T A$ ]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 5x - 3y = 8x$$

$$-3x + 5y = 8y$$

$$\Rightarrow -3x - 3y = 0$$

$$-3x - 3y = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow \cancel{x} + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular  $(2-1)=1$  free variable, which is  $y$ .

Let  $y = 1 \therefore x = -1$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}; \text{ It is not unit vector or normal vector. Since the length of } \mathbf{v}_1$$

is  $|\mathbf{v}_1| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$ ; this vector  $\mathbf{v}_1$  has to be converted into unit vector by dividing its magnitude or its length.

$$\text{We have, } \mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} -1 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

[Dividing by its magnitude or its length  $\sqrt{2}$  ]

$$\mathbf{v}_1 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

[Multiplying by  $\sqrt{2}$  on numerator and denominator]

Similarly,

We have,  $\mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

$$\text{Let } \mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore \mathbf{A}^T \mathbf{A} \mathbf{v}_2 = \lambda \mathbf{v}_2$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For  $\lambda = 2$  & from (iv) putting the value of  $\mathbf{A}^T \mathbf{A}$  ]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 3x - 3y = 0$$

$$-3x + 3y = 0$$

$$\Rightarrow x - y = 0$$

$$x - y = 0$$

$$\Rightarrow x = y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) = 1 free variable, which is y.

$$\text{Let } y = 1 \therefore x = 1$$

$$\therefore \mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ It is not unit vector or normal vector. Since the length of } \mathbf{v}_2$$

is  $|\mathbf{v}_2| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$ ; this vector  $\mathbf{v}_2$  has to be converted into unit vector by dividing its magnitude or its length.

We have,

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{Thus, } \mathbf{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \quad [\text{Dividing by its magnitude or its length } \sqrt{2}]$$

$$\Rightarrow \mathbf{v}_2 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} \quad [\text{Multiplying by } \sqrt{2} \text{ on numerator and denominator}]$$

Now we need to construct the augmented orthogonal matrix V,

$$\mathbf{V} = (\mathbf{v}_1 \quad \mathbf{v}_2) = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\therefore \mathbf{V}^T = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \quad \text{-----(vi)}$$

From (ii), (iii) & (vi), Putting the values of  $\mathbf{U}, \mathbf{S}, \mathbf{V}^T$  in the relation  $\mathbf{A} = \mathbf{USV}^T$

Therefore  $\mathbf{A} = \mathbf{USV}^T$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{Answer}$$

**Justification Test:**

$$\mathbf{A} = \mathbf{USV}^T$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \quad [\text{Putting the value of } \mathbf{U}, \mathbf{V}^T, \mathbf{S}]$$

$$\mathbf{A} = \mathbf{USV}^T$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 & 2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 \\ 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} & 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 \times (-2) + 0 & (-1) \times 2 + 0 \\ 0 + 1 & 0 + 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \text{ [Given Matrix] Proved}$$

**Example 44: Compute Singular Value Decomposition (SVD) for the given matrix**

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find U, In order to find U, Now we need to find the eigenvectors for  $AA^T$

Given,

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\text{So, } AA^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \text{-----(i)}$$

Now we need find the eigenvalues of  $AA^T$

$$|AA^T - \lambda I| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

Therefore,

$$\begin{vmatrix} 11-\lambda & 1-0 \\ 1-0 & 11-\lambda \end{vmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\Rightarrow 121 - 11\lambda - 11\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 121 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 120 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda - 12\lambda + 120 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 10, 12$$

Therefore eigenvalues are 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 12 & 0 \\ 0 & 10 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values  $S = \sqrt{D} = \begin{pmatrix} \sqrt{12} & 0 \\ 0 & \sqrt{10} \end{pmatrix}$  -----(ii)

Now we need to find the eigenvector of  $AA^T$

We have,  $A\mathbf{u}_1 = \lambda\mathbf{u}_1$

$[AV = \lambda V]$

Let  $\mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore AA^T\mathbf{u}_1 = \lambda\mathbf{u}_1$

$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$

$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 12 \begin{pmatrix} x \\ y \end{pmatrix}$

[For  $\lambda = 12$  & from (i) putting the value of  $AA^T$ ]

$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \end{pmatrix}$

$\begin{pmatrix} 11x + y \\ x + 11y \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \end{pmatrix}$

$\Rightarrow 11x + y = 12x$

$x + 11y = 12y$

$\Rightarrow -x + y = 0$

$x - y = 0$

$\Rightarrow x - y = 0$

$x - y = 0$

$\Rightarrow x - y = 0$

Let  $y = 1 \therefore x = 1$

$\therefore \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$  It is not unit vector or normal vector. Since the length of  $\mathbf{u}_1$

is  $|\mathbf{u}_1| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$ ; this vector  $\mathbf{u}_1$  has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis,  $\mathbf{u}_1$  needs to be of length one. We divide  $\mathbf{u}_1$  by its magnitude to accomplish this,

$$\text{Thus, } \therefore \mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

[Dividing by its magnitude or its length  $\sqrt{2}$  ]

Again,

We have,  $\mathbf{A}\mathbf{u}_2 = \lambda\mathbf{u}_2$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

$$\therefore \mathbf{A}\mathbf{A}^T \mathbf{u}_2 = \lambda\mathbf{u}_2$$

$$\text{Let } \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

For  $\lambda = 10$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 10 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 10 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For  $\lambda = 10$  & from (i) putting the value of  $\mathbf{A}\mathbf{A}^T$  ]

$$\begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \end{pmatrix}$$

$$\begin{pmatrix} 11x + y \\ x + 11y \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \end{pmatrix}$$

$$\Rightarrow 11x + y = 10x$$

$$x + 11y = 10y$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x + y = 0$$

Let  $y = -1 \therefore x = 1$

$$\therefore \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ It is not unit vector or normal vector. Since the length of } \mathbf{u}_2$$

is  $|\mathbf{u}_2| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$  ; this vector  $\mathbf{u}_2$  has to be converted into unit vector by dividing its magnitude or its length.

Since U has an orthonormal basis,  $\mathbf{u}_2$  needs to be of length one. We divide  $\mathbf{u}_2$  by its magnitude to accomplish this,

Thus,  $\therefore \mathbf{u}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$  [Dividing by its magnitude or its length  $\sqrt{2}$ ]

Since  $\mathbf{u}_2$  is already of length one, dividing by the magnitude we get the same vector back,

Setting up the augmented matrix U

$$\mathbf{U} = (\mathbf{u}_1 \quad \mathbf{u}_2) = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \text{-----(iii)}$$

Now we need to find the eigenvectors of  $\mathbf{A}^T \mathbf{A}$  which are the columns of V.  
Start with the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find V, we have to start with  $\mathbf{A}^T \mathbf{A}$  the transpose of A is

$$\mathbf{A}^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\begin{aligned} \therefore \mathbf{A}^T \mathbf{A} &= \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \\ &= \begin{pmatrix} 3 \times 3 + 1 & 3 - 3 & 3 - 1 \\ 3 - 3 & 1 + 9 & 1 + 3 \\ 3 - 1 & 1 + 3 & 1 + 1 \end{pmatrix} \\ &= \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \text{-----(iv)} \end{aligned}$$

Now we need to find the eigenvalue of  $\mathbf{A}^T \mathbf{A}$

$$\begin{aligned} |\mathbf{A}^T \mathbf{A} - \lambda \mathbf{I}| &= 0 \\ \left| \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| &= 0 \\ \left| \begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| &= 0 \end{aligned}$$

Therefore,

$$\begin{aligned} & \begin{vmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{vmatrix} = 0 \\ & \begin{vmatrix} 10-\lambda & 0 & 2 \\ 0 & 10-\lambda & 4 \\ 2 & 4 & 2-\lambda \end{vmatrix} = 0 \\ & (10-\lambda)((10-\lambda)(2-\lambda)-16) + 2\{0-2(10-\lambda)\} = 0 \\ & \Rightarrow (10-\lambda)(20-10\lambda-2\lambda+\lambda^2-16)-4(10-\lambda) = 0 \\ & \Rightarrow (10-\lambda)(20-10\lambda-2\lambda+\lambda^2-16-4) = 0 \\ & \Rightarrow (10-\lambda)(\lambda^2-12\lambda) = 0 \\ & \Rightarrow \lambda(10-\lambda)(\lambda-12) = 0 \\ & \Rightarrow \lambda = 0, 10, 12 \end{aligned}$$

Therefore eigenvalues are 0, 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 12 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } S = \sqrt{D} = \begin{pmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{pmatrix} \text{-----(v)}$$

Now we need to find the eigenvector of  $A^T A$  which, are the columns of V.

We have,  $A v_1 = \lambda v_1$

$$[A V = \lambda V]$$

$$\therefore A^T A v_1 = \lambda v_1$$

$$\text{Let } v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$A^T A v_1 = \lambda v_1$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 12 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad [\text{For } \lambda = 12 \text{ \& from (iv) putting the value of } A^T A]$$

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 12x \\ 12y \\ 12z \end{pmatrix}$$

$$\Rightarrow 10x + 2z = 12x$$

$$10y + 4z = 12y$$

$$2x + 4y + 2z = 12z$$

$$\Rightarrow 2x + 4y + 2z = 12z$$

$$10x + 2z = 12x$$

$$10y + 4z = 12y$$

$$\Rightarrow 2x + 4y - 10z = 0$$

$$-2x + 2z = 0$$

$$-2y + 4z = 0$$

$$\Rightarrow 2x + 4y - 10z = 0$$

$$-2x + 0.y + 2z = 0$$

$$-2y + 4z = 0$$

We have,

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

i = 2,

$$L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\rightarrow -(-2)(2x + 4y - 10z) + 2(-2x + 0.y + 2z)$$

$$\rightarrow 4x + 8y - 20z - 4x + 0 + 4z$$

$$\rightarrow 8y - 16z$$

Thus we obtain a new system,

$$2x + 4y - 10z = 0$$

$$8y - 16z = 0$$

$$-2y + 4z = 0$$

$$\Rightarrow 2x + 4y - 10z = 0$$

$$y - 2z = 0$$

$$y - 2z = 0$$

Since the 1<sup>st</sup> and 2<sup>nd</sup> equations are identical. We can disregard one of them

$$\begin{array}{l} 2x + 4y - 10z = 0 \\ y - 2z = 0 \end{array}$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2)=1 free variable, which is z.

Here free variable is z, Let z = 1,  $\therefore y = 2$ , then  $2x + 4 \times 2 - 10 \times 1 = 0$

$$\therefore x = 1$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$|\mathbf{v}_1| = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{6}$$

Since  $V$  has an orthonormal basis,  $\mathbf{v}_1$  needs to be of length one. We divide  $\mathbf{v}_1$  by its

magnitude to accomplish this, Thus,  $\therefore \mathbf{v}_1 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$

**Again,**

$$\text{We have, } \mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

$$\therefore \mathbf{A}^T \mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$$

$$\text{Let } \mathbf{v}_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A}^T \mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 10 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

[For  $\lambda = 10$  & from (iv) putting the value of  $\mathbf{A}^T \mathbf{A}$ ]

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 10x \\ 10y \\ 10z \end{pmatrix}$$

$$\Rightarrow 10x + 2z = 10x$$

$$10y + 4z = 10y$$

$$2x + 4y + 2z = 10z$$

$$\Rightarrow 2x + 4y + 2z = 10z$$

$$10x + 2z = 10x$$

$$10y + 4z = 10y$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$2z = 0$$

$$4z = 0$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$2z = 0$$

$$2z = 0$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$2z = 0$$

$$\Rightarrow 2x + 4y - 8z = 0$$

$$z = 0$$

$$\therefore z = 0$$

$$\text{Then } \Rightarrow 2x + 4y - 8 \times 0 = 0$$

$$\Rightarrow 2x + 4y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular  $(2-1)=1$  free variable, which is  $y$ .

Here free variable is  $y$ , Let  $y = -1$ , then  $x = 2$

$$\therefore \mathbf{v}_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$$

$$|\mathbf{v}_2| = \sqrt{(2)^2 + (-1)^2 + (0)^2} = \sqrt{5}$$

Since  $V$  has an orthonormal basis,  $\mathbf{v}_2$  needs to be of length one. We divide  $\mathbf{v}_2$  by it's

magnitude to accomplish this, Thus,  $\therefore \mathbf{v}_2 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{-1}{\sqrt{5}} \\ 0 \end{pmatrix}$

**Again,**

$$\text{We have, } \mathbf{A}\mathbf{v}_3 = \lambda\mathbf{v}_3$$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

$$\text{We have, } \mathbf{A}^T\mathbf{A}\mathbf{v}_3 = \lambda\mathbf{v}_3$$

$$\text{Let } \mathbf{v}_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A}^T\mathbf{A}\mathbf{v}_3 = \lambda\mathbf{v}_3$$

$$\begin{pmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

[For  $\lambda = 0$  & from (iv) putting the value of  $A^T A$ ]

$$\begin{pmatrix} 10x + 0 + 2z \\ 0 + 10y + 4z \\ 2x + 4y + 2z \end{pmatrix} = \begin{pmatrix} 0x \\ 0y \\ 0z \end{pmatrix}$$

$$\Rightarrow 10x + 2z = 0$$

$$10y + 4z = 0$$

$$2x + 4y + 2z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$

$$10x + 2z = 0$$

$$10y + 4z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$

$$10x + 0.y + 2z = 0$$

$$10y + 4z = 0$$

$$L_i \rightarrow -a_{i1}L_1 + a_{i1}L_i$$

$$i = 2,$$

$$L_2 \rightarrow -a_{21}L_1 + a_{11}L_2$$

$$\rightarrow -10(2x + 4y + 2z) + 2(10x + 0.y + 2z)$$

$$\rightarrow -20x - 40y - 20z + 20x + 0 + 4z$$

$$\rightarrow -40y - 16z$$

Thus we obtain the following system:

$$\Rightarrow 2x + 4y + 2z = 0$$

$$-40y - 16z = 0$$

$$10y + 4z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$

$$10y + 4z = 0$$

$$10y + 4z = 0$$

$$\Rightarrow 2x + 4y + 2z = 0$$

$$10y + 4z = 0$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular  $(3-2)=1$  free variable, which is  $z$

Here free variable is  $z$ , Let  $z = -5$ , then  $y = 2$ , then  $x = 1$

$$\therefore \mathbf{v}_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix}$$

$$|\mathbf{v}_3| = \sqrt{(1)^2 + (2)^2 + (-5)^2} = \sqrt{30}$$

Since V has an orthonormal basis,  $\mathbf{v}_3$  needs to be of length one. We divide  $\mathbf{v}_3$  by it's

magnitude to accomplish this, Thus,  $\therefore \mathbf{v}_3 = \begin{pmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ \frac{-5}{\sqrt{30}} \end{pmatrix}$

Now we need to construct the augmented orthogonal matrix V,

$$\mathbf{V} = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3)$$

All this to give us

$$\mathbf{V} = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

when we really want its transpose

$$\mathbf{V}^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

For S we take the square roots of the non-zero eigenvalues and populate the diagonal with them, putting the largest in  $s_{11}$ , the next largest in  $s_{22}$  and so on until the smallest value ends up in  $s_{mn}$ . The non-zero eigenvalues of U and V are always the same, so that's why it doesn't matter which one we take them from. Because we are doing full SVD, instead of reduced SVD (next section), we have to add a zero column vector to S so that it is of the proper dimensions to allow multiplication between U and V. The diagonal entries in S are the singular values of A, the columns in U are called left singular vectors, and the columns in V are called right singular vectors.

$$\mathbf{S} = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix}$  [Ignoring third row of the matrix as all elements are zero in the last row]

Now we have all the pieces of the puzzle

$$A = USV^T$$

Putting the value of  $U, V^T, S$  in  $A = USV^T$

$$A = USV^T$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & -\frac{5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{12}}{\sqrt{2}} & \frac{\sqrt{10}}{\sqrt{2}} & 0 \\ \frac{\sqrt{12}}{\sqrt{2}} & -\frac{\sqrt{10}}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & -\frac{5}{\sqrt{30}} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \text{ [Proved]}$$

### Home Task:

Find the singular value decomposition of the matrix

i.  $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$  ii.  $A = \begin{bmatrix} 2 & 1 & -2 \end{bmatrix}$

iii.  $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

Hints:

$$U = \begin{bmatrix} 0.82 & -0.58 & 0 & 0 \\ 0.58 & 0.82 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.40 & -0.91 \\ 0.91 & 0.40 \end{bmatrix}$$

$$S = \begin{bmatrix} 5.47 & 0 \\ 0 & 0.37 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

### Singular Value Decomposition:

The Singular Value Decomposition is an important factorization of a real or complex matrix, with several applications in signal processing and statistics. Applications, which employ the SVD, include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of Eigen vectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

### Algorithm of Singular Value Decomposition:

To find the Singular Value Decomposition of the matrix:

1. Find the eigenvalues of the matrix  $\mathbf{A}^T \mathbf{A}$  and arrange them in descending order
2. Find the number of nonzero eigenvalues of the matrix  $\mathbf{A}^T \mathbf{A}$
3. Find the orthogonal eigenvectors of the matrix  $\mathbf{A}^T \mathbf{A}$  corresponding to the obtained eigenvalues, and arrange them in the same order to form the column-vectors of the matrix  $V$
4. Form a diagonal matrix  $S = \sum = \sigma$  placing on the leading diagonal of it the square roots  $\sigma_i = \sqrt{\lambda_i}$  of  $\mathbf{p} = \min\{m, n\}$  first eigenvalues of the matrix  $\mathbf{A}^T \mathbf{A}$  got in  $I$  in descending order
5. Find the first column-vectors of the matrix  $U$ ,  $\mathbf{u}_i = \sigma_i^{-1} \mathbf{A} \mathbf{v}_i (i = 1 : r)$
6. Add to the matrix  $U$  the rest of  $m-r$  vectors using the Gram-Schmidt Orthogonalization process

### Image Processing with SVD

We want to approximate the  $m \times n$  matrix  $A$  by using far fewer entries than in the original matrix.

$$\mathbf{A} = \mathbf{U} \mathbf{S} \mathbf{V}^T = \mathbf{S} \mathbf{U} \mathbf{V}^T$$

$$\Rightarrow \mathbf{A} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T + \mathbf{0} \cdot \mathbf{u}_{r+1} \mathbf{v}_{r+1}^T + \dots$$

Since the singular values are always greater than zero. Adding on the dependent terms where the singular values are equal to zero does not effect the image. The terms at the end of the equation zero out leaving us with:

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$$

**Conclusion:** We can further approximate the matrix (original image) by leaving off more singular terms of the matrix A. Since the singular values are arranged in descending order, the last terms will have the least effect on the overall image.

Doing this (leaving more singular values) reduces the amount of space required to store the image on a computer.

**Justification using above example:**

The original image is:  $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$S = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{ i.e. } \sigma_1 = 2\sqrt{2}, \sigma_2 = \sqrt{2}$$

$$U = (u_1 \ u_2) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$V = (v_1 \ v_2) = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{aligned} \Rightarrow A &= \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T \\ &= 2\sqrt{2} \begin{pmatrix} -1 \\ 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \sqrt{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \\ &= \begin{pmatrix} -2\sqrt{2} \\ 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \\ &= \begin{pmatrix} 2 & -2 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} = A \text{ (Original Image can be obtained)} \end{aligned}$$

We can get approximate image by using less singular values. Say number of singular values are 10 i.e.  $\sigma_1, \sigma_2, \dots, \sigma_{10}$ .

Then,

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_{10} u_{10} v_{10}^T$$

**If we use 5 singular values then**

$$\Rightarrow A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_5 u_5 v_5^T$$

**In this result we can get approximate image of the original image (A)**



### Cartesian product

Cartesian product is an operator on sets (tuples, bags, ...). The result from the Cartesian product of  $n$  sets is a set of all possible ordered tuples, containing on  $i$ -th place an element from  $i$ -th set.

For example, the cartesian product of  $A = \{a, b, c\}$  and  $B = \{c, d, e\}$  is  
 $A \times B = \{(a, c), (a, d), (a, e), (b, c), (b, d), (b, e), (c, c), (c, d), (c, e)\}$ .

Formally, the Cartesian product of two sets  $A$  and  $B$  (written as  $A \times B$ ) is the set of tuples  $(a, b)$  with  $a$  element of  $A$  and  $b$  element of  $B$ .

**LCS Slide:**

### B. LU Decomposition

A procedure for decomposing a  $N \times N$  matrix  $A$  into a product of a lower triangular matrix  $L$  and an upper triangular matrix  $U$ ,

$$LU = A$$

LU decomposition is implemented in *Mathematica* as `LU Decomposition[m]`.

Written explicitly for a  $3 \times 3$  matrix, the decomposition is

$$\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$
$$\begin{bmatrix} l_{11} u_{11} & l_{11} u_{12} & l_{11} u_{13} \\ l_{21} u_{11} & l_{21} u_{12} + l_{22} u_{22} & l_{21} u_{13} + l_{22} u_{23} \\ l_{31} u_{11} & l_{31} u_{12} + l_{32} u_{22} & l_{31} u_{13} + l_{32} u_{23} + l_{33} u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

### Lower/Upper Triangular (LU) Decomposition

As the title suggests, this method breaks down a matrix into two separate matrices. These matrices are called the upper and lower triangular matrices. The upper triangular matrix results from reducing the original matrix into row echelon form by way of elementary row operations (excluding row interchange). The lower triangular matrix is created by storing a representative of each elementary row operation as follows:

- If multiplying a diagonal entry to convert it to a 1, store the reciprocal of the multiplier in the respective diagonal location in the new matrix (lower triangular matrix).
- When introducing zeros to positions beneath a leading 1, store the opposite of these numbers in their respective positions in the new matrix.

If a matrix A can be decomposed into lower and upper triangular form, then the matrix A = LU.

LU decomposition is used to find the solutions of linear systems. Though this method may seem more cumbersome, it takes fewer steps than the Guass-Jordan method to determine the solutions. Before discussing how to use LU decomposition to solve a linear system, we need to first see how to decompose a matrix into an LU representation. The following is an example of where to store the operations in the lower matrix. Notice that this is a square matrix and is not yet augmented with the vector of constants from the equations.

**Example 41:** Given the following system of linear equations, determine the value of each of the variables using the LU decomposition method.

$$\begin{aligned} 6x_1 - 2x_2 &= 14 \\ 9x_1 - x_2 + x_3 &= 21 \\ 3x_1 - 7x_2 + 5x_3 &= 9 \end{aligned}$$

| Upper<br>Triangular                                                     | Explanation of Step                                                                                                                                                                                                                             | Lower<br>Triangular                                                 |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| $\begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}$   | ← Beginning Matrix<br>Matrix Storing Elementary Row Operations →                                                                                                                                                                                | $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ |
| $\begin{bmatrix} 1 & -1/3 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}$ | In order to force a value of 1 at position (1,1), we must multiply row 1 by 1/6. Thus storing its reciprocal, 6, in position (1,1) in the lower matrix. $r_1' = r_1 \times 1/6$                                                                 | $\begin{bmatrix} 6 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ |
| $\begin{bmatrix} 1 & -1/3 & 0 \\ 0 & 2 & 1 \\ 0 & 8 & 5 \end{bmatrix}$  | Introducing zeros to positions (2,1) and (3,1) require multiplications by -9 and -3 respectively. So we will store the opposite of these numbers in their respective locations. $r_2' = r_1 \times (-9) + r_2$ , $r_3' = r_1 \times (-3) + r_3$ | $\begin{bmatrix} 6 & 0 & 0 \\ 9 & 0 & 0 \\ 3 & 0 & 0 \end{bmatrix}$ |

On to the next position in the main diagonal, (2,2). To replace the value in this position with a 1, multiply row 2 by 1/2, thus storing a 2 (the reciprocal) in position (2,2) in the lower triangular matrix.  $\mathbf{r}_2' = \mathbf{r}_2 \times (1/2)$

$$\begin{bmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 1/2 \\ 0 & 8 & 5 \end{bmatrix} \quad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix}$$

Replacing the position under the leading 1, position (3,2), with a zero can be done with a multiplication of -8. We will then store 8, the opposite of -8, in the lower matrix at that position.  $\mathbf{r}_3' = \mathbf{r}_2 \times (-8) + \mathbf{r}_3$

$$\begin{bmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 0 \end{bmatrix}$$

Only a multiplication of 1 is necessary to introduce a 1 to the next diagonal position. In fact nothing is being done to the upper triangular matrix, but we need the 1 in the lower matrix to show that.  $\mathbf{r}_3' = \mathbf{r}_3 \times 1$

$$\begin{bmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix}$$

If a matrix A can be decomposed into an LU representation, then A is equal to the product of the lower and upper triangular matrices. This can be shown with one matrix multiplication.

$$\begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

### Solving Systems of Equations using the LU decomposition.

Systems of linear equations can be represented in a number of ways. In the Gauss-Jordan elimination method, the system was represented as an augmented matrix. In this method, we will represent the system as a matrix equation.

1. Rewrite the system  $\mathbf{Ax} = \mathbf{b}$  using the LU representation for A. Making the system  $\mathbf{LUx} = \mathbf{b}$ .

$$\begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 21 \\ 9 \end{bmatrix}$$

2. Define a new column matrix  $\mathbf{y}$  so that  $\mathbf{Ux} = \mathbf{y}$ .

$$\begin{bmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

3. Rewrite step one with the substitution from step two yielding  $Ly = b$ .

$$\begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 21 \\ 9 \end{bmatrix}$$

4. Solve step three for  $y$  using **forward substitution**.

$$y_1=7/3, y_2=29/6, y_3=33/2$$

5. Using the results from step four, solve for  $x$  in step two using **back substitution**.

$$\begin{bmatrix} 1 & -1/3 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$$

$$x_1=43/36, x_2=-41/12, x_3=33/2$$

Though it seems like a complex set of equations is being substituted for a simple one line  $Ax = b$ , the cofactors of  $A$  (meaning LU) are in such a simple form that the complex set of equations is really not so complex. There is one restriction to this method. Since there is no way to store a row interchange in the lower matrix, row interchanges are not allowed. There is an adjustment that can be made in order to accomplish an LU decomposition is a row interchange is necessary though we will not discuss it here.

### **QR Decomposition**

Given a matrix  $A$ , its Q-R-decomposition is a matrix decomposition of the form

$$A=QR$$

Where  $R$  is an upper triangular matrix and  $Q$  is an orthogonal matrix, i.e., one satisfying

$$Q^T Q = I,$$

Where  $Q^T$  is the transpose of  $Q$  and  $I$  is the identity matrix. This matrix decomposition can be used to solve linear systems of equations.

### **QR-Decomposition**

**Example 42** Find the QR-decomposition for the matrix,

$$A = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & 7 \\ 0 & -1 & -1 \end{bmatrix}$$

**Solution**

The columns from A are,

$$\mathbf{c}_1 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \quad \mathbf{c}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \mathbf{c}_3 = \begin{bmatrix} 3 \\ 7 \\ -1 \end{bmatrix}$$

We performed Gram-Schmidt on these vectors in [Example 3](#) of the previous section. So, the orthonormal vectors that we'll use for Q are,

$$\mathbf{u}_1 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{5}} \\ 0 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{30}} \\ -\frac{5}{\sqrt{30}} \end{bmatrix} \quad \mathbf{u}_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

and the matrix Q is,

$$Q = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{30}} & \frac{2}{\sqrt{6}} \\ 0 & -\frac{5}{\sqrt{30}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

The matrix R is,

$$R = \begin{bmatrix} \langle \mathbf{c}_1, \mathbf{u}_1 \rangle & \langle \mathbf{c}_2, \mathbf{u}_1 \rangle & \langle \mathbf{c}_3, \mathbf{u}_1 \rangle \\ 0 & \langle \mathbf{c}_2, \mathbf{u}_2 \rangle & \langle \mathbf{c}_3, \mathbf{u}_2 \rangle \\ 0 & 0 & \langle \mathbf{c}_3, \mathbf{u}_3 \rangle \end{bmatrix} = \begin{bmatrix} \sqrt{5} & \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ 0 & \frac{6}{\sqrt{30}} & \frac{22}{\sqrt{30}} \\ 0 & 0 & \frac{16}{\sqrt{6}} \end{bmatrix}$$

So, the QR-Decomposition for this matrix is,

$$\begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & 7 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} & \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{30}} & \frac{2}{\sqrt{6}} \\ 0 & -\frac{5}{\sqrt{30}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \sqrt{5} & \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \\ 0 & \frac{6}{\sqrt{30}} & \frac{22}{\sqrt{30}} \\ 0 & 0 & \frac{16}{\sqrt{6}} \end{bmatrix}$$

D.

**Note:**

**Method # 01: SVD** is based on a theorem from linear algebra which says that a rectangular matrix  $A$  can be broken down into the product of three matrices- an orthogonal matrix  $U$ , a diagonal matrix  $S$ , and the transpose of an orthogonal matrix  $V$ . The theorem is usually presented something like this:

$$\mathbf{A}_{mn} = \mathbf{U}_{mm} \mathbf{S}_{mn} \mathbf{V}_{nn}^T$$

Where  $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ ,  $\mathbf{V}^T \mathbf{V} = \mathbf{I}$ ; the columns of  $U$  are orthonormal eigenvectors of  $\mathbf{A}\mathbf{A}^T$ , the columns of  $V$  are orthonormal eigenvectors of  $\mathbf{A}^T \mathbf{A}$ , and  $S$  is a diagonal matrix containing the square roots of eigen values from  $U$  or  $V$  in descending order.

The following example merely applies this definition to a small matrix in order to compute its SVD. In the next section, I attempt to interpret the application of SVD to document classification.

Start with the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find  $U$ , we have to start with  $\mathbf{A}\mathbf{A}^T$ . the transpose of  $A$  is

$$\mathbf{A}^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

so

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}$$

Next, we have to find the eigenvalues and corresponding eigenvectors of  $\mathbf{A}\mathbf{A}^T$ . We know that eigenvectors are defined by the equation  $\mathbf{A}\vec{\mathbf{u}} = \lambda\vec{\mathbf{u}}$ , and applying this to  $\mathbf{A}\mathbf{A}^T$  gives us

$$\begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

We rewrite this as the set of equations

$$\begin{aligned} 11x_1 + x_2 &= \lambda x_1 \\ x_1 + 11x_2 &= \lambda x_2 \end{aligned}$$

and rearrange to get

$$\begin{aligned} (11-\lambda)x_1 + x_2 &= 0 \\ x_1 + (11-\lambda)x_2 &= 0 \end{aligned}$$

Solve for  $\lambda$  by setting the determinant of the coefficient matrix to zero,

$$\begin{vmatrix} (11-\lambda) & 1 \\ 1 & (11-\lambda) \end{vmatrix} = 0$$

Which works out as

$$\begin{aligned} (11-\lambda)(11-\lambda) - 1 &= 0 \\ (\lambda-10)(\lambda-12) &= 0 \\ \lambda &= 10, \lambda = 12 \end{aligned}$$

to give us our two eigenvalues  $\lambda = 10, \lambda = 12$ . Plugging  $\lambda$  back in to the original equations gives us our eigenvectors. For  $\lambda = 10$  we get

$$\begin{aligned} (11-10)x_1 + x_2 &= 0 \\ x_1 &= x_2 \end{aligned}$$

which is true for lots of values, so we'll pick  $x_1 = 1$  and  $x_2 = -1$  since those are small and easier to work with. Thus, we have the eigenvector  $[1, -1]$  corresponding to the eigenvalue  $\lambda = 10$ . for  $\lambda = 12$  we have

$$\begin{aligned} (11-12)x_1 + x_2 &= 0 \\ x_1 &= x_2 \end{aligned}$$

and for the same reason as before we'll take  $\mathbf{x}_1 = \mathbf{1}$  and  $\mathbf{x}_2 = \mathbf{1}$ . Now, for  $\lambda = 12$  we have the eigenvector  $[1, 1]$ . These eigenvectors become column vectors in a matrix ordered by the size of the corresponding eigenvalue. In other words, the eigenvector of the largest eigenvalue is column one, the eigenvector of the next large eigenvalue is column two, and so forth and so on until we have the eigenvector of the smallest eigenvalue as the last column of our matrix. In the matrix below, the eigenvector for  $\lambda = 12$  is column one, and the eigenvector for  $\lambda = 10$  is column two.

Finally, we have to convert this matrix into an orthogonal matrix, which we do by applying the Gram-Schmidt orthonormalization process to the column vectors. Begin by normalizing  $\mathbf{v}_1$ .

$$\vec{\mathbf{u}}_1 = \frac{\vec{\mathbf{v}}_1}{\left| \vec{\mathbf{v}}_1 \right|} = \frac{[1, 1]}{\sqrt{1^2 + 1^2}} = \frac{[1, 1]}{\sqrt{2}} = \left[ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

Compute

$$\begin{aligned} \vec{\omega}_2 &= \vec{\mathbf{v}}_2 - \vec{\mathbf{u}}_1 \cdot \vec{\mathbf{v}}_2 * \vec{\mathbf{u}}_1 = \\ [1, -1] - \left[ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] \cdot [1, -1] * \left[ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] &= \\ [1, -1] - 0 * \left[ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right] &= [1, -1] - [0, 0] = [1, -1] \end{aligned}$$

and normalize

$$\vec{\mathbf{u}}_2 = \frac{\vec{\omega}_2}{\left| \vec{\omega}_2 \right|} = \left[ \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

to give

$$\mathbf{U} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix}$$

The calculation of  $\mathbf{V}$  is similar.  $\mathbf{V}$  is based on  $\mathbf{A}^T \mathbf{A}$ , so we have

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

Find the eigenvalues of  $\mathbf{A}^T \mathbf{A}$  by

$$\begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

which represents the system of equations

$$10x_1 + 2x_3 = \lambda x_1$$

$$10x_2 + 4x_3 = \lambda x_2$$

$$2x_1 + 4x_2 + 2x_3 = \lambda x_2$$

which rewrite as

$$(10 - \lambda)x_1 + 2x_3 = 0$$

$$(10 - \lambda)x_2 + 4x_3 = 0$$

$$2x_1 + 4x_2 + (2 - \lambda)x_3 = 0$$

which are solved by setting

$$\begin{vmatrix} (10 - \lambda) & 0 & 2 \\ 0 & (10 - \lambda) & 4 \\ 2 & 4 & (2 - \lambda) \end{vmatrix} = 0$$

This works out as

$$\begin{aligned} (10 - \lambda) \begin{vmatrix} (10 - \lambda) & 4 \\ 4 & (2 - \lambda) \end{vmatrix} + 2 \begin{vmatrix} 0 & (10 - \lambda) \\ 2 & 4 \end{vmatrix} = \\ (10 - \lambda)[(10 - \lambda)(2 - \lambda) - 16] + 2[0 - (20 - 2\lambda)] = \\ \lambda(\lambda - 10)(\lambda - 12) = 0, \end{aligned}$$

so for  $\lambda = 12, \vec{v}_1 = [1, 2, 1]$ . For  $\lambda = 10$  we have

$$(10 - 10)x_1 + 2x_3 = 2x_3 = 0$$

$$x_3 = 0$$

$$2x_1 + 4x_2 = 0$$

$$x_1 = -2x_2$$

$$x_1 = 2, x_2 = -1$$

which means for  $\lambda = 10, \vec{v}_2 = [2, -1, 0]$ . For  $\lambda = 0$

$$10x_1 + 2x_3 = 0$$

$$x_3 = -5$$

$$10x_1 - 20 = 0$$

$$x_1 = 2$$

$$2x_1 + 8 - 10 = 0$$

$$x_1 = 1$$

which means for  $\lambda = 0, \vec{v}_3 = [1, 2, -5]$ . Order  $\vec{v}_1, \vec{v}_2$ , and  $\vec{v}_3$  as column vectors in a matrix according to the size of the eigenvalue to get

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 5 \end{bmatrix}$$

and use the Gram-Schmidt orthonormalization process to convert that to an orthonormal matrix.

$$\vec{u}_1 = \frac{\vec{v}_1}{|\vec{v}_1|} = \left[ \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right]$$

$$\vec{\omega}_2 = \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 = [2, -1, 0]$$

$$\vec{u}_2 = \frac{\vec{\omega}_2}{|\vec{\omega}_2|} = \left[ \frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}, 0 \right]$$

$$\vec{\omega}_3 = \vec{v}_3 - \vec{u}_1 \cdot \vec{v}_3 * \vec{u}_1 - \vec{u}_2 \cdot \vec{v}_3 * \vec{u}_2 = \left[ \frac{-2}{3}, \frac{-4}{3}, \frac{10}{3} \right]$$

$$\vec{u}_3 = \frac{\vec{\omega}_3}{|\vec{\omega}_3|} = \left[ \frac{1}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{-5}{\sqrt{30}} \right]$$

All this to give us

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{6}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

when we really want its transpose

$$V^T = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{30}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix}$$

For S we take the square roots of the non-zero eigenvalues and populate the diagonal with them, putting the largest in  $s_{11}$ , the next largest in  $s_{22}$  and so on until the smallest value ends up in  $s_{mn}$ . The non-zero eigenvalues of U and V are always the same, so that's why it doesn't matter which one we take them from. Because we are doing full SVD, instead of reduced SVD (next section), we have to add a zero column vector to S so that it is of the proper dimensions to allow multiplication between U and V. The diagonal entries in S are the singular values of A, the columns in U are called left singular vectors, and the columns in V are called right singular vectors.

$$S = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix}$$

Now we have all the pieces of the puzzle

$$\begin{aligned} A_{mn} = U_{mn} S_{mn} V_{nn}^T &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & \sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix} = \\ & \begin{bmatrix} \frac{\sqrt{12}}{\sqrt{2}} & \frac{\sqrt{10}}{\sqrt{2}} & 0 \\ \frac{\sqrt{12}}{\sqrt{2}} & \frac{-\sqrt{10}}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 0 \\ \frac{1}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{-5}{\sqrt{30}} \end{bmatrix} = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \end{aligned}$$

### Singular Value Decomposition:

The Singular Value Decomposition is an important factorization of a real or complex matrix, with several applications in signal processing and statistics. Applications, which employ the SVD, include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of Eigen vectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

### Algorithm of Singular Value Decomposition:

To find the Singular Value Decomposition of the matrix:

7. Find the eigenvalues of the matrix  $A^T A$  and arrange them in descending order
8. Find the number of nonzero eigenvalues of the matrix  $A^T A$
9. Find the orthogonal eigenvectors of the matrix  $A^T A$  corresponding to the obtained eigenvalues, and arrange them in the same order to form the column-vectors of the matrix V

10. Form a diagonal matrix  $S = \sum = \sigma$  placing on the leading diagonal of it the square roots  $\sigma_i = \sqrt{\lambda_i}$  of  $p = \min\{m, n\}$  first eigenvalues of the matrix  $A^T A$  got in I in descending order
11. Find the first column-vectors of the matrix U,  $u_i = \sigma_i^{-1} A v_i (i = 1 : r)$
12. Add to the matrix U the rest of m-r vectors using the Gram-Schmidt Orthogonalization process

**Example 46: Compute Singular Value Decomposition (SVD) for the given matrix**

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

**Answer:**

$$\text{Given, } A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \therefore A^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\therefore AA^T = \begin{bmatrix} 1+1 & 0+1 & 1+0 \\ 0+1 & 0+1 & 0+0 \\ 1+0 & 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}_{3 \times 3 \text{ order}}$$

**Again,**

$$A^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Now, } A^T A = \begin{bmatrix} 1+0+1 & 1+0+0 \\ 1+0+0 & 1+1+0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2 \text{ order}}$$

**Rule#01: Find the eigenvalues of the matrix**

$$|A^T A - \lambda I| = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)(2-\lambda) - 1 = 4 - 2\lambda - 2\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow 3 - 4\lambda + \lambda^2 = 0$$

$$\Rightarrow \lambda^2 - 3\lambda - \lambda + 3 = 0$$

$$\Rightarrow \lambda(\lambda - 3) - 1(\lambda - 3) = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 3) = 0$$

$$\Rightarrow \lambda_1 = 3, \lambda_2 = 1 \text{ [Descending order]}$$

**Rule#02:** Find the number of nonzero eigenvalues of the matrix  $A^T A : r = 2$

**Rule#03:** Find the orthogonal eigenvectors of the matrix  $A^T A$  corresponding to the obtained eigenvalues  $\Rightarrow \lambda_1 = 3, \lambda_2 = 1$ , and arrange them in the same order to form the column-vectors of the matrix  $V$

$$\text{Now, } A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2 \text{ order}}$$

$$\text{First normalize } \vec{v}_1 = [2, 1]$$

$$\vec{u}_1 = \left[ \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right]$$

Next, let

$$\begin{aligned} \vec{w}_2 &= \vec{v}_2 - \vec{u}_1 \cdot \vec{v}_2 * \vec{u}_1 = [1, 2] - \left[ \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right] \cdot [1, 2] * \left[ \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right] \\ &= [1, 2] - \left( \frac{2}{\sqrt{5}} + \frac{2}{\sqrt{5}} \right) * \left[ \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right] \\ &= [1, 2] - \left( \frac{4}{\sqrt{5}} \right) * \left[ \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right] \\ &= [1, 2] - \left( \frac{8}{5}, \frac{4}{5} \right) \\ &= \left( \frac{-3}{5}, \frac{6}{5} \right) \\ \sqrt{\left( \frac{-3}{5} \right)^2 + \left( \frac{6}{5} \right)^2} &= \sqrt{\frac{9}{25} + \frac{36}{25}} = \sqrt{\frac{45}{25}} = \sqrt{\frac{9}{5}} = \frac{3}{\sqrt{5}} \end{aligned}$$

**Source Code (Using Programming Language C) to represent the above adjacency Matrix**

```
#include <stdio.h>
int matrix[10][10]={0};
void main()
```

```

{
    int node1,node2,total_node;
    FILE *fp;
    fp = fopen("input.txt","rt");
    fscanf(fp,"%d",&total_node);
    while(!feof(fp))
    {
        fscanf(fp,"%d",&node1);
        fscanf(fp,"%d",&node2);
        matrix[node1][node2] = 1;
    }
    for(int i=1;i<=total_node;i++)
    {
        printf("node%d: ",i);
        for (int j=1;j<=total_node;j++)
        {
            if (matrix[i][j] == 1)
                printf(" %d",j);
        }
        printf("\n");
    }
}

```

|     |
|-----|
| 4   |
| 1 2 |
| 2 3 |
| 1 3 |
| 4 3 |

### Longest Common Subsequence (LCS):

- Biologists need to measure similarity between DNA and thus determine how closely related an organism is to another.
- They do this by considering DNA as strings of letters A, C, G, and T and then comparing similarities in the strings.
- Formally, they look at common subsequences in the strings.
- Example X = ABCBDAB, Y=BDCABA
- Subsequences may be: ABA, BCA, BCB, BBA BCBA BDAB etc.
- But the Longest Common Subsequences (LCS) is **BCBA** and **BDAB**.

How to find LCS efficiently?

### LCS recurrence:

Notice the issues...this recurrence is exponential.

We don't know max ahead of time.

The sub problems overlap, to find LCS we need to find LCS of c [i, j-1] and of c [i-1, j]

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i-1, j-1] + 1 & \text{if } i, j > 0 \text{ and } x_i = y_j, \\ \max(c[i, j-1], c[i-1, j]) & \text{if } i, j > 0 \text{ and } x_i \neq y_j \end{cases}$$

### LCS Algorithm:

First we'll find the length of LCS. Later we'll modify the algorithm to find LCS itself. Recall we want to let  $X_i$ ,  $Y_j$  to be the prefixes of X and Y of length  $i$  and  $j$  respectively. And that Define  $c[i, j]$  to be the length of LCS of  $X_i$  and  $Y_j$ . Then the length of LCS of X and Y will be  $c[m, n]$ .

$$c[i, j] = \begin{cases} c[i-1, j-1] + 1 & \text{if } x[i] = y[j], \\ \max(c[i, j-1], c[i-1, j]) & \text{otherwise} \end{cases}$$

### LCS Recursive Solution:

1)

We start with  $i = j = 0$  (empty sub strings of x and y)

Since  $X_0$  and  $Y_0$  are empty strings, their LCS is always empty (i.e.  $c[0, 0] = 0$ )

LCS of empty string and any other string is empty, so for every  $i$  and  $j$ :  $c[0, j] = c[i, 0] = 0$

2) When we calculate  $c[i, j]$ , we consider two cases:

**First case:**  $x[i] = y[j]$ : one more symbol in strings X and Y matches, so the length of LCS  $X_i$  and  $Y_j$  equals to the length of LCS of smaller strings  $X_{i-1}$  and  $Y_{j-1}$ , plus 1

**3) Second case:**  $x[i] \neq y[j]$

As symbols don't match, our solution is not improved, and the length of LCS ( $X_i$ ,  $Y_j$ ) is the same as before (i.e. maximum of LCS ( $X_i$ ,  $Y_{j-1}$ ) and LCS ( $X_{i-1}$ ,  $Y_j$ ))

### LCS Example:

We'll see how LCS algorithm works on the following Example:

X = ABCB

Y = BDCAB

What is the Longest Common Subsequence?  
Of X and Y?

LCS (X, Y) = BCB

X = A **B**    **C**    **B**    Y = B D **C** A **B**

### //LCS Coding:

```
#include<stdio.h>
#include <conio.h>
#include<string.h>

void lcs_length(char x[], char y[]);
void print_lcs(char x[],int i,int j);
int b[10][10];
void main(){
    clrscr();
    char x[10]="0", y[10]="0",x1[11],y1[11];
    printf("take 1st string as input:");
    gets(x1);//take input string
    strcat(x,x1); //add x and x1
    printf("take 2nd string as input:");
    gets(y1);
    strcat(y,y1);
    lcs_length(x,y);
    getch();
}

void lcs_length(char x[], char y[]){

    int i,j,c[10][10]; //10 row,10 column

    int m = strlen(x)-1;
    int n = strlen (y)-1;

    printf("the length of 1st string is: %d\n",m);
    printf("the length of 2nd string is: %d\n",n);
    for (i=1;i<=m;i++)
        c[i][0]=0;
    for (j=0;j<=n;j++)
        c[0][j]=0;

    for (i=1;i<=m;i++)
        for(j=1;j<=n;j++)
        {
            if(x[i]==y[j])
            {
                c[i][j]=c[i-1][j-1]+1;
                b[i][j]=1;
            }
        }
    }
```

```

        }
        else if(c[i-1][j]>=c[i][j-1])
        {
            c[i][j]=c[i-1][j];
            b[i][j]=2;
        }
        else
        {
            c[i][j]=c[i][j-1];
            b[i][j]=3;
        }
    }

    for (i=0;i<=m;i++)
    {
        for(j=0;j<=n;j++)
            printf("%d",c[i][j]);
        printf("\n");
    }
    printf("\n");
    for (i=0;i<=m;i++)
    {
        for(j=0;j<=n;j++)
            printf("%d",b[i][j]);
        printf("\n");
    }
    print_lcs(x,m,n);
}

void print_lcs(char x[],int i, int j)
{
    if(i==0||j==0) //m=i=length of 1st string,n=j=length of 2nd string
    //i=0,j=0 means length of string is zero that is no element exists
        return ;
    if (b[i][j]==1)//here i=m,j=n,that is the last bottom element
    {
        print_lcs(x,i-1,j-1);//recursion call, since we will calculate from
        bottom,here from 3
        printf("%c",x[i]);
    }
    else if (b[i][j]==2)
        print_lcs(x,i-1,j);
    else print_lcs(x,i,j-1);
}

```

## Business Applications of Matrices

1. A manufacturing unit produces three types of television sets A, B & C. The sales of television sets in two different cities are, 400, 300, 200 & 300, 200, 100 respectively. The cost price of each TV set A, B, C is 1000, 2000 & 3000 and selling price is 1500, 300 & 3500 dollars respectively. Represent the information's in the form of matrices and then find the total profit of the company using matrix algebra.

2. Three persons buy cold drinks of different brands A, B & C. The 1st person buys 12 bottles of A, 5 bottles of B, 3 bottles of C. The second person buys 4 bottles of A, 6 bottles of B and 10 bottles of C. The third person buys 6 bottles of A, 7 bottles of B and 9 bottles of C. The cost of each bottle of brand A TK.4, brand B TK.5 and brand C TK.6. Represent the information's in the form of matrices and then using matrix operation find the total sum of money spent individually by the three persons for cold drinks.

3. A manufacturer produces three products A, B & C, which he sells in the market. Annual sale volumes are indicated as follows:

| Market | Products |        |        |
|--------|----------|--------|--------|
|        | A        | B      | C      |
| I      | 8,000    | 10,000 | 15,000 |
| II     | 10,000   | 2,000  | 20,000 |

- (i) If unit sale prices of A, B & C are TK .8, TK .10 & TK .5 respectively, find the total revenue in each market with the help of matrix algebra.
- (ii) If unit cost prices of A, B & C are TK, 6, TK, 9 & TK, 4 respectively, find the gross profit with the help of matrix algebra.

4. A man buys 8 dozens of Mangoes, 10 dozens of Bananas and 5 dozens of Apples. Mango, Banana and Apple cost TK. 35, TK. 20 & TK. 60, per dozen respectively. Represent the information's in the form of matrices and hence, obtain the total costs of three kinds of foods.

5. A store has in stock 30 dozen shirts, 15 dozen Trousers and 25 dozen pair of Shoes. If

The selling prices are TK, 250 per shirt, TK, 80 per trousers and TK, 50 per pair of shoes, using matrix algebra, find the total amount the storeowner will get after selling all the items in the stock.

6. In a development of a city, a contractor has taken a contract certain houses for which he needs building materials like stone, sand etc. there are three firms A, B and C that can supply him three materials. At one time, these firms A, B and C supplied him 40, 35 and 25 trucks load of stone and 10, 5 and 8 trucks load of sand respectively. If the cost of one truck load of stone and sand are TK.1200 and TK. 800 respectively, find the total amount paid by the contractor to each of these firms A, B & C by matrix algebra.

7. A Finance company has the offices located in every Division, District and Thana in a certain State. Assume that there are 5 Divisions, 65 Districts and 400 Thana in the state. Each office has one head clerk, one cashier, one clerk and one peon. A divisional office has, in addition, one office supervisor, two clerks and one peon. A district office has, in addition, one clerk one typist and one peon. The basic monthly salaries are as follows: supervisor TK. 5000, Head clerk TK. 3500. Cashier TK, 3000, Clerk TK, 2000, Typist TK, 1500 and peon TK, 1000. Using matrix algebra. Find

- (i) The total numbers of posts of each kind in all the offices taken together.
- (ii) The total basic monthly salary of each kind of office.
- (iii) The total basic monthly salary of all the offices taken together.

8. The per unit prices of three commodities I, II & III are  $x$ ,  $y$  &  $z$  respectively. 'A' purchases 4 units of commodity III and sells 3 units of I and 5 units of II. 'B' purchases 3 of II and sells 2 units of I and 1 unit of III. 'C' purchases 1 unit of I and sells 4 unit of II and 6 units of III. In the process A, B & C earn TK. 6000, 5000 & 13000 respectively. Represent the information in the system of linear equations and find the per unit prices of the three commodities by Cramer's rule method.

9. Three persons A, B & C have \$480, \$760 & \$710 respectively. They utilized the amount to purchase three types of shares P, Q, and R, 'A' purchases 2 shares of P, 5 shares of Q and 3 shares of R, 'B' purchases 4 shares of P, 3 shares of Q and 6 shares of R, 'C' purchases 1 share P, 4 shares of Q and 10 shares of R. Find the per unit price of three types of shares using matrix.

10. To control a certain crop disease it is necessary to use 8 units of chemical A, 14 units of chemical B and 13 units of chemical C. One barrel of spray P contains one unit of A, 2 units of B and 3 units of C. One barrel of spray Q contains 2 unit of A, 3 units of B and 2 units of C. One barrel of spray R contains one unit of A, 2 units of B and 2 units of C. How many barrels of each type of spray should be used to control the disease?

11. A mixture is to be made of three foods A, B & C. One kg of food A contains 1 gm of Nutrient P, 1 gm of nutrient Q and 2 Gms of nutrient R. One kg of food B contains 1 gm of Nutrient P, 2 Gms of nutrient Q and 3 Gms of nutrient R. One kg of food C contains 1 gm of Nutrient P, 2 Gms of nutrient Q and 8 Gms of nutrient R. How to form a mixture which will have 7 Gms of P, 16 gms of Q and 22 gms of R.

Exa: 1

$$X = \begin{pmatrix} A & B & C \\ 1000 & 2000 & 3000 \end{pmatrix} \rightarrow \text{Cost} \quad 1 \times 3$$

$$Y = \begin{pmatrix} 1500 & 3000 & 3500 \end{pmatrix} \rightarrow \text{Selling Price} \quad 1 \times 3$$

$$\begin{matrix} & \text{City1} & \text{City2} \\ A & \begin{pmatrix} 400 & 300 \end{pmatrix} & \text{No.of unit produced} \\ \& = B & \begin{pmatrix} 300 & 200 \end{pmatrix} & \text{No.of unit sold.} \\ & C & \begin{pmatrix} 200 & 100 \end{pmatrix} & 3 \times 2 \end{matrix}$$

For determining total cost, we have – (cö\_g Matrix Gi Column Gi mv†\_ 2q Matrix Gi row mv†\_ ,Y n†e|)

$$Z = \begin{pmatrix} 1000 & 2000 & 3000 \end{pmatrix} 1 \times 3 \begin{pmatrix} 400 & 300 \\ 300 & 200 \\ 200 & 100 \end{pmatrix} 3 \times 2$$

$$\begin{aligned} &= 1000.400 + 2000.300 + 3000.200 & 1000.300 + 2000.200 + 3000.100 \\ &= 4,00,000 + 6,00,000 + 6,00,000 & 3,00,000 + 4,00,000 + 3,00,000 \\ &= (16,00,000 \quad 10,00,000) \rightarrow \text{Cost Matrix} \end{aligned}$$

For determining total selling price, we have –

$$YZ = (1500 \quad 3000 \quad 3500) 1 \times 3 \begin{pmatrix} 400 & 300 \\ 300 & 200 \\ 200 & 100 \end{pmatrix} 3 \times 2$$

$$\begin{aligned} &= 1500.400 + 3000.300 + 3500.200 & 1500.300 + 3000.200 + 3500.100 \\ &= 6,00,000 + 9,00,000 + 7,00,000 & 4,50,000 + 6,00,000 + 3,50,000 \\ &= (22,00,000 + 14,00,000) \rightarrow \text{Selling price} \end{aligned}$$

Total profit matrix will be –

$$\begin{aligned} YZ - XZ &= (22,00,000 \quad 14,00,000) - (16,00,000 \quad 10,00,000) \\ &= (22,00,000 - 16,00,000 \quad 14,00,000 - 10,00,000) \\ &= (6,00,000 \quad 4,00,000) \end{aligned}$$

$\therefore$  Total profit will be,

$$\begin{aligned} &6,00,000 + 4,00,000 \\ &= 10,00,000 \end{aligned}$$

Exa – 2

$$X = \begin{pmatrix} 12 & 5 & 3 \\ 4 & 6 & 10 \\ 6 & 7 & 3 \end{pmatrix} \begin{matrix} \text{quantity} \\ \text{1st person} \\ \text{2nd person} \\ \text{3rd person} \end{matrix}$$

$1 \times 3$

$$Y = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \begin{matrix} A \\ B \\ C \end{matrix}$$

$3 \times 1$

$$\text{For 1}^{\text{st}} \text{ person } XY = (12 \quad 5 \quad 3) 1 \times 3 \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} 3 \times 1$$

$$\begin{aligned} &= 12.4 + 5.5 + 3.6 \\ &= 48 + 25 + 18 \\ &= 91 \end{aligned}$$

$$\text{For 2}^{\text{nd}} \text{ person } XY = (4 \quad 6 \quad 10) \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

$$\begin{aligned}
&= 4.4 + 6.5 + 10.6 \\
&= 16 + 30 + 60 \\
&= 106
\end{aligned}$$

$$\text{For 3}^{\text{rd}} \text{ person XY} = \begin{pmatrix} 6 & 7 & 9 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

$$\begin{aligned}
&= 6.4 + 7.5 + 9.6 \\
&= 24 + 35 + 54 \\
&= 113
\end{aligned}$$

$$\begin{aligned}
\text{Total money spent} &= (91 + 106 + 113) \\
&= 310
\end{aligned}$$

Exa-3

A B C

$$X = \begin{pmatrix} 6 & 9 & 4 \end{pmatrix} \xrightarrow{\text{Cost}} \begin{matrix} 1 \times 3 \end{matrix}$$

$$Y = \begin{pmatrix} 8 & 10 & 5 \end{pmatrix} \xrightarrow{\text{Sales price}} \begin{matrix} 1 \times 3 \end{matrix}$$

$$Z = \begin{matrix} \text{Market-I} & \text{Market-II} \\ \begin{matrix} A \\ B \\ C \end{matrix} \end{matrix} \begin{pmatrix} 8,000 & 10,000 \\ 10,000 & 2,000 \\ 15,000 & 20,000 \end{pmatrix} \begin{matrix} 3 \times 2 \\ 1 \times 3 \end{matrix}$$

for determining the total cost, we have

$$\begin{aligned}
XZ &= \begin{pmatrix} 6 & 9 & 4 \end{pmatrix} \begin{pmatrix} 8,000 & 10,000 \\ 10,000 & 2,000 \\ 15,000 & 20,000 \end{pmatrix} \begin{matrix} 1 \times 3 \\ 3 \times 2 \end{matrix} \\
&= (6 \times 8,00 + 9 \times 10,000 + 4 \times 15,000 \quad 6 \times 10,000 + 9 \times 2,000 + 4 \times 20,00) \\
&= (48,00 + 90,000 + 60,000 \quad 60,000 + 18,000 + 80,00) \\
&= (1,98,000 \quad 1,58,000)
\end{aligned}$$

for determining the total selling price

$$\begin{aligned}
 YZ &= (8 \ 10 \ 5) \begin{pmatrix} 8,000 & 10,000 \\ 10,000 & 2,000 \\ 15,000 & 20,000 \end{pmatrix} \\
 &= (8.8000 + 10.10,000 + 5.15,000 \quad 8.10,000 + 10.2,000 + 5.20,00) \\
 &= (64,00 + 1,00,000 + 75,000 \quad 80,000 + 20,000 + 1,00,00) \\
 &= (2,39,000 \quad 2,00,000)
 \end{aligned}$$

Total profit matrix will be

$$\begin{aligned}
 YZ - XZ &= (2,39,00 \quad 2,00,000) - (1,98,000 \ 1,58,000) \\
 &= (2,39,00 \quad 1,98,000 \quad 2,00,000 - 1,58,000) \\
 &= (41,000 \quad 42,000)
 \end{aligned}$$

$$\begin{aligned}
 \text{Gross profit will be, } &(41,000 + 42,000) \\
 &= 83,000
 \end{aligned}$$

Exa: - 4

Mango Banana Apples

$$A = (8 \ 10 \ 5) \xrightarrow[\text{1} \times \text{3}]{\text{quantity}}$$

$$B = \begin{pmatrix} 35 \\ 20 \\ 60 \end{pmatrix} \xrightarrow[\text{3} \times \text{1}]{\text{Cost}} \begin{matrix} \text{Mango} \\ \text{Banana} \\ \text{Apple} \end{matrix}$$

For determining the total cost we have,

$$\begin{aligned}
 12 &\dots\dots\dots 35 \\
 \therefore &\dots\dots\dots \frac{35}{12} \\
 \therefore 96 &\dots\dots\dots \frac{35 \times 96}{12} = 280
 \end{aligned}$$

$$\begin{aligned}
 AB &= (8 \ 10 \ 5) \begin{pmatrix} 35 \\ 20 \\ 60 \end{pmatrix} \\
 &= (8.35 + 10.20 + 5.60) \\
 &= (280 + 200 + 300) \\
 &= 780
 \end{aligned}$$

Exa: - 5

$$A = 12 (30 \quad 15 \quad 25) \xrightarrow{\text{Quantity}}$$

$$= (360 \quad 180 \quad 300)$$

$$B = \begin{pmatrix} 250 \\ 80 \\ 50 \end{pmatrix} \xrightarrow{\text{Cost}}$$

For determining the total cost – we have,

$$\begin{aligned} AB &= (360 \quad 180 \quad 300) \begin{pmatrix} 250 \\ 80 \\ 50 \end{pmatrix} \\ &= (360 \times 250 + 180 \times 80 + 300 \times 50) \\ &= (90,000 + 14,400 + 15,000) \\ &= 1,19,400 \end{aligned}$$

Exa: - 6

$$X = \begin{pmatrix} A & B & C \\ 40 & 35 & 25 \\ 10 & 5 & 8 \end{pmatrix} \begin{matrix} \nearrow \text{Quantity} \\ \text{Stone} \\ \text{Sand} \end{matrix}$$

$2 \times 3$

$$Y = \begin{pmatrix} \text{Stone} & \text{Sand} \\ 1200 & 800 \end{pmatrix} \xrightarrow{\text{Cost}}$$

$1 \times 2$

$$YX = \begin{pmatrix} 1200 & 800 \end{pmatrix} \begin{pmatrix} 40 & 35 & 25 \\ 10 & 5 & 8 \end{pmatrix}$$

$1 \times 2 \qquad 2 \times 3$

$$\begin{aligned} &= (1200.40 + 800.10 \quad 1200.35 + 800.5 \quad 1200.25 + 800.8) \\ &= (48,00 + 8,000 \quad 2,000 + 4000 \quad 30,000 + 6400) \\ &= (56,000 \quad 46,000 \quad 36400) \end{aligned}$$

Total amount paid by the contrator will be,

$$\begin{aligned} &56,000 + 46,000 + 36400 \\ &= 1,38,400 \end{aligned}$$

Exa:-7 **(Blank)**

Exq-8

$$\begin{array}{lcl} & \text{I} & \text{II} & \text{III} \\ & -4z + 3x + 5y = 6000 \\ \text{A 1 or} & 3x + 5y - 4z = 6000 \\ & -3y + 2x + z = 5000 \end{array}$$

$\mu q \rightarrow (-)$   
 $w e \mu q \rightarrow (+)$

$$\text{B} \quad 2 \text{ or } 2x - 3y + z = 5000$$

$$\text{C} \quad 3 - x + 4y + 6z = 13000$$

$$\begin{pmatrix} (+) & (-) & (+) \\ 3 & 5 & -4 \\ 2 & -3 & 1 \\ -1 & 4 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6000 \\ 5000 \\ 13000 \end{pmatrix} \quad \begin{array}{l} 3x \ 5y \ 4z \ x \\ 2x \ 3y \ z \ y \\ 1x \ 4y \ 6z \ z \end{array}$$

$$\begin{aligned} \text{Here } \Delta &= \begin{vmatrix} (+) & (-) & (+) \\ 3 & 5 & -4 \\ 2 & -3 & 1 \\ -1 & 4 & 6 \end{vmatrix} = +3 \begin{vmatrix} -3 & 1 \\ 4 & 6 \end{vmatrix} - 5 \begin{vmatrix} 2 & 1 \\ -1 & 6 \end{vmatrix} - 4 \begin{vmatrix} 2 & -3 \\ -1 & 4 \end{vmatrix} \\ &= 3(-18 - 4) - 5(12 + 1) - 4(8 - 3) \\ &= -54 - 12 - 60 - 5 - 32 + 12 \\ &= -151 \end{aligned}$$

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} (+) & (-) & (+) \\ 6000 & 5 & -4 \\ 5000 & -3 & 1 \\ 13000 & 4 & 6 \end{vmatrix} = +6000 \begin{vmatrix} -3 & 1 \\ 4 & 6 \end{vmatrix} - 5 \begin{vmatrix} 5000 & 1 \\ 13000 & 6 \end{vmatrix} - 4 \begin{vmatrix} 5000 & -3 \\ 13000 & 4 \end{vmatrix} \\ &= 6000(-18 - 4) - 5(30000 - 13000) - 4(20000 + 39000) \\ &= (108000 - 24000 - 150000 + 65000 - 80000 - 156000) \\ &= 173000 - 410000 \\ &= -453000 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 3 & 6000 & -4 \\ 2 & 5000 & 1 \\ -1 & 13000 & 6 \end{vmatrix} = +3 \begin{vmatrix} -3 & 5000 \\ 4 & 13000 \end{vmatrix} - 6000 \begin{vmatrix} 2 & 1 \\ -1 & 6 \end{vmatrix} - 4 \begin{vmatrix} 2 & 5000 \\ -1 & 13000 \end{vmatrix} \\ &= 3(30000 - 13000) - 6000(12 + 1) - 4(26000 + 5000) \\ &= 9000 - 39000 - 72000 - 6000 - 104000 - 20000 \\ &= 90000 - 241000 \\ &= -151000 \end{aligned}$$

$$\Delta 3 = \begin{vmatrix} 3 & 5 & 6000 \\ 2 & -3 & 5000 \\ -1 & 4 & 13000 \end{vmatrix} = +3 \begin{vmatrix} -3 & 5000 \\ 4 & 13000 \end{vmatrix} - 5 \begin{vmatrix} 2 & 5000 \\ -1 & 13000 \end{vmatrix} + 6 \begin{vmatrix} 2 & -3 \\ -1 & 4 \end{vmatrix}$$

$$= 3(-39000 - 20000) - 5(26000 + 5000) + 6000(8 - 3)$$

$$= 3(-59000) - 5(31000) + 6000(5)$$

$$= -155000 + 30000$$

$$= -332000 + 30000$$

$$\text{Here, } X = \frac{\Delta 1}{\Delta} = \frac{-453000}{-151} = 3000$$

$$Y = \frac{\Delta 2}{\Delta} = \frac{-15100}{-151} = 1000$$

$$Z = \frac{\Delta 3}{\Delta} = \frac{-302000}{-151} = 2000$$

#### Exq-9

Let,

X = per unit price of P share

Y = " " " " Q "

Z = " " " " R "

$$(A) 2x + 5y + 3z = 480$$

$$(B) 4x + 3y + 6z = 760$$

$$(C) x + 4y + 10z = 710$$

$$\begin{pmatrix} 2 & 5 & 3 \\ 4 & 3 & 6 \\ 1 & 4 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 480 \\ 760 \\ 710 \end{pmatrix}$$

$$\text{Here, } \Delta = \begin{vmatrix} (+) & (-) & (+) \\ 2 & 5 & 3 \\ 4 & 3 & 6 \\ 1 & 4 & 10 \end{vmatrix} = +2 \begin{vmatrix} 3 & 6 \\ 4 & 10 \end{vmatrix} - 5 \begin{vmatrix} 4 & 6 \\ 1 & 10 \end{vmatrix} + 3 \begin{vmatrix} 4 & 3 \\ 1 & 4 \end{vmatrix}$$

$$= 2(30 - 24) - 5(40 - 6) + 3(16 - 3)$$

$$= 60 - 48 - 200 + 30 + 48 - 9$$

$$= 138 - 257$$

$$= -119$$

$$\Delta 1 = \begin{vmatrix} (+) & (-) & (+) \\ 480 & 5 & 3 \\ 760 & 3 & 6 \\ 710 & 4 & 10 \end{vmatrix} = +480 \begin{vmatrix} 3 & 6 \\ 4 & 10 \end{vmatrix} - 5 \begin{vmatrix} 760 & 6 \\ 710 & 10 \end{vmatrix} + 3 \begin{vmatrix} 760 & 3 \\ 710 & 4 \end{vmatrix}$$

$$= 480(30-24) - 5(7600-4260) + 3(3040-2130)$$

$$= 14400 - 11520 - 38000 + 21300 + 9120 - 6390$$

$$= 44820 - 55910 = -11,090$$

$$\Delta 2 = \begin{vmatrix} (+) & (-) & (+) \\ 2 & 480 & 3 \\ 4 & 760 & 6 \\ 1 & 710 & 10 \end{vmatrix} = +2 \begin{vmatrix} 760 & 6 \\ 710 & 10 \end{vmatrix} - 480 \begin{vmatrix} 4 & 6 \\ 1 & 10 \end{vmatrix} + 3 \begin{vmatrix} 4 & 760 \\ 1 & 710 \end{vmatrix}$$

$$= 2(7600 - 4260) - 4800(40 - 6) - 3(2840 - 760)$$

$$= (15200 - 8520 - 19200 + 2880 + 8520 - 2280)$$

$$= 26600 - 30000$$

$$= -3400$$

$$\Delta 3 = \begin{vmatrix} (+) & (-) & (+) \\ 2 & 5 & 480 \\ 4 & 3 & 760 \\ 1 & 4 & 710 \end{vmatrix} = +2 \begin{vmatrix} 3 & 760 \\ 4 & 710 \end{vmatrix} - 5 \begin{vmatrix} 4 & 760 \\ 1 & 710 \end{vmatrix} + 480 \begin{vmatrix} 4 & 3 \\ 1 & 4 \end{vmatrix}$$

$$= 2(2130 - 3040) - 5(2840 - 760) + 480(16 - 3)$$

$$= 4260 - 6080 - 14200 + 3800 + 7680 - 1440$$

$$= 15740 - 21720 = -5980$$

Here,

$$X = \frac{\Delta 1}{\Delta} = \frac{-11090}{-119} = 93.193$$

$$Y = \frac{\Delta 2}{\Delta} = \frac{3400}{-119} = \frac{200}{7} = 28.571$$

$$Z = \frac{\Delta 3}{\Delta} = \frac{-5980}{-119} = 50.252$$

Exq: - 10

Let, X barrels of spray P

Y    "    "    "    Q  
Z    "    "    "    R

P   Q   R  
A    $x + 2y + z = 8$

B    $2x + 3y + 2z = 14$

C    $3x + 2y + 2z = 13$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ 14 \\ 13 \end{pmatrix}$$

Here,

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 3 & 2 & 2 \end{vmatrix} = +1 \begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix} + 1 \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} \\ &= 1(6-4) - 2(4-6) + 1(4-9) \\ &= 6-4-8+12+4-9 \\ &= 22-21 = 1 \end{aligned}$$

$$\begin{aligned} \Delta 1 &= \begin{vmatrix} 8 & 2 & 1 \\ 14 & 3 & 2 \\ 13 & 2 & 2 \end{vmatrix} = +8 \begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix} - 2 \begin{vmatrix} 14 & 2 \\ 13 & 2 \end{vmatrix} + 1 \begin{vmatrix} 14 & 3 \\ 13 & 2 \end{vmatrix} \\ &= 8(6-4) - 2(28-26) + 1(28-39) \\ &= 48-32-56+52+28-39 \\ &= 128-127 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \Delta 2 &= \begin{vmatrix} 1 & 8 & 1 \\ 2 & 14 & 2 \\ 3 & 13 & 2 \end{vmatrix} = +1 \begin{vmatrix} 14 & 2 \\ 13 & 2 \end{vmatrix} - 8 \begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix} + 1 \begin{vmatrix} 2 & 14 \\ 3 & 13 \end{vmatrix} \\ &= 1(28-26) - 8(4-6) + 1(26-42) \\ &= 28-26-32+48+26-42 \\ &= 102-100 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \Delta 3 &= \begin{vmatrix} 1 & 2 & 8 \\ 2 & 3 & 14 \\ 3 & 2 & 13 \end{vmatrix} = +1 \begin{vmatrix} 3 & 14 \\ 2 & 13 \end{vmatrix} - 2 \begin{vmatrix} 2 & 14 \\ 3 & 13 \end{vmatrix} + 8 \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} \\ &= 1(39-28) - 2(26-42) + 8(4-9) \\ &= 39-28-52+34+32-72 \\ &= 155-152 \\ &= 3 \end{aligned}$$

Now,

$$X = \frac{\Delta 1}{\Delta} = \frac{1}{1} = 1$$

$$Y = \frac{\Delta 2}{\Delta} = \frac{2}{1} = 2$$

$$Z = \frac{\Delta_3}{\Delta} = \frac{3}{1} = 3$$

Exq:- 11

|          | <u>Nut. P</u> |   | <u>Nut .Q</u> |   | <u>Nut .R</u> |
|----------|---------------|---|---------------|---|---------------|
| <u>A</u> | x             | + | y             | + | $2z = .7$     |
| <u>B</u> | x             | + | $2y$          | + | $3z = 16$     |
| <u>C</u> | x             | + | $2y$          | + | $8z = 22$     |

$$\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 3 \\ 1 & 2 & 8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 \\ 16 \\ 22 \end{pmatrix}$$

Here,

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 2 & 8 \end{vmatrix} = +1 \begin{vmatrix} 2 & 3 \\ 2 & 8 \end{vmatrix} - 1 \begin{vmatrix} 1 & 3 \\ 1 & 8 \end{vmatrix} + 2 \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} \\ &= 1(16-6) - 1(8-3) + 2(2-2) \\ &= 16-6-8+3+4-4 \\ &= 23-18 \\ &= 5 \end{aligned}$$

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 7 & 1 & 2 \\ 16 & 2 & 3 \\ 22 & 2 & 8 \end{vmatrix} = +7 \begin{vmatrix} 2 & 3 \\ 2 & 8 \end{vmatrix} - 1 \begin{vmatrix} 16 & 3 \\ 22 & 8 \end{vmatrix} + 2 \begin{vmatrix} 16 & 2 \\ 22 & 2 \end{vmatrix} \\ &= 7(16-6) - 1(128-66) + 2(32-44) \\ &= 112-42-128+66+64-44 \\ &= 242-258 \\ &= 16 \end{aligned}$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 1 & 7 & 2 \\ 1 & 16 & 3 \\ 1 & 22 & 8 \end{vmatrix} = +1 \begin{vmatrix} 16 & 3 \\ 22 & 8 \end{vmatrix} - 7 \begin{vmatrix} 1 & 3 \\ 1 & 8 \end{vmatrix} + 2 \begin{vmatrix} 1 & 16 \\ 1 & 8 \end{vmatrix} \\ &= 1(128-66) - 7(8-3) + 2(8-16) \\ &= 128-66-56+21+16-32 \\ &= 165-154 \\ &= 11 \end{aligned}$$

$$\begin{aligned}
 \Delta 3 &= \begin{vmatrix} 1 & 1 & 7 \\ 1 & 2 & 16 \\ 1 & 2 & 22 \end{vmatrix} = +1 \begin{vmatrix} 2 & 16 \\ 2 & 22 \end{vmatrix} - 1 \begin{vmatrix} 1 & 16 \\ 1 & 22 \end{vmatrix} + 7 \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} \\
 &= 1(44-32) - 1(22-16) + 7(2-2) \\
 &= 44-32-22+16+14-14 \\
 &= 74-64 \\
 &= 6
 \end{aligned}$$

Now,

$$X = \frac{\Delta 1}{\Delta} = \frac{16}{5} = 3.2$$

$$Y = \frac{\Delta 2}{\Delta} = \frac{11}{5} = 2.2$$

$$Z = \frac{\Delta 3}{\Delta} = \frac{6}{5} = 1.2$$

# Vector Analysis



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## Contents

1. **Vector analysis:** Scalar and vectors, operation of vectors, vector addition and multiplication - their applications.
2. Vector components in spherical and cylindrical systems, Dot Product, Cross Product, Scalar Field, Vector Field
3. Derivative of vectors and problems
4. **Del operator:** Del operator, gradient, divergence and curl and their physical significance.
5. **Vector Integration:** Line Integrals, physical significance of Vector integration and Problems
6. **Vector's Theorem :** Greens, Gauss & Stocks theorem and their applications

## Introduction

**Physical quantities** can be divided into two main groups, scalar quantities and vector quantities

**Scalar Quantities:** A Physical Quantity which has magnitude only is called as a Scalar.

**Example:** Time, Temperature, Mass, Volume are examples of scalars.

That is, the measurement of years, months, weeks, days, hours, minutes, seconds, and even milliseconds, A temperature of 15°C, A mass of 0.2 kg, etc

**Vectors:** A Physical Quantity which has both magnitude and direction is called as Vector  
Examples: velocity, displacement, acceleration, force etc.

### Some Examples:

01. A speed of 10 km/h is a scalar quantity but a velocity of 10 km/h due north is a vector quantity.

02. A temperature of 100°C is a scalar quantity.

03. The weight of a 7 kg mass is a vector quantity.  $[w = mg]$

**Vector Notation:** Typical notation to designate a vector  $\vec{AB}$  is a boldfaced character or a character with an arrow on it, or a character with a line under it (i.e.,  $\vec{AB}$ , AB, AB).

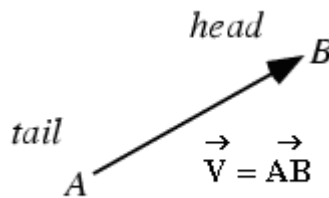


Figure: 01

**Magnitude of a vector:** The magnitude of a vector  $\vec{OP}$  or  $\vec{V}$  is its length and is normally denoted by  $|\vec{V}|$  or V. Given a vector  $\vec{V}$  with tail at the origin O and head at  $P(x, y)$ , what's its length?

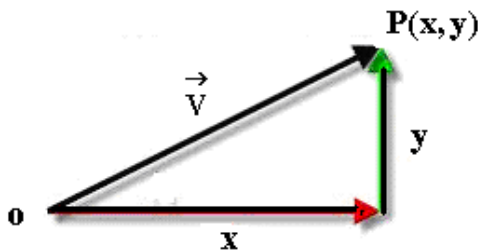


Figure: 02

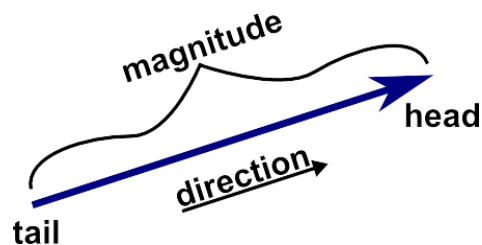


Figure: 03

According to Pythagoras, the length of the hypotenuse OP is the square root of  $x^2 + y^2$ .

That is, **Magnitude of a vector**  $\vec{V}$  = its length =  $|\vec{OP}| = |\vec{V}| = \sqrt{x^2 + y^2}$

The **zero vectors** are the vector with zero magnitude that is vector's length is zero.

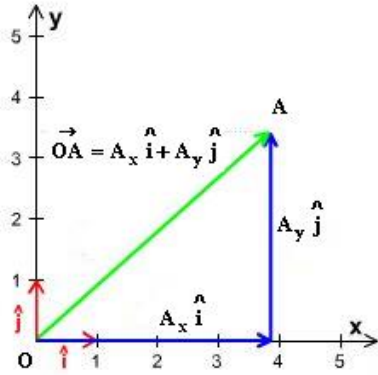


Figure # 04

From figure # 04

$$\vec{OA} = A_x \hat{i} + A_y \hat{j}$$

The length of the vector  $\vec{OA} = \sqrt{A_x^2 + A_y^2}$

From figure # 05

$$\vec{OA} = \vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

The length of the vector  $\vec{OA} = |\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$

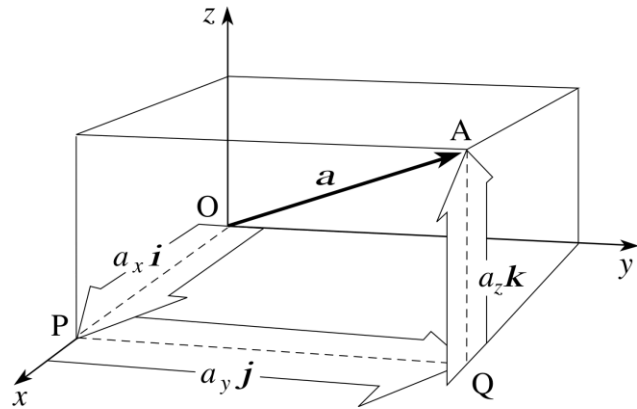


Figure # 05

### Graphical Vector Addition

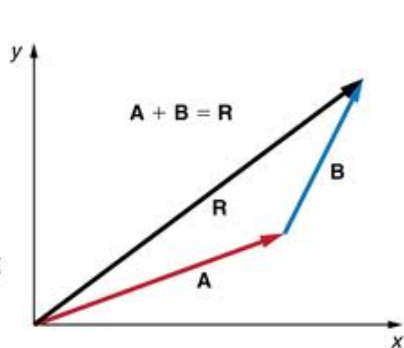


Figure 06

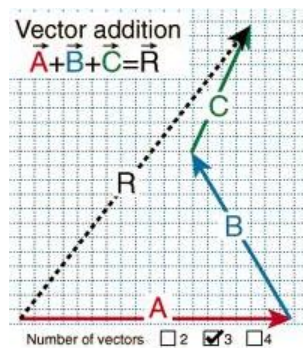


Figure 07

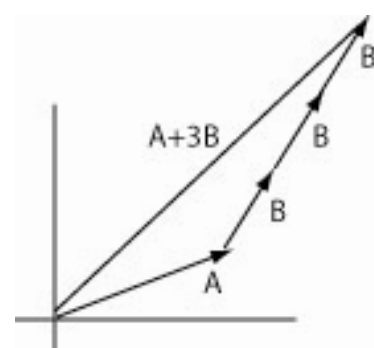


Figure: 08

**Position vector:**

In geometry, a position or position vector, also known as location vector or radius vector. The origin is (0, 0) and the position vector is basically just a straight line drawn from the origin to some other point.

The position vector is the vector from the origin of the coordinate system  $O(0,0)$  to the point  $P(x,y)$ . It is shown as the vector  $op$  (Figure 09)

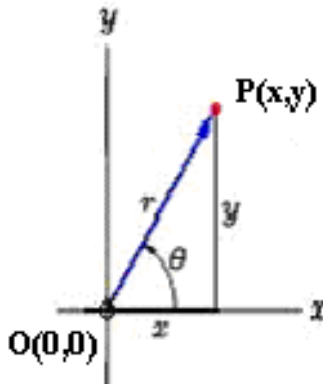


Figure: 09

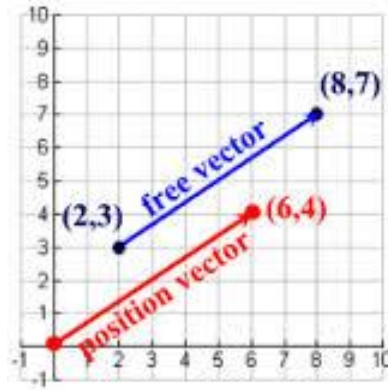


Figure: 10

**Q# 01:**

Draw the curves:

i)  $\vec{G}(t) = t\hat{i} + 2t\hat{j}, \quad t \text{ in } [0,1]$

ii)  $\vec{H}(t) = t^2\hat{i} + 2t^2\hat{j}, \quad t \text{ in } [0,1]$

iii)  $\vec{J}(t) = (1 - \frac{t}{2})\hat{i} + (2 - t)\hat{j}, \quad t \text{ in } [0,2]$

iv) Compute  $\int_C xy \, dx$ , where C is the curve given by

$$\vec{J}(t) = (1 - \frac{t}{2})\hat{i} + (2 - t)\hat{j}, \quad t \text{ in } [0,2]$$

Answer Q # 01:

i)  $\vec{G}(t) = t\hat{i} + 2t\hat{j}, \quad t \text{ in } [0,1]$

$$\vec{G}(0) = 0\hat{i} + 0\hat{j},$$

$$\vec{G}(1) = 1\hat{i} + 2\hat{j},$$

$$\therefore \vec{OP} = \vec{G}(t) = t\hat{i} + 2t\hat{j}, \quad t \text{ in } [0,1] \text{ [Figure \# 11]}$$

ii)  $\vec{H}(t) = t^2\hat{i} + 2t^2\hat{j}, \quad t \text{ in } [0,1]$

$$\vec{H}(0) = 0\hat{i} + 0\hat{j},$$

$$\vec{H}(1) = 1\hat{i} + 2\hat{j},$$

$$\therefore \vec{OP} = \vec{H}(t) = t^2\hat{i} + 2t^2\hat{j}, \quad t \text{ in } [0,1] \text{ [Figure \# 11]}$$

$$\text{iii) } \vec{J}(t) = \left(1 - \frac{t}{2}\right)\hat{i} + (2-t)\hat{j}, \quad t \text{ in } [0,2]$$

$$\vec{J}(0) = 1\hat{i} + 2\hat{j},$$

$$\vec{J}(2) = \left(1 - \frac{2}{2}\right)\hat{i} + (2-2)\hat{j},$$

$$\vec{J}(2) = 0\hat{i} + 0\hat{j},$$

$$\therefore \vec{PO} = \vec{J}(t) = \left(1 - \frac{t}{2}\right)\hat{i} + (2-t)\hat{j}, \quad t \text{ in } [0,2] \text{ [Figure \# 12]}$$

iv) Compute  $\int_C xy \, dx$ , where C is the curve given by

$$\vec{J}(t) = \left(1 - \frac{t}{2}\right)\hat{i} + (2-t)\hat{j}, \quad t \text{ in } [0,2]$$

$$\text{Here, } x = \left(1 - \frac{t}{2}\right) \text{ and } y = (2-t)$$

$$\Rightarrow \frac{dx}{dt} = -\frac{1}{2}$$

$$\begin{aligned} \therefore \int_C xy \, dx &= \int_0^2 \left(1 - \frac{t}{2}\right)(2-t) \frac{dx}{dt} dt = \int_0^2 \left(1 - \frac{t}{2}\right)(2-t) \left(-\frac{1}{2}\right) dt = \int_0^2 \left(2-t-t+\frac{t^2}{2}\right) \left(-\frac{1}{2}\right) dt \\ &= -\frac{1}{2} \int_0^2 \left(2-2t+\frac{t^2}{2}\right) dt = -\frac{1}{2} \left(2t-t^2+\frac{t^3}{6}\right)_0^2 = -\frac{1}{2} \left(4-4+\frac{8}{6}\right) = -\frac{2}{3} \text{ Answer} \end{aligned}$$

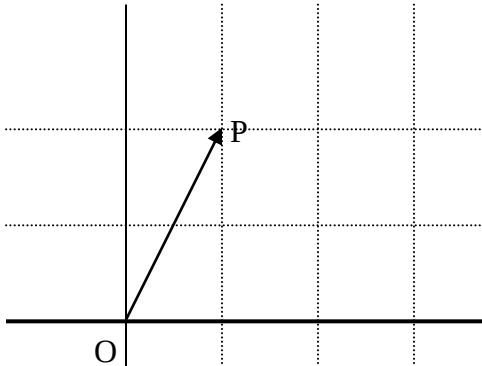


Figure # 11 for G (t) and H (t)

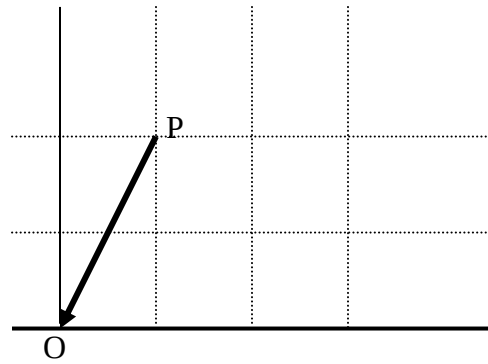


Figure # 12 for J (t)

### Unit vectors:

A unit vector  $\hat{\mathbf{e}}$  is a vector of unit length. A unit vector is sometimes denoted by replacing the arrow on a vector with a "^" on a boldfaced character (i.e.,  $\hat{\mathbf{e}}$ ). Therefore,

$$|\hat{\mathbf{e}}| = 1$$

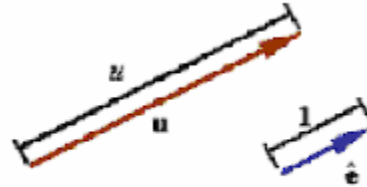


Figure #13

Any vector can be made into a unit vector by dividing it by its length.

$$\hat{\mathbf{e}} = \frac{\vec{\mathbf{u}}}{|\vec{\mathbf{u}}|}$$

$$\Rightarrow \vec{\mathbf{u}} = |\vec{\mathbf{u}}| \times \hat{\mathbf{e}}$$

Any vector  $\vec{\mathbf{u}}$  can be fully represented by providing its magnitude and a unit vector along its direction.  $\vec{\mathbf{u}} = |\vec{\mathbf{u}}| \times \hat{\mathbf{e}}$  [That is, Any Vector = Length of this Vector  $\times$  Unit Vector]

### Vector components:

For example,  $\vec{\mathbf{u}} = \vec{\mathbf{u}}_1 + \vec{\mathbf{u}}_2 + \vec{\mathbf{u}}_3$

Where,  $\vec{\mathbf{u}}_1 = |\vec{\mathbf{u}}_1| \hat{\mathbf{e}}_1$  [ Any Vector = Length of this Vector  $\times$  Unit Vector]

$$\vec{\mathbf{u}}_2 = |\vec{\mathbf{u}}_2| \hat{\mathbf{e}}_2$$

$$\vec{\mathbf{u}}_3 = |\vec{\mathbf{u}}_3| \hat{\mathbf{e}}_3$$

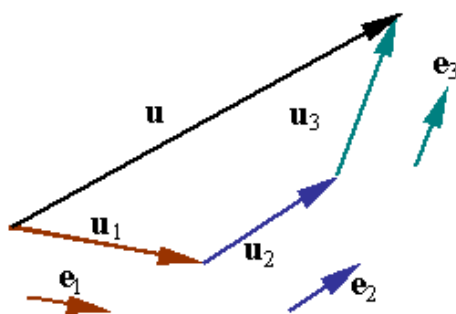


Figure #14

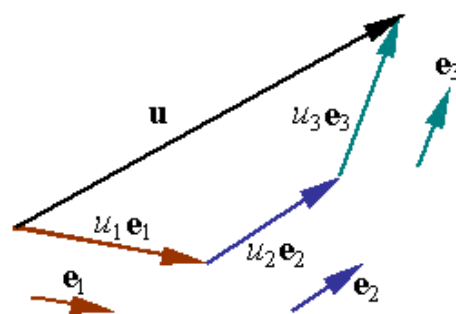


Figure #15

The original vector  $\vec{u}$  can now be written as

$$\therefore \vec{u} = \vec{u}_1 + \vec{u}_2 + \vec{u}_3$$

$$\therefore \vec{u} = |\vec{u}_1| \hat{e}_1 + |\vec{u}_2| \hat{e}_2 + |\vec{u}_3| \hat{e}_3$$

Vectors  $\hat{e}_1, \hat{e}_2, \hat{e}_3$  are unit vectors and  $|\vec{u}_1|, |\vec{u}_2|, |\vec{u}_3|$  are length of the vectors

$\vec{u}_1, \vec{u}_2, \vec{u}_3$  respectively.

**As for Example:** Here,  $\hat{e}_1$  is a unit vector of  $\vec{AB}$

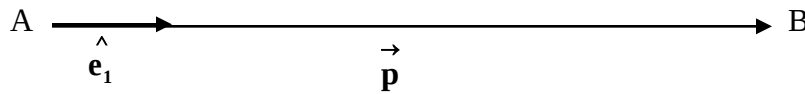


Figure 16

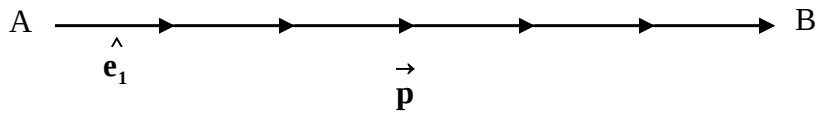


Figure 17

$$|\vec{AB}| = 6 \quad [From Figure 17]$$

$$\therefore \vec{AB} = \vec{P} = |\vec{AB}| \hat{e}_1 = 6 \hat{e}_1 \quad [Any Vector = Length of this Vector \times Unit Vector]$$

$$\therefore \text{Unit Vector } \hat{e}_1 = \frac{\vec{AB}}{|\vec{AB}|} = \frac{6 \hat{e}_1}{6} = \hat{e}_1$$

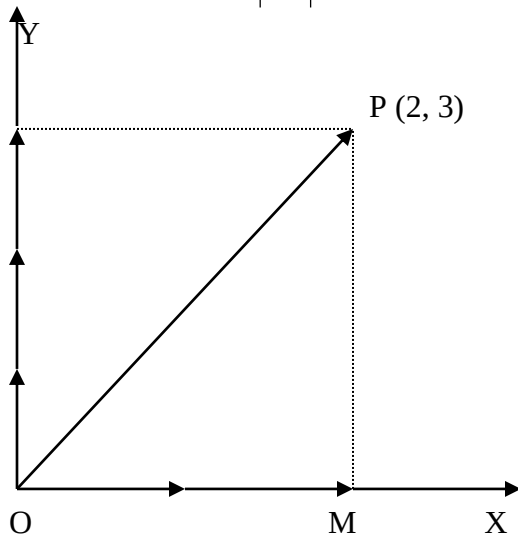


Figure # 18

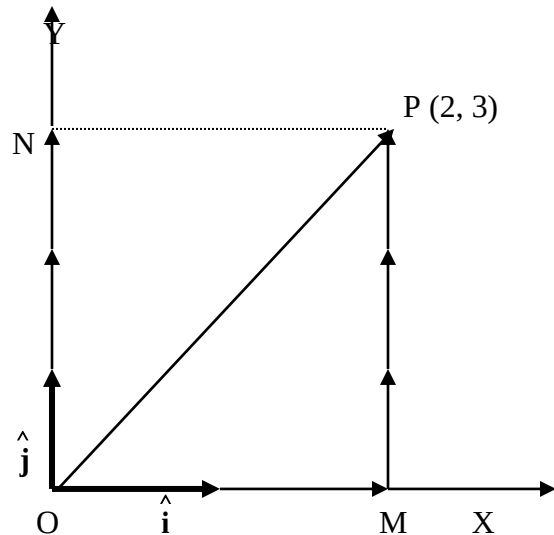


Figure # 19

From figure 18

$$\vec{OP} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

From figure 19

$$\vec{OM} = 2\hat{i} \quad [\vec{OM} = 2\hat{i} = \text{Length of this Vector} \times \text{Unit Vector}]$$

$$\vec{ON} = \vec{MP} = 3\hat{j} \quad [\vec{ON} = \vec{MP} = 3\hat{j} = \text{Length of this Vector} \times \text{Unit Vector}]$$

From,  $\triangle OMP$ ,

$$\vec{OP} = \vec{OM} + \vec{MP} = 2\hat{i} + 3\hat{j}$$

$$\text{Here, } |\vec{OM}| = 2, \quad |\vec{MP}| = 3$$

$$\therefore |\vec{OP}| = \sqrt{OM^2 + MP^2} = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$\therefore \text{Unit vector of } \vec{OM} = \frac{\vec{OM}}{|\vec{OM}|} = \frac{2\hat{i}}{2} = \hat{i}$$

$$\therefore \text{Unit vector of } \vec{ON} = \vec{MP} = \frac{\vec{ON}}{|\vec{ON}|} = \frac{3\hat{j}}{3} = \hat{j}$$

$$\therefore \text{Unit vector of } \vec{OP} = \hat{e} = \frac{\vec{OP}}{|\vec{OP}|} = \frac{2\hat{i} + 3\hat{j}}{\sqrt{13}} = \frac{2}{\sqrt{13}}\hat{i} + \frac{3}{\sqrt{13}}\hat{j}$$

$$\therefore \text{Magnitude of Unit vector of } \vec{OP} = |\hat{e}| = \sqrt{\left(\frac{2}{\sqrt{13}}\right)^2 + \left(\frac{3}{\sqrt{13}}\right)^2} = \sqrt{\frac{4}{13} + \frac{9}{13}} = \sqrt{\frac{13}{13}} = 1$$

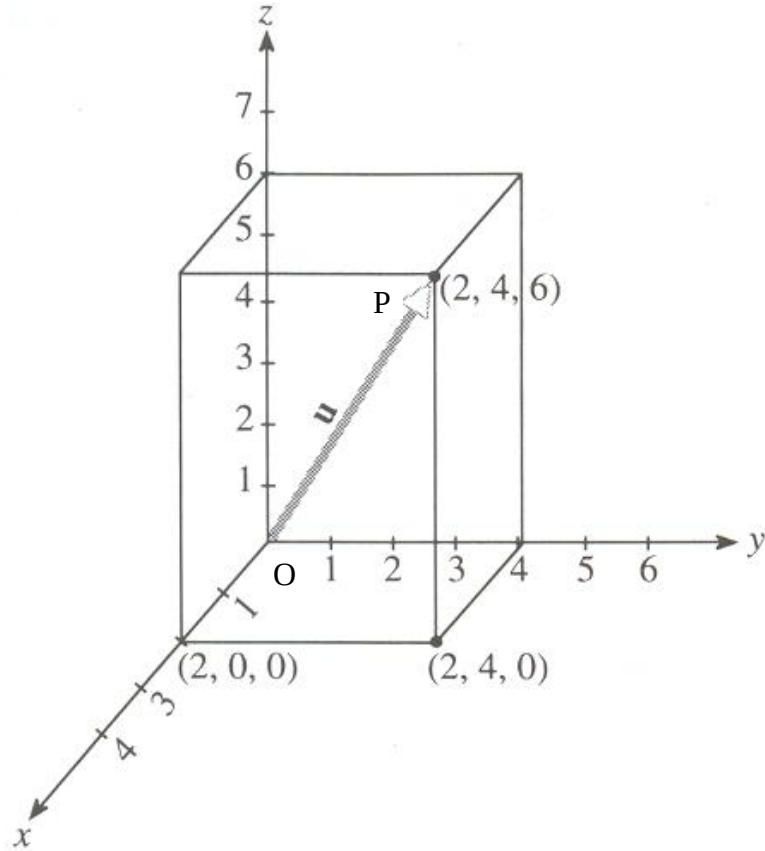


Figure 20

From figure 20

$$\vec{OP} = 2\hat{i} + 4\hat{j} + 6\hat{k}$$

$$\therefore |\vec{OP}| = \sqrt{2^2 + 4^2 + 6^2} = \sqrt{4 + 16 + 36} = \sqrt{56}$$

$$\therefore \text{Unit vector of } \vec{OP} = \hat{e} = \frac{\vec{OP}}{|\vec{OP}|} = \frac{2\hat{i} + 4\hat{j} + 6\hat{k}}{\sqrt{56}} = \frac{2}{\sqrt{56}}\hat{i} + \frac{4}{\sqrt{56}}\hat{j} + \frac{6}{\sqrt{56}}\hat{k}$$

$\therefore$  Magnitude of Unit vector of  $\vec{OP}$

$$= |\hat{e}| = \sqrt{\left(\frac{2}{\sqrt{56}}\right)^2 + \left(\frac{4}{\sqrt{56}}\right)^2 + \left(\frac{6}{\sqrt{56}}\right)^2} = \sqrt{\frac{4}{56} + \frac{16}{56} + \frac{36}{56}} = \sqrt{\frac{56}{56}} = 1$$

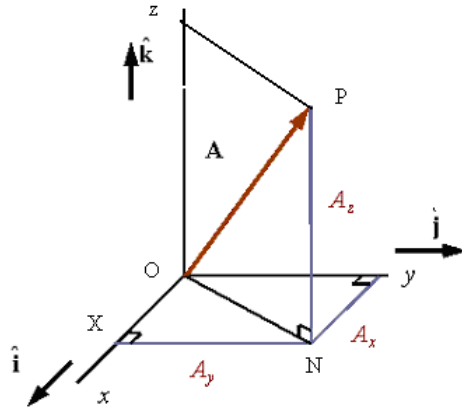


Figure 21

From figure 21

$\Delta ONP$ ,

$$\begin{aligned}\vec{OP} = \vec{A} &= \vec{ON} + \vec{NP} \\ &= (\vec{OX} + \vec{XN}) + \vec{NP} \\ &= (\vec{OX} + \vec{OY}) + \vec{NP} \\ &= (\vec{OX} + \vec{OY}) + \vec{OZ} \\ &= A_x \hat{i} + A_y \hat{j} + A_z \hat{k}\end{aligned}$$

and

$$\begin{aligned}|\vec{A}| = |\vec{OP}| &= \sqrt{|\vec{OP}|^2} = \sqrt{|\vec{ON}|^2 + |\vec{NP}|^2} = \sqrt{|\vec{OX}|^2 + |\vec{XN}|^2 + |\vec{OZ}|^2} = \sqrt{A_x^2 + A_y^2 + A_z^2} \\ \vec{A} &= A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \quad \text{and} \quad |\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}\end{aligned}$$

**Q # 02:** Find the unit tangent vector to the graph of  $\vec{r}(t) = t^2 \hat{i} + t^3 \hat{j}$  at the point where  $t = 2$

**Answer # 02**

Given,  $\vec{r}(t) = t^2 \hat{i} + t^3 \hat{j}$  -----(i)

$\therefore \vec{r}(2) = 4 \hat{i} + 8 \hat{j}$  -----(ii)

Here  $x = 4, y = 8$

$\therefore \vec{OP} = \vec{r}(2) = 4 \hat{i} + 8 \hat{j}$  [See figure no 22]

The tangent vector will be drawn at  $(4, 8)$  which is  $\vec{T}(2)$

From (i),

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = 2t\hat{i} + 3t^2\hat{j} \text{-----(iii)}$$

$$\therefore \vec{T}(t) = \vec{r}'(t) = 2t\hat{i} + 3t^2\hat{j} \text{-----(iv)}$$

$$\therefore \vec{T}(2) = \vec{r}'(2) = 2 \times 2\hat{i} + 3 \times 2^2\hat{j}$$

$$\therefore \text{Tangent vector: } \vec{PQ} = \vec{T}(2) = \vec{r}'(2) = 4\hat{i} + 12\hat{j} \text{-----(v)}$$

$\therefore$  Unit tangent vector:

$$\begin{aligned} \frac{\vec{T}(2)}{|\vec{T}(2)|} &= \frac{\vec{r}'(2)}{|\vec{r}'(2)|} = \frac{4\hat{i} + 12\hat{j}}{\sqrt{4^2 + 12^2}} = \frac{4}{\sqrt{160}}\hat{i} + \frac{12}{\sqrt{160}}\hat{j} = \frac{4}{\sqrt{16 \times 10}}\hat{i} + \frac{12}{\sqrt{16 \times 10}}\hat{j} \\ &= \frac{4}{4\sqrt{10}}\hat{i} + \frac{12}{4\sqrt{10}}\hat{j} = \frac{1}{\sqrt{10}}\hat{i} + \frac{3}{\sqrt{10}}\hat{j} \end{aligned}$$

Figure is following:

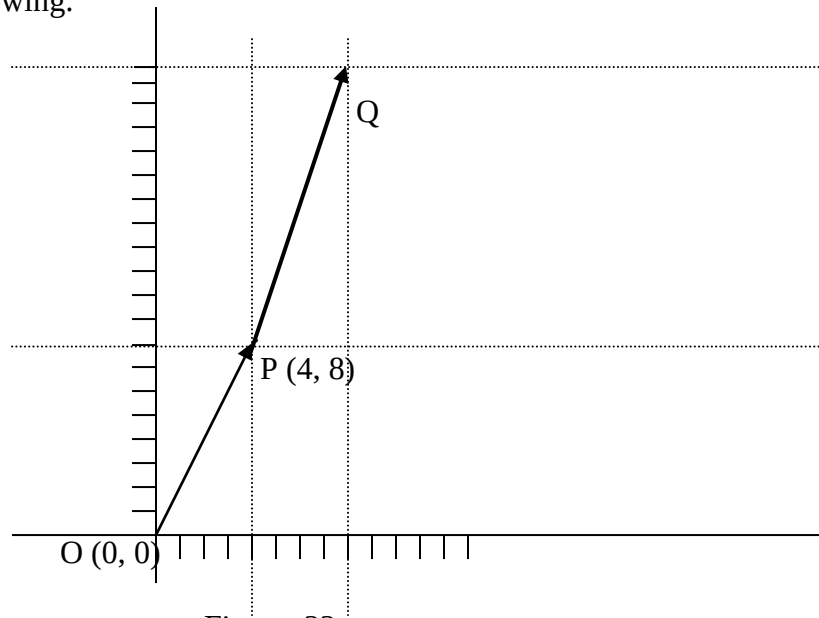


Figure: 22

When  $t = 1$ ,  $\vec{r}(1) = 1.\hat{i} + 1.\hat{j}$

When  $t = 2$ ,  $\vec{r}(2) = 4.\hat{i} + 8.\hat{j}$

When  $t = 3$ ,  $\vec{r}(3) = 9.\hat{i} + 27.\hat{j}$

From (iii),

$$\vec{r}'(t) = 2t\hat{i} + 3t^2\hat{j}$$

When  $t = 2$ ,  $\vec{r}'(2) = 2 \times 2\hat{i} + 3 \times 2^2\hat{j} = 4\hat{i} + 12\hat{j}$  Answer

### Vector Components:

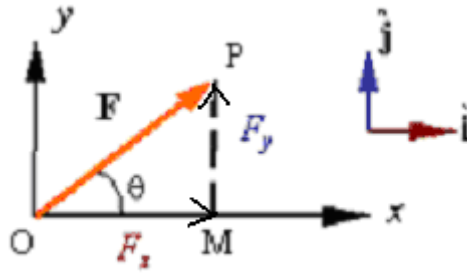


Figure 23

Here,

$$OM = F_x, MP = F_y, OP = |\vec{F}|$$

$$\vec{OM} = F_x \hat{i} \quad [\vec{OM} = F_x \hat{i} = \text{Length of this Vector} \times \text{Unit Vector}]$$

$$\vec{MP} = F_y \hat{j} \quad [\vec{MP} = F_y \hat{j} = \text{Length of this Vector} \times \text{Unit Vector}]$$

From,  $\triangle OMP$ ,

$$\vec{OP} = \vec{F} = \vec{OM} + \vec{MP} = F_x \hat{i} + F_y \hat{j} \text{-----(i)}$$

From,  $\triangle OMP$ ,

$$\frac{OM}{OP} = \cos\theta$$

$$OM = OP \cos\theta$$

$$F_x = |\vec{F}| \cos\theta \text{-----(ii)}$$

From,  $\triangle OMP$ ,

$$\frac{MP}{OP} = \sin\theta$$

$$MP = OP \sin\theta$$

$$F_y = |\vec{F}| \sin\theta \text{-----(iii)}$$

**Projection:** Imagine parallel rays of light shining vertically downwards on to the  $x$ -axis. The quantity  $|\vec{b} \cos \theta|$ ; which gives the size of the ‘shadow’ of vector  $\vec{b}$  on the  $x$ -axis, is often termed the **projection** of  $\vec{b}$  on to the  $x$ -axis

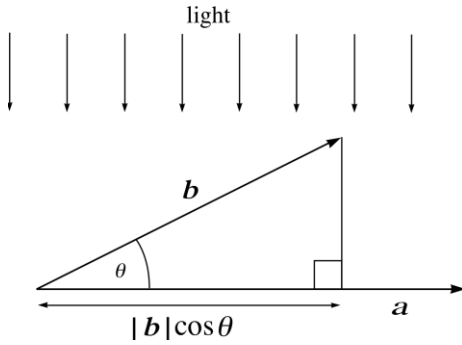


Figure: 24

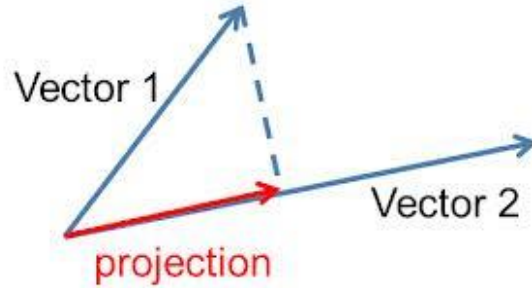


Figure: 25

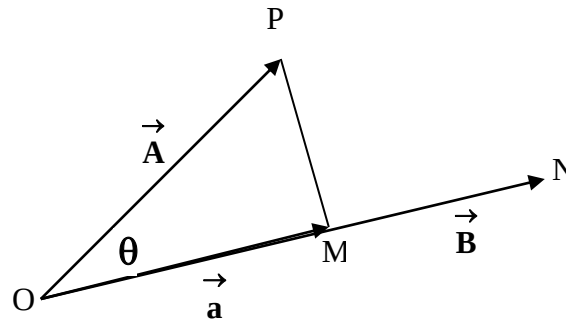


Figure: 26

Projection of  $\vec{A}$  on  $\vec{B}$  is  $\vec{a}$

Here,  $\frac{OM}{OP} = \cos \theta$

$$\therefore \frac{|\vec{a}|}{|\vec{A}|} = \cos \theta$$

$\therefore$  Projection of  $\vec{A}$  on  $\vec{B}$  is  $|\vec{a}| = |\vec{A}| \cos \theta$  -----(i)

$$\therefore \frac{|\vec{a}|}{|\vec{A}|} = \cos \theta$$
 -----(ii)

Again,

$$\mathbf{OP} = \left| \vec{A} \right|$$

$$\mathbf{ON} = \left| \vec{B} \right|$$

Again,

$$\vec{A} \cdot \vec{B} = \left| \vec{A} \right| \left| \vec{B} \right| \cos \theta$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{A} \right| \left| \vec{B} \right|} \text{-----(iii)}$$

From (ii) and (iii), we can write,

$$\therefore \frac{\left| \vec{a} \right|}{\left| \vec{A} \right|} = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{A} \right| \left| \vec{B} \right|}$$

$$\therefore \left| \vec{a} \right| = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{A} \right| \left| \vec{B} \right|} \left| \vec{A} \right|$$

$$\therefore \left| \vec{a} \right| = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|} \text{-----(iv)}$$

Again,

$$\vec{a} = \left| \vec{a} \right| \times \text{unit vector of } \vec{B}$$

$$\vec{a} = \left| \vec{a} \right| \times \frac{\vec{B}}{\left| \vec{B} \right|} \text{-----(v)}$$

Putting the value of  $\left| \vec{a} \right|$  from (iv) in (v)

$$\vec{a} = \left| \vec{a} \right| \times \frac{\vec{B}}{\left| \vec{B} \right|}$$

$$\vec{a} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} \times \frac{\vec{B}}{|\vec{B}|}$$

$$\text{The projection } \vec{a} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|^2} \vec{B}$$

### Work done by dot products of two vectors:

If you have taken physics class, you have probably encountered the notion of work in mechanics.



Figure: 27

If a constant force of  $F$  (in the direction of motion) is applied to move an object a distance  $d$  in a straight line, then the work exerted is

$$\text{Work} = \text{Force} \times \text{distance}$$

$$W = F \times d$$

The unit for force is N (newton) and the unit for distance is m (meter). The unit for work is joule=(newton)(meter).

Now suppose that there is an angle  $\theta$  between direction in which the constant force is applied and the direction of motion.

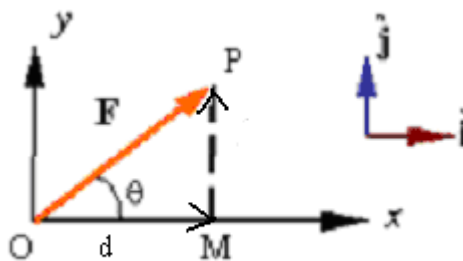


Figure: 28

In this case the work is given by:

Force Components in the direction of X-axis  $\times$  distance in the direction of Force

$$\text{components} = \left| \vec{F} \right| \cos \theta \times OM = \left| \vec{F} \right| \cos \theta \times \left| \vec{d} \right|$$

**Physical Significance of The scalar or dot product:**

The man is pulling the block with a constant force  $\mathbf{a}$  so that it moves along the horizontal ground. The **work done** in moving the block through a distance  $\mathbf{b}$  is then given by the distance moved through multiplied by the magnitude of the component of the force in the direction of motion

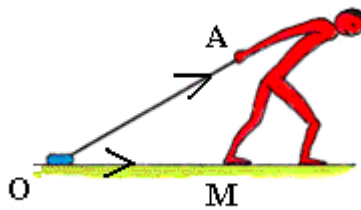


Figure 29

One important physical application of the scalar product is the calculation of work:

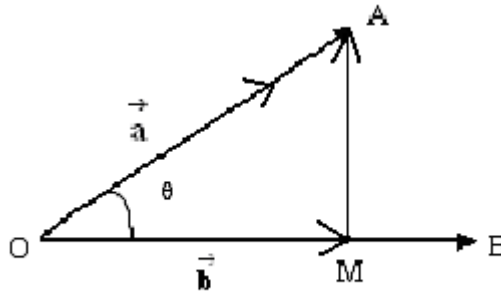


Figure 30

Here,  
From Figure 30

From,  $\Delta OMA$ ,

$$\frac{OM}{OA} = \cos \theta$$

$$OM = OA \cos \theta$$

$$OM = \left| \vec{a} \right| \cos \theta$$

$\therefore$  The force or vector components of the vector  $\vec{a}$  in the direction of OB is  $\left| \vec{a} \right| \cos \theta$

The scalar or dot product as

$$\text{Work done} = w = \vec{a} \cdot \vec{b} = |\vec{a}| \cos \theta \times |\vec{b}|$$

That is, Force Components in the direction of OB  $\times$  distance (OM) in the direction (OB) of Force or vector components =  $|\vec{a}| \cos \theta \times |\vec{b}|$

The scalar or dot product as

$$\text{Work done} = w = \vec{a} \cdot \vec{b} = |\vec{a}| \cos \theta \times |\vec{b}|$$

$$\Rightarrow \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

[Vector Gi Dot Product gv#i g#a" GKwU Vector hw` Force Vector nq, Aci Vector hw` displacement Vector nq Zvn#j Force Vector KZ@,,K #mLv#b #h KvR msNwUZ nq #mB Kv#Ri cwigvbB n#"Q Dot Product, that is, Work done =  $w = \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$ ;

$$\text{GLv#b Kv#Ri cwigvb : } w = |\vec{a}| |\vec{b}| \cos \theta ]$$

**Summary:** From the physical interpretation of the dot product, the work done in moving an object a distance  $d$  by a force of magnitude  $F$  in the same direction (Figure

28) as the force is  $W = \vec{F} \cdot \vec{d}$

When a constant force  $\vec{F}$  is applied to a body acting at an angle  $\theta$  to the direction of motion (Figure 28), then the work done by  $\vec{F}$  is defined to be  $W = \vec{F} \cdot \vec{d} = |\vec{F}| \cos \theta \times |\vec{d}| = |\vec{F}| |\vec{d}| \cos \theta$

**Q # 03:** A block of mass "m" moves from point A to B along a smooth plane surface under the action of force as shown in the figure. Find the work done.

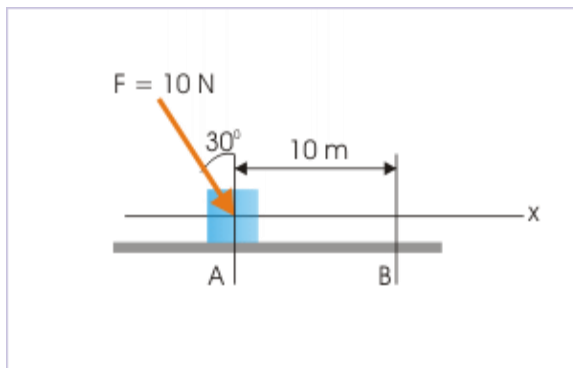


Figure 31

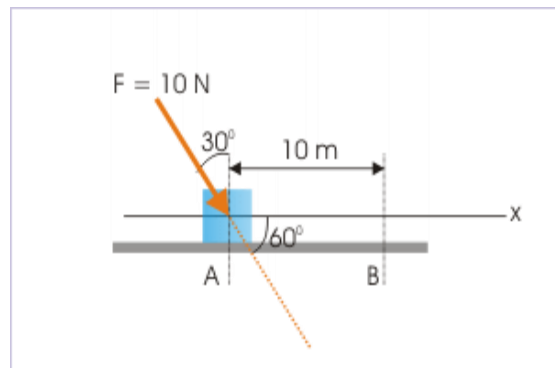


Figure 32

Answer:

$$\text{We have, Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = |\vec{F}| |\vec{d}| \cos\theta$$

$$\text{Here, } |\vec{F}| = 10\text{N}, |\vec{d}| = AB = 10 \text{ meter and } \theta = -60^\circ$$

$$\text{Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = |\vec{F}| |\vec{d}| \cos\theta$$

$$\text{Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = 10 \times 10 \times \cos(-60^\circ)$$

$$\text{Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = 10 \times 10 \times \cos 60^\circ \quad [ \cos(-\theta) = \cos\theta ]$$

$$\text{Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = 10 \times 10 \times \frac{1}{2}$$

$$\text{Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = 10 \times 5$$

$$\text{Work done} = w = \vec{F} \cdot \vec{d} = |\vec{F}| \cos\theta \times |\vec{d}| = 50 \text{ Joule}$$

### Laws of vector operation:

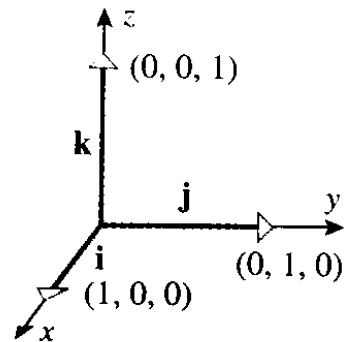
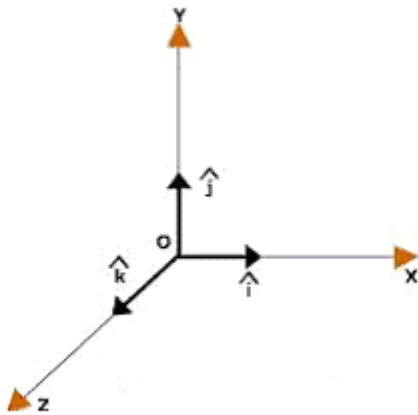


Figure: 33

We have, from figure 30

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos\theta$$

$$\therefore \hat{i} \cdot \hat{j} = \|\hat{i}\| \|\hat{j}\| \cos \theta$$

$$\therefore \hat{i} \cdot \hat{j} = \|\hat{i}\| \|\hat{j}\| \cos 90 = 1.1.0 = 0$$

Similarly,

$$\therefore \hat{k} \cdot \hat{i} = \|\hat{k}\| \|\hat{i}\| \cos 90 = 1.1.0 = 0$$

$$\therefore \hat{i} \cdot \hat{i} = \|\hat{i}\| \|\hat{i}\| \cos 0 = 1.1.1 = 1 \quad [\text{The length or magnitude of unit vector is 1}]$$

Similarly

$$\therefore \hat{j} \cdot \hat{j} = \|\hat{j}\| \|\hat{j}\| \cos 0 = 1.1.1 = 1$$

$$\therefore \hat{k} \cdot \hat{k} = \|\hat{k}\| \|\hat{k}\| \cos 0 = 1.1.1 = 1$$

**Q # 04:** If  $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$  and  $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$ , Find  $\vec{A} \cdot \vec{B}$

**Answer 04:**

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= (A_x B_x + A_y B_y + A_z B_z) \end{aligned}$$

$$[\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

Answer

### Direction cosines:

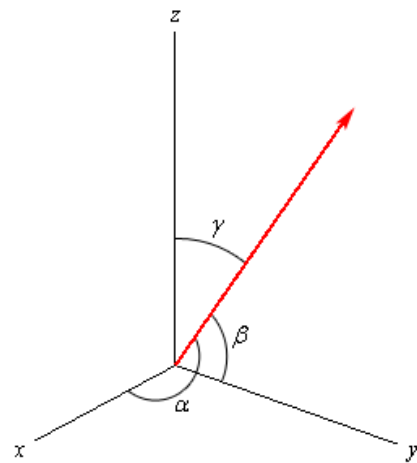
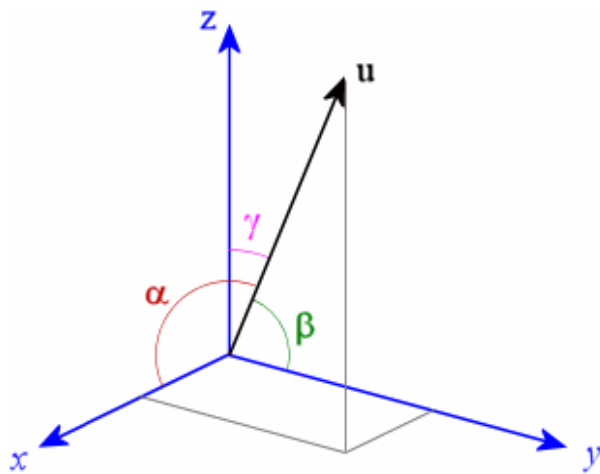


Figure # 34

Direction cosines are defined as

$$l = \cos\alpha$$

$$m = \cos\beta$$

$$n = \cos\gamma$$

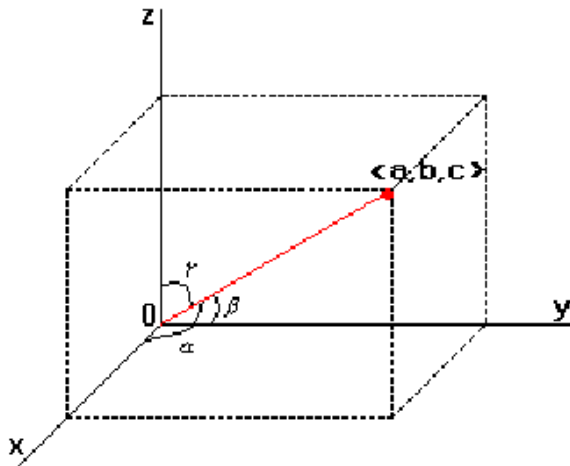


Figure # 35

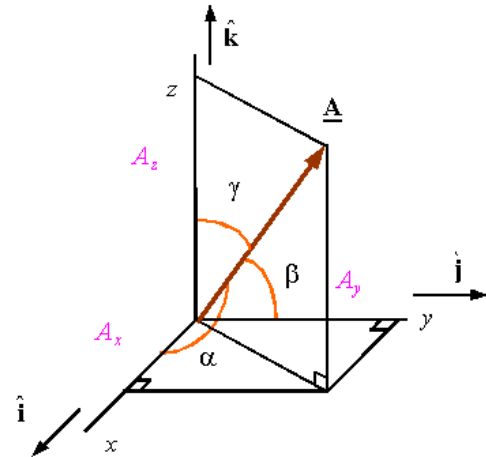


Figure # 36

Where the angles  $\alpha, \beta$  and  $\gamma$  are the angles shown in the figure. As shown in the figure, the direction cosines represent the cosines of the angles made between the vector and the three coordinate directions.

The direction cosines can be calculated from the components of the vector and its magnitude through the relations

$$l = \cos\alpha = \frac{A_x}{A}, m = \cos\beta = \frac{A_y}{A}, n = \cos\gamma = \frac{A_z}{A} \text{ [from figure 36]}$$

The three direction cosines are not independent and must satisfy the relation

$$l^2 + m^2 + n^2 = 1 \text{-----(i)}$$

This results from the fact that

$$l^2 + m^2 + n^2 = 1$$

$$\Rightarrow \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\Rightarrow \left(\frac{A_x}{A}\right)^2 + \left(\frac{A_y}{A}\right)^2 + \left(\frac{A_z}{A}\right)^2 = 1$$

$$\Rightarrow \frac{A_x^2}{A^2} + \frac{A_y^2}{A^2} + \frac{A_z^2}{A^2} = 1 \text{-----(ii)}$$

Since from figure 36

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\therefore |\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

A unit vector can be constructed along a vector using the direction cosines as its components along the x, y, and z directions. For example, the unit-vector  $\hat{e}$  along the

$$\begin{aligned} \text{vector } \vec{A} \text{ is obtained from } \hat{e} &= \frac{\vec{A}}{|\vec{A}|} = \frac{A_x \hat{i} + A_y \hat{j} + A_z \hat{k}}{|\vec{A}|} \\ \Rightarrow \hat{e} &= \frac{\vec{A}}{|\vec{A}|} = \frac{A_x \hat{i}}{|\vec{A}|} + \frac{A_y \hat{j}}{|\vec{A}|} + \frac{A_z \hat{k}}{|\vec{A}|} \\ \Rightarrow \hat{e} &= \frac{\vec{A}}{|\vec{A}|} = \frac{A_x}{|\vec{A}|} \hat{i} + \frac{A_y}{|\vec{A}|} \hat{j} + \frac{A_z}{|\vec{A}|} \hat{k} \\ \Rightarrow \hat{e} &= \frac{\vec{A}}{|\vec{A}|} = \cos\alpha \hat{i} + \cos\beta \hat{j} + \cos\gamma \hat{k} \text{ -----(iii)} \\ \Rightarrow \hat{e} &= \frac{\vec{A}}{|\vec{A}|} = l \hat{i} + m \hat{j} + n \hat{k} \text{ -----(iv)} \end{aligned}$$

Therefore,  $\vec{A} = |\vec{A}| \times \hat{e}$

$$\Rightarrow \vec{A} = |\vec{A}| \times \hat{e}$$

$$\Rightarrow \vec{A} = |\vec{A}| \times (l \hat{i} + m \hat{j} + n \hat{k}) \quad [\text{From (iv)}]$$

$$\Rightarrow \vec{A} = |\vec{A}| \times (\cos\alpha \hat{i} + \cos\beta \hat{j} + \cos\gamma \hat{k}) \quad [\text{From (iii)}]$$

**Parallel Vectors:**

When  $\vec{A}$  and  $\vec{B}$  are parallel to each other, their Dot Product is identical to the ordinary multiplication of their sizes, that is  $\vec{A} \cdot \vec{B} = AB$  since  $\theta = 0^\circ$  and  $\cos 0^\circ = 1$ .

**Perpendicular Vectors:**

When  $\vec{A}$  and  $\vec{B}$  are perpendicular to each other, their Dot Product is always Zero that is  $\vec{A} \cdot \vec{B} = 0$ , since  $\theta = 90^\circ$  and  $\cos 90^\circ = 0$

**Q# 05: Determine whether  $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$  and  $\vec{B} = 2\hat{i} - 2\hat{j} - 2\hat{k}$  are perpendicular**

**Answer:**

Given,  $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$  and  $\vec{B} = 2\hat{i} - 2\hat{j} - 2\hat{k}$

$$\therefore \vec{A} \cdot \vec{B} = (3\hat{i} + 5\hat{j} - 2\hat{k}) \cdot (2\hat{i} - 2\hat{j} - 2\hat{k})$$

$$\therefore \vec{A} \cdot \vec{B} = 3 \times 2 + 5 \times (-2) + (-2) \times (-2)$$

$$[ \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0 ]$$

$$\therefore \vec{A} \cdot \vec{B} = 6 - 10 + 4$$

$$\therefore \vec{A} \cdot \vec{B} = 0$$

Since  $\vec{A} \cdot \vec{B} = 0$  Then  $\vec{A}$  and  $\vec{B}$  are perpendicular to each other

**Q# 06: Find the angle between  $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$  and  $\vec{B} = 4\hat{i} + \hat{j} - 3\hat{k}$**

**Answer: We have,**

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

Given,  $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$  and  $\vec{B} = 4\hat{i} + \hat{j} - 3\hat{k}$

$$\therefore \vec{A} \cdot \vec{B} = (2\hat{i} - 3\hat{j} + \hat{k}) \cdot (4\hat{i} + \hat{j} - 3\hat{k})$$

$$\therefore \vec{A} \cdot \vec{B} = 2 \times 4 + (-3) \times 1 + 1 \times (-3)$$

$$[ \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0 ]$$

$$\therefore \vec{A} \cdot \vec{B} = 8 - 3 - 3$$

$$\therefore \vec{A} \cdot \vec{B} = 8 - 6$$

$$\therefore \vec{A} \cdot \vec{B} = 2$$

Again, Given,  $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$  and  $\vec{B} = 4\hat{i} + \hat{j} - 3\hat{k}$

$$\therefore |\vec{A}| = \sqrt{2^2 + (-3)^2 + 1^2} = \sqrt{4+9+1} = \sqrt{14}$$

$$\text{and } \therefore |\vec{B}| = \sqrt{4^2 + (1)^2 + (-3)^2} = \sqrt{16+1+9} = \sqrt{26}$$

$$\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$\cos\theta = \frac{2}{\sqrt{14} \times \sqrt{26}}$$

$$\theta = \cos^{-1}\left(\frac{2}{\sqrt{14} \times \sqrt{26}}\right) \text{ Answer}$$

**Q# 07:** A particle acted on by constant forces  $\vec{F}_1 = 4\hat{i} + \hat{j} - 3\hat{k}$  and  $\vec{F}_2 = 3\hat{i} + \hat{j} - \hat{k}$  (both measured in Newton), is displaced from the point  $(1, 2, 3)$  to the point  $(5, 4, 1)$  (measured in meters). Find the total work done by the forces.

**Answer:** Figure 30 shows the displacement of the particle and the forces acting on it. Although the forces are shown acting at the initial point A, they are assumed to act on the particle throughout the displacement from A to B. The resultant force is:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 = 4\hat{i} + \hat{j} - 3\hat{k} + 3\hat{i} + \hat{j} - \hat{k} = 7\hat{i} + 2\hat{j} - 4\hat{k} \text{ -----(i)}$$

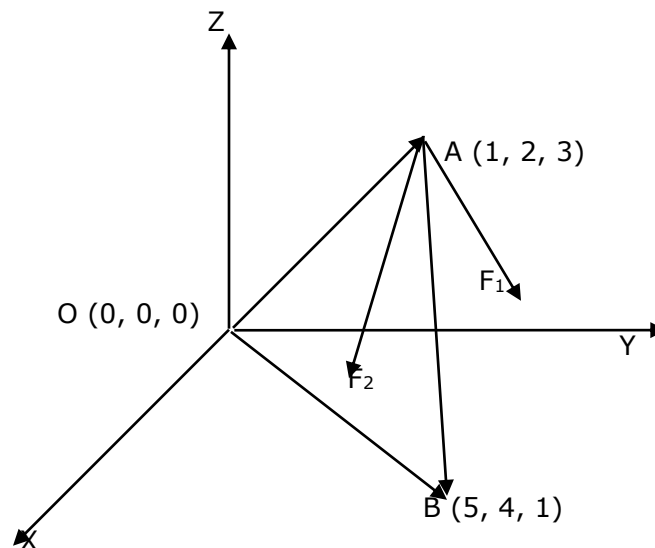


Figure 37

The displacement is the vector  $\vec{d} = \vec{AB}$ , but,

From  $\Delta OAB$ ,

$$\vec{OA} + \vec{AB} = \vec{OB}$$

$$\vec{AB} = \vec{OB} - \vec{OA} = (5\hat{i} + 4\hat{j} + \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{d} = \vec{AB} = 4\hat{i} + 2\hat{j} - 2\hat{k} \text{-----(ii)}$$

The work done,  $W$ , is given by

$$\vec{F} \cdot \vec{d} = (7\hat{i} + 2\hat{j} - 4\hat{k}) \cdot (4\hat{i} + 2\hat{j} - 2\hat{k}) = 7 \times 4 + 2 \times 2 + (-4) \times (-2) = 28 + 4 + 8 = 40 \text{ Joul}$$

**Answer**

**Q# 08:** A rope is attached to a 100-lb block (mass) on a ramp that is inclined at an angle of  $30^\circ$  with the ground (Figure no 38). How much force does the block exert against the ramp and how much force must be applied to the rope in a direction parallel to the ramp (slope) to prevent the block from sliding down the ramp? (Assume that the ramp is smooth, that is, exerts no frictional (NI©Y ej) forces)

Solution:

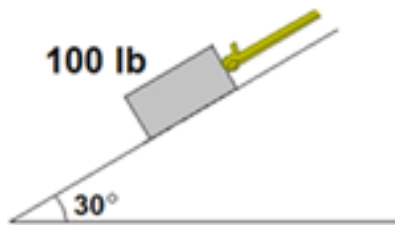


Figure 38

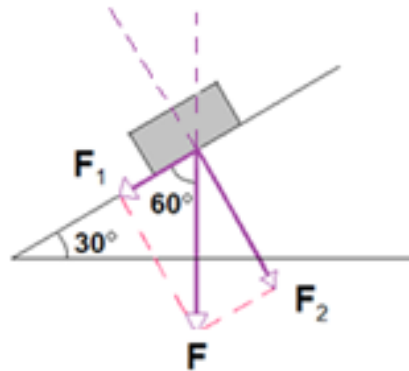


Figure 39

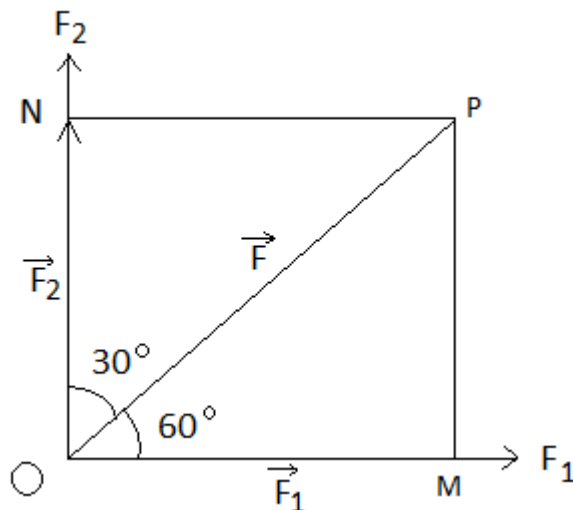


Figure 40

Let  $\vec{F}$  denote the downward force of gravity on the block. So  $|\vec{F}| = 100 \text{ lb}$  and let

$\vec{F}_1$  and  $\vec{F}_2$  be the vector components of  $\vec{F}$  parallel and perpendicular to the ramp (Figure no 39).

From,  $\triangle OMP$ ,

$$\frac{OM}{OP} = \cos\theta$$

$$OM = OP \cos\theta$$

$$|\vec{F}_1| = |\vec{F}| \cos\theta \text{-----(i)}$$

From,  $\triangle OMP$ ,

$$\frac{MP}{OP} = \sin\theta$$

$$MP = OP \sin\theta$$

$$|\vec{F}_2| = |\vec{F}| \sin\theta \text{-----(ii)}$$

From (i),

$$|\vec{F}_1| = |\vec{F}| \cos\theta$$

$$|\vec{F}_1| = |\vec{F}| \cos 60^\circ = 100 \times \frac{1}{2} = 50 \text{ lb}$$

and from (ii),

$$|\vec{F}_2| = |\vec{F}| \sin\theta$$

$$|\vec{F}_2| = |\vec{F}| \sin 60^\circ = 100 \times \frac{\sqrt{3}}{2} = 50\sqrt{3} \text{ lb}$$

Thus the block exerts a force of approximately  $50\sqrt{3}$  -lb against the ramp, and it requires a force of 50-lb to prevent the block from sliding down the ramp.

**Q# 09:** A wagon is pulled horizontally by exerting a constant force of 10 lb on the handle at an angle of  $60^\circ$  with the horizontal. How much work is done in moving the wagon 50 ft?

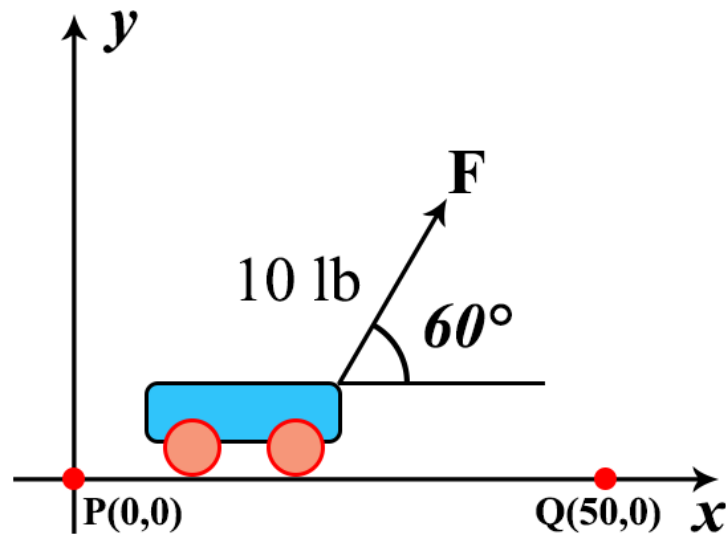


Figure 41

Answer: Introduce an xy-co-ordinate system so that the wagon moves from  $\mathbf{P(0,0)}$  to  $\mathbf{Q(50,0)}$  along the x-axis (Figure no 40). In the co-ordinate system  $\vec{\mathbf{PQ}} = 50\hat{\mathbf{i}}$

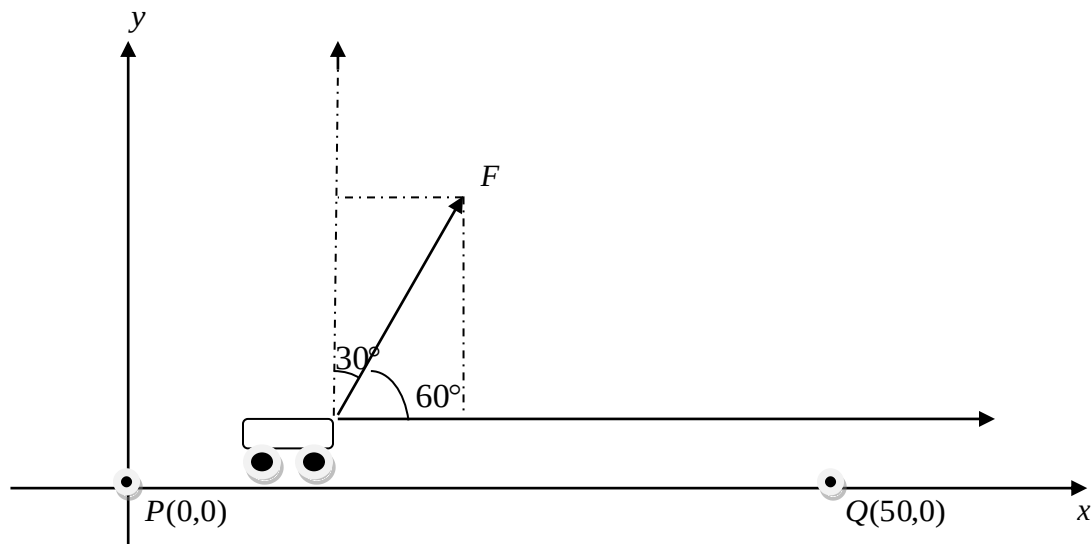


Figure 42

and  $\vec{\mathbf{F}} = (10\cos 60^\circ)\hat{\mathbf{i}} + (10\sin 60^\circ)\hat{\mathbf{j}}$

$$\vec{F} = (10 \times \frac{1}{2}) \hat{i} + (10 \times \frac{\sqrt{3}}{2}) \hat{j}$$

$$\vec{F} = 5 \hat{i} + 5\sqrt{3} \hat{j}$$

So, the work done is:

$$W = \vec{F} \cdot \vec{PQ} = (5 \hat{i} + 5\sqrt{3} \hat{j}) \cdot (50 \hat{i})$$

$$W = \vec{F} \cdot \vec{PQ} = (5 \hat{i} + 5\sqrt{3} \hat{j}) \cdot (50 \hat{i} + 0 \hat{j})$$

$$W = \vec{F} \cdot \vec{PQ} = 250 \text{ 1b Answer}$$

Or

Component of Force  $F$  in the direction of X axis is,  $\left| \vec{F} \right| \cos 60^\circ = (10 \times \frac{1}{2}) = 5$

Work done = force  $\times$  displacement =  $5 \times 50 = 250$

## The cross product:

The cross product of vectors  $\vec{a}$  and  $\vec{b}$  is a vector perpendicular to both  $\vec{a}$  and  $\vec{b}$  and has a magnitude equal to the area of the parallelogram generated from  $\vec{a}$  and  $\vec{b}$ . The direction of the cross product is given by the right-hand rule. The cross product is denoted by a "×" between the vectors

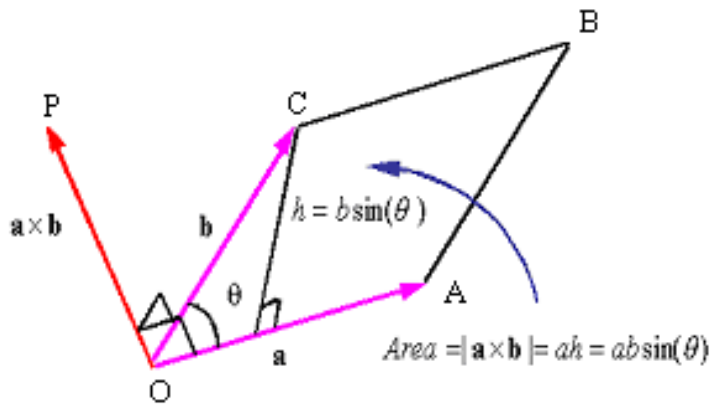


Figure # 43

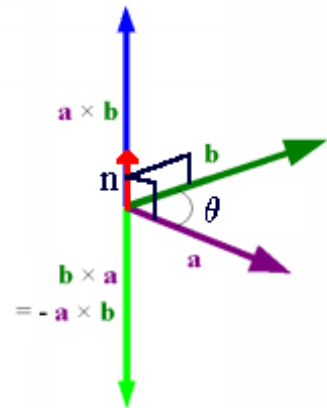


Figure # 44

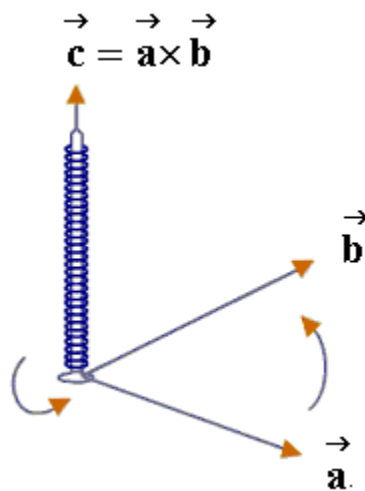


Figure # 45: The direction of  $\vec{c}$  is that in which a right handed screw advances when turned from  $\vec{a}$  to  $\vec{b}$

Area of Parallelogram OABC = base×height

Area of Parallelogram =  $\vec{OA} \times \vec{h}$  [from figure 41]

Area of Parallelogram =  $|\vec{a} \times \vec{h}|$  -----(i)

We have,

$$\frac{h}{OC} = \sin\theta \text{ [Figure 41]}$$

$$\frac{h}{\left| \vec{b} \right|} = \sin\theta \text{ [Figure 41]}$$

$$\Rightarrow h = \left| \vec{b} \right| \sin\theta$$

From (i),

$$\therefore \text{Area of Parallelogram} = \left| \vec{a} \right| h = \left| \vec{a} \right| \left| \vec{b} \right| \sin\theta \quad [ \quad h = \left| \vec{b} \right| \sin\theta ]$$

We have,

**Any Vector = Length of this Vector  $\times$  Unit Vector**

$$\therefore \vec{OP} = \text{Length of this Vector } \vec{OP} \times \text{Unit Vector of } \vec{OP}$$

$$\therefore \vec{OP} = \vec{a} \times \vec{b} = \left| \vec{a} \right| \left| \vec{b} \right| \sin\theta \hat{n} \text{-----(ii)}$$

Here,  $\left| \vec{a} \right| \left| \vec{b} \right| \sin\theta$  is the magnitude (length) of the vector  $\vec{OP}$  or  $\vec{a} \times \vec{b}$  and

$\hat{n}$  is the unit vector of  $\vec{OP}$  or  $\vec{a} \times \vec{b}$

where  $\theta$  is the measure of the angle between  $\vec{a}$  and  $\vec{b}$  ( $0^\circ \leq \theta \leq 180^\circ$ ) on the plane defined by the span of the vectors, and  $\hat{n}$  is a unit vector perpendicular to both  $\vec{a}$  and  $\vec{b}$ .

Order is important in the cross product. If the order of operations changes in a cross

product the direction of the resulting vector is reversed. That is,  $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

[If  $\vec{a}$  and  $\vec{b}$  are two vectors, then the cross product  $\vec{a} \times \vec{b}$  is a vector perpendicular to the plane containing  $\vec{a}$  and  $\vec{b}$ . The magnitude of  $\vec{a} \times \vec{b}$  is  $|\vec{a}| |\vec{b}| \sin\theta$ , where  $\theta$  is the angle between  $\vec{a}$  and  $\vec{b}$ . The direction of  $\vec{a} \times \vec{b}$  is given by the right-hand rule.]

[If  $\vec{a}$  and  $\vec{b}$  are two vectors, then the cross product  $\vec{a} \times \vec{b}$  is a vector perpendicular to the plane containing  $\vec{a}$  and  $\vec{b}$ . The magnitude of  $\vec{a} \times \vec{b}$  is  $|\vec{a}| |\vec{b}| \sin\theta$ , where  $\theta$  is the angle between  $\vec{a}$  and  $\vec{b}$ . The direction of  $\vec{a} \times \vec{b}$  is given by the right-hand rule.]

is,  $\vec{a} \times \vec{b} = \left| \vec{a} \right| \left| \vec{b} \right| \sin\theta \hat{n}$

$$\vec{a} \times \vec{b} = \left| \vec{a} \right| \left| \vec{b} \right| \sin\theta \hat{n}$$

[If  $\vec{a}$  and  $\vec{b}$  are two vectors, then the cross product  $\vec{a} \times \vec{b}$  is a vector perpendicular to the plane containing  $\vec{a}$  and  $\vec{b}$ . The magnitude of  $\vec{a} \times \vec{b}$  is  $|\vec{a}| |\vec{b}| \sin\theta$ , where  $\theta$  is the angle between  $\vec{a}$  and  $\vec{b}$ . The direction of  $\vec{a} \times \vec{b}$  is given by the right-hand rule.]

**Laws of vector operation:**

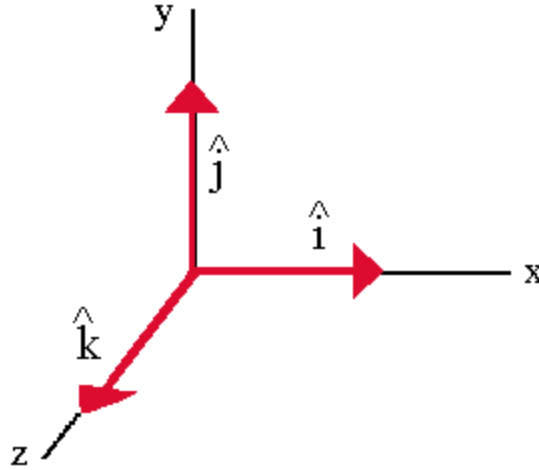


Figure # 46

We have,

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$$

From Figure 44,

$$\hat{i} \times \hat{j} = |\hat{i}| |\hat{j}| \sin \theta \hat{n} = |\hat{i}| |\hat{j}| \sin 90^\circ \hat{n} = 1 \times 1 \times 1 \times \hat{n} = \hat{n}$$

$$\hat{i} \times \hat{j} = \hat{k} \text{ [say } \hat{n} = \hat{k}] \text{-----(i)}$$

Similarly,

$$\hat{j} \times \hat{k} = |\hat{j}| |\hat{k}| \sin \theta \hat{n} = |\hat{j}| |\hat{k}| \sin 90^\circ \hat{n} = 1 \times 1 \times 1 \times \hat{n} = \hat{n} \text{ [here } \hat{n} = \hat{i}] \text{-----(ii)}$$

$$\hat{k} \times \hat{i} = |\hat{k}| |\hat{i}| \sin \theta \hat{n} = |\hat{k}| |\hat{i}| \sin 90^\circ \hat{n} = 1 \times 1 \times 1 \times \hat{n} = \hat{n} \text{ [here } \hat{n} = \hat{j}] \text{-----(iii)}$$

Again,

$$\hat{j} \times \hat{i} = |\hat{j}| |\hat{i}| \sin(-\theta) \hat{n} = -|\hat{j}| |\hat{i}| \sin \theta \hat{n} = -|\hat{j}| |\hat{i}| \sin 90^\circ \hat{n} = -1 \times 1 \times 1 \times \hat{n} = -\hat{n}$$

$$[ \sin(-\theta) = -\sin \theta ]$$

$$\hat{j} \times \hat{i} = -\hat{k} \text{ [say } \hat{n} = \hat{k}] \text{-----(iv)}$$

Similarly,

$$\hat{i} \times \hat{k} = \left| \begin{array}{cc} \hat{i} & \hat{k} \\ \sin(-\theta) & \hat{\eta} \end{array} \right| = - \left| \begin{array}{cc} \hat{i} & \hat{k} \\ \sin\theta & \hat{\eta} \end{array} \right| = - \left| \begin{array}{cc} \hat{i} & \hat{k} \\ \sin 90^\circ & \hat{\eta} \end{array} \right| = -1 \times 1 \times 1 \times \hat{\eta} = -\hat{\eta}$$

$$[ \sin(-\theta) = -\sin\theta ]$$

$$\hat{i} \times \hat{k} = -\hat{j} \text{ [say } \hat{\eta} = \hat{j}] \text{-----(v)}$$

$$\hat{k} \times \hat{j} = \left| \begin{array}{cc} \hat{k} & \hat{j} \\ \sin(-\theta) & \hat{\eta} \end{array} \right| = - \left| \begin{array}{cc} \hat{k} & \hat{j} \\ \sin\theta & \hat{\eta} \end{array} \right| = - \left| \begin{array}{cc} \hat{k} & \hat{j} \\ \sin 90^\circ & \hat{\eta} \end{array} \right| = -1 \times 1 \times 1 \times \hat{\eta} = -\hat{\eta}$$

$$[ \sin(-\theta) = -\sin\theta ]$$

$$\hat{k} \times \hat{j} = -\hat{i} \text{ [say } \hat{\eta} = \hat{i}] \text{-----(vi)}$$

Again,

$$\hat{k} \times \hat{k} = \left| \begin{array}{cc} \hat{k} & \hat{k} \\ \sin\theta & \hat{\eta} \end{array} \right|$$

$$\hat{k} \times \hat{k} = \left| \begin{array}{cc} \hat{k} & \hat{k} \\ \sin\theta & \hat{\eta} \end{array} \right| = \left| \begin{array}{cc} \hat{k} & \hat{k} \\ \sin 0^\circ & \hat{\eta} \end{array} \right| = 1 \times 1 \times 0 \times \hat{\eta} = 0 \text{-----(vii)}$$

Similarly,

$$\hat{j} \times \hat{j} = \left| \begin{array}{cc} \hat{j} & \hat{j} \\ \sin\theta & \hat{\eta} \end{array} \right| = \left| \begin{array}{cc} \hat{j} & \hat{j} \\ \sin 0^\circ & \hat{\eta} \end{array} \right| = 1 \times 1 \times 0 \times \hat{\eta} = 0 \text{-----(viii)}$$

$$\hat{i} \times \hat{i} = \left| \begin{array}{cc} \hat{i} & \hat{i} \\ \sin\theta & \hat{\eta} \end{array} \right| = \left| \begin{array}{cc} \hat{i} & \hat{i} \\ \sin 0^\circ & \hat{\eta} \end{array} \right| = 1 \times 1 \times 0 \times \hat{\eta} = 0 \text{-----(ix)}$$

**Scalar triple product:**  $\vec{A} \cdot (\vec{B} \times \vec{C})$  or  $\vec{B} \cdot (\vec{C} \times \vec{A})$  or  $\vec{C} \cdot (\vec{A} \times \vec{B})$  are known as a scalar triple product. It is symbolically denoted by  $[ABC]$  or  $[BCA]$  or  $[CAB]$

$$\text{We know, } (\vec{A} \times \vec{B}) = -(\vec{B} \times \vec{A})$$

$$\text{Hence, } \vec{A} \cdot (\vec{B} \times \vec{C}) = -\vec{A} \cdot (\vec{C} \times \vec{B})$$

$$\text{That is, } [ABC] = -[ACB]$$

**Q # 10:** If  $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$  and  $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$ , Find  $\vec{A} \times \vec{B}$

Answer: 08:

$$\vec{A} \times \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$= A_x B_x (\hat{i} \times \hat{i}) + A_x B_y (\hat{i} \times \hat{j}) + A_x B_z (\hat{i} \times \hat{k}) + A_y B_x (\hat{j} \times \hat{i}) + A_y B_y (\hat{j} \times \hat{j}) + A_y B_z (\hat{j} \times \hat{k}) \\ + A_z B_x (\hat{k} \times \hat{i}) + A_z B_y (\hat{k} \times \hat{j}) + A_z B_z (\hat{k} \times \hat{k})$$

$$\begin{aligned}
&= \mathbf{A}_x \mathbf{B}_x \times \mathbf{0} + \mathbf{A}_x \mathbf{B}_y (\hat{\mathbf{k}}) + \mathbf{A}_x \mathbf{B}_z (-\hat{\mathbf{j}}) + \mathbf{A}_y \mathbf{B}_x (-\hat{\mathbf{k}}) + \mathbf{A}_y \mathbf{B}_y \times \mathbf{0} + \mathbf{A}_y \mathbf{B}_z (\hat{\mathbf{i}}) \\
&\quad + \mathbf{A}_z \mathbf{B}_x (\hat{\mathbf{j}}) + \mathbf{A}_z \mathbf{B}_y (-\hat{\mathbf{i}}) + \mathbf{A}_z \mathbf{B}_z \times \mathbf{0} \\
&= \mathbf{A}_x \mathbf{B}_y \hat{\mathbf{k}} - \mathbf{A}_x \mathbf{B}_z \hat{\mathbf{j}} - \mathbf{A}_y \mathbf{B}_x \hat{\mathbf{k}} + \mathbf{A}_y \mathbf{B}_z \hat{\mathbf{i}} + \mathbf{A}_z \mathbf{B}_x \hat{\mathbf{j}} - \mathbf{A}_z \mathbf{B}_y \hat{\mathbf{i}} \\
&= \mathbf{A}_x \mathbf{B}_y \hat{\mathbf{k}} - \mathbf{A}_x \mathbf{B}_z \hat{\mathbf{j}} - \mathbf{A}_y \mathbf{B}_x \hat{\mathbf{k}} + \mathbf{A}_y \mathbf{B}_z \hat{\mathbf{i}} + \mathbf{A}_z \mathbf{B}_x \hat{\mathbf{j}} - \mathbf{A}_z \mathbf{B}_y \hat{\mathbf{i}} \\
&= \mathbf{A}_y \mathbf{B}_z \hat{\mathbf{i}} - \mathbf{A}_z \mathbf{B}_y \hat{\mathbf{i}} - \mathbf{A}_x \mathbf{B}_z \hat{\mathbf{j}} + \mathbf{A}_z \mathbf{B}_x \hat{\mathbf{j}} + \mathbf{A}_x \mathbf{B}_y \hat{\mathbf{k}} - \mathbf{A}_y \mathbf{B}_x \hat{\mathbf{k}} \\
&= \hat{\mathbf{i}}(\mathbf{A}_y \mathbf{B}_z - \mathbf{A}_z \mathbf{B}_y) - \hat{\mathbf{j}}(\mathbf{A}_x \mathbf{B}_z - \mathbf{A}_z \mathbf{B}_x) + \hat{\mathbf{k}}(\mathbf{A}_x \mathbf{B}_y - \mathbf{A}_y \mathbf{B}_x) \text{-----(i)} \\
&= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \mathbf{A}_x & \mathbf{A}_y & \mathbf{A}_z \\ \mathbf{B}_x & \mathbf{B}_y & \mathbf{B}_z \end{vmatrix} \text{-----(ii)}
\end{aligned}$$

**Q # 11:** Find a unit vector perpendicular to the vectors  $\vec{\mathbf{a}} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}}$  and  $\vec{\mathbf{b}} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

**Answer:** A vector perpendicular to  $\vec{\mathbf{a}}$  and  $\vec{\mathbf{b}}$  is  $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$

$$\vec{\mathbf{a}} \times \vec{\mathbf{b}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} = \hat{\mathbf{i}}(2-0) - \hat{\mathbf{j}}(6-0) + \hat{\mathbf{k}}(6+1) = 2\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 7\hat{\mathbf{k}}$$

A unit vector, perpendicular to  $\vec{\mathbf{a}}$  and  $\vec{\mathbf{b}}$ , in this direction is obtained by simply dividing  $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$  by its magnitude. Thus

$$\hat{\mathbf{e}} = \frac{2\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 7\hat{\mathbf{k}}}{\sqrt{2^2 + (-6)^2 + 7^2}} = \frac{2\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 7\hat{\mathbf{k}}}{\sqrt{89}} \text{ is the required vector.}$$

**Q#12:** Show that  $\vec{\mathbf{A}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ ,  $\vec{\mathbf{B}} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$  and  $\vec{\mathbf{C}} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$  are coplanar.

**Answer:**

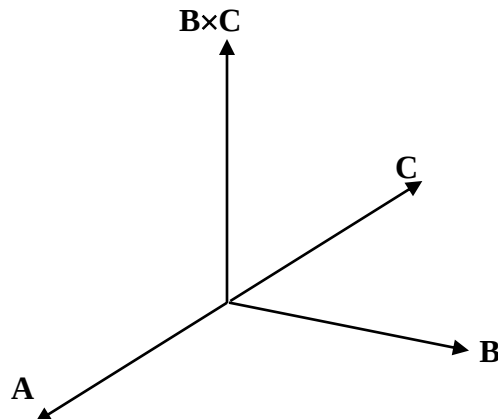


Figure # 47

If **A** is a third vector perpendicular to **(B×C)**, then **A**, **B** and **C** are coplanar and **A. (B×C) = 0**

Therefore, three vectors **A**, **B**, **C** are coplanar if **A. (B×C) = 0**

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{vmatrix} = \hat{i}(1-2) - \hat{j}(-2-6) + \hat{k}(2+3) = -\hat{i} + 8\hat{j} + 5\hat{k}$$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\hat{i} + 2\hat{j} - 3\hat{k}) \cdot (-\hat{i} + 8\hat{j} + 5\hat{k}) = -1 + 16 - 15 = 0$$

Therefore A, B, C are coplanar.

**Q# 13:** Figure no 46 shows a force  $\vec{F}$  of 100 N applied in the positive z-direction at the point **Q(1,1,1)** of a cube whose sides have a length of 1 m. assuming that the cube is free to rotate about the point **P(0,0,0)** (the origin), find the scalar moment of the force about **P** and describe the direction of rotation.

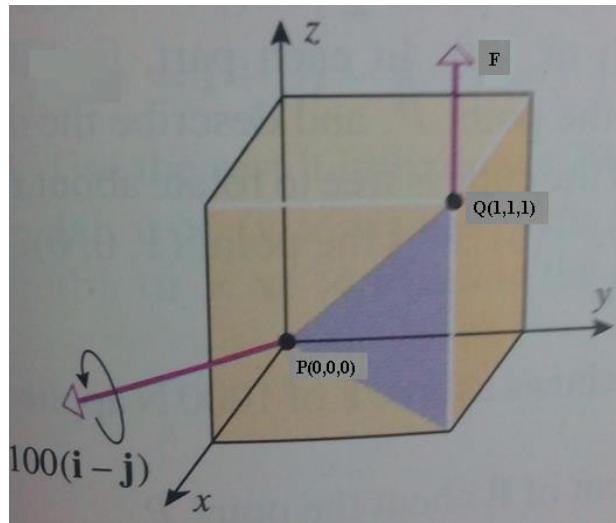


Figure 48

**Answer:** The force vector  $\vec{F} = 0.\hat{i} + 0.\hat{j} + 100\hat{k}$  and the vector from **P** to **Q** is

$\vec{PQ} = \hat{i} + \hat{j} + \hat{k}$ , so the vector moment of  $\vec{F}$  about **P** is

$$\vec{PQ} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & 100 \end{vmatrix} = \hat{i}(100-0) - \hat{j}(100-0) + \hat{k}(0-0)$$

$$= 100\hat{i} - 100\hat{j}$$

Thus the scalar moment of  $\vec{F}$  about  $\mathbf{P}$  is

$$|\vec{PQ} \times \vec{F}| = \sqrt{(100)^2 + (-100)^2} = \sqrt{10000 + 10000} = \sqrt{20000} = \sqrt{2 \times (100)^2} = 100\sqrt{2} \text{ N.m}$$

and the direction of rotation is counterclockwise looking along the vector  $100\hat{i} - 100\hat{j}$   
 $= 100(\hat{i} - \hat{j})$  towards its initial point (**Figure no 46**)

### Scalar Fields:

A scalar field is a map over some space of scalar values. That is it is a map of values with no direction.

### **Examples:**

1. A simple example of a scalar field is a map of the temperature distribution in a room. The function that gives the temperature of any point in the room you are sitting is a scalar field
  - Some parts of it, maybe near the door or windows, will probably be cooler, while other parts, maybe near a heater, will be warmer. And in between these regions of course there must be a continuous smooth change in temperature.
  - This quantity "temperature", let's call it  $T$ , therefore has various different values throughout that three-dimensional space that you're sitting in. Let's describe position by the three Cartesian coordinates  $x$ ,  $y$  and  $z$ .
  - So at any given position  $(x, y, z)$  the temperature  $T$  has a particular value, and if we change that position then  $T$  will probably change too. In other words  $T$  is a function of  $x$ ,  $y$  and  $z$  and we can write  $T(x, y, z)$ . For example:  $T(x, y, z)$  can be used to represent the temperature at the point  $(x, y, z)$
  - This means that  $T$  is a **scalar field**.

**As for example:**

$$T(x, y, z) = x^2 + yz$$

$$T(2, 5, 6) = 2^2 + 5 \times 6 = 34^0$$

$$T(4, 2, 8) = 4^2 + 2 \times 8 = 32^0$$

$$T(5, 4, 2) = 5^2 + 4 \times 2 = 33^0$$

.....

.....

- The temperature at that position just has a value,  $34^0$  degrees say, there is only one piece of information. There is no direction associated with that temperature.

2. To indicate the temperature distribution throughout space, or the air pressure
3. The temperature of a swimming pool is a scalar field: to each point we associate a scalar value of temperature.
4. In this course the most important example is the electromagnetic potential field.
5. A scalar valued function is a function that takes one or more values but returns a single value.  $f(x, y, z) = x^2 + 2yz^5$  is an example of a scalar valued function.

Zwor Pz $\alpha$ K $\hat{\eta}$  wefe (Electromagnetic Potential Field): $\hat{K}vb$  GKK abPvR $\odot$  $\hat{K}$  Amxg  $\hat{\_}$  $\hat{K}$  Zwor  $\hat{\eta}$  $\hat{\eta}$   $\hat{K}vb$  we $\rangle$ y $\hat{Z}$  Avb $\hat{Z}$   $\hat{h}$  cwigvb KvR Ki $\hat{Z}$  nq Zv $\hat{K}$  H  $\hat{\eta}$  $\hat{\eta}$  H we $\rangle$ yi Zwor  $\hat{\eta}$  $\hat{\eta}$  wefe e $\hat{j}$  | KvR  $\hat{\_}$ jvi ivwk |

Zwor Pz $\alpha$ K $\hat{\eta}$  (Electromagnetic Field): $\hat{K}vb$  GKK abPvR $\odot$  $\hat{K}$  Zwor  $\hat{\eta}$  $\hat{\eta}$   $\hat{K}vb$  we $\rangle$ y $\hat{Z}$   $\hat{\_}$ vcb Ki $\hat{j}$   $\hat{t}m$   $\hat{h}$  ej Ab $\sim$ fe K $\hat{i}$  Zv $\hat{K}$  Zwor Pz $\alpha$ K $\hat{\eta}$  e $\hat{j}$  |  $\hat{h}gb$  GKwU abPv $\hat{R}$  $\odot$ i Kv $\hat{Q}$  Avi GKwU ab PvR $\odot$  ivL $\hat{j}$   $\hat{t}m$  weKl $\odot$ b ej Abyfe Ki $\hat{e}$  | ZLb Zvi wW $\hat{i}$ Kkb n $\hat{e}$  ewn:gywL | GKwU FYPv $\hat{R}$  $\odot$ i Kv $\hat{Q}$  Avi GKwU ab PvR $\odot$  ivL $\hat{j}$   $\hat{t}m$  AvKl $\odot$ b ej Abyfe Ki $\hat{e}$  | ZLb Zvi wW $\hat{i}$ Kkb n $\hat{e}$  A $\hat{S}$ :gywL |

Newton's Gravitational Field: GKK f $\hat{ii}$   $\hat{K}vb$  e $\hat{\_}$  $\hat{K}$  c,,w\_ex  $\hat{h}$  e $\hat{j}$  AvKl $\odot$ b K $\hat{i}$  |

mg $\hat{y}$  $\hat{\_}$ ai c,, $\hat{o}$ , cvnv $\hat{t}oi$  cv $\hat{\_}$  $\hat{t}$  $\hat{k}$ , cvnv $\hat{t}oi$  Dc $\hat{i}$  wewfb $\odot$ e RvqMvq evZv $\hat{t}mi$  Pvc wewfb $\odot$ e |

**Vector space:** Basically, a vector space is the set of all vectors that can be created by Linear combinations of a given set of vectors. If you take a vector and multiply it by any real number, and take another vector and multiply it by any real number, and then add them together, this new vector is a linear combination of the first two. So a vector space is all the possible linear combinations of the set of basis vectors. The basis vectors are said to "span" the vector space. You can find different sets of basis vectors that span the same vector space.

### Vector fields:

A vector field can be considered a map of vectors over some space. . For example if one were to show wind vectors on a weather map; that would be a vector field. The electric field surrounding a charge is a vector field. A vector field in the plane, for instance, can be visualized as a collection of arrows with a given magnitude and direction each attached to a point in the plane

### Examples:

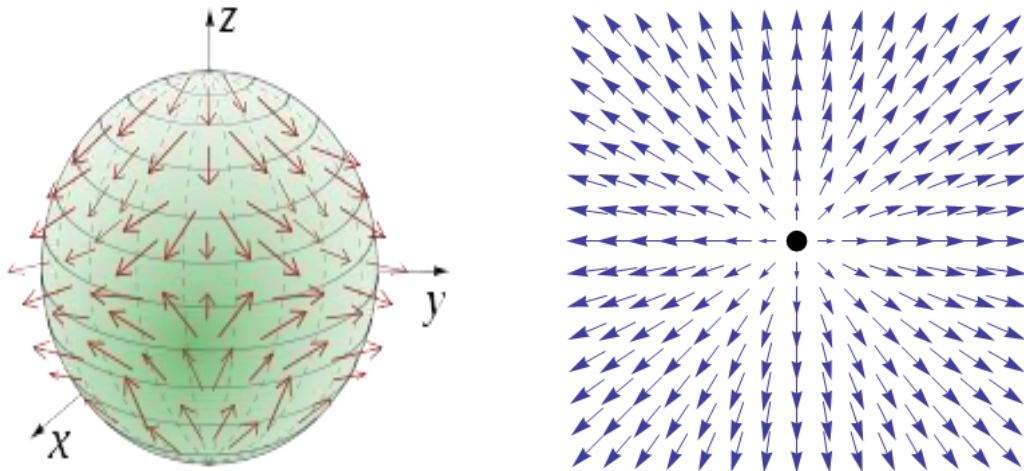


Figure # 49

- Now imagine the air moving around in that room you're in. In some parts it will be moving quickly, above the heater maybe, or near an open window, or near your nose, while in other parts it will be moving slowly.
  - The quantity describing that air movement is "velocity", let's call it  $\mathbf{v}$ . That quantity  $\mathbf{v}$  also has a different value at different positions, so we can write  $\mathbf{v}(\mathbf{x}, \mathbf{y}, \mathbf{z})$  and this quantity too is a field.
  - At any position  $(\mathbf{x}, \mathbf{y}, \mathbf{z})$  the air at that point is moving in a particular **direction**, with a particular **speed**.
- The water flow in the same pool is a vector field
- The speed and direction of a moving fluid throughout space, or the strength and direction of some force, such as the magnetic or gravitational force, as it changes from point to point.
- Examples are movement of a fluid, or the force generated by a magnetic or gravitational field, or atmospheric models, where both the strength (speed) and the direction of winds are recorded.
- Wind vectors on a weather map; that would be a vector field. The electric field surrounding a charge is a vector field
- Examples of vector fields include the electromagnetic field and the Newtonian gravitational field.
- Three vector fields are shown below. Which represents the electric field emanating from a positive point charge in the middle? (Note that vectors of similar magnitude are colored similarly in these plots)

Three vector fields are shown below. Which represents the electric field emanating from a positive point charge in the middle? (Note that vectors of similar magnitude are colored similarly in these plots)

Let  $\mathbf{v}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = x^2y\hat{i} + yz\hat{j} + \hat{k}$

Let  $\mathbf{v}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = x^2y\hat{i} + yz\hat{j} + \hat{k}$

**Visualize Scalar Field on a Surface:** Surface is colored using the value of a scalar function defined on each vertex.



Figure # 50

**Visualize Vector Field on a Surface:**

1. Imagine what happens when you **throw** a **stone into** the **water**
2. Imagine what happens when you **throw** a **stone to** the honey-comb

**Differentiation of Vectors:**

In many practical problems, we often deal with vectors that change with time, e.g.

Velocity, acceleration etc

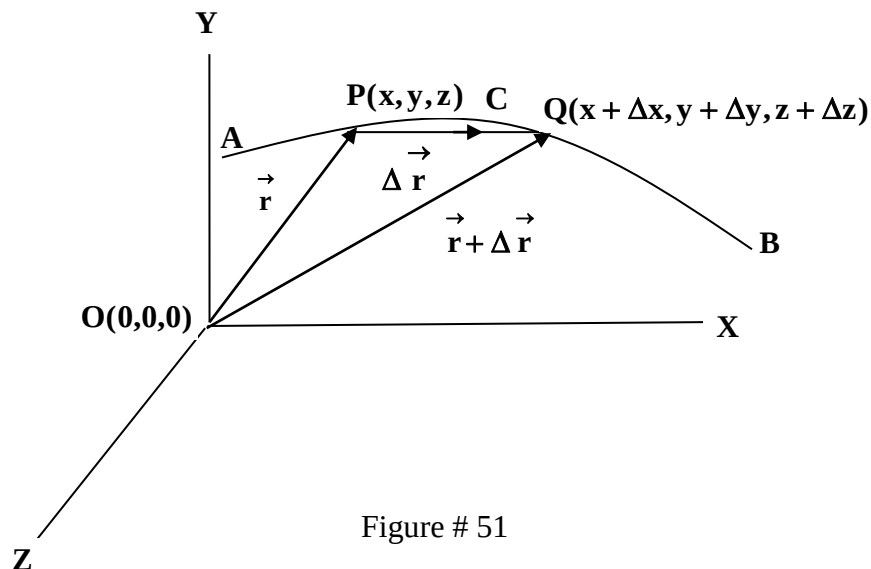


Figure # 51

We consider a position vector  $\vec{OP} = \vec{r}$ , which is drawn from O to P then  $\vec{OP}$  moves from P to Q. Then  $\Delta \vec{r}$  is small increment from P to Q. So,  $\vec{OQ} = \vec{r} + \Delta \vec{r}$  is a new

**vector drawn** from O to Q. Assume that  $\vec{r}$  is a vector function of  $x, y, z$  and depends on a scalar variable  $t$ .

From figure,

$$\vec{OP} = \vec{r}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k} \text{-----(i)}$$

$$\therefore \Delta \vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k} \text{-----(ii)}$$

$$\vec{OQ} = (x + \Delta x)\hat{i} + (y + \Delta y)\hat{j} + (z + \Delta z)\hat{k}$$

$$\vec{OQ} = (x\hat{i} + y\hat{j} + z\hat{k}) + (\Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k})$$

$$\vec{OQ} = \vec{r} + \Delta \vec{r} \text{-----(iii)}$$

From

$\Delta OPQ$ ,

$$\vec{OP} + \vec{PQ} = \vec{OQ}$$

$$\vec{r} + \Delta \vec{r} = \vec{r} + \Delta \vec{r}$$

$$\Delta \vec{r} = \vec{r} + \Delta \vec{r} - \vec{r} \text{-----(iv)}$$

Then  $\frac{\Delta \vec{r}}{\Delta t}$  is the average rate of change of  $\vec{r}$  with respect to time  $t$ .

$$\text{i.e. } \frac{\Delta \vec{r}}{\Delta t} = \frac{(\vec{r} + \Delta \vec{r}) - \vec{r}}{\Delta t} \text{-----(v)}$$

When  $Q \rightarrow P$  then PQ will be tangent

So, then  $\Delta t \rightarrow 0$ , then  $\Delta \vec{r} \rightarrow d\vec{r}$

*[Note:  $d\vec{r}$  is a tangent vector to any point to the curve]*

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{(\vec{r} + \Delta \vec{r}) - \vec{r}}{\Delta t}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{(\vec{r} + \Delta \vec{r}) - \vec{r}}{\Delta t} \text{-----(vi)}$$

$\frac{d\vec{r}}{dt}$  represents the velocity  $\vec{v}$

This is the derivative of  $\vec{r}$  with respect to the scalar variable  $t$ .  
Again,

$$\vec{a} = \frac{d\vec{v}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left( \frac{d\vec{r}}{dt} \right) = \frac{d^2 \vec{r}}{dt^2} \text{-----(vii)} \quad \left[ \vec{v} = \frac{d\vec{r}}{dt} \right]$$

$\frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2}$  represents the acceleration  $\vec{a}$  along the curve.

Again,

$$\vec{OP} = \vec{r}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore \vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \text{-----(viii)}$$

**Remark:** If  $\vec{r}(t)$  is the position function of a particle moving along a curve in 2-space (2-dimensional space) or 3-space, then the instantaneous velocity, instantaneous acceleration and instantaneous speed of the particle at time  $t$  are defined by

$$\text{Velocity: } \vec{v}(t) = \frac{d\vec{r}}{dt}$$

$$\text{Acceleration: } \vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left( \frac{d\vec{r}}{dt} \right) = \frac{d^2 \vec{r}}{dt^2}$$

$$\text{Speed: } \left| \vec{v}(t) \right| = \frac{ds}{dt}$$

**Theorem:**

If  $C$  is the graph in 2-space or 3-space of a smooth vector-valued function  $\vec{r}(t)$ , then its arc length  $L$  from  $t = a$  to  $t = b$  is

$$L = \int_a^b \left| \frac{d\vec{r}}{dt} \right| dt$$

### Displacement Vector and Distance Traveled:

If a particle travels along a curve  $C$  in 2-space or 3-space, the displacement of the particle over the time interval  $t_1 \leq t \leq t_2$  is commonly denoted by  $\Delta \vec{r}$  and is defined by

$$\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1) \text{-----(i)}$$

The displacement vector, which describes the change in position of the particle during the time interval, can be obtained by integrating the velocity function from  $t_1$  to  $t_2$ .

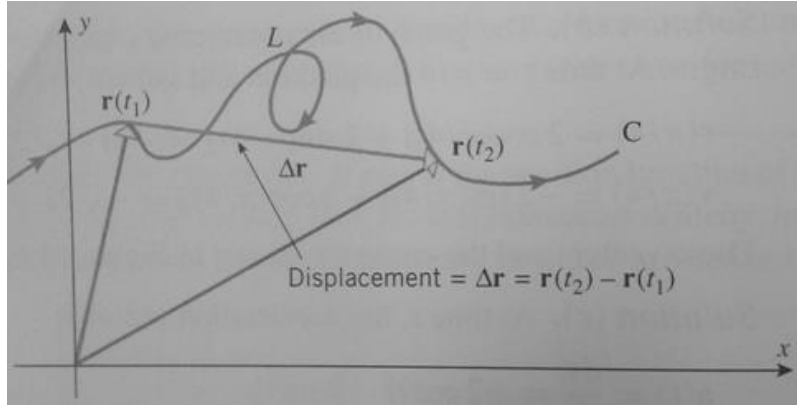


Figure 52

$$\Delta \vec{r} = \int_{t_1}^{t_2} \vec{v}(t) dt = \int_{t_1}^{t_2} \frac{d\vec{r}}{dt} dt = \left[ \vec{r}(t) \right]_{t_1}^{t_2} = \vec{r}(t_2) - \vec{r}(t_1) \text{ -----(ii)}$$

The distance travelled by the particle over the time interval  $t_1 \leq t \leq t_2$  is:

$$s = \int_{t_1}^{t_2} \left| \frac{d\vec{r}}{dt} \right| dt = \int_{t_1}^{t_2} |\vec{v}(t)| dt \text{ -----(iii)}$$

**Q#14:** A particle moves along a curve whose parametric equations are  $x = e^{-t}, y = 2\cos 3t, z = 2\sin 3t$ , Where  $t$  is the time.

- Determine its velocity and acceleration at any time
- Find the magnitudes of the velocity and acceleration at  $t = 0$ .

Answer: The position vector  $\vec{r}$  of the particle is  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\Rightarrow \vec{r} = e^{-t}\hat{i} + 2\cos 3t\hat{j} + 2\sin 3t\hat{k}$$

Then the velocity is  $\vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(e^{-t})\hat{i} + \frac{d}{dt}(2\cos 3t)\hat{j} + \frac{d}{dt}(2\sin 3t)\hat{k}$

$$\frac{d\vec{r}}{dt} = -e^{-t}\hat{i} - 6\sin 3t\hat{j} + 6\cos 3t\hat{k}$$

and the acceleration is:  $\vec{a} = \frac{d\vec{V}}{dt} = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d}{dt}(-e^{-t}\hat{i} - 6\sin 3t\hat{j} + 6\cos 3t\hat{k})$

$$\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = e^{-t}\hat{i} - 18\cos 3t\hat{j} - 18\sin 3t\hat{k}$$

b) At  $t = 0$ ,

Then the velocity is  $\vec{V} = \frac{d\vec{r}}{dt} = -e^{-0}\hat{i} - 6\sin 3 \times 0\hat{j} + 6\cos 3 \times 0\hat{k}$

$$\vec{V} = \frac{d\vec{r}}{dt} = -e^{-0}\hat{i} - 6\sin 0 + 6\cos 0$$

$$\vec{V} = \frac{d\vec{r}}{dt} = -\hat{i} + 6\hat{k} \quad [e^{-0} = \frac{1}{e^0} = 1; \sin 0 = 0; \cos 0 = 1]$$

and the acceleration is:  $\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = e^{-0}\hat{i} - 18\cos 3 \times 0\hat{j} - 18\sin 3 \times 0\hat{k}$

$$\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = \hat{i} - 18\hat{j}$$

Magnitude of the velocity is  $|\vec{V}| = \sqrt{(-1)^2 + (6)^2} = \sqrt{37}$

Magnitude of the acceleration is  $|\vec{a}| = \sqrt{(1)^2 + (-18)^2} = \sqrt{325}$  Answer

**Q#15:** A particle moves along a curve whose parametric equations are  $x = 2t^2$ ,  $y = t^2 - 4t$ ,  $z = 3t - 5$ , where  $t$  is time. Find the component of the velocity at time  $t = 1$  in the direction  $\vec{a} = \hat{i} - 3\hat{j} + 2\hat{k}$

Answer:

The position vector  $\vec{r}$  of the particle is  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$   
 $\Rightarrow \vec{r} = 2t^2\hat{i} + (t^2 - 4t)\hat{j} + (3t - 5)\hat{k}$

Then the velocity is  $\vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}[2t^2\hat{i} + (t^2 - 4t)\hat{j} + (3t - 5)\hat{k}]$   
 $\vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}[2t^2\hat{i}] + \frac{d}{dt}[(t^2 - 4t)\hat{j}] + \frac{d}{dt}[(3t - 5)\hat{k}]$   
 $\vec{V} = \frac{d\vec{r}}{dt} = 4t\hat{i} + (2t - 4)\hat{j} + 3\hat{k}$  -----(i)

The Velocity at  $t = 1$ ;

$$\vec{V} = \frac{d\vec{r}}{dt} = 4t\hat{i} + (2t - 4)\hat{j} + 3\hat{k}$$

$$\vec{V} = \frac{d\vec{r}}{dt} = 4.1\hat{i} + (2.1 - 4)\hat{j} + 3\hat{k}$$

$$\vec{V} = \frac{d\vec{r}}{dt} = 4\hat{i} - 2\hat{j} + 3\hat{k}$$
 -----(ii)

The unit vector of  $\vec{a}$  is  $\hat{e} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{1^2 + (-3)^2 + 2^2}} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{1+9+4}} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}$

$$= \frac{\hat{i}}{\sqrt{14}} - \frac{3\hat{j}}{\sqrt{14}} + \frac{2\hat{k}}{\sqrt{14}} \text{-----(iii)}$$

The component of the velocity in the given direction  $\vec{a} = \hat{i} - 3\hat{j} + 2\hat{k}$  is  $\vec{V} \cdot \hat{e}$ , where  $\hat{e}$  is a unit vector in the direction of a.

$$\vec{V} \cdot \hat{e} = (4\hat{i} - 2\hat{j} + 3\hat{k}) \cdot \left( \frac{\hat{i}}{\sqrt{14}} - \frac{3\hat{j}}{\sqrt{14}} + \frac{2\hat{k}}{\sqrt{14}} \right)$$

$$\vec{V} \cdot \hat{e} = \left( \frac{4}{\sqrt{14}} + \frac{6}{\sqrt{14}} + \frac{6}{\sqrt{14}} \right)$$

$$\vec{V} \cdot \hat{e} = \frac{16}{\sqrt{14}} \text{ Answer}$$

**Q#16:** A particle moves so that its position vector is given by  $\vec{r} = \cos\omega t \hat{i} + \sin\omega t \hat{j}$ , where  $\omega$  is a constant. Show that (a) the velocity  $\vec{V}$  of the particle is perpendicular to  $\vec{r}$ , (b) The acceleration  $\vec{a}$  is directed toward the origin and has magnitude proportional to the distance from the origin (c)  $\vec{r} \times \vec{V} = \text{a constant vector}$

**Answer: Given,**  $\vec{r} = \cos\omega t \hat{i} + \sin\omega t \hat{j}$

a) Then the velocity is  $\vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(\cos\omega t)\hat{i} + \frac{d}{dt}(\sin\omega t)\hat{j}$

$$\vec{V} = \frac{d\vec{r}}{dt} = -\omega \sin\omega t \hat{i} + \omega \cos\omega t \hat{j}$$

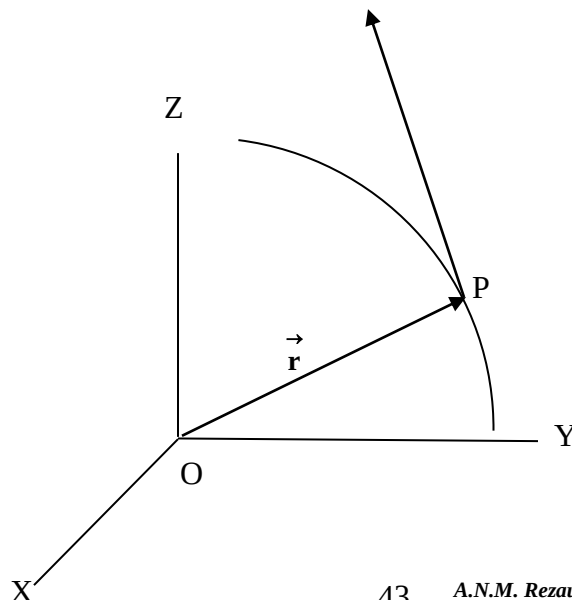


Figure # 53

$$\text{Then } \vec{r} \cdot \vec{V} = (\cos\omega t \hat{i} + \sin\omega t \hat{j}) \cdot (-\omega \sin\omega t \hat{i} + \omega \cos\omega t \hat{j})$$

$$\text{Then } \vec{r} \cdot \vec{V} = (\cos\omega t)(-\omega \sin\omega t) + (\sin\omega t)(\omega \cos\omega t)$$

$$\text{Then } \vec{r} \cdot \vec{V} = -\omega \sin\omega t \cos\omega t + \omega \sin\omega t \cos\omega t$$

$$\text{Then } \vec{r} \cdot \vec{V} = 0$$

Hence  $\vec{r}$  and  $\vec{V}$  are perpendicular.

$$\text{b) The acceleration is: } \vec{a} = \frac{d\vec{V}}{dt} = \frac{d}{dt} (-\omega \sin\omega t \hat{i} + \omega \cos\omega t \hat{j})$$

$$\text{the acceleration is: } \vec{a} = \frac{d\vec{V}}{dt} = -\omega^2 \cos\omega t \hat{i} - \omega^2 \sin\omega t \hat{j}$$

$$\text{the acceleration is: } \vec{a} = \frac{d\vec{V}}{dt} = -\omega^2 (\cos\omega t \hat{i} + \sin\omega t \hat{j}) = -\omega^2 \vec{r}$$

$$\vec{a} = -\omega^2 \vec{r} \text{-----(i)}$$

$$\vec{a} \propto \vec{r}$$

$$|\vec{a}| \propto |\vec{r}| \text{-----(ii)}$$

From equation (1), the acceleration is opposite to the direction of  $\vec{r}$ . i.e. it is directed toward the origin, Its magnitude is proportional to  $|\vec{r}|$  which is the distance from the origin.

$$\text{c) Here } \vec{r} = \cos\omega t \hat{i} + \sin\omega t \hat{j} \text{ and } \vec{V} = -\omega \sin\omega t \hat{i} + \omega \cos\omega t \hat{j}$$

$$\vec{r} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\omega t & \sin\omega t & 0 \\ -\omega \sin\omega t & \omega \cos\omega t & 0 \end{vmatrix}$$

$$= \hat{i}(\sin\omega t \times 0 - 0 \times \omega \cos\omega t) - \hat{j}(\cos\omega t \times 0 - (-\omega \sin\omega t) \times 0) + \hat{k}(\cos\omega t \times \omega \cos\omega t - (-\omega \sin\omega t) \times \sin\omega t)$$

$$\begin{aligned}
&= \hat{i} \times \mathbf{0} - \hat{j} \times \mathbf{0} + \hat{k}(\omega \cos^2 \omega t + \omega \sin^2 \omega t) \\
&= \hat{k} \omega (\cos^2 \omega t + \sin^2 \omega t) \\
&= \hat{k} \omega \cdot 1 \quad [\cos^2 \omega t + \sin^2 \omega t = 1] \\
&= \hat{k} \omega
\end{aligned}$$

**Q#17:** A particle moves along a circular path in such a way that its x- and y-coordinates at time t are  $x = 2\cos t$ ,  $y = 2\sin t$

- Find the instantaneous velocity and speed of the particle at time t.
- Sketch the path of the particle and show the position and velocity vectors at time  $t = \frac{\pi}{4}$  with the velocity vector drawn so that its initial point is at the tip of the position vector
- Show that at each instant the acceleration vector is perpendicular to the velocity vector

**Answer:**

**a)**

Let the position vector at any time t is:

$$\vec{OP} = \vec{r}(t) = x\hat{i} + y\hat{j}$$

At time t, the position vector is:

$$\vec{OP} = \vec{r}(t) = 2\cos t \hat{i} + 2\sin t \hat{j} \quad [\text{Given, } x = 2\cos t, y = 2\sin t] \quad \text{-----(i)}$$

So the instantaneous velocity is:

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t \hat{i} + 2\cos t \hat{j} \quad \text{-----(ii)}$$

So the instantaneous speed is:

$$|\vec{V}(t)| = \sqrt{(-2\sin t)^2 + (2\cos t)^2} = \sqrt{4\sin^2 t + 4\cos^2 t} = 2$$

Answer (b):

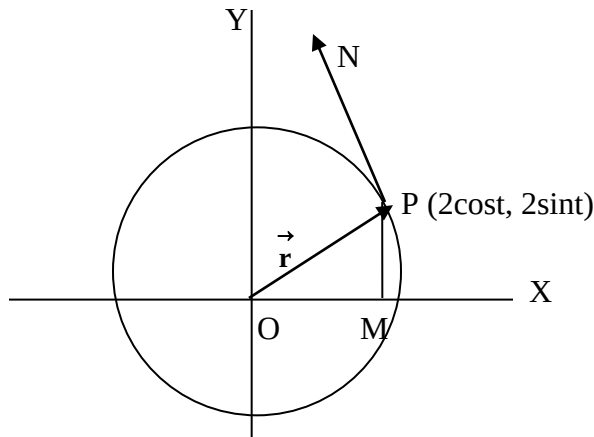


Figure #: 54

$$\vec{OP} = \vec{r}(t) = 2\cos t \hat{i} + 2\sin t \hat{j}$$

Here,  $OM = x = 2\cos t$  and  $PM = y = 2\sin t$

$$\angle POM = t$$

$$\frac{PM}{OP} = \sin t$$

$$\frac{2\sin t}{OP} = \sin t \quad [\text{Given } PM = y = 2\sin t]$$

$$OP \sin t = 2\sin t$$

$$OP = 2 \text{ -----(iii)}$$

Similarly,

$$\frac{OM}{OP} = \cos t$$

$$\frac{2\cos t}{OP} = \cos t$$

$$OP \cos t = 2\cos t$$

$$OP = 2 \text{ -----(iv)}$$

Hence the radius of the circle is  $OP = 2$ .

At time  $t = \frac{\pi}{4}$ , the position and velocity vector of the particles are:

$$\vec{OP} = \vec{r}(t) = 2\cos t \hat{i} + 2\sin t \hat{j} \quad [\text{From (i)}]$$

$$\vec{r}\left(\frac{\pi}{4}\right) = 2\cos\frac{\pi}{4} \hat{i} + 2\sin\frac{\pi}{4} \hat{j}$$

$$\vec{r}\left(\frac{\pi}{4}\right) = 2\frac{1}{\sqrt{2}} \hat{i} + 2\frac{1}{\sqrt{2}} \hat{j}$$

$$\vec{OP} = \vec{r}\left(\frac{\pi}{4}\right) = \sqrt{2} \hat{i} + \sqrt{2} \hat{j} \quad \text{When } t = \frac{\pi}{4}.$$

From (ii),

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t \hat{i} + 2\cos t \hat{j}$$

$$\vec{V}\left(\frac{\pi}{4}\right) = -2\sin\frac{\pi}{4} \hat{i} + 2\cos\frac{\pi}{4} \hat{j}$$

$$\vec{PN} = \vec{V}\left(\frac{\pi}{4}\right) = -2\frac{1}{\sqrt{2}} \hat{i} + 2\frac{1}{\sqrt{2}} \hat{j} = \vec{V}\left(\frac{\pi}{4}\right) = -\sqrt{2} \hat{i} + \sqrt{2} \hat{j}$$

**Answer (c):**

We have,

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t \hat{i} + 2\cos t \hat{j} \quad [\text{From (ii)}]$$

At time  $t$ , the acceleration vector is:

$$\therefore \vec{a}(t) = \frac{d\vec{v}}{dt} = -2\cos t \hat{i} - 2\sin t \hat{j} \text{------(v)}$$

Test: From (ii) & (v),

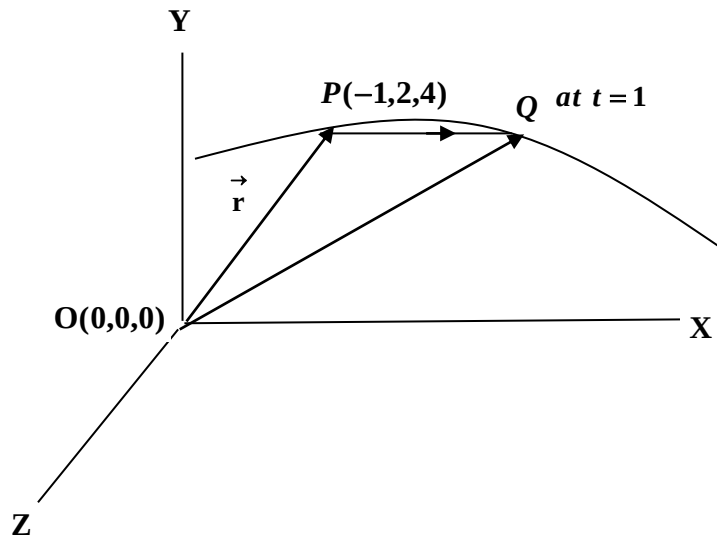
$$\vec{V}(t) \cdot \vec{a}(t) = (-2\sin t \hat{i} + 2\cos t \hat{j}) \cdot (-2\cos t \hat{i} - 2\sin t \hat{j}) = 4\sin t \cos t - 4\sin t \cos t = 0$$

Since the dot product of Velocity Vector (ii) and acceleration Vector (v) is Zero, Hence acceleration vector is perpendicular to the velocity vector.

*(Proved)*

**Q# 18:** A particle moves through 3-space in such a way that its velocity is  $\vec{v}(t) = \hat{i} + t \hat{j} + t^2 \hat{k}$ . Find the co-ordinates of the particle at time  $t=1$  given that the particle is at the point  $(-1,2,4)$  at time  $t=0$

Answer: We have,



**Figure 55**

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \quad \text{Where } \vec{r} \text{ is a position vector.}$$

$$\text{Given, } \vec{v}(t) = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \hat{i} + t\hat{j} + t^2\hat{k}$$

$$\frac{d\vec{r}(t)}{dt} = \hat{i} + t\hat{j} + t^2\hat{k}$$

$$d\vec{r}(t) = (\hat{i} + t\hat{j} + t^2\hat{k})dt \text{ -----(i)}$$

Integrate (i) both sides, we get,

$$\int d\vec{r}(t) = \int (\hat{i} + t\hat{j} + t^2\hat{k})dt$$

$$\int d\vec{r}(t) = \int \hat{i} dt + \int t\hat{j} dt + \int t^2\hat{k} dt$$

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C \text{ -----(ii)}$$

Where C is a vector constant of integration. Since the co-ordinates of the particle at time  $t = 0$  are  $(-1, 2, 4)$ , the position vector at time  $t = 0$  is

We have the position vector

$$\vec{r}(t) = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{r}(0) = (-1)\hat{i} + 2\hat{j} + 4\hat{k} \quad [\text{at time } t = 0, \text{ the position vector is at } (-1, 2, 4)]$$

$$\vec{r}(0) = (-1)\hat{i} + 2\hat{j} + 4\hat{k} \text{ -----(iii)}$$

Again putting  $t = 0$  in (ii), we get,

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C$$

$$\vec{r}(0) = 0\hat{i} + \frac{0^2}{2}\hat{j} + \frac{0^3}{3}\hat{k} + C$$

$$\vec{r}(0) = 0 + 0 + 0 + C$$

$$\vec{r}(0) = C \text{ -----(iv)}$$

Comparing (iii) and (iv), we get,

$$\vec{r}(0) = (-1)\hat{i} + 2\hat{j} + 4\hat{k} = C$$

$$C = -\hat{i} + 2\hat{j} + 4\hat{k} \text{ -----(v)}$$

Putting the value of C in (ii), we get,

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C$$

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} - \hat{i} + 2\hat{j} + 4\hat{k}$$

$$\vec{r}(t) = (t-1)\hat{i} + \left(\frac{t^2}{2} + 2\right)\hat{j} + \left(\frac{t^3}{3} + 4\right)\hat{k} \text{ -----(vi)}$$

Thus, at time  $t = 1$ , the position vector of the particle is

From (vi),

$$\vec{r}(t) = (t-1)\hat{i} + \left(\frac{t^2}{2} + 2\right)\hat{j} + \left(\frac{t^3}{3} + 4\right)\hat{k}$$

$$\vec{r}(1) = (1-1)\hat{i} + \left(\frac{1^2}{2} + 2\right)\hat{j} + \left(\frac{1^3}{3} + 4\right)\hat{k}$$

$$\vec{r}(1) = 0\hat{i} + \frac{5}{2}\hat{j} + \frac{13}{3}\hat{k}.$$

So, the coordinates of the particle at time  $t = 1$  is  $(0, \frac{5}{2}, \frac{13}{3})$  Answer

**Q# 19:** Suppose that a particle moves along a circular helix (**figure 53**) in 3-space so that its position vector at time  $t$  is  $\vec{r}(t) = (4\cos\pi t)\hat{i} + (4\sin\pi t)\hat{j} + t\hat{k}$ . Find the distance traveled and the displacement of the particle during the time interval  $1 \leq t \leq 5$

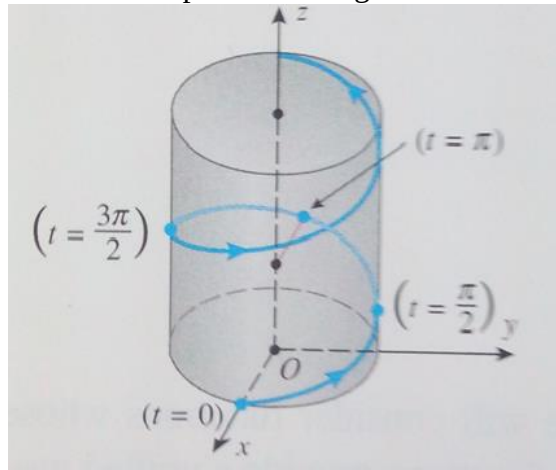


Figure 56

Answer: Given,

$$\vec{r}(t) = (4\cos\pi t)\hat{i} + (4\sin\pi t)\hat{j} + t\hat{k} \quad \text{-----(i)}$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}}{dt} = \frac{d}{dt}(4\cos\pi t\hat{i} + 4\sin\pi t\hat{j} + t\hat{k})$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}}{dt} = -4\sin\pi t \cdot \frac{d}{dt}(\pi t)\hat{i} + 4\cos\pi t \cdot \frac{d}{dt}(\pi t)\hat{j} + \hat{k}$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}}{dt} = -4\sin\pi t.(\pi)\hat{i} + 4\cos\pi t.(\pi)\hat{j} + \hat{k}$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}}{dt} = -4\pi\sin\pi t\hat{i} + 4\pi\cos\pi t\hat{j} + \hat{k} \quad \text{-----(ii)}$$

$$\therefore \left| \vec{v}(t) \right| = \sqrt{(-4\pi\sin\pi t)^2 + (4\pi\cos\pi t)^2 + 1^2}$$

$$\therefore \left| \vec{v}(t) \right| = \sqrt{16\pi^2 \sin^2 \pi t + 16\pi^2 \cos^2 \pi t + 1^2}$$

$$\therefore \left| \vec{v}(t) \right| = \sqrt{16\pi^2 (\sin^2 \pi t + \cos^2 \pi t) + 1}$$

$$\therefore \left| \vec{v}(t) \right| = \sqrt{16\pi^2 \cdot 1 + 1} \quad [\sin^2 t + \cos^2 t = 1]$$

$$\therefore \left| \vec{v}(t) \right| = \sqrt{16\pi^2 + 1} \quad \text{-----(iii)}$$

The displacement vector, **(figure 54)** which describes the change in position of the particle during the time interval, can be obtained by integrating the velocity function from  $t_1$  to  $t_2$ .

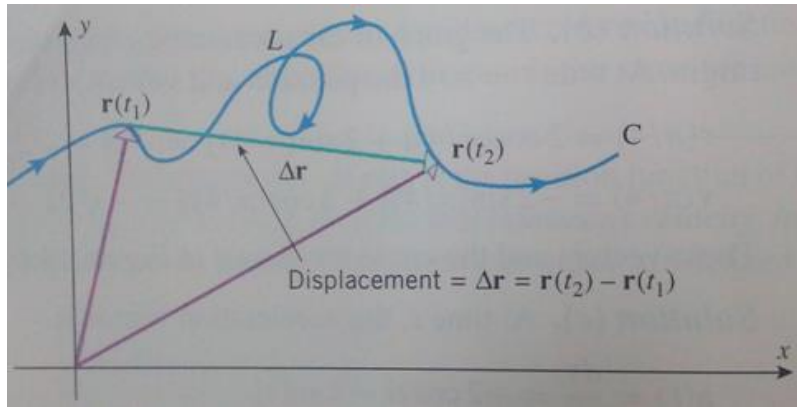


Figure 57

The distance travelled by the particle from time  $t = 1$  to  $t = 5$  is:

$$s = \int_1^5 \left| \frac{d\vec{r}}{dt} \right| dt = \int_1^5 \left| \vec{v}(t) \right| dt$$

$$s = \int_1^5 \left| \frac{d\vec{r}}{dt} \right| dt = \int_1^5 \sqrt{16\pi^2 + 1} dt$$

$$s = \int_1^5 \left| \frac{d\vec{r}}{dt} \right| dt = \sqrt{16\pi^2 + 1} [t]_1^5$$

$$s = \int_1^5 \left| \frac{d\mathbf{r}}{dt} \right| dt = \sqrt{16\pi^2 + 1} [5 - 1]$$

$$s = \int_1^5 \left| \frac{d\mathbf{r}}{dt} \right| dt = \sqrt{16\pi^2 + 1} [4]$$

$$s = \int_1^5 \left| \frac{d\mathbf{r}}{dt} \right| dt = 4\sqrt{16\pi^2 + 1}$$

From (i)

$$\vec{\mathbf{r}}(t) = (4\cos\pi t)\hat{\mathbf{i}} + (4\sin\pi t)\hat{\mathbf{j}} + t\hat{\mathbf{k}}$$

$$\therefore \vec{\mathbf{r}}(5) = (4\cos 5\pi)\hat{\mathbf{i}} + (4\sin 5\pi)\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$$

$$\therefore \vec{\mathbf{r}}(1) = (4\cos\pi)\hat{\mathbf{i}} + (4\sin\pi)\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

Moreover, the displacement over the time interval is:

$$\vec{\Delta\mathbf{r}} = \vec{\mathbf{r}}(5) - \vec{\mathbf{r}}(1)$$

$$\vec{\Delta\mathbf{r}} = (4\cos 5\pi)\hat{\mathbf{i}} + (4\sin 5\pi)\hat{\mathbf{j}} + 5\hat{\mathbf{k}} - (4\cos\pi\hat{\mathbf{i}} + 4\sin\pi\hat{\mathbf{j}} + \hat{\mathbf{k}})$$

$$\vec{\Delta\mathbf{r}} = 4(-1)\hat{\mathbf{i}} + 4(0)\hat{\mathbf{j}} + 5\hat{\mathbf{k}} - [4(-1)\hat{\mathbf{i}} + 4(0)\hat{\mathbf{j}} + \hat{\mathbf{k}}]$$

$$\vec{\Delta\mathbf{r}} = -4\hat{\mathbf{i}} + 5\hat{\mathbf{k}} + 4\hat{\mathbf{i}} - \hat{\mathbf{k}}$$

$$\vec{\Delta\mathbf{r}} = 4\hat{\mathbf{k}}$$

Which tells us that the change in the position of the particle over the time interval was 4 units straight up. Answer

## Gradient, Divergence & Curl

### Gradient:

- Measures the rate and direction of change in a scalar field; the gradient of a scalar field is a vector field. The derivative of a scalar field results in a vector field called the gradient.
- Computes the gradient of a scalar function. That is, it finds the Gradient, the slope, how fast you change, in any given direction.
- A gradient is applied to a scalar quantity that is a function of a 3D vector field: position. The gradient measures the direction in which the scalar quantity changes the most, as well as the rate of change with respect to position. A common example of this is height as a function of latitude and longitude, often applied to mountain ranges. A measure of the slope, and direction of the slope, is often called the gradient.

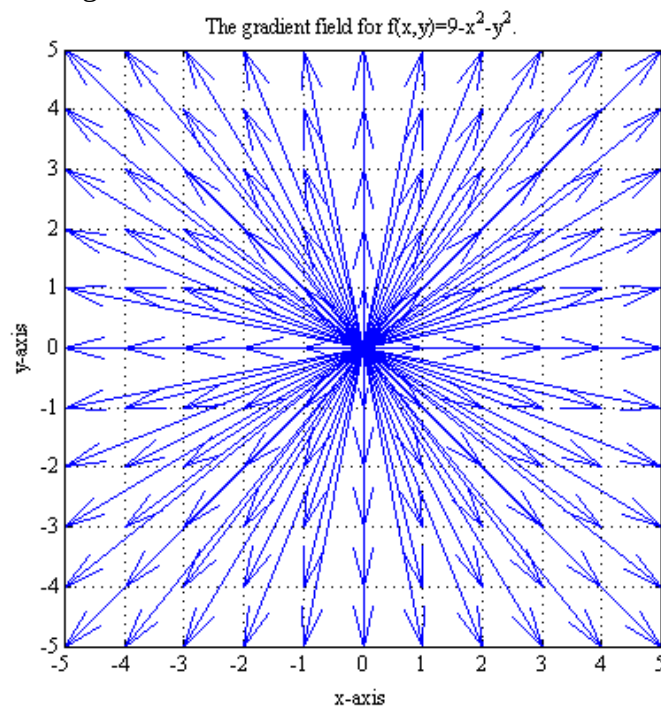


Figure # 58

### Divergence:

- Measures a vector field's tendency to originate from or converge upon a given point.
- Computes the divergence of a vector function. That is, it finds how much "stuff" is leaving a point in space.
- A divergence is applied to a vector as a function of position, yielding a scalar. The divergence actually measures how much the vector function is spreading out. If you are at a location from which the vector field tends to point away in all directions, you will definitely have a positive divergence. If the field points inward toward a point, the divergence in and near that point is negative. If just as much of the vector field points in as out, the divergence will be approximately zero.

- If we again think of  $\vec{F}$  as the velocity field of a flowing fluid then  $\text{div} \vec{F}$  represents the net rate of change of the mass of the fluid flowing from the point  $(x, y, z)$  per unit volume. This can also be thought of as the tendency of a fluid to diverge from a point.

The **divergence** of a vector field is relatively easy to understand intuitively. Imagine that the vector field  $\vec{F}$  below gives the velocity of some fluid flow. It appears that the fluid is exploding outward from the origin

This expansion of fluid flowing with velocity field  $\vec{F}$  is captured by the divergence of  $\vec{F}$ , which we denote  $\text{div} \vec{F}$ . The divergence of the above vector field is positive since the flow is expanding.

In contrast, the below vector field represents fluid flowing so that it compresses as it moves toward the origin. Since this compression of fluid is the opposite of expansion, the divergence of this vector field is negative.

A three-dimensional vector field  $\vec{F}$  showing expansion of fluid flow is shown below.

Again, because of the expansion, we can conclude that  $\text{div} \vec{F} > 0$ .

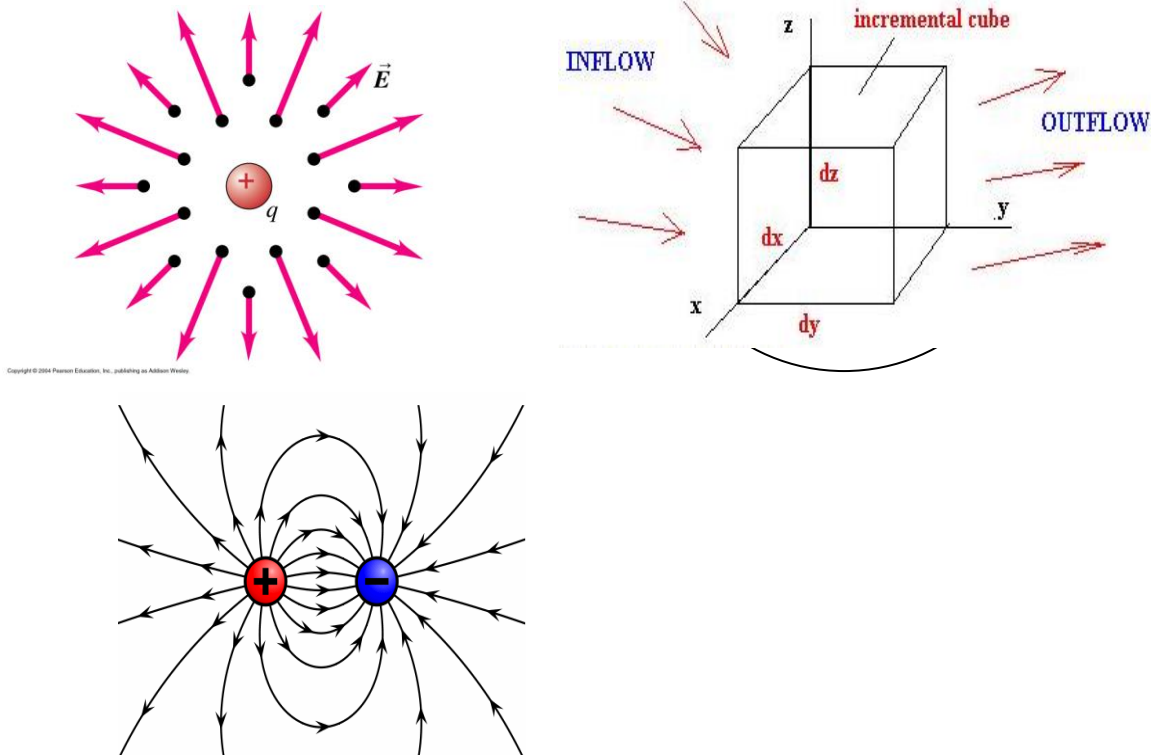


Figure # 59: Divergence of vectors flow field

[†Kvb GKwU point G Pv†R©i intensity/effect n†”Q divergence †hgb: †Kvb GKwU point G heat w`jvg| †hgb: †Kvb GKwU point p G heat w`†j Zv Pvw iw`†K Qwo†q co†e| q point G Zvi intensity/effect KZ? GUvB divergence]

### Curl:

- In vector calculus, the **curl** (or **rotor**) is a vector operator that describes the infinitesimal rotation of a 3-dimensional vector field. At every point in the field, the curl is represented by a vector. The attributes of this vector (length and direction) characterize the rotation at that point.
- The direction of the curl is the axis of rotation, as determined by the right-hand rule, and the magnitude of the curl is the magnitude of rotation. If the vector field represents the flow velocity of a moving fluid, then the curl is the **circulation density** of the fluid. A vector field whose curl is zero is called irrotational. The curl is a form of differentiation for vector fields
- measures a vector field's tendency to rotate about a point; the curl of a vector field is another vector field.
- It computes the rotational aspects of a vector function, maybe people thought how vectors "curl" around a center point, like wind curling around a low pressure on a weather map.
- A curl measures just that, the curl of a vector field. Unlike the divergence, a curl yields a vector. A vector field that tends to point around an axis, such as vectors pointing tangential to a circle, will yield a non-zero curl with the axis around which the curl occurs as the direction. Another example is the velocity field of motion spiraling in or out, such as a whirlpool. Point your right-hand thumb along the direction of the curl. Curl your fingers around this axis. They will curl in the same direction as the vector field. I do not know the names of the texts, but I know there are books available with vector fields to illustrate both divergence and curl.

[GKwU field G wK cwigvb twist/wrapping (c`uvP) Av†Q Zv measurement KivB curling]

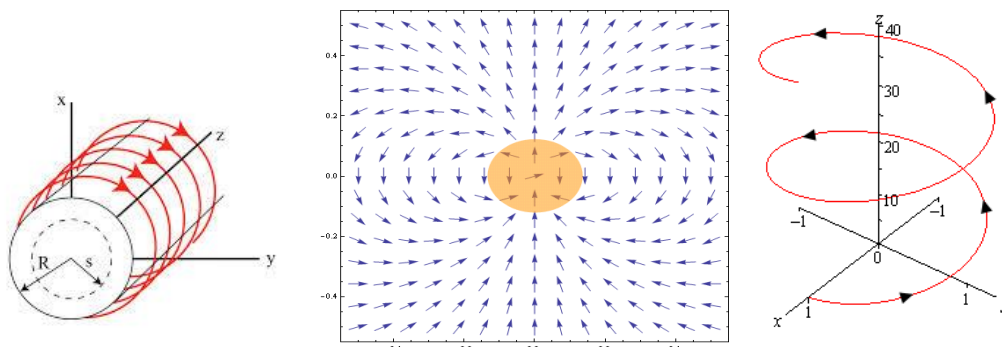


Figure # 60

## Mathematical Expression of Gradient, divergence, curl of a Vector Field

**Vector differential operator:**  $\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$  and is denoted by the symbol  $\vec{\nabla}$  (pronounced 'del' or sometimes 'Nabla')

That is  $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$  -----(i)

Beware!  $\vec{\nabla}$  Cannot exist alone: it is an operator and must operate on a stated scalar function  $\phi(x, y, z)$ .

If F is a vector,  $\vec{\nabla} F$  has no meaning

### **Grad (gradient of a scalar function)**

If a scalar function  $\phi(x, y, z)$  is continuously differentiable with respect to its variables  $x, y, z$ , throughout the region, then the *gradient* of  $\phi$ , Written  $\text{grad } \phi$ , is defined as the vector

$$\text{grad } \phi = \vec{\nabla} \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \text{ -----(ii) where } \phi \text{ is a function of } x, y, z$$

Note that, while  $\phi$  is a scalar function,  $\text{grad } \phi$  is a vector function

**Divergence of a vector field:** If we form the scalar (dot) product of  $\vec{\nabla}$  with a vector function  $\vec{A}(x, y, z) = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$  we get a scalar result called the divergence of  $\vec{A}$ :

$$\text{div } \vec{A} \equiv \vec{\nabla} \cdot \vec{A} \equiv \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_x \hat{i} + A_y \hat{j} + A_z \hat{k})$$

$$\text{div } \vec{A} \equiv \vec{\nabla} \cdot \vec{A} \equiv \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \text{ -----(iii)}$$

**Curl of a vector field:** The curl of a vector field  $\vec{A}(x, y, z) = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$  is defined by

$$\vec{\nabla} \times \vec{A} = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \text{ -----(iv)}$$

**Q# 20:** If  $\phi(x, y, z) = 3x^2y - y^3z^2$ , find  $\vec{\nabla} \phi$  (or  $\text{grad } \phi$ ) at the point (1, -2, -1).

$$\text{Answer: } \vec{\nabla} \phi = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (3x^2y - y^3z^2)$$

$$= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2)$$

$$\begin{aligned}
&= 6xy\hat{i} + (3x^2 - 3y^2z^2)\hat{j} - 2y^3z\hat{k} \\
&= 6(1)(-2)\hat{i} + \{3(1)^2 - 3(-2)^2(-1)^2\}\hat{j} - 2(-2)^3(-1)\hat{k} \\
&= -12\hat{i} - 9\hat{j} - 16\hat{k} \text{ (Answer).}
\end{aligned}$$

Q# 21: Find  $\vec{\nabla} \phi$  if (a)  $\phi = \ln \left| \vec{r} \right|$  (b)  $\phi = \frac{1}{\left| \vec{r} \right|}$

(a) Let  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$  Then  $\left| \vec{r} \right| = \sqrt{x^2 + y^2 + z^2}$  and  $\left| \vec{r} \right|^2 = x^2 + y^2 + z^2$

$$\begin{aligned}
\text{Then } \phi &= \ln \left| \vec{r} \right| = \ln \sqrt{x^2 + y^2 + z^2} = \ln (x^2 + y^2 + z^2)^{1/2} = \frac{1}{2} \ln(x^2 + y^2 + z^2) \\
\therefore \vec{\nabla} \phi &= \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi + \hat{j} \frac{\partial}{\partial y} \phi + \hat{k} \frac{\partial}{\partial z} \phi \\
&= \frac{1}{2} \hat{i} \frac{\partial}{\partial x} \ln(x^2 + y^2 + z^2) + \frac{1}{2} \hat{j} \frac{\partial}{\partial y} \ln(x^2 + y^2 + z^2) + \frac{1}{2} \hat{k} \frac{\partial}{\partial z} \ln(x^2 + y^2 + z^2) \\
&= \frac{1}{2} \hat{i} \left( \frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial x} (x^2 + y^2 + z^2) \right\} + \frac{1}{2} \hat{j} \left( \frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial y} (x^2 + y^2 + z^2) \right\} + \frac{1}{2} \\
&\quad \hat{k} \left( \frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \right\} \\
&= \frac{1}{2} \hat{i} \left( \frac{1}{x^2 + y^2 + z^2} \right) (2x + 0 + 0) + \frac{1}{2} \hat{j} \left( \frac{1}{x^2 + y^2 + z^2} \right) (0 + 2y + 0) + \frac{1}{2} \\
&\quad \hat{k} \left( \frac{1}{x^2 + y^2 + z^2} \right) (0 + 0 + 2z) \\
&= \frac{1}{2} \hat{i} \left( \frac{2x}{x^2 + y^2 + z^2} \right) + \frac{1}{2} \hat{j} \left( \frac{2y}{x^2 + y^2 + z^2} \right) + \frac{1}{2} \hat{k} \left( \frac{2z}{x^2 + y^2 + z^2} \right) \\
&= \frac{1}{2} \times 2 \left\{ \frac{x\hat{i} + y\hat{j} + z\hat{k}}{x^2 + y^2 + z^2} \right\} = \left\{ \frac{x\hat{i} + y\hat{j} + z\hat{k}}{x^2 + y^2 + z^2} \right\} = \frac{\vec{r}}{\left| \vec{r} \right|^2} \text{ Answer}
\end{aligned}$$

(b) Given,  $\phi = \frac{1}{\left| \vec{r} \right|} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}$

$$\begin{aligned}
\vec{\nabla} \phi &= \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi + \hat{j} \frac{\partial}{\partial y} \phi + \hat{k} \frac{\partial}{\partial z} \phi \\
&= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2}
\end{aligned}$$

$$\begin{aligned}
&= \hat{i}(-\frac{1}{2})(x^2 + y^2 + z^2)^{-\frac{1}{2}-1} \frac{\delta}{\delta x}(x^2 + y^2 + z^2) + \hat{j}(-\frac{1}{2})(x^2 + y^2 + z^2)^{-\frac{1}{2}-1} \frac{\delta}{\delta y}(x^2 + y^2 + z^2) \\
&\quad + \hat{k}(-\frac{1}{2})(x^2 + y^2 + z^2)^{-\frac{1}{2}-1} \frac{\delta}{\delta z}(x^2 + y^2 + z^2) \\
&= \hat{i}(-\frac{1}{2})(x^2 + y^2 + z^2)^{-\frac{1}{2}-1}(2x + 0 + 0) + \hat{j}(-\frac{1}{2})(x^2 + y^2 + z^2)^{-\frac{1}{2}-1}(0 + 2y + 0) \\
&\quad + \hat{k}(-\frac{1}{2})(x^2 + y^2 + z^2)^{-\frac{1}{2}-1}(0 + 0 + 2z) \\
&= \hat{i}(-x)(x^2 + y^2 + z^2)^{-\frac{3}{2}} + \hat{j}(-y)(x^2 + y^2 + z^2)^{-\frac{3}{2}} + \hat{k}(-z)(x^2 + y^2 + z^2)^{-\frac{3}{2}} \\
&= \frac{-x\hat{i}}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{-y\hat{j}}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} + \frac{-z\hat{k}}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \\
&= -\frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} = -\frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^1 \cdot (x^2 + y^2 + z^2)^{\frac{1}{2}}} \\
&= -\frac{\vec{r}}{|\vec{r}|^2 \cdot |\vec{r}|} = -\frac{\vec{r}}{|\vec{r}|^3} \text{ Answer}
\end{aligned}$$

**Q#22:** Find the level curve of  $f(x, y) = -x^2 + y^2$  passing through (2, 3). Draw Graph the gradient at the point

Answer: Given,  $f(x, y) = -x^2 + y^2$

$$f(2, 3) = -2^2 + 3^2 = -4 + 9 = 5$$

Hence the level curve is the hyperbola, i.e.

$$f(x, y) = -x^2 + y^2 = 5$$

$$\text{i.e. } -x^2 + y^2 = 5$$

$$\text{i.e. } x^2 - y^2 = -5$$

$$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1 \text{ [This is the equation of hyperbola, i. e. } \frac{x^2}{a} - \frac{y^2}{b} = 1 \text{]} \text{-----(i)}$$

From (i),

$$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1 \quad \Rightarrow x^2 - y^2 = -5 \quad \Rightarrow y^2 = 5 + x^2 \quad \Rightarrow y = \pm\sqrt{5+x^2}$$

$$\Rightarrow y = \pm\sqrt{5+x^2} \text{-----(ii)}$$

| x                     | 0             | -1            | -2      | -3             | 1             | 2       | 3              | -4             | 4              |  |
|-----------------------|---------------|---------------|---------|----------------|---------------|---------|----------------|----------------|----------------|--|
| $y = \pm\sqrt{5+x^2}$ | $\pm\sqrt{5}$ | $\pm\sqrt{6}$ | $\pm 3$ | $\pm\sqrt{14}$ | $\pm\sqrt{6}$ | $\pm 3$ | $\pm\sqrt{14}$ | $\pm\sqrt{19}$ | $\pm\sqrt{19}$ |  |
|                       | $= \pm 2.23$  | $\pm 2.44$    | $\pm 3$ | $\pm 3.74$     | $\pm 2.44$    | $\pm 3$ | $\pm 3.74$     | $\pm 4.35$     | $\pm 4.35$     |  |

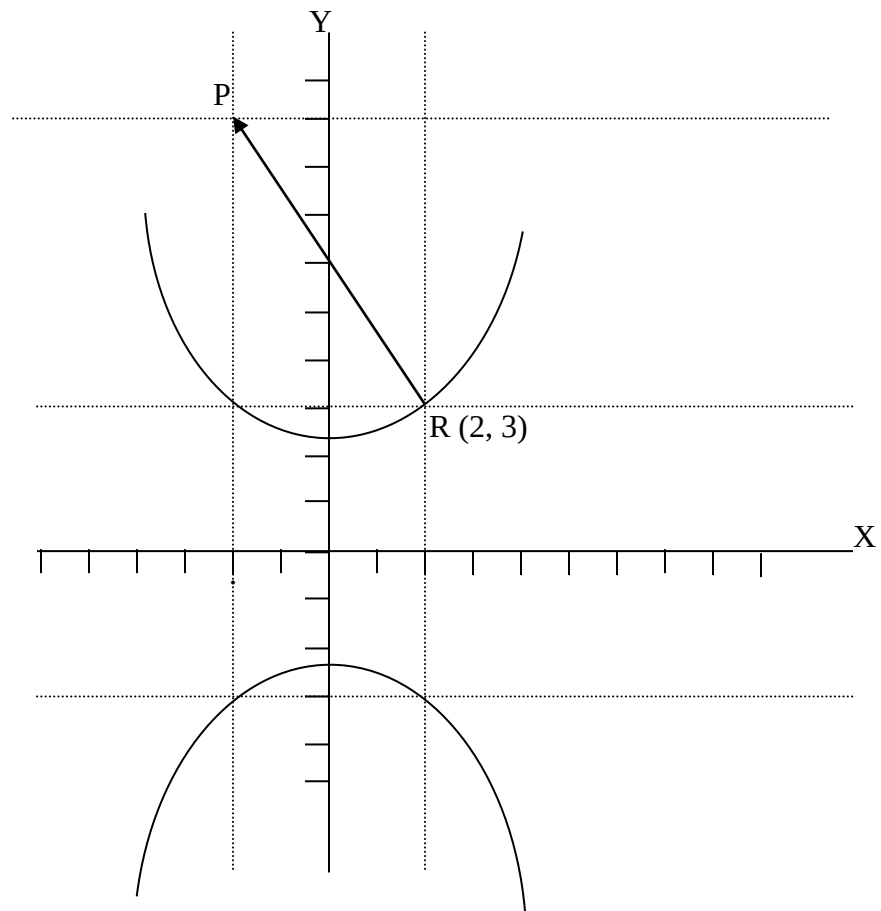


Figure # 61

Now, Gradient of the function, i.e.

$$\vec{\nabla} f(x, y) = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (-x^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (-x^2 + y^2) + \hat{k} \frac{\partial}{\partial z} (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (-x^2) + \hat{i} \frac{\partial}{\partial x} (y^2) + \hat{j} \frac{\partial}{\partial y} (-x^2) + \hat{j} \frac{\partial}{\partial y} (y^2) + \hat{k} \frac{\partial}{\partial z} (-x^2) + \hat{k} \frac{\partial}{\partial z} (y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i}(-2x) + \hat{i} \times 0 + \hat{j} \times 0 + \hat{j}(2y) + \hat{k} \times 0 + \hat{k} \times 0$$

$$\vec{\nabla} f(x, y) = -2x \hat{i} + 2y \hat{j}$$

$$\vec{\nabla} f(2, 3) = -2 \times 2 \hat{i} + 2 \times 3 \hat{j}$$

$$\vec{\nabla} f(2, 3) = -4 \hat{i} + 6 \hat{j}$$

Hence the gradient Vector is  $\vec{RP} = \vec{\nabla} f(2, 3) = -4 \hat{i} + 6 \hat{j}$  Answer

**Q# 23:** Sketch the level curve for the function  $f(x, y) = x^2 + y^2$  through the point (3, 4) and draw the gradient vector at this point.

**Answer:** Given, the function  $f(x, y) = x^2 + y^2$  through the point (3, 4),

$$f(3, 4) = 3^2 + 4^2$$

$$f(3, 4) = 9 + 16 = 25$$

Since  $f(3, 4) = 25$ , the level curve through the point (3, 4) has the equation

$f(x, y) = x^2 + y^2 = 25$ , which is the circle. That is  $x^2 + y^2 = 25$  whose centre (0, 0) and radius 5.

Now,

$$f(x, y) = x^2 + y^2$$

Now, Gradient of the function,

$$\vec{\nabla} f(x, y) = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (x^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (x^2) + \hat{i} \frac{\partial}{\partial x} (y^2) + \hat{j} \frac{\partial}{\partial y} (x^2) + \hat{j} \frac{\partial}{\partial y} (y^2) + \hat{k} \frac{\partial}{\partial z} (x^2) + \hat{k} \frac{\partial}{\partial z} (y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i}(2x) + \hat{i} \times 0 + \hat{j} \times 0 + \hat{j}(2y) + \hat{k} \times 0 + \hat{k} \times 0$$

$$\vec{\nabla} f(x, y) = 2x \hat{i} + 2y \hat{j} \quad \text{-----(i)}$$

The gradient vector at (3,4) is

$$\therefore \vec{\nabla} f(3,4) = 2 \times 3 \hat{i} + 2 \times 4 \hat{j}$$

$$\vec{\nabla} f(3,4) = 6 \hat{i} + 8 \hat{j} \quad \text{-----(ii)}$$

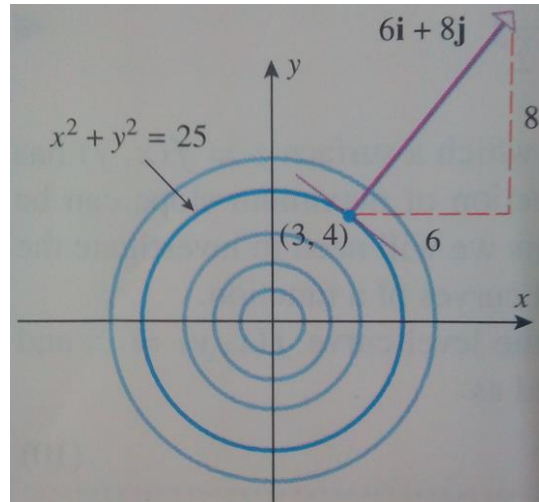


Figure # 62

Hence the gradient vector is perpendicular to the circle at (3,4) .

**Q# 24:** Sketch the gradient field of  $\phi(x, y) = x + y$

**Answer:** Now, the gradient of the function  $\phi(x, y) = x + y$

$$\vec{\nabla} \phi(x, y) = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x + y)$$

$$\vec{\nabla} \phi(x, y) = \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x + y)$$

$$\vec{\nabla} \phi(x, y) = \hat{i} \frac{\partial}{\partial x} (x + y) + \hat{j} \frac{\partial}{\partial y} (x + y) + \hat{k} \frac{\partial}{\partial z} (x + y)$$

$$\vec{\nabla} \phi(x, y) = \hat{i}(1 + 0) + \hat{j}(0 + 1) + \hat{k}(0 + 0)$$

$$\vec{\nabla} \phi(x, y) = \hat{i} + \hat{j}$$

This is the same at each point. A portion of the vector field is sketched in figure below:

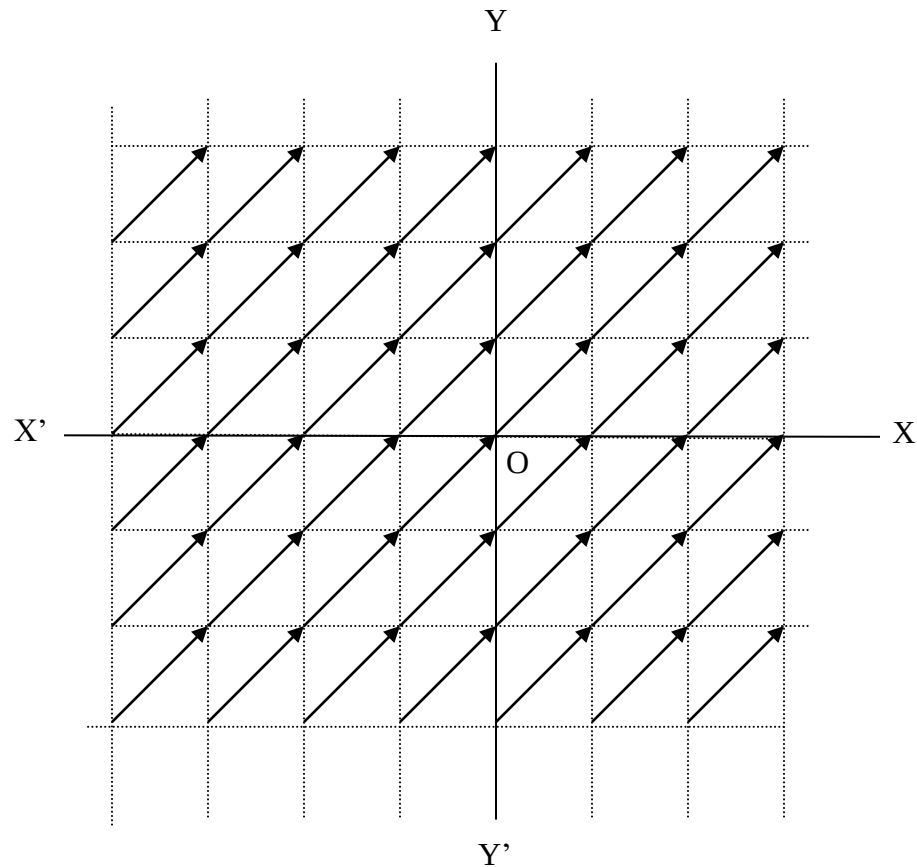


Figure: 63

Q# 25:

Given,  $\Phi(x, y) = x^2 y$

$$\frac{\partial \Phi}{\partial x} = 2xy \dots \dots \dots (i)$$

$$\frac{\partial \Phi}{\partial y} = x^2 \times 1 \dots \dots \dots (ii)$$

$$\therefore \frac{d\Phi}{dx} = \frac{d}{dx}(x^2 y)$$

$$= x^2 \frac{d}{dx}(y) + y \frac{d}{dx}(x^2) \left[ \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u) \right]$$

$$= x^2 \frac{dy}{dx} + y \times 2x \times 1$$

$$= x^2 \frac{dy}{dx} + 2xy$$

$$\begin{aligned}
\therefore \frac{d\Phi}{dy} &= \frac{d}{dy}(x^2 y) \\
&= x^2 \frac{d}{dx}(y) + y \frac{d}{dy}(x^2) [\therefore \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)] \\
&= x^2 \times 1 + y \times 2x \frac{dx}{dy} \\
&= x^2 + 2xy \frac{dx}{dy}
\end{aligned}$$

We know,

$$\begin{aligned}
d\Phi &= \frac{\partial \Phi}{\partial x} \times dx + \frac{\partial \Phi}{\partial y} \times dy \dots \dots \dots (iii) \\
\Rightarrow d\Phi &= 2xy \times dx + x^2 \times dy \dots \dots \dots (iv) [ \text{from equ(i) \& (ii)} ] \\
\Rightarrow \frac{d\Phi}{dx} &= 2xy \times \frac{dx}{dx} + x^2 \frac{dy}{dx} [ \text{Divide both sides by } dx ] \\
\therefore \frac{d\Phi}{dx} &= 2xy + x^2 \frac{dy}{dx}
\end{aligned}$$

from equ (iv) ,

$$\begin{aligned}
d\Phi &= 2xy \times dx + x^2 dy \\
\Rightarrow \frac{d\Phi}{dy} &= 2xy \frac{dx}{dy} + x^2 \frac{dy}{dy} [ \text{Divide both sides by } dy ] \\
\therefore \frac{d\Phi}{dy} &= 2xy \frac{dx}{dy} + x^2
\end{aligned}$$

**Q#26: Show that  $\vec{\nabla} \phi$  is a vector perpendicular to the surface  $\phi(x, y, z) = c$ , where  $c$  is a constant.**

**Answer:** Let  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$  be the position vector to any point  $P(x, y, z)$  on the surface.

$\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$  lies in the tangent plane to the surface at P.

Given,  $\phi(x, y, z) = c$

$$\Rightarrow d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = d(c)$$

$$\Rightarrow d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = 0$$

$$\Rightarrow \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = 0 \dots \dots \dots (i)$$

$$\Rightarrow \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0 \dots \dots \dots (ii)$$

$$\Rightarrow \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k}) = 0$$

$$\Rightarrow \nabla \phi \cdot d\vec{r} = 0 \quad \left[ \nabla \phi = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} ; d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k} \right]$$

So that  $\nabla \phi$  perpendicular to  $d\vec{r}$  and therefore to the surface.

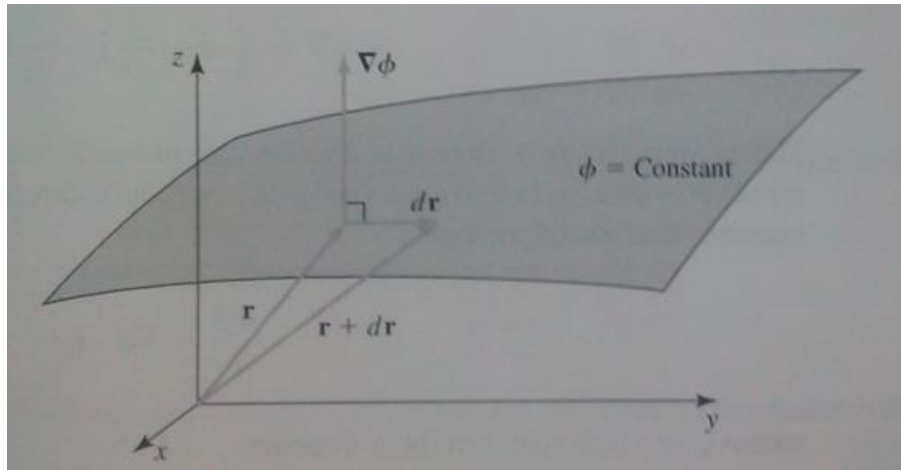


Figure # 64

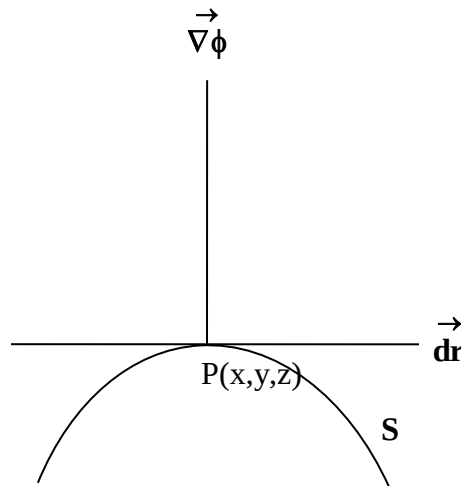


Figure # 65

It is clear that the vector  $\nabla \phi$  is perpendicular (normal) to the tangent vector  $d\vec{r}$  at point  $P(x, y, z)$  that is  $\nabla \phi \cdot d\vec{r} = 0$

So we conclude that  $\nabla \phi$  is normal (perpendicular) to the surface  $\phi(x, y, z) = c$  at  $(x, y, z)$ .

[N.B. We always remember that  $\vec{\nabla} \phi$  is perpendicular to the tangent to the surface but not with surface directly and  $\vec{\nabla} \phi$  is normal to the surface  $\phi(x, y, z) = c$ , Where  $\vec{\nabla} \phi$  is a vector component that is  $\vec{\nabla} \phi = \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right)$ ]

Q# 27: Find a unit normal to the surface  $x^2y + 2xz = 4$  at the point  $(2, -2, 3)$

Answer: Given,  $\phi(x, y, z) = x^2y + 2xz = 4$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \phi = \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \\ &= \hat{i} \frac{\partial}{\partial x} (x^2y + 2xz) + \hat{j} \frac{\partial}{\partial y} (x^2y + 2xz) + \hat{k} \frac{\partial}{\partial z} (x^2y + 2xz) \\ &= (2xy + 2z) \hat{i} + x^2 \hat{j} + 2x \hat{k} \\ &= (2 \times 2 \times (-2) + 2 \times 3) \hat{i} + 2^2 \hat{j} + 2 \times 2 \hat{k} \text{ at the point } (2, -2, 3) \\ &= -2 \hat{i} + 4 \hat{j} + 4 \hat{k} \end{aligned}$$

$$\text{Then a unit normal to the surface} = \frac{-2 \hat{i} + 4 \hat{j} + 4 \hat{k}}{\sqrt{(-2)^2 + (4)^2 + (4)^2}} = -\frac{1}{3} \hat{i} + \frac{2}{3} \hat{j} + \frac{2}{3} \hat{k} \text{ Answer}$$

Q# 28:

Find the level surface of  $F(x, y, z) = x^2 + y^2 + z^2$  passing through  $(1, 1, 1)$ . Graph the gradient at the point.

Answer: Given,  $F(x, y, z) = x^2 + y^2 + z^2$

$$\therefore F(1, 1, 1) = 1^2 + 1^2 + 1^2 = 3$$

$$\text{Hence } F(x, y, z) = x^2 + y^2 + z^2 = 3$$

Because  $F(1, 1, 1) = 3$ , the level surface passing through  $(1, 1, 1)$  is the sphere  $x^2 + y^2 + z^2 = 3$ .

The gradient of the function is

$$F(x, y, z) = x^2 + y^2 + z^2$$

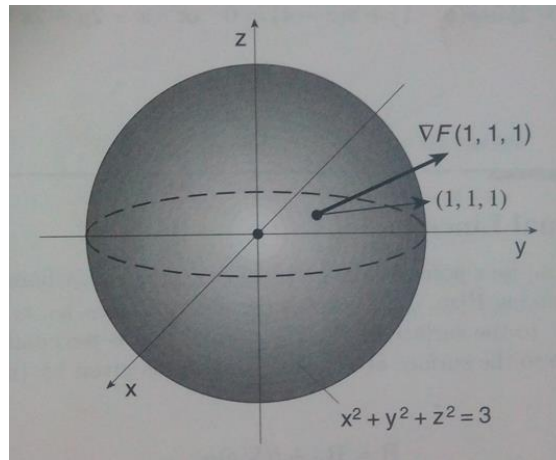
$$F(x, y, z) = x^2 + y^2 + z^2$$

$$\therefore \nabla F(x, y, z) = \left( \frac{\partial F}{\partial x} \hat{i} + \frac{\partial F}{\partial y} \hat{j} + \frac{\partial F}{\partial z} \hat{k} \right)$$

$$\therefore \nabla F(x, y, z) = \left( \frac{\partial F}{\partial x} \hat{i} + \frac{\partial F}{\partial y} \hat{j} + \frac{\partial F}{\partial z} \hat{k} \right) (x^2 + y^2 + z^2)$$

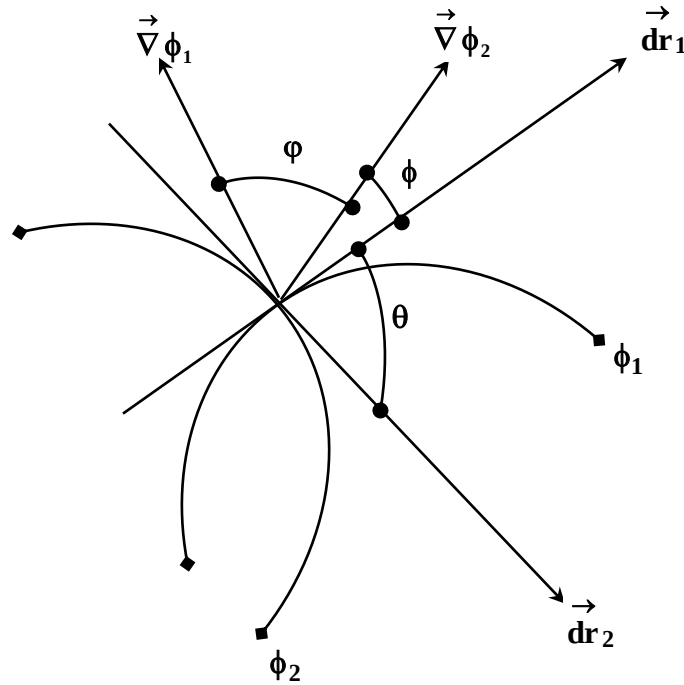
$$\begin{aligned}
&\therefore \nabla F(x, y, z) = \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \\
&\therefore \nabla F(x, y, z) = \hat{i} \frac{\partial}{\partial x} (2x + 0 + 0) + \hat{j} \frac{\partial}{\partial y} (0 + 2y + 0) + \hat{k} \frac{\partial}{\partial z} (0 + 0 + 2z) \\
&\therefore \nabla F(x, y, z) = \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z) \\
&\therefore \nabla F(x, y, z) = 2x\hat{i} + 2y\hat{j} + 2z\hat{k} \\
&\text{And so, at the given point} \\
&\therefore \nabla F(1, 1, 1) = 2\hat{i} + 2\hat{j} + 2\hat{k} \\
&\therefore \nabla F(1, 1, 1) = 2\hat{i} + 2\hat{j} + 2\hat{k}
\end{aligned}$$

The level surface and  $\nabla F(1, 1, 1)$  are illustrated in figure no 63



**Figure # 66**

**Q#29:** Prove that the angle between the surfaces at the point is equal to the angle between the normals to the surfaces at the point.



**Figure # 67**

Here,  $\hat{\eta}_1$  is unit vector of  $\vec{\nabla}\phi_1$  and  $\hat{\eta}_2$  is unit vector of  $\vec{\nabla}\phi_2$

We can write,

$$\hat{\eta}_1 = \frac{\vec{\nabla}\phi_1}{|\vec{\nabla}\phi_1|} \text{ and } \hat{\eta}_2 = \frac{\vec{\nabla}\phi_2}{|\vec{\nabla}\phi_2|}$$

We have,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos\theta$$

$$\therefore \hat{\eta}_1 \cdot \hat{\eta}_2 = |\hat{\eta}_1| |\hat{\eta}_2| \cos\varphi \quad [ \varphi \text{ be the angle between the normals to the surfaces } \phi_1 \text{ and } \phi_2 ]$$

$$\therefore \hat{\eta}_1 \cdot \hat{\eta}_2 = 1 \cdot 1 \cos\varphi \quad [ \text{The length or magnitude of unit vector is 1} ]$$

$$\Rightarrow \frac{\vec{\nabla}\phi_1}{|\vec{\nabla}\phi_1|} \cdot \frac{\vec{\nabla}\phi_2}{|\vec{\nabla}\phi_2|} = 1 \cdot 1 \cos\varphi$$

$$\Rightarrow \vec{\nabla} \phi_1 \cdot \vec{\nabla} \phi_2 = \left| \vec{\nabla} \phi_1 \right| \left| \vec{\nabla} \phi_2 \right| \cos \phi \text{-----(i)}$$

Again, from figure # 64

$$\phi + \theta = 90$$

$$\pm \theta \pm \phi = \pm 90$$

$$\phi - \theta = 0$$

$$\therefore \phi = \theta \text{-----(ii) (Proved)}$$

**Q# 30:** Find the angle between the surfaces  $x^2 + y^2 + z^2 = 9$  and  $z = x^2 + y^2 - 3$  at the point (2,-1, 2)

**Answer:**

$$\text{Given, } z = x^2 + y^2 - 3$$

$$\Rightarrow x^2 + y^2 - z = 3$$

$$\text{Let, } \phi_1(x, y, z) = x^2 + y^2 + z^2 = 9 \text{ and } \phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\begin{aligned} \therefore \nabla \phi_1 &= \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi_1 = \left( \frac{\partial \phi_1}{\partial x} \hat{i} + \frac{\partial \phi_1}{\partial y} \hat{j} + \frac{\partial \phi_1}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi_1 + \hat{j} \frac{\partial}{\partial y} \phi_1 + \hat{k} \frac{\partial}{\partial z} \phi_1 \\ &= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \\ &= 2x \hat{i} + 2y \hat{j} + 2z \hat{k} \\ &= 4 \hat{i} - 2 \hat{j} + 4 \hat{k} \text{ at the point } (2, -1, 2) \end{aligned}$$

$$\text{A normal to } x^2 + y^2 + z^2 = 9 \text{ at } (2, -1, 2) \text{ is } \nabla \phi_1 = 4 \hat{i} - 2 \hat{j} + 4 \hat{k}$$

Again,

$$\begin{aligned} \therefore \nabla \phi_2 &= \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi_2 = \left( \frac{\partial \phi_2}{\partial x} \hat{i} + \frac{\partial \phi_2}{\partial y} \hat{j} + \frac{\partial \phi_2}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi_2 + \hat{j} \frac{\partial}{\partial y} \phi_2 + \hat{k} \frac{\partial}{\partial z} \phi_2 \\ &= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 - z) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 - z) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 - z) \\ &= 2x \hat{i} + 2y \hat{j} - \hat{k} \\ &= 4 \hat{i} - 2 \hat{j} - \hat{k} \text{ at the point } (2, -1, 2) \end{aligned}$$

$$\text{A normal to } x^2 + y^2 - z = 3 \text{ at } (2, -1, 2) \text{ is } \nabla \phi_2 = 4 \hat{i} - 2 \hat{j} - \hat{k}$$

Let  $\phi$  be the angle between the normals to the surfaces at the point (2,-1,2)

Then we have,

$$\nabla \phi_1 \cdot \nabla \phi_2 = |\nabla \phi_1| |\nabla \phi_2| \cos \phi, \quad \text{[from equation no (i), page no 48]}$$

$$\Rightarrow (4 \hat{i} - 2 \hat{j} + 4 \hat{k}) \cdot (4 \hat{i} - 2 \hat{j} - \hat{k}) = \left| 4 \hat{i} - 2 \hat{j} + 4 \hat{k} \right| \left| 4 \hat{i} - 2 \hat{j} - \hat{k} \right| \cos \phi$$

$$\Rightarrow 16 + 4 - 4 = \sqrt{(4)^2 + (-2)^2 + (4)^2} \sqrt{(4)^2 + (-2)^2 + (-1)^2} \cos \phi$$

$$\Rightarrow \cos \phi = \frac{16}{6\sqrt{21}}$$

$$\therefore \phi = \cos^{-1}\left(\frac{16}{6\sqrt{21}}\right) \text{ Answer [ } \phi = \theta \text{ ]}$$

Q# 31: Prove that  $\nabla \cdot \left[ \frac{\vec{f(r)}}{r} \right] = \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)]$

Here  $\left| \vec{r} \right| = r$

L.H.S.

$$\begin{aligned} & \nabla \cdot \left[ \frac{\vec{f(r)}}{r} \right] \\ &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left[ \frac{\vec{f(r)}}{r} (x\hat{i} + y\hat{j} + z\hat{k}) \right] \quad \left[ \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \right] \\ &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left( \hat{i} \frac{f(r)}{r} x + \hat{j} \frac{f(r)}{r} y + \hat{k} \frac{f(r)}{r} z \right) \\ &= \frac{\partial}{\partial x} \frac{f(r)}{r} x + \frac{\partial}{\partial y} \frac{f(r)}{r} y + \frac{\partial}{\partial z} \frac{f(r)}{r} z \text{-----(i)} \\ & \left[ \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1 \right] \end{aligned}$$

$$\begin{aligned} \text{Now, } & \frac{\partial}{\partial x} \frac{f(r)}{r} x \\ &= \frac{\partial}{\partial x} \left\{ \frac{f(r)}{r} x \right\} \\ &= \frac{f(r)}{r} \frac{\partial}{\partial x} (x) + x \frac{\partial}{\partial x} \frac{f(r)}{r} \\ &= \frac{f(r)}{r} \cdot 1 + x \frac{\partial}{\partial x} \frac{f(r)}{r} \\ &= \frac{f(r)}{r} + x \frac{\partial}{\partial x} \frac{f(r)}{r} \\ &= \frac{f(r)}{r} + x \frac{\partial}{\partial x} \{ f(r) r^{-1} \} \\ &= \frac{f(r)}{r} + x \left[ f(r) \frac{\partial}{\partial x} (r^{-1}) + r^{-1} \frac{\partial}{\partial x} \{ f(r) \} \right] \\ &= \frac{f(r)}{r} + x \left[ f(r) (-1) (r^{-2}) \frac{\partial r}{\partial x} + r^{-1} \frac{\partial}{\partial x} \{ f(r) \} \right] \\ &= \frac{f(r)}{r} + x \left[ f(r) (-1) (r^{-2}) \frac{\partial r}{\partial x} + r^{-1} \{ f'(r) \} \frac{\partial r}{\partial x} \right] \\ &= \frac{f(r)}{r} + x \left[ -f(r) (r^{-2}) \frac{\partial r}{\partial x} + r^{-1} \{ f'(r) \} \frac{\partial r}{\partial x} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{f(r)}{r} + x[r^{-1}\{f'(r)\}\frac{\delta r}{\delta x} - f(r)(r^{-2})\frac{\delta r}{\delta x}] \\
&= \frac{f(r)}{r} + x[\frac{1}{r}\{f'(r)\}\frac{\delta r}{\delta x} - f(r)\frac{1}{r^2}\frac{\delta r}{\delta x}] \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{\delta r}{\delta x} \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{\delta}{\delta x}(x^2 + y^2 + z^2)^{1/2} \\
&[ \left| \vec{r} \right| = r = \sqrt{x^2 + y^2 + z^2} = (x^2 + y^2 + z^2)^{1/2}] \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \cdot \frac{\delta}{\delta x}(x^2 + y^2 + z^2) \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \cdot (2x) \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}](x^2 + y^2 + z^2)^{-1/2} \cdot (x) \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}] \frac{x}{(x^2 + y^2 + z^2)^{1/2}} \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}] \frac{x}{\sqrt{x^2 + y^2 + z^2}} \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}] \frac{x}{r} \\
&= \frac{f(r)}{r} + \frac{x^2 f'(r)}{r^2} - \frac{x^2 f(r)}{r^3} \text{-----(ii)}
\end{aligned}$$

Similarly,

$$\frac{\delta f(r)}{\delta y} \frac{\vec{r}}{r} y = \frac{f(r)}{r} + \frac{y^2 f'(r)}{r^2} - \frac{y^2 f(r)}{r^3} \text{-----(iii)}$$

and

$$\frac{\delta f(r)}{\delta z} \frac{\vec{r}}{r} z = \frac{f(r)}{r} + \frac{z^2 f'(r)}{r^2} - \frac{z^2 f(r)}{r^3} \text{-----(iv)}$$

Putting the value of (ii), (iii) and (iv) in (i)

$$\begin{aligned}
\nabla \cdot \left[ \frac{f(r) \vec{r}}{r} \right] &= \frac{\delta f(r)}{\delta x} \frac{\vec{r}}{r} x + \frac{\delta f(r)}{\delta y} \frac{\vec{r}}{r} y + \frac{\delta f(r)}{\delta z} \frac{\vec{r}}{r} z \\
&= \frac{f(r)}{r} + \frac{x^2 f'(r)}{r^2} - \frac{x^2 f(r)}{r^3} + \frac{f(r)}{r} + \frac{y^2 f'(r)}{r^2} - \frac{y^2 f(r)}{r^3} + \frac{f(r)}{r} + \frac{z^2 f'(r)}{r^2} - \frac{z^2 f(r)}{r^3}
\end{aligned}$$

$$\begin{aligned}
&= 3 \frac{f(r)}{r} + \frac{x^2 f'(r)}{r^2} - \frac{x^2 f(r)}{r^3} + \frac{y^2 f'(r)}{r^2} - \frac{y^2 f(r)}{r^3} + \frac{z^2 f'(r)}{r^2} - \frac{z^2 f(r)}{r^3} \\
&= 3 \frac{f(r)}{r} + \frac{f'(r)}{r^2} (x^2 + y^2 + z^2) - \frac{f(r)}{r^3} (x^2 + y^2 + z^2) \\
&= 3 \frac{f(r)}{r} + \frac{f'(r)}{r^2} r^2 - \frac{f(r)}{r^3} r^2 \\
&= 3 \frac{f(r)}{r} + f'(r) - \frac{f(r)}{r} \\
&= 2 \frac{f(r)}{r} + f'(r) \\
&= \frac{1}{r^2} \cdot r^2 \left[ 2 \frac{f(r)}{r} + f'(r) \right] \\
&= \frac{1}{r^2} [2rf(r) + r^2 f'(r)] \\
&= \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)] \quad \left[ \frac{d}{dr} [r^2 f(r)] = 2rf(r) + r^2 f'(r) \right] \\
&\text{(Proved)}
\end{aligned}$$

### Directional derivatives

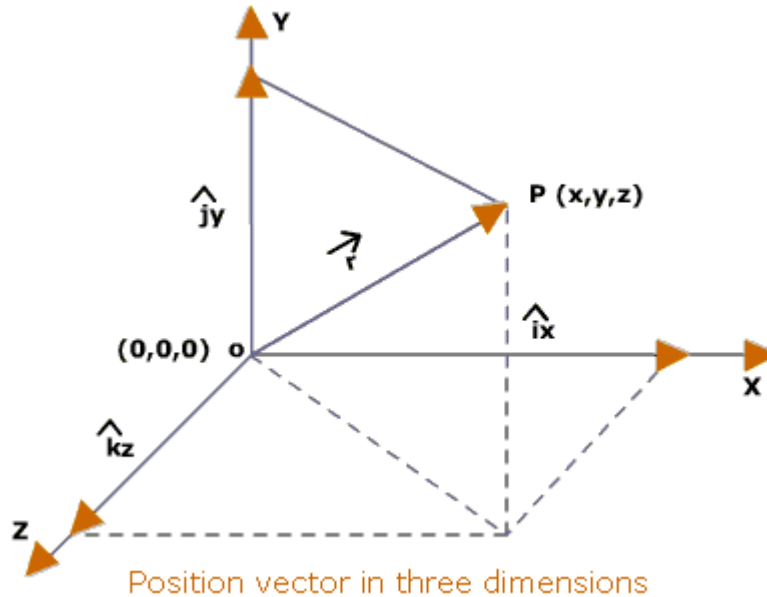


Figure # 68

Let  $\vec{OP}$  is a position vector  $\vec{r}$  where  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$  and  $d\vec{r}$  is a small displacement corresponding to changes  $dx, dy, dz$  in  $x, y, z$  respectively, then

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} \dots\dots\dots(i)$$

If  $\phi(x, y, z)$  is a scalar function at P, then the *gradient* of  $\phi$

$$\text{grad } \phi = \vec{\nabla} \phi = \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \dots\dots\dots(ii)$$

$$\begin{aligned} \text{Then } \text{grad } \phi \cdot d\vec{r} &= \vec{\nabla} \phi \cdot d\vec{r} = \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k}) \\ &= \left( \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k}) \\ &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \\ &= d\phi \\ &= \text{The total differential } d\phi \text{ of } \phi \end{aligned}$$

$$\text{grad } \phi \cdot d\vec{r} = d\phi$$

$$d\phi = \text{grad } \phi \cdot d\vec{r}$$

$$d\phi = \vec{\nabla} \phi \cdot d\vec{r} \dots\dots\dots(iii)$$

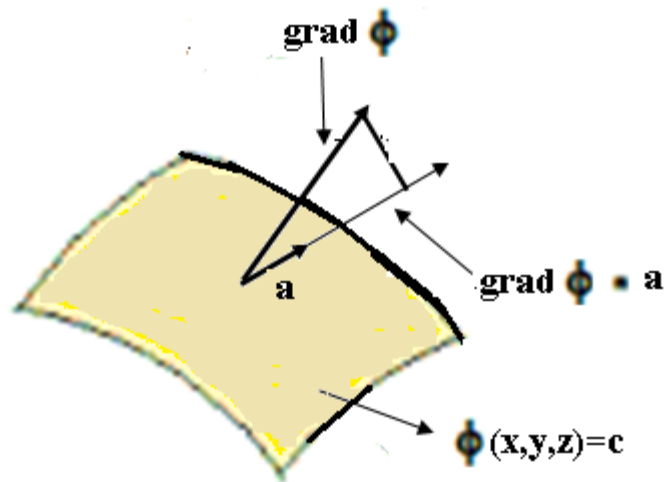


Figure # 69

We have just established that

$$d\phi = dr \cdot \text{grad } \phi$$

If  $ds$  is the small element of arc between  $P(r)$  and  $Q(r+dr)$  then  $ds = \left| \frac{d\vec{r}}{ds} \right|$

$$\frac{d\vec{r}}{ds} = \frac{d\vec{r}}{\left| d\vec{r} \right|}$$

and  $\frac{d\vec{r}}{ds}$  is thus a unit vector in the direction of  $d\vec{r}$ .

$$\therefore \frac{d\phi}{ds} = \frac{d\vec{r}}{ds} \cdot \text{Grad } \phi$$

If we denote this unit vector by  $\hat{a}$ , i.e.  $\frac{d\vec{r}}{ds} = \hat{a}$ , the result becomes

$$\frac{d\phi}{ds} = \hat{a} \cdot \text{Grad } \phi$$

$\frac{d\phi}{ds}$  is the projection of  $\text{grad } \phi$  on the unit vector  $\hat{a}$  is called the directional derivative of  $\phi$  in the direction of  $\hat{a}$ . It gives the rate of change of  $\phi$  with distance measured in the direction of  $\hat{a}$  and  $\frac{d\phi}{ds} = \hat{a} \cdot \text{Grad } \phi$  will be a maximum when  $\hat{a}$  and  $\text{grad } \phi$  have the same direction, since then,  $\hat{a} \cdot \text{grad } \phi = \left| \hat{a} \right| \left| \text{grad } \phi \right| \cos \theta$  and  $\theta$  will be zero

Thus direction of  $\text{grad } \phi$  gives the direction in which the maximum rate of change of  $\phi$  occurs.

**Q# 32:** Find the directional derivative of the function  $\phi = x^2z + 2xy^2 + yz^2$  at the point of (1, 2, -1) in the direction of the vector  $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$ .

We start off with  $\phi = x^2z + 2xy^2 + yz^2$

$$\therefore \nabla \phi = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\therefore \nabla \phi = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla \phi =$$

$$\hat{i} \frac{\partial}{\partial x} (x^2z + 2xy^2 + yz^2) + \hat{j} \frac{\partial}{\partial y} (x^2z + 2xy^2 + yz^2) + \hat{k} \frac{\partial}{\partial z} (x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla \phi = \hat{i}(2xz + 2.1.y^2 + 0) + \hat{j}(0 + 2x.2y + 1.z^2) + \hat{k}(x^2.1 + 0 + y.2z)$$

$$\therefore \nabla \phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

At the point (1, 2, -1)

$$\nabla \phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

$$\therefore \nabla \phi = \hat{i}[2.1.(-1) + 2(2^2)] + \hat{j}[4.1.2 + (-1)^2] + \hat{k}[(1^2 + 2.2.(-1))]$$

$$\therefore \nabla \phi = \hat{i}[-2 + 8] + \hat{j}[8 + 1] + \hat{k}[1 - 4]$$

$$\therefore \nabla \phi = \hat{i}[6] + \hat{j}[9] + \hat{k}(-3)$$

$$\therefore \nabla \phi = 6\hat{i} + 9\hat{j} - 3\hat{k}$$

Next we have to find out the unit vector  $\hat{a}$  where  $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|2\hat{i} + 3\hat{j} - 4\hat{k}|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|4 + 9 + 16|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|}$$

Hence  $\frac{d\phi}{ds} = \hat{a} \cdot \vec{\nabla} \phi$

$$\frac{d\phi}{ds} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|} \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (2\hat{i} + 3\hat{j} - 4\hat{k}) \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (12 + 27 + 12)$$

$$\frac{d\phi}{ds} = \frac{51}{|29|} \text{ Answer}$$

Q#33: Find the directional derivative of the function  $\phi = x^2y + y^2z + z^2x$  at the point of (1, -1, 2) in the direction of the vector  $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$ .

We have  $\phi = x^2y + y^2z + z^2x$

$$\begin{aligned}
\therefore \nabla \phi &= \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi \\
\therefore \nabla \phi &= \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) (x^2 y + y^2 z + z^2 x) \\
\therefore \nabla \phi &= \hat{i} \frac{\delta}{\delta x} (x^2 y + y^2 z + z^2 x) + \hat{j} \frac{\delta}{\delta y} (x^2 y + y^2 z + z^2 x) + \hat{k} \frac{\delta}{\delta z} (x^2 y + y^2 z + z^2 x) \\
\therefore \nabla \phi &= \hat{i} (2xy + 0 + z^2 \cdot 1) + \hat{j} (x^2 \cdot 1 + 2yz + 0) + \hat{k} (0 + y^2 \cdot 1 + 2zx) \\
\therefore \nabla \phi &= \hat{i} (2xy + z^2) + \hat{j} (x^2 + 2yz) + \hat{k} (y^2 + 2zx)
\end{aligned}$$

At the point (1, -1, 2)

$$\begin{aligned}
\therefore \nabla \phi &= \hat{i} (2xy + z^2) + \hat{j} (x^2 + 2yz) + \hat{k} (y^2 + 2zx) \\
\therefore \nabla \phi &= \hat{i} [2 \cdot 1 \cdot (-1) + 2^2] + \hat{j} [1^2 + 2(-1) \cdot 2] + \hat{k} [(-1)^2 + 2 \cdot 2 \cdot 1] \\
\therefore \nabla \phi &= \hat{i} [-2 + 4] + \hat{j} [1 - 4] + \hat{k} [1 + 4] \\
\therefore \nabla \phi &= 2\hat{i} - 3\hat{j} + 5\hat{k}
\end{aligned}$$

Next we have to find out the unit vector  $\hat{a}$  where  $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$

$$\begin{aligned}
\therefore \hat{a} &= \frac{\vec{A}}{|\vec{A}|} \\
\therefore \hat{a} &= \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{4^2 + 2^2 + (-5)^2}} \\
\therefore \hat{a} &= \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{16 + 4 + 25}} \\
\therefore \hat{a} &= \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{45}}
\end{aligned}$$

Hence  $\frac{d\phi}{ds} = \hat{a} \cdot \vec{\nabla} \phi$

$$\begin{aligned}
\frac{d\phi}{ds} &= \frac{(4\hat{i} + 2\hat{j} - 5\hat{k})}{\sqrt{45}} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) \\
\frac{d\phi}{ds} &= \frac{1}{\sqrt{45}} (4\hat{i} + 2\hat{j} - 5\hat{k}) \cdot (2\hat{i} - 3\hat{j} + 5\hat{k}) \\
\frac{d\phi}{ds} &= \frac{1}{\sqrt{45}} (8 - 6 - 25)
\end{aligned}$$

$$\frac{d\phi}{ds} = \frac{1}{|45|} (-23) \text{ Answer}$$

**Q# 34:** If  $\vec{V}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$  Find divergence of  $\vec{V}$  that is  $\text{div}\vec{V}$

$$\begin{aligned} \text{Answer: } \text{div}\vec{V} &= \nabla \cdot \vec{V} = \left( \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) \\ &= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(xyz) - \frac{\partial}{\partial z}(y^2) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{k} \cdot \hat{i} = 0 \text{ etc.}] \\ &= z + xz \text{ Answer} \end{aligned}$$

**Q# 35:** Let  $\vec{V}$  be a constant vector field. Show that  $\text{div}\vec{V} = 0$

Answer: Let,  $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ , where  $a, b, c$  are constants, Then

$$\begin{aligned} \text{div}\vec{V} &= \nabla \cdot \vec{V} = \left( \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (a\hat{i} + b\hat{j} + c\hat{k}) \\ &= \frac{\partial}{\partial x}(a) + \frac{\partial}{\partial y}(b) + \frac{\partial}{\partial z}(c) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{k} \cdot \hat{i} = 0 \text{ etc.}] \\ &= 0 \end{aligned}$$

**Q# 36:** what is solenoidal?

Answer: If  $\vec{A}$  is solenoidal then  $\nabla \cdot \vec{A} = 0$

**Q# 37:** Let  $\vec{V}$  be a constant vector field. Show that  $\text{curl}\vec{V} = 0$

Answer Let,  $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ , where  $a, b, c$  are constants, Then

$$\begin{aligned} \text{curl}\vec{V} &= \nabla \times \vec{V} = \left( \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (a\hat{i} + b\hat{j} + c\hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a & b & c \end{vmatrix} \\ &= \hat{i} \left( \frac{\partial c}{\partial y} - \frac{\partial b}{\partial z} \right) - \hat{j} \left( \frac{\partial c}{\partial x} - \frac{\partial a}{\partial z} \right) + \hat{k} \left( \frac{\partial b}{\partial x} - \frac{\partial a}{\partial y} \right) \\ &= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-0) \\ &= 0 \end{aligned}$$

**Q# 38:** If  $\vec{v}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$  Find  $\text{curl}\vec{V}$

Answer: The curl of a vector field  $\vec{v}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$  is defined by

$$\begin{aligned}\nabla \times \vec{v} &= \left( \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} \\ &= \hat{i} \left[ \frac{\partial}{\partial y}(-y^2) - \frac{\partial}{\partial z}(xyz) \right] - \hat{j} \left[ \frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial z}(xz) \right] + \hat{k} \left[ \frac{\partial}{\partial x}(xyz) - \frac{\partial}{\partial y}(xz) \right] \\ &= \hat{i}[-2y - xy] - \hat{j}[0 - x] + \hat{k}[yz - 0] \\ &= -[2y + xy]\hat{i} + \hat{j}x + \hat{k}yz \\ &= -[2y + xy]\hat{i} + x\hat{j} + yz\hat{k}\end{aligned}$$

**Q# 39:** What is irrotational Field or conservative vector field?

A vector field  $\vec{V}$  for which the curl vanishes, that is:  $\nabla \times \vec{V} = \mathbf{0}$

**Q# 40:** Determine  $\vec{F}$  is a conservative vector field or not where

$$\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$$

Answer:

So all that we need to do is compute the curl and see if we get the zero vector or not.

The curl of a vector field  $\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$  is defined by

$$\begin{aligned}\nabla \times \vec{F} &= \left( \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix} \\ &= \hat{i} \left[ \frac{\partial}{\partial y}(-x^2y^2) - \frac{\partial}{\partial z}(xyz) \right] - \hat{j} \left[ \frac{\partial}{\partial x}(-x^2y^2) - \frac{\partial}{\partial z}(x^2y) \right] \\ &\quad + \hat{k} \left[ \frac{\partial}{\partial x}(xyz) - \frac{\partial}{\partial y}(x^2y) \right] \\ &= \hat{i}[-2x^2y - xy] - \hat{j}[-2xy^2 - 0] + \hat{k}[yz - x^2] \\ &\neq \mathbf{0}\end{aligned}$$

So, the curl isn't the zero vectors and so this vector field is not conservative.

**Q# 41:** If  $\vec{A} = (xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k})$ , find  $\nabla \times \vec{A}$  (or curl A) at the point (1, -1, 1).

Answer:

$$\nabla \times \vec{A} = \left( \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \hat{i} \left[ \frac{\partial}{\partial y} (2yz^4) - \frac{\partial}{\partial z} (-2x^2yz) \right] - \hat{j} \left[ \frac{\partial}{\partial x} (2yz^4) - \frac{\partial}{\partial z} (xz^3) \right] + \hat{k} \left[ \frac{\partial}{\partial x} (-2x^2yz) - \frac{\partial}{\partial y} (xz^3) \right]$$

$$= \hat{i} [2z^4 + 2x^2y] - \hat{j} [0 - 3xz^2] + \hat{k} [-4xyz - 0]$$

$$= \hat{i} [2.1^4 + 2.1^2 \cdot (-1)] - \hat{j} [0 - 3 \cdot 1 \cdot 1^2] + \hat{k} [-4 \cdot 1 \cdot (-1) \cdot 1 - 0]$$

$$= \hat{i} [2 - 2] - \hat{j} [0 - 3] + \hat{k} [4]$$

$$= 3\hat{j} + 4\hat{k}$$

**Q# 42:** Prove that;  $\text{curl}(\phi \vec{F}) = \text{grad} \phi \times \vec{F}$ ; if  $\vec{F}$  is irrotational and  $\phi(x, y, z)$  is a Scalar function.

Answer: Let,  $\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$

$$\therefore \text{curl}(\phi \vec{F}) = \nabla \times (\phi \vec{F})$$

$$= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times [\phi (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k})]$$

$$= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (\phi F_1 \hat{i} + \phi F_2 \hat{j} + \phi F_3 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \phi F_1 & \phi F_2 & \phi F_3 \end{vmatrix}$$

$$= \hat{i} \left[ \frac{\partial}{\partial y} (\phi F_3) - \frac{\partial}{\partial z} (\phi F_2) \right] - \hat{j} \left[ \frac{\partial}{\partial x} (\phi F_3) - \frac{\partial}{\partial z} (\phi F_1) \right] + \hat{k} \left[ \frac{\partial}{\partial x} (\phi F_2) - \frac{\partial}{\partial y} (\phi F_1) \right]$$

$$\begin{aligned}
&= \hat{i}[\phi \frac{\delta}{\delta y}(F_3) + F_3 \frac{\delta \phi}{\delta y} - \phi \frac{\delta}{\delta z}(F_2) - F_2 \frac{\delta \phi}{\delta z}] - \hat{j}[\phi \frac{\delta}{\delta x}(F_3) + F_3 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta z}(F_1) - F_1 \frac{\delta \phi}{\delta z}] \\
&+ \hat{k}[\phi \frac{\delta}{\delta x}(F_2) + F_2 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta y}(F_1) - F_1 \frac{\delta \phi}{\delta y}] \\
&= \phi[\hat{i}\{\frac{\delta}{\delta y}(F_3) - \frac{\delta}{\delta z}(F_2)\} - \hat{j}\{\frac{\delta}{\delta x}(F_3) - \frac{\delta}{\delta z}(F_1)\} + \hat{k}\{\frac{\delta}{\delta x}(F_2) - \frac{\delta}{\delta y}(F_1)\}] \\
&+ \hat{i}[F_3 \frac{\delta \phi}{\delta y} - F_2 \frac{\delta \phi}{\delta z}] - \hat{j}[F_3 \frac{\delta \phi}{\delta x} - F_1 \frac{\delta \phi}{\delta z}] + \hat{k}[F_2 \frac{\delta \phi}{\delta x} - F_1 \frac{\delta \phi}{\delta y}] \\
&= \phi \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ F_1 & F_2 & F_3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta \phi}{\delta x} & \frac{\delta \phi}{\delta y} & \frac{\delta \phi}{\delta z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\
&= \phi(\nabla \times \underline{F}) + \nabla \phi \times \underline{F} \\
&= \phi \cdot \underline{0} + \nabla \phi \times \underline{F} \quad [(\nabla \times \underline{F}) = \underline{0} \text{ for irrotational}] \\
&= \nabla \phi \times \underline{F} \\
&= \underline{\text{grad}} \phi \times \underline{F} \text{ (Proved)}
\end{aligned}$$

**Q# 43:** Prove that  $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

Answer: Let  $\vec{a}, \vec{b}, \vec{c}$  are three vectors

Then  $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$

$\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$

$\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$

$$\begin{aligned}
(\vec{b} \times \vec{c}) &= (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \times (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}) \\
&= \underline{0} + b_1 c_2 \hat{k} + b_1 c_3 (-\hat{j}) + b_2 c_1 (-\hat{k}) + \underline{0} + b_2 c_3 \hat{i} + b_3 c_1 \hat{j} + b_3 c_2 (-\hat{i}) + \underline{0} \\
&= b_1 c_2 \hat{k} - b_1 c_3 \hat{j} - b_2 c_1 \hat{k} + b_2 c_3 \hat{i} + b_3 c_1 \hat{j} - b_3 c_2 \hat{i} \\
&= \hat{i}(b_2 c_3 - b_3 c_2) + \hat{j}(b_3 c_1 - b_1 c_3) + \hat{k}(b_1 c_2 - b_2 c_1)
\end{aligned}$$

Now,

$\vec{a} \times (\vec{b} \times \vec{c})$

$$= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times \{\hat{i}(b_2 c_3 - b_3 c_2) + \hat{j}(b_3 c_1 - b_1 c_3) + \hat{k}(b_1 c_2 - b_2 c_1)\}$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ (b_2c_3 - b_3c_2) & (b_3c_1 - b_1c_3) & (b_1c_2 - b_2c_1) \end{vmatrix} \\
&= \hat{i}\{a_2(b_1c_2 - b_2c_1) - a_3(b_3c_1 - b_1c_3)\} - \hat{j}\{a_1(b_1c_2 - b_2c_1) - a_3(b_2c_3 - b_3c_2)\} \\
&\quad + \hat{k}\{a_1(b_3c_1 - b_1c_3) - a_2(b_2c_3 - b_3c_2)\} \\
&= \hat{i}(a_2b_1c_2 - a_2b_2c_1 - a_3b_3c_1 + a_3b_1c_3) + \hat{j}(-a_1b_1c_2 + a_1b_2c_1 + a_3b_2c_3 - a_3b_3c_2) \\
&\quad + \hat{k}(a_1b_3c_1 - a_1b_1c_3 - a_2b_2c_3 + a_2b_3c_2) \\
&= \hat{i}(a_1b_1c_1 - a_1b_1c_1 + a_2b_1c_2 - a_2b_2c_1 - a_3b_3c_1 + a_3b_1c_3) + \\
&\quad \hat{j}(a_2b_2c_2 - a_2b_2c_2 - a_1b_1c_2 + a_1b_2c_1 + a_3b_2c_3 - a_3b_3c_2) \\
&\quad + \hat{k}(a_3b_3c_3 - a_3b_3c_3 + a_1b_3c_1 - a_1b_1c_3 - a_2b_2c_3 + a_2b_3c_2) \\
&= \hat{i}\{(a_1c_1 + a_2c_2 + a_3c_3)b_1 - (a_1b_1 + a_2b_2 + a_3b_3)c_1\} + \hat{j}\{(a_2c_2 + a_1c_1 + a_3c_3)b_2 \\
&\quad - (a_2b_2 + a_1b_1 + a_3b_3)c_2\} + \hat{k}\{(a_3c_3 + a_1c_1 + a_2c_2)b_3 - (a_3b_3 + a_1b_1 + a_2b_2)c_3\} \\
&= \hat{i}\{(a_1c_1 + a_2c_2 + a_3c_3)b_1 - (a_1b_1 + a_2b_2 + a_3b_3)c_1\} + \hat{j}\{(a_1c_1 + a_2c_2 + a_3c_3)b_2 \\
&\quad - (a_1b_1 + a_2b_2 + a_3b_3)c_2\} + \hat{k}\{(a_1c_1 + a_2c_2 + a_3c_3)b_3 - (a_1b_1 + a_2b_2 + a_3b_3)c_3\} \\
&= \hat{i}(a_1c_1 + a_2c_2 + a_3c_3)b_1 + \hat{j}(a_1c_1 + a_2c_2 + a_3c_3)b_2 + \hat{k}(a_1c_1 + a_2c_2 + a_3c_3)b_3 \\
&\quad - \hat{i}(a_1b_1 + a_2b_2 + a_3b_3)c_1 - \hat{j}(a_1b_1 + a_2b_2 + a_3b_3)c_2 - \hat{k}(a_1b_1 + a_2b_2 + a_3b_3)c_3 \\
&= \hat{i}(a_1c_1 + a_2c_2 + a_3c_3)b_1 + \hat{j}(a_1c_1 + a_2c_2 + a_3c_3)b_2 + \hat{k}(a_1c_1 + a_2c_2 + a_3c_3)b_3 \\
&\quad - \{ \hat{i}(a_1b_1 + a_2b_2 + a_3b_3)c_1 + \hat{j}(a_1b_1 + a_2b_2 + a_3b_3)c_2 + \hat{k}(a_1b_1 + a_2b_2 + a_3b_3)c_3 \} \\
&= (a_1c_1 + a_2c_2 + a_3c_3)(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) - (a_1b_1 + a_2b_2 + a_3b_3)(c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) \\
&= \{(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot (c_1\hat{i} + c_2\hat{j} + c_3\hat{k})\} (b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) - \{(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot \\
&\quad (b_1\hat{i} + b_2\hat{j} + b_3\hat{k})\} (c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) \\
&= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \quad \text{Proved}
\end{aligned}$$

Q# 44:

a) Prove that  $\nabla \times (\nabla \times \vec{A}) = -\nabla^2 \vec{A} + \nabla(\nabla \cdot \vec{A})$

b) Prove that  $\vec{\nabla} \times (\phi \vec{A}) = (\vec{\nabla} \phi) \times \vec{A} + \phi (\vec{\nabla} \times \vec{A})$

Answer a)

Let,  $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$

$$\begin{aligned}
 \therefore \vec{\nabla} \times \vec{A} &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
 &= \hat{i} \left[ \frac{\partial}{\partial y} (A_3) - \frac{\partial}{\partial z} (A_2) \right] - \hat{j} \left[ \frac{\partial}{\partial x} (A_3) - \frac{\partial}{\partial z} (A_1) \right] + \hat{k} \left[ \frac{\partial}{\partial x} (A_2) - \frac{\partial}{\partial y} (A_1) \right] \\
 &= \hat{i} \left[ \frac{\partial}{\partial y} (A_3) - \frac{\partial}{\partial z} (A_2) \right] + \hat{j} \left[ \frac{\partial}{\partial z} (A_1) - \frac{\partial}{\partial x} (A_3) \right] + \hat{k} \left[ \frac{\partial}{\partial x} (A_2) - \frac{\partial}{\partial y} (A_1) \right] \\
 \therefore \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \left\{ \hat{i} \left[ \frac{\partial}{\partial y} (A_3) - \frac{\partial}{\partial z} (A_2) \right] + \hat{j} \left[ \frac{\partial}{\partial z} (A_1) - \frac{\partial}{\partial x} (A_3) \right] \right. \\
 &\quad \left. + \hat{k} \left[ \frac{\partial}{\partial x} (A_2) - \frac{\partial}{\partial y} (A_1) \right] \right\} \\
 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} & \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} & \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \end{vmatrix} \\
 &= \hat{i} \left[ \frac{\partial}{\partial y} \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) - \frac{\partial}{\partial z} \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \right] - \hat{j} \left[ \frac{\partial}{\partial x} \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \right. \\
 &\quad \left. - \frac{\partial}{\partial z} \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \right] + \hat{k} \left[ \frac{\partial}{\partial x} \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) - \frac{\partial}{\partial y} \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \right] \\
 &= \hat{i} \left[ \frac{\partial^2 A_2}{\partial y \partial x} - \frac{\partial^2 A_1}{\partial y^2} \right] - \left[ \frac{\partial^2 A_1}{\partial z^2} - \frac{\partial^2 A_3}{\partial z \partial x} \right] - \hat{j} \left[ \left( \frac{\partial^2 A_2}{\partial x^2} - \frac{\partial^2 A_1}{\partial x \partial y} \right) - \left( \frac{\partial^2 A_3}{\partial z \partial y} - \frac{\partial^2 A_2}{\partial z^2} \right) \right] \\
 &\quad + \hat{k} \left[ \left( \frac{\partial^2 A_1}{\partial x \partial z} - \frac{\partial^2 A_3}{\partial x^2} \right) - \left( \frac{\partial^2 A_3}{\partial y^2} - \frac{\partial^2 A_2}{\partial y \partial z} \right) \right] \\
 &= \hat{i} \left[ -\frac{\partial^2 A_1}{\partial y^2} - \frac{\partial^2 A_1}{\partial z^2} \right] + \hat{j} \left[ -\frac{\partial^2 A_2}{\partial x^2} - \frac{\partial^2 A_2}{\partial z^2} \right] + \hat{k} \left[ -\frac{\partial^2 A_3}{\partial x^2} - \frac{\partial^2 A_3}{\partial y^2} \right] + \hat{i} \left[ \frac{\partial^2 A_2}{\partial y \partial x} \right. \\
 &\quad \left. + \frac{\partial^2 A_3}{\partial z \partial x} \right] + \hat{j} \left[ \frac{\partial^2 A_1}{\partial x \partial y} + \frac{\partial^2 A_3}{\partial z \partial y} \right] + \hat{k} \left[ \frac{\partial^2 A_1}{\partial x \partial z} + \frac{\partial^2 A_2}{\partial y \partial z} \right]
 \end{aligned}$$

$$\begin{aligned}
&= \hat{i} \left[ -\frac{\delta^2 A_1}{\delta x^2} - \frac{\delta^2 A_1}{\delta y^2} - \frac{\delta^2 A_1}{\delta z^2} \right] + \hat{j} \left[ -\frac{\delta^2 A_2}{\delta y^2} - \frac{\delta^2 A_2}{\delta x^2} - \frac{\delta^2 A_2}{\delta z^2} \right] \\
&+ \hat{k} \left[ -\frac{\delta^2 A_3}{\delta z^2} - \frac{\delta^2 A_3}{\delta x^2} - \frac{\delta^2 A_3}{\delta y^2} \right] + \hat{i} \left[ \frac{\delta^2 A_1}{\delta x^2} + \frac{\delta^2 A_2}{\delta y \delta x} + \frac{\delta^2 A_3}{\delta z \delta x} \right] + \hat{j} \left[ \frac{\delta^2 A_1}{\delta x \delta y} \right. \\
&\quad \left. + \frac{\delta^2 A_2}{\delta y^2} + \frac{\delta^2 A_3}{\delta z \delta y} \right] + \hat{k} \left[ \frac{\delta^2 A_3}{\delta z^2} + \frac{\delta^2 A_1}{\delta x \delta z} + \frac{\delta^2 A_2}{\delta y \delta z} \right] \\
&= -\left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_1 \hat{i} - \left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_2 \hat{j} - \left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} \right. \\
&\quad \left. + \frac{\delta^2}{\delta z^2} \right) A_3 \hat{k} + \hat{i} \frac{\delta}{\delta x} \left[ \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] + \hat{j} \frac{\delta}{\delta y} \left[ \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] \\
&\quad + \hat{k} \frac{\delta}{\delta z} \left[ \frac{\delta A_3}{\delta z} + \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} \right] \\
&= -\left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_1 \hat{i} - \left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_2 \hat{j} - \left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} \right. \\
&\quad \left. + \frac{\delta^2}{\delta z^2} \right) A_3 \hat{k} + \hat{i} \frac{\delta}{\delta x} \left[ \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] + \hat{j} \frac{\delta}{\delta y} \left[ \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] \\
&\quad + \hat{k} \frac{\delta}{\delta z} \left[ \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] \\
&= -\left( \frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left( \hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \left( \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} \right. \\
&\quad \left. + \frac{\delta A_3}{\delta z} \right) \\
&= -\nabla^2 (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \nabla \left( \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right) \\
&= -\nabla^2 \vec{A} + \nabla \left[ \left( \hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \right] \\
&= -\nabla^2 \vec{A} + \nabla (\nabla \cdot \vec{A}) \text{ Proved}
\end{aligned}$$

Answer b)

$$\text{Let, } \vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$$

$$\phi \vec{A} = \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k})$$

$$\phi \vec{A} = \phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k}$$

$$\text{L.H.S. } \vec{\nabla} \times (\phi \vec{A}) = \left( \hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k})$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ \phi A_1 & \phi A_2 & \phi A_3 \end{vmatrix} \\
&= \hat{i} \left[ \frac{\delta}{\delta y} (\phi A_3) - \frac{\delta}{\delta z} (\phi A_2) \right] - \hat{j} \left[ \frac{\delta}{\delta x} (\phi A_3) - \frac{\delta}{\delta z} (\phi A_1) \right] + \hat{k} \left[ \frac{\delta}{\delta x} (\phi A_2) - \frac{\delta}{\delta y} (\phi A_1) \right] \\
&= \hat{i} \left[ \phi \frac{\delta}{\delta y} (A_3) + A_3 \frac{\delta \phi}{\delta y} - \phi \frac{\delta}{\delta z} (A_2) - A_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[ \phi \frac{\delta}{\delta x} (A_3) + A_3 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta z} (A_1) - A_1 \frac{\delta \phi}{\delta z} \right] \\
&\quad + \hat{k} \left[ \phi \frac{\delta}{\delta x} (A_2) + A_2 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta y} (A_1) - A_1 \frac{\delta \phi}{\delta y} \right] \\
&= \phi \left[ \hat{i} \left\{ \frac{\delta}{\delta y} (A_3) - \frac{\delta}{\delta z} (A_2) \right\} - \hat{j} \left\{ \frac{\delta}{\delta x} (A_3) - \frac{\delta}{\delta z} (A_1) \right\} + \hat{k} \left\{ \frac{\delta}{\delta x} (A_2) - \frac{\delta}{\delta y} (A_1) \right\} \right] \\
&\quad + \hat{i} \left[ A_3 \frac{\delta \phi}{\delta y} - A_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[ A_3 \frac{\delta \phi}{\delta x} - A_1 \frac{\delta \phi}{\delta z} \right] + \hat{k} \left[ A_2 \frac{\delta \phi}{\delta x} - A_1 \frac{\delta \phi}{\delta y} \right] \\
&= \phi \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ A_1 & A_2 & A_3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta \phi}{\delta x} & \frac{\delta \phi}{\delta y} & \frac{\delta \phi}{\delta z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
&= \phi \left( \hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left( \hat{i} \frac{\delta \phi}{\delta x} + \hat{j} \frac{\delta \phi}{\delta y} + \hat{k} \frac{\delta \phi}{\delta z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&= \phi \left( \hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left( \hat{i} \frac{\delta \phi}{\delta x} + \hat{j} \frac{\delta \phi}{\delta y} + \hat{k} \frac{\delta \phi}{\delta z} \right) \phi \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&= \phi (\vec{\nabla} \times \vec{A}) + (\vec{\nabla} \phi) \times \vec{A} \quad (\text{Proved})
\end{aligned}$$

**Q# 45:** If  $r^2 = x^2 + y^2 + z^2$  then find  $\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial r}{\partial z}$

**We have,**  $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$

$$\therefore \left| \vec{r} \right| = r = \sqrt{x^2 + y^2 + z^2} = (x^2 + y^2 + z^2)^{1/2}$$

$$\therefore \left| \vec{r} \right|^2 = r^2 = \left\{ (x^2 + y^2 + z^2)^{1/2} \right\}^2 = x^2 + y^2 + z^2$$

$$r^2 = x^2 + y^2 + z^2 \text{-----(i)}$$

Differentiating (i) with respect to x partially,

$$\therefore 2r \frac{\partial r}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2 + z^2)$$

$$\therefore 2r \frac{\partial r}{\partial x} = 2x + 0 + 0$$

$$\therefore \frac{\partial r}{\partial x} = \frac{x}{r}$$

similarly

$$\frac{\partial r}{\partial y} = \frac{y}{r},$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

**Q# 46:** Show that  $\nabla \cdot \mathbf{r} = 3$

$$\begin{aligned} \text{Also, } \nabla \cdot \mathbf{r} &= \left( \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3 \end{aligned}$$

$$[ \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0 ]$$

**Q# 47:** Show that  $\mathbf{r} \cdot \mathbf{r} = r^2$

$$\text{Also, } \mathbf{r} \cdot \mathbf{r} = (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$$

$$\begin{aligned} \mathbf{r} \cdot \mathbf{r} &= x^2 + y^2 + z^2 [ \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1 ] \\ &= r^2 [ r^2 = x^2 + y^2 + z^2 ] \end{aligned}$$

Similarly,

$$\vec{A} \cdot \vec{A} = A^2$$

$$\vec{\nabla} \cdot \vec{\nabla} = \nabla^2$$

**Q# 48:** Show that,  $\nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r}$

Proof: **L.H.S** =  $\nabla$

$$\begin{aligned} &= \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \\ &= \hat{i} \frac{\partial}{\partial x} \frac{\partial r}{\partial r} + \hat{j} \frac{\partial}{\partial y} \frac{\partial r}{\partial r} + \hat{k} \frac{\partial}{\partial z} \frac{\partial r}{\partial r} \\ &= \hat{i} \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \hat{j} \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \hat{k} \frac{\partial r}{\partial z} \frac{\partial}{\partial r} \\ &= \left( \hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right) \frac{\partial}{\partial r} \\ &= \left( \hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right) \frac{\partial}{\partial r} [ \frac{\partial r}{\partial x} = \frac{x}{r}; \frac{\partial r}{\partial y} = \frac{y}{r}; \frac{\partial r}{\partial z} = \frac{z}{r} ] \\ &= \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} \frac{\partial}{\partial r} \end{aligned}$$

$$= \frac{\mathbf{r}}{r} \frac{\partial}{\partial r}$$

**Q# 49:** Show that  $\nabla \cdot (\phi \vec{A}) = \phi(\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A}$

Solution

Let  $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$  is a vector and  $\phi$  is a function of a variable or variables

$$\begin{aligned} \text{L.H.S } \nabla \cdot (\phi \vec{A}) &= \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (\phi \vec{A}) \\ &= \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &= \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k}) \\ &= \frac{\delta}{\delta x} (\phi A_1) + \frac{\delta}{\delta y} (\phi A_2) + \frac{\delta}{\delta z} (\phi A_3) \quad [ \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1 ] \\ &= \phi \frac{\delta}{\delta x} (A_1) + A_1 \frac{\delta \phi}{\delta x} + \phi \frac{\delta}{\delta y} (A_2) + A_2 \frac{\delta \phi}{\delta y} + \phi \frac{\delta}{\delta z} (A_3) + A_3 \frac{\delta \phi}{\delta z} \\ &\quad [ \frac{d}{dx} (uv) = u \frac{d}{dx} v + v \frac{d}{dx} u ] \\ &= A_1 \frac{\delta}{\delta x} (\phi) + A_2 \frac{\delta}{\delta y} (\phi) + A_3 \frac{\delta}{\delta z} (\phi) + \phi \frac{\delta}{\delta x} (A_1) + \phi \frac{\delta}{\delta y} (A_2) + \phi \frac{\delta}{\delta z} (A_3) \\ &= A_1 \frac{\delta}{\delta x} (\phi) + A_2 \frac{\delta}{\delta y} (\phi) + A_3 \frac{\delta}{\delta z} (\phi) + \phi \left\{ \frac{\delta}{\delta x} (A_1) + \frac{\delta}{\delta y} (A_2) + \frac{\delta}{\delta z} (A_3) \right\} \\ &= A_1 \frac{\delta \phi}{\delta x} + A_2 \frac{\delta \phi}{\delta y} + A_3 \frac{\delta \phi}{\delta z} + \phi \left\{ \frac{\delta}{\delta x} (A_1) + \frac{\delta}{\delta y} (A_2) + \frac{\delta}{\delta z} (A_3) \right\} \\ &= \frac{\delta \phi}{\delta x} A_1 + \frac{\delta \phi}{\delta y} A_2 + \frac{\delta \phi}{\delta z} A_3 + \phi \left\{ \frac{\delta}{\delta x} (A_1) + \frac{\delta}{\delta y} (A_2) + \frac{\delta}{\delta z} (A_3) \right\} \\ &= \phi \left\{ \frac{\delta}{\delta x} (A_1) + \frac{\delta}{\delta y} (A_2) + \frac{\delta}{\delta z} (A_3) \right\} + \frac{\delta \phi}{\delta x} A_1 + \frac{\delta \phi}{\delta y} A_2 + \frac{\delta \phi}{\delta z} A_3 \\ &= \phi \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left( \frac{\delta \phi}{\delta x} \hat{i} + \frac{\delta \phi}{\delta y} \hat{j} + \frac{\delta \phi}{\delta z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &\quad [ \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1 ] \\ &= \phi \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &= \phi(\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} \quad [ \vec{\nabla} = \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} ] \end{aligned}$$

$$\therefore \nabla \cdot (\phi \vec{A}) = \phi(\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A}$$

**Q# 50:** Show that  $\nabla^2(\ln r) = \frac{1}{r^2}$

$$\text{L.H.S} = \nabla^2(\ln r)$$

$$= \nabla \cdot \nabla(\ln r)$$

$$[ \quad \vec{\nabla} \cdot \vec{\nabla} = \nabla^2 ]$$

$$= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} \ln r \right]$$

$$[ \quad \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ]$$

$$= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} (\ln r) \right]$$

$$= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \frac{1}{r} \right]$$

$$[ \quad \frac{\partial}{\partial r} (\ln r) = \frac{1}{r} ]$$

$$= \nabla \cdot \left[ \frac{\mathbf{r}}{r^2} \right]$$

$$= \nabla \cdot \left[ \frac{1}{r^2} \vec{r} \right]$$

$$= \frac{1}{r^2} [\nabla \cdot \mathbf{r}] + \left[ \nabla \left( \frac{1}{r^2} \right) \right] \cdot \mathbf{r}$$

$$[ \quad \nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ]$$

$$= \frac{3}{r^2} + \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} \left( \frac{1}{r^2} \right) \right] \cdot \mathbf{r}$$

$$[ \quad \nabla \cdot \mathbf{r} = 3 \quad \& \quad \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ]$$

$$= \frac{3}{r^2} + \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right] \cdot \mathbf{r}$$

$$= \frac{3}{r^2} + \frac{\mathbf{r}}{r} (-2r^{-2-1}) \cdot \mathbf{r}$$

$$= \frac{3}{r^2} + \frac{\mathbf{r}}{r} \left( -\frac{2}{r^3} \right) \cdot \mathbf{r}$$

$$= \frac{3}{r^2} - \frac{2}{r^4} (\vec{r} \cdot \vec{r})$$

$$= \frac{3}{r^2} - \frac{2}{r^4} \times r^2 \quad [ \quad \vec{r} \cdot \vec{r} = r^2 ]$$

$$= \frac{3}{r^2} - \frac{2}{r^2}$$

$$= \frac{1}{r^2}.$$

**Q# 51:** Show that  $\nabla^2 r^n = n(n+1)r^{n-2}$

$$\text{L.H.S} = \nabla^2 r^n = \nabla \cdot \nabla(r^n)$$

$$[ \quad \vec{\nabla} \cdot \vec{\nabla} = \nabla^2 ]$$

$$\begin{aligned}
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} r^n \right] & [ \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} n r^{n-1} \right] \\
&= \nabla \cdot \left[ n \mathbf{r} r^{n-1} r^{-1} \right] \\
&= \nabla \cdot \left[ n \mathbf{r} r^{n-1-1} \right] \\
&= \nabla \cdot \left[ n \mathbf{r} r^{n-2} \right] \\
&= n \left[ \nabla \cdot (r r^{n-2}) \right] \\
&= n \left[ \nabla \cdot (r^{n-2} \mathbf{r}) \right] \\
&= n \left[ r^{n-2} (\nabla \cdot \mathbf{r}) + \bar{\nabla} (r^{n-2}) \cdot \mathbf{r} \right] & [ \nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ] \\
&= n \left[ 3r^{n-2} + \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} (r^{n-2}) \cdot \mathbf{r} \right] & [ \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ] \& [ \nabla \cdot \mathbf{r} = 3 ] \\
&= n \left[ 3r^{n-2} + \frac{1}{r} \frac{\partial}{\partial r} (r^{n-2}) (r \cdot \mathbf{r}) \right] \\
&= n \left[ 3r^{n-2} + \frac{1}{r} (n-2) (r^{n-2-1}) (r \cdot \mathbf{r}) \right] \\
&= n \left[ 3r^{n-2} + \frac{1}{r} (n-2) r^{n-3} r^2 \right] & [ r \cdot \mathbf{r} = r^2 ] \\
&= n \left[ 3r^{n-2} + (n-2) r^{n-3} \mathbf{r} \right] \\
&= n \left[ 3r^{n-2} + (n-2) r^{n-3+1} \right] \\
&= n \left[ 3r^{n-2} + (n-2) r^{n-2} \right] \\
&= n[3+n-2]r^{n-2} \\
&= n(n+1)r^{n-2}
\end{aligned}$$

**Q# 52:** Show that  $\nabla \cdot (r^3 \mathbf{r}) = 6r^3$

**L.H.S** =  $\nabla \cdot (r^3 \mathbf{r})$

$$\begin{aligned}
&= r^3 (\nabla \cdot \mathbf{r}) + \nabla (r^3) \cdot \mathbf{r} & [ \nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ] \\
&= 3r^3 + \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} r^3 \right] \cdot \mathbf{r} & [ \nabla \cdot \mathbf{r} = 3 \& \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ] \\
&= 3r^3 + \left[ \frac{\mathbf{r}}{r} 3r^{3-1} \right] \cdot \mathbf{r} \\
&= 3r^3 + \frac{1}{r} 3r^2 (r \cdot \mathbf{r}) \\
&= 3r^3 + 3r(r^2) & [ r \cdot \mathbf{r} = r^2 ] \\
&= 3r^3 + 3r^3
\end{aligned}$$

$$= 6r^3$$

**Q# 53:** Show that  $\nabla \cdot \left[ r \nabla \left( \frac{1}{r^3} \right) \right] = \frac{3}{r^4}$ .

$$\text{L.H.S} = \nabla \cdot \left[ r \nabla \left( \frac{1}{r^3} \right) \right]$$

$$= \nabla \cdot \left[ r \left\{ \frac{r}{r} \frac{\partial}{\partial r} \left( \frac{1}{r^3} \right) \right\} \right]$$

$$[ \nabla = \frac{r}{r} \frac{\partial}{\partial r} ]$$

$$= \nabla \cdot \left[ r \left\{ \frac{r}{r} \frac{\partial}{\partial r} r^{-3} \right\} \right]$$

$$= \nabla \cdot \left[ r \frac{\partial}{\partial r} r^{-3} \right]$$

$$= \nabla \cdot [r(-3)r^{-3-1}]$$

$$= \nabla \cdot [r(-3)r^{-4}]$$

$$= \nabla \cdot [-3r^{-4}r]$$

$$= -3r^{-4}(\nabla \cdot r) + \{ \nabla(-3r^{-4}) \} \cdot r$$

$$[ \nabla \cdot (\phi \vec{A}) = \phi(\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ]$$

$$= -3r^{-4}(\nabla \cdot r) - 3 \{ \nabla(r^{-4}) \} \cdot r$$

$$= -9r^{-4} - 3 \left\{ \frac{r}{r} \frac{\partial}{\partial r} r^{-4} \right\} \cdot r$$

$$[ \nabla \cdot r = 3 ] \& [ \nabla = \frac{r}{r} \frac{\partial}{\partial r} ]$$

$$= -9r^{-4} - 3 \left\{ \frac{r}{r} \frac{\partial}{\partial r} r^{-4} \right\} \cdot r$$

$$= -9r^{-4} - 3 \left\{ \frac{r}{r} (-4)r^{-4-1} \right\} \cdot r$$

$$= -9r^{-4} - 3 \left\{ (-4)r^{-4-1} \times \frac{1}{r} \right\} (r \cdot r)$$

$$= -\frac{9}{r^4} - 3(-4)r^{-5}r^{-1}(r \cdot r)$$

$$= -\frac{9}{r^4} - 3(-4)r^{-6}(r \cdot r)$$

$$= -\frac{9}{r^4} + \frac{12}{r^6} \cdot r^2$$

$$[ r \cdot r = r^2 ]$$

$$= -\frac{9}{r^4} + \frac{12}{r^4}$$

$$= \frac{3}{r^4}$$

**Q# 54:** Show that  $\nabla^2 \left[ \nabla \cdot \left( \frac{\vec{r}}{r^2} \right) \right] = \frac{2}{r^4}$ .

$$\begin{aligned}
 \text{L.H.S} &= \nabla^2 \left[ \nabla \cdot \left( \frac{\vec{r}}{r^2} \right) \right] \\
 &= \nabla^2 \left[ \nabla \cdot \left( \frac{1}{r^2} \vec{r} \right) \right] \\
 &= \nabla^2 \left[ \frac{1}{r^2} (\nabla \cdot \vec{r}) + \left\{ \nabla \left( \frac{1}{r^2} \right) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left( \frac{1}{r^2} \right) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} (-2)(r^{-2-1}) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} (-2)(r^{-3}) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \left\{ \vec{r} \cdot (-2)(r^{-3} \cdot r^{-1}) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \left\{ \vec{r} \cdot (-2)(r^{-4}) \right\} \cdot \vec{r} \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \frac{-2}{r^4} (\vec{r} \cdot \vec{r}) \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} + \frac{-2}{r^4} r^2 \right] \\
 &= \nabla^2 \left[ \frac{3}{r^2} - \frac{2}{r^2} \right] \\
 &= \nabla^2 \left( \frac{1}{r^2} \right) \\
 &= \nabla \cdot \nabla \left( \frac{1}{r^2} \right) \\
 &= \nabla \cdot \left[ \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left( \frac{1}{r^2} \right) \right] \\
 &= \nabla \cdot \left[ \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right]
 \end{aligned}$$

$$[ \nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ]$$

$$[ \nabla \cdot \vec{r} = 3 ] \& [ \nabla = \frac{\vec{r}}{r} \frac{\partial}{\partial r} ]$$

$$[ \vec{r} \cdot \vec{r} = r^2 ]$$

$$[ \vec{\nabla} \cdot \vec{\nabla} = \nabla^2 ]$$

$$[ \nabla = \frac{\vec{r}}{r} \frac{\partial}{\partial r} ]$$

$$\begin{aligned}
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} (-2)(r^{-2-1}) \right] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} (-2)(r^{-3}) \right] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \left( -\frac{2}{r^3} \right) \right] \\
&= \nabla \cdot \left[ r \left( -\frac{2}{r^4} \right) \right] \\
&= \nabla \cdot \left[ \left( -\frac{2}{r^4} \right) \mathbf{r} \right] \\
&= -\frac{2}{r^4} (\nabla \cdot \mathbf{r}) + \left\{ \nabla \left( -\frac{2}{r^4} \right) \right\} \cdot \mathbf{r} \quad [ \nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ] \\
&= -\frac{2}{r^4} (\nabla \cdot \mathbf{r}) - 2 \left\{ \nabla (r^{-4}) \right\} \cdot \mathbf{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} (r^{-4}) \right\} \cdot \mathbf{r} \quad [ \nabla \cdot \mathbf{r} = 3 ] [ \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ] \\
&= -\frac{6}{r^4} - 2 \left\{ \frac{\mathbf{r}}{r} (-4)(r^{-4-1}) \right\} \cdot \mathbf{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \frac{\mathbf{r}}{r} (-4)(r^{-5}) \right\} \cdot \mathbf{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \mathbf{r} (-4)(r^{-5} r^{-1}) \right\} \cdot \mathbf{r} \\
&= -\frac{6}{r^4} + 8 \left\{ \mathbf{r} (r^{-6}) \right\} \cdot \mathbf{r} \\
&= -\frac{6}{r^4} + 8 \frac{1}{r^6} (\mathbf{r} \cdot \mathbf{r}) \\
&= -\frac{6}{r^4} + 8 \frac{1}{r^6} r^2 \quad [ \mathbf{r} \cdot \mathbf{r} = r^2 ] \\
&= -\frac{6}{r^4} + \frac{8}{r^4} \\
&= \frac{2}{r^4} .
\end{aligned}$$

**Q# 55:** Show that  $\text{grad div } \vec{A} = -2\mathbf{r}^{-3}\mathbf{r}$  ; Where,  $\mathbf{A} = \frac{\mathbf{r}}{r}$

**Answer:**  $\text{grad div } \vec{A}$

$$\begin{aligned}
&= \text{grad}(\vec{\nabla} \cdot \vec{A}) \\
&= \vec{\nabla}(\vec{\nabla} \cdot \vec{A})
\end{aligned}$$

Now, **L.H.S** = **grad div A**

$$\begin{aligned}
&= \text{grad}(\vec{\nabla} \cdot \vec{A}) \\
&= \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) \\
&= \vec{\nabla}\left(\vec{\nabla} \cdot \frac{\vec{r}}{r}\right) \quad \left[\text{Given, } \vec{A} = \frac{\vec{r}}{r}\right] \\
&= \vec{\nabla}\left[\vec{\nabla} \cdot \left(\frac{1}{r}\vec{r}\right)\right] \\
&= \vec{\nabla}\left[\frac{1}{r}(\vec{\nabla} \cdot \vec{r}) + \left\{\vec{\nabla}\left(\frac{1}{r}\right)\right\} \cdot \vec{r}\right] \quad \left[\vec{\nabla} \cdot (\phi \vec{A}) = \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}\right] \\
&= \vec{\nabla}\left[\frac{3}{r} + \left\{\frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-1})\right\} \cdot \vec{r}\right] \quad \left[\vec{\nabla} \cdot \vec{r} = 3\right] \text{ \& } \left[\vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}\right] \\
&= \vec{\nabla}\left[\frac{3}{r} + \left\{\frac{\vec{r}}{r}(-1)(r^{-1-1})\right\} \cdot \vec{r}\right] \\
&= \vec{\nabla}\left[\frac{3}{r} + \left\{\frac{\vec{r}}{r}(-1)(r^{-2})\right\} \cdot \vec{r}\right] \\
&= \vec{\nabla}\left[\frac{3}{r} + \left\{r(-1)(r^{-2}r^{-1})\right\} \cdot \vec{r}\right] \\
&= \vec{\nabla}\left[\frac{3}{r} + \left\{r(-1)(r^{-3})\right\} \cdot \vec{r}\right] \\
&= \vec{\nabla}\left[\frac{3}{r} + \left\{(-1)(r^{-3})\right\}(r \cdot r)\right] \\
&= \vec{\nabla}\left[\frac{3}{r} - \frac{1}{r^3}r^2\right] \quad \left[r \cdot r = r^2\right] \\
&= \vec{\nabla}\left[\frac{3}{r} - \frac{1}{r}\right] \\
&= \vec{\nabla}\left(\frac{2}{r}\right) \\
&= 2\frac{\vec{r}}{r} \frac{\partial}{\partial r}\left(\frac{1}{r}\right) \quad \left[\vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}\right] \\
&= 2\frac{\vec{r}}{r} \frac{\partial}{\partial r}(r^{-1}) \\
&= 2\frac{\vec{r}}{r}(-1)r^{-1-1}
\end{aligned}$$

$$\begin{aligned}
&= 2 \frac{\mathbf{r}}{r} (-1) r^{-2} \\
&= 2r(-1)r^{-2}r^{-1} \\
&= 2r(-1)r^{-3} \\
&= -2r^{-3}\mathbf{r}
\end{aligned}$$

Q# 56: i. Show that  $\nabla^2 f(\mathbf{r}) = f''(\mathbf{r}) + \frac{2}{r} f'(\mathbf{r})$

$$\text{L.H.S.} = \nabla^2 f(\mathbf{r})$$

$$\begin{aligned}
&= \nabla \cdot \nabla f(\mathbf{r}) & [\vec{\nabla} \cdot \vec{\nabla} = \nabla^2] \\
&= \nabla \cdot \left( \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} f(\mathbf{r}) \right) & [\nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r}] \\
&= \nabla \cdot \left( \frac{\mathbf{r}}{r} f'(\mathbf{r}) \right) & [\frac{\partial}{\partial r} f(\mathbf{r}) = f'(\mathbf{r})] \\
&= \nabla \cdot \left[ \frac{f'(\mathbf{r})}{r} \mathbf{r} \right] \\
&= \frac{f'(\mathbf{r})}{r} (\nabla \cdot \mathbf{r}) + \left\{ \nabla \left( \frac{f'(\mathbf{r})}{r} \right) \right\} \cdot \mathbf{r} & [\nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A}] \\
&= 3 \frac{f'(\mathbf{r})}{r} + \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} \left\{ \frac{f'(\mathbf{r})}{r} \right\} \right] \cdot \mathbf{r} & [\nabla \cdot \mathbf{r} = 3] \\
&= 3 \frac{f'(\mathbf{r})}{r} + \left[ \frac{1}{r} \left\{ \frac{r f''(\mathbf{r}) - f'(\mathbf{r}) \cdot 1}{r^2} \right\} \right] (\mathbf{r} \cdot \mathbf{r}) & [\frac{d}{dx} \left( \frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}] \\
&= 3 \frac{f'(\mathbf{r})}{r} + \left[ \left\{ \frac{r f''(\mathbf{r}) - f'(\mathbf{r}) \cdot 1}{r^3} \right\} \right] (\mathbf{r} \cdot \mathbf{r}) \\
&= 3 \frac{f'(\mathbf{r})}{r} + \left[ \left\{ \frac{r f''(\mathbf{r}) - f'(\mathbf{r}) \cdot 1}{r^3} \right\} \right] r^2 & [\mathbf{r} \cdot \mathbf{r} = r^2] \\
&= 3 \frac{f'(\mathbf{r})}{r} + \left[ \left\{ \frac{r f''(\mathbf{r}) - f'(\mathbf{r}) \cdot 1}{r} \right\} \right] \\
&= 3 \frac{f'(\mathbf{r})}{r} + f''(\mathbf{r}) - \frac{f'(\mathbf{r})}{r} \\
&= f''(\mathbf{r}) + \frac{2}{r} f'(\mathbf{r})
\end{aligned}$$

ii. If  $\nabla^2 f(\mathbf{r}) = 0$

$$f''(\mathbf{r}) + \frac{2}{r} f'(\mathbf{r}) = 0 \quad [\nabla^2 f(\mathbf{r}) = f''(\mathbf{r}) + \frac{2}{r} f'(\mathbf{r})]$$

$$\frac{\partial^2}{\partial r^2}(f(r)) + \frac{2}{r} \frac{\partial}{\partial r}(f(r)) = 0 \quad [ \quad \frac{\partial}{\partial r}(f(r)) = f'(r) ] \& [ \quad \frac{\partial^2}{\partial r^2}(f(r)) = f''(r) ]$$

$$\frac{\partial}{\partial r} \left\{ \frac{\partial}{\partial r}(f(r)) \right\} + \frac{2}{r} \frac{\partial}{\partial r}(f(r)) = 0$$

$$\frac{\partial}{\partial r}(p) + \frac{2}{r}p = 0, \quad [ \quad p = \frac{\partial}{\partial r}(f(r)) = f'(r) ]$$

$$\frac{\partial p}{\partial r} \times \frac{\partial r}{p} + \frac{2p}{r} \times \frac{\partial r}{p} = 0 \quad [ \text{Multiplying by } \frac{\partial r}{p} \text{ on both sides} ]$$

$$\frac{\partial p}{p} + \frac{2\partial r}{r} = 0$$

$$\int \frac{\partial p}{p} + \int \frac{2\partial r}{r} = \int 0 \quad [ \text{Integrating} ]$$

$$\ln p + 2 \ln r = \ln A$$

$$\ln p + \ln r^2 = \ln A$$

$$\ln pr^2 = \ln A \quad [ \quad \ln ab = \ln a + \ln b ]$$

$$pr^2 = A$$

$$p = Ar^{-2}$$

$$\frac{\partial f(r)}{\partial r} = Ar^{-2}$$

$$\int \frac{\partial f(r)}{\partial r} \partial r = \int Ar^{-2} \partial r$$

$$\int \frac{\partial}{\partial r} \{f(r)\} \partial r = \int Ar^{-2} \partial r$$

Integrating

$$f(r) = A \frac{r^{-2+1}}{-2+1} + B$$

$$f(r) = A \frac{r^{-1}}{-1} + B$$

$$f(r) = -A \frac{1}{r} + B$$

$$f(r) = \frac{-A}{r} + B$$

$$f(r) = B + \frac{c}{r}, \quad \text{Where, } c = -A$$

Q# 57: Show that  $\nabla^2 \left( \frac{1}{r} \right) = 0$

$$\begin{aligned}
\text{L.H.S} &= \nabla^2 \left( \frac{1}{r} \right) \\
&= \nabla \cdot \nabla \left( \frac{1}{r} \right) & [ \quad \vec{\nabla} \cdot \vec{\nabla} = \nabla^2 ] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} \left( \frac{1}{r} \right) \right] & [ \quad \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} (r^{-1}) \right] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} (-1) r^{-1-1} \right] \\
&= \nabla \cdot \left[ \frac{\mathbf{r}}{r} (-1) r^{-2} \right] \\
&= \nabla \cdot \left[ r(-1) r^{-2} r^{-1} \right] \\
&= \nabla \cdot \left[ r(-1) r^{-3} \right] \\
&= \nabla \cdot \left[ -r^{-3} \mathbf{r} \right] \\
&= \nabla \cdot \left[ -\frac{\mathbf{r}}{r^3} \right] \\
&= \nabla \cdot \left[ -r r^{-3} \right] \\
&= \nabla \cdot \left[ -r^{-3} \mathbf{r} \right] \\
&= -r^{-3} (\nabla \cdot \mathbf{r}) - (\nabla r^{-3}) \cdot \mathbf{r} & [ \quad \nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A} ] \\
&= -3r^{-3} - \left\{ \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} (r^{-3}) \right\} \cdot \mathbf{r} & [ \quad \nabla \cdot \mathbf{r} = 3 ] \& [ \quad \nabla = \frac{\mathbf{r}}{r} \frac{\partial}{\partial r} ] \\
&= -\frac{3}{r^3} - \left\{ \frac{1}{r} (-3) r^{-3-1} \right\} (\mathbf{r} \cdot \mathbf{r}) \\
&= -\frac{3}{r^3} - \left\{ \frac{1}{r} (-3) r^{-4} \right\} (\mathbf{r} \cdot \mathbf{r}) \\
&= -\frac{3}{r^3} - \left\{ (-3) r^{-4} r^{-1} \right\} (\mathbf{r} \cdot \mathbf{r}) \\
&= -\frac{3}{r^3} - \left\{ (-3) r^{-4-1} \right\} (\mathbf{r} \cdot \mathbf{r}) \\
&= -\frac{3}{r^3} - \left\{ (-3) r^{-5} \right\} (\mathbf{r} \cdot \mathbf{r}) \\
&= -\frac{3}{r^3} - \left\{ \frac{-3}{r^5} \right\} (\mathbf{r} \cdot \mathbf{r})
\end{aligned}$$

$$= -\frac{3}{r^3} - \left\{ \frac{-3}{r^5} \right\} r^2 \quad [ \quad r \cdot r = r^2 ]$$

$$= -\frac{3}{r^3} + \frac{3}{r^3} = 0$$

**Q# 58: Find  $\vec{\nabla} \phi$  if (a)  $\phi = \ln \left| \vec{r} \right|$  (b)  $\phi = \frac{1}{\left| \vec{r} \right|}$**

Answer:

a) Given  $\phi = \ln \left| \vec{r} \right|$

$$\vec{\nabla} \phi = \vec{\nabla} \ln \left| \vec{r} \right|$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r} \frac{\partial}{\partial r} \ln \left| \vec{r} \right| \quad [ \quad \nabla = \frac{r}{r} \frac{\partial}{\partial r} ]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r} \frac{1}{r}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r^2} \text{ Answer}$$

b) Given,  $\phi = \frac{1}{\left| \vec{r} \right|}$

$$\vec{\nabla} \phi = \vec{\nabla} \frac{1}{\left| \vec{r} \right|}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r} \frac{\partial}{\partial r} \frac{1}{\left| \vec{r} \right|} \quad [ \quad \nabla = \frac{r}{r} \frac{\partial}{\partial r} ]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r} \frac{\partial}{\partial r} \left| \vec{r} \right|^{-1}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r} (-1) \left| \vec{r} \right|^{-1-1} \quad [ \quad \frac{d}{dx} (x^n) = nx^{n-1} ]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{r}{r} (-1) \left| \vec{r} \right|^{-2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \left| \frac{\vec{r}}{r} \right|^{-2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \frac{1}{\left| \frac{\vec{r}}{r} \right|^2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \frac{1}{r^2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r^3} \text{ Answer}$$

c) Prove that i.  $\vec{\nabla} \cdot \left[ \frac{\vec{f}(r)}{r} \right] = f'(r) + \frac{2}{r} f(r)$

ii.  $\vec{\nabla} \cdot \left[ \frac{\vec{f}(r)}{r} \right] = \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)]$

Answer: i.  $\vec{\nabla} \cdot \left[ \frac{\vec{f}(r)}{r} \right]$

$$= \frac{\vec{f}(r)}{r} (\vec{\nabla} \cdot \vec{r}) + \left\{ \vec{\nabla} \frac{\vec{f}(r)}{r} \right\} \cdot \vec{r} \quad [ \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A} ]$$

$$= \frac{3\vec{f}(r)}{r} + \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left( \frac{\vec{f}(r)}{r} \right) \right\} \cdot \vec{r} \quad [ \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r} ]$$

$$= \frac{3\vec{f}(r)}{r} + \left\{ \frac{\vec{r}}{r} \left( \frac{r f'(r) - f(r) \cdot 1}{r^2} \right) \right\} \cdot \vec{r}$$

$$= \frac{3\vec{f}(r)}{r} + \left\{ \left( \frac{r f'(r) - f(r) \cdot 1}{r^3} \right) \right\} (\vec{r} \cdot \vec{r})$$

$$= \frac{3\vec{f}(r)}{r} + \left\{ \left( \frac{r f'(r) - f(r) \cdot 1}{r^3} \right) \right\} r^2 \quad [ \vec{r} \cdot \vec{r} = r^2 ]$$

$$= \frac{3\vec{f}(r)}{r} + \left\{ \left( \frac{r f'(r) - f(r) \cdot 1}{r} \right) \right\}$$

$$= \frac{3\vec{f}(r) + r f'(r) - f(r) \cdot 1}{r}$$

$$= \frac{3\vec{f}(r) + r f'(r) - f(r)}{r}$$

$$\begin{aligned}
&= \frac{2f(r) + rf'(r)}{r} \\
&= \frac{2f(r)}{r} + f'(r) \\
&= f'(r) + \frac{2f(r)}{r} \\
&= f'(r) + \frac{2f(r)}{r} \quad (\text{Proved}) \\
&= \frac{1}{r^2} [r^2 f'(r) + 2rf(r)] \\
&= \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)] \quad (\text{Proved})
\end{aligned}$$

**Q# 59:** Show that:  $\vec{\nabla} r^n = nr^{n-2} \vec{r}$

Answer: L.H.S.

$$\begin{aligned}
&\vec{\nabla} r^n \\
&= \frac{\vec{r}}{r} \frac{\partial}{\partial r} r^n \quad \left[ \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \right] \\
&= \frac{\vec{r}}{r} nr^{n-1} \\
&= r \cdot nr^{n-1} \cdot r^{-1} \\
&= r \cdot nr^{n-2} \\
&= nr^{n-2} \vec{r} \quad (\text{Proved})
\end{aligned}$$

**Q# 60:** Evaluate  $\vec{\nabla} \cdot (\vec{A} \times \vec{r})$  if  $\vec{\nabla} \times \vec{A} = \vec{0}$

Answer:

$$\begin{aligned}
\vec{\nabla} \cdot (\vec{A} \times \vec{r}) &= \vec{r} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{r}) \quad \left[ \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \right] \\
&= \vec{r} \cdot \vec{0} - \vec{A} \cdot (\vec{\nabla} \times \vec{r}) \\
&= \vec{r} \cdot \vec{0} - \vec{A} \cdot \left\{ \vec{\nabla} \times (x\hat{i} + y\hat{j} + z\hat{k}) \right\} \\
&= \vec{0} - \vec{A} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
&= \vec{A} \cdot \hat{i} \left[ \frac{\delta}{\delta y}(z) - \frac{\delta}{\delta z}(y) \right] - \hat{j} \left[ \frac{\delta}{\delta x}(z) - \frac{\delta}{\delta z}(x) \right] + \hat{k} \left[ \frac{\delta}{\delta x}(y) - \frac{\delta}{\delta y}(x) \right] \\
&= \vec{A} \cdot \hat{i} [0 - 0] - \hat{j} [0 - 0] + \hat{k} [0 - 0] \\
&= 0 \text{ Answer}
\end{aligned}$$

Q# 61: If  $\vec{v} = \vec{\omega} \times \vec{r}$ , Prove  $\vec{\omega} = \frac{1}{2} \text{curl } \vec{v}$ , Where  $\vec{\omega}$  is a constant vector.

Answer:

$$\text{Let } \vec{r} = x \hat{i} + y \hat{j} + z \hat{k}, \vec{\omega} = \omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}$$

$$\text{Then } \text{curl } \vec{v} = \nabla \times \vec{v}$$

$$\text{curl } \vec{v} = \nabla \times (\vec{\omega} \times \vec{r})$$

$$\text{curl } \vec{v} = (\vec{r} \cdot \nabla) \vec{\omega} - \vec{r} (\nabla \cdot \vec{\omega}) - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\text{curl } \vec{v} = (x \hat{i} + y \hat{j} + z \hat{k}) \cdot \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} (\nabla \cdot \vec{\omega}) - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} (\nabla \cdot \vec{\omega}) - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\begin{aligned} \text{curl } \vec{v} &= \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} \left( \frac{\delta}{\delta x} \omega_1 + \frac{\delta}{\delta y} \omega_2 + \frac{\delta}{\delta z} \omega_3 \right) \\ &\quad - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r}) \end{aligned}$$

$$\begin{aligned} \text{curl } \vec{v} &= \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} \left( \frac{\delta}{\delta x} \omega_1 + \frac{\delta}{\delta y} \omega_2 + \frac{\delta}{\delta z} \omega_3 \right) \\ &\quad - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r}) \end{aligned}$$

$$\text{curl } \vec{v} = \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} (0 + 0 + 0) - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - 0 - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\vec{\omega} \cdot \nabla) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\begin{aligned} \text{curl } \vec{v} &= \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) \cdot \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \vec{r} \\ &\quad + \vec{\omega} (\nabla \cdot \vec{r}) \end{aligned}$$

$$\text{curl } \vec{v} = \left( x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \left( \omega_1 \frac{\delta}{\delta x} + \omega_2 \frac{\delta}{\delta y} + \omega_3 \frac{\delta}{\delta z} \right) \vec{r} + \vec{\omega} (\nabla \cdot \vec{r})$$

$$\text{curl } \vec{v} = (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \frac{\delta}{\delta x} + \omega_2 \frac{\delta}{\delta y} + \omega_3 \frac{\delta}{\delta z})(x \hat{i} + y \hat{j} + z \hat{k}) + \omega(\vec{\nabla} \cdot \vec{r})$$

$$\text{curl } \vec{v} = (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) + \omega 3 \quad [\quad (\vec{\nabla} \cdot \vec{r}) = 3]$$

$$\text{curl } \vec{v} = (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \omega + 3 \omega$$

$$\text{curl } \vec{v} = (x.0 + y.0 + z.0) + 2 \omega$$

$$\text{curl } \vec{v} = 0 + 2 \omega$$

$$\text{curl } \vec{v} = 2 \omega$$

$$\text{Therefore, } \omega = \frac{1}{2} \text{curl } \vec{v} \quad (\text{Proved})$$

**Q# 62:** Show that  $\phi(x, y, z)$  is any solution of Laplace's equation. Then  $\nabla \phi$  is a vector which both solenoidal and irrotational.

**Answer:**

We have, A solenoidal vector field satisfies  $\vec{\nabla} \cdot \vec{B} = 0$

A vector field  $\vec{V}$  is said to be *irrotational* if its curl is zero. That is, if  $\vec{\nabla} \times \vec{v} = 0$ .

A conservative vector field is also **irrotational**.

Since  $\phi(x, y, z)$  satisfies the Laplace's equation hence,  $\nabla^2 \phi = 0$  or  $\nabla \cdot \nabla \phi = 0$

Therefore,  $\vec{\nabla} \phi$  is solenoidal.

$$\text{and also } \text{curl } \vec{v} = \vec{\nabla} \times (\vec{\nabla} \phi) = (\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z}) \times (\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z}) \phi$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ \frac{\delta \phi}{\delta x} & \frac{\delta \phi}{\delta y} & \frac{\delta \phi}{\delta z} \end{vmatrix}$$

$$= \hat{i} \left( \frac{\delta^2 \phi}{\delta y \delta z} - \frac{\delta^2 \phi}{\delta z \delta y} \right) - \hat{j} \left( \frac{\delta^2 \phi}{\delta x \delta z} - \frac{\delta^2 \phi}{\delta z \delta x} \right) + \hat{k} \left( \frac{\delta^2 \phi}{\delta x \delta y} - \frac{\delta^2 \phi}{\delta y \delta x} \right)$$

$$= \hat{i} \times 0 - \hat{j} \times 0 + \hat{k} \times 0$$

$$= 0$$

**Hence**  $\vec{\nabla} \phi$  is also irrotational. (Proved)

**Q# 63:** If  $\vec{A}$  and  $\vec{B}$  are irrotational then prove that  $\vec{A} \times \vec{B}$  is solenoidal.

Answer: Since  $\vec{A}$  and  $\vec{B}$  are irrotational, hence  $\vec{\nabla} \times \vec{A} = \vec{0}$  and  $\vec{\nabla} \times \vec{B} = \vec{0}$

and if  $\vec{A} \times \vec{B}$  is solenoidal then  $\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = 0$

$$\begin{aligned} \text{L.H.S. } \vec{\nabla} \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \quad [ \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) ] \\ &= \vec{B} \cdot \vec{0} - \vec{A} \cdot \vec{0} \\ &= \vec{0} \text{ (Proved)} \end{aligned}$$

Hence  $\vec{A} \times \vec{B}$  is solenoidal. (Proved)

**Q# 64:** Prove that  $(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{C}) \times (\vec{C} \times \vec{A}) = [\vec{A} \vec{B} \vec{C}]^2$

Solution:

L.H.S

$$(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{C}) \times (\vec{C} \times \vec{A})$$

let,  $\vec{B} \times \vec{C} = \vec{X}$

$$\begin{aligned} \therefore (\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{C}) \times (\vec{C} \times \vec{A}) &= (\vec{A} \times \vec{B}) \cdot (\vec{X}) \times (\vec{C} \times \vec{A}) \\ &= (\vec{A} \times \vec{B}) \cdot [(\vec{X} \cdot \vec{A})\vec{C} - (\vec{X} \cdot \vec{C})\vec{A}] \end{aligned}$$

$$[\text{From Q \# 43, } \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}]$$

$$\therefore (\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{C}) \times (\vec{C} \times \vec{A}) = (\vec{A} \times \vec{B}) \cdot [(\vec{B} \times \vec{C} \cdot \vec{A})\vec{C} - (\vec{B} \times \vec{C} \cdot \vec{C})\vec{A}] \text{ --- (i)}$$

$$[ \vec{B} \times \vec{C} = \vec{X} ]$$

Now,  $\vec{B} \times \vec{C}$

$$= (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \times (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= \hat{i}(b_2 c_3 - b_3 c_2) - \hat{j}(b_1 c_3 - b_3 c_1) + \hat{k}(b_1 c_2 - b_2 c_1)$$

$\therefore \vec{B} \times \vec{C}$

$$= [\hat{i}(b_2 c_3 - b_3 c_2) - \hat{j}(b_1 c_3 - b_3 c_1) + \hat{k}(b_1 c_2 - b_2 c_1)] \cdot (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k})$$

$$= c_1(b_2 c_3 - b_3 c_2) - c_2(b_1 c_3 - b_3 c_1) + c_3(b_1 c_2 - b_2 c_1) [ \because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1 ]$$

$$= c_1 b_2 c_3 - c_1 b_3 c_2 - c_2 b_1 c_3 + c_2 b_3 c_1 + c_3 b_1 c_2 - c_3 b_2 c_1$$

$$\therefore \mathbf{B} \times \mathbf{C} \cdot \mathbf{C} = 0 \text{ -----(ii)}$$

From (i)

$$\begin{aligned} \therefore (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) &= (\mathbf{A} \times \mathbf{B}) \cdot [(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A})\mathbf{C} - (\mathbf{B} \times \mathbf{C} \cdot \mathbf{C})\mathbf{A}] \\ &= (\mathbf{A} \times \mathbf{B}) \cdot [(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A})\mathbf{C} - 0] \quad [\mathbf{B} \times \mathbf{C} \cdot \mathbf{C} = 0; \text{ from (ii)}] \\ &= (\mathbf{A} \times \mathbf{B}) \cdot [(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A})\mathbf{C}] \\ &= [\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}][\mathbf{B} \times \mathbf{C} \cdot \mathbf{A}] \\ &= [\mathbf{ABC}][\mathbf{ABC}] \end{aligned}$$

**We have, Scalar triple product:**  $\vec{\mathbf{A}} \cdot (\vec{\mathbf{B}} \times \vec{\mathbf{C}})$  or  $\vec{\mathbf{B}} \cdot (\vec{\mathbf{C}} \times \vec{\mathbf{A}})$  or  $\vec{\mathbf{C}} \cdot (\vec{\mathbf{A}} \times \vec{\mathbf{B}})$  are known as a scalar triple product. It is symbolically denoted by  $[\mathbf{ABC}]$  or  $[\mathbf{BCA}]$  or  $[\mathbf{CAB}]$

$$= [\mathbf{ABC}]^2$$

(Proved)

## Line integrals

Categories of curves:

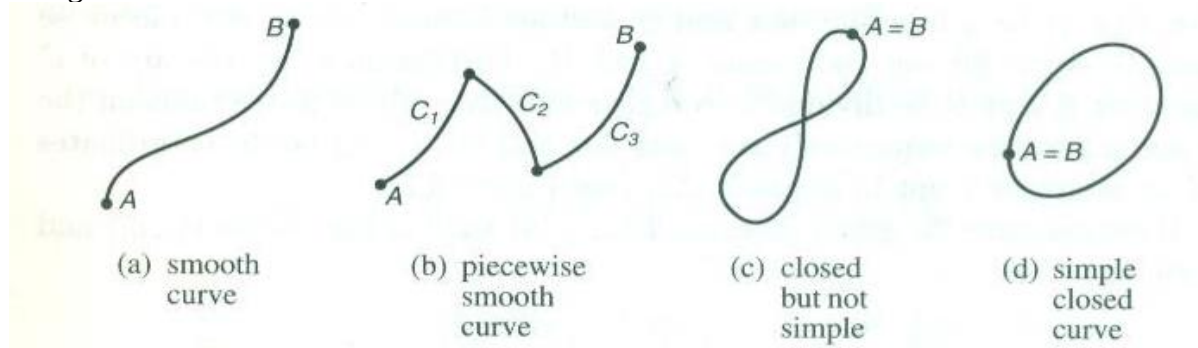


Figure # 70

1.  $C$  is smooth curve
2.  $C$  is piecewise smooth if it consists of a finite number of smooth curves  $C_1, C_2, \dots, C_n$  joined end to end i.e.  $C = C_1 \cup C_2 \cup \dots \cup C_n$
3.  $C$  is a closed curve if  $A = B$
4.  $C$  is a simple closed curve if  $A = B$  and the curve does not cross itself

First, we approximate the curve  $C$  by a polygonal path - a path made up of straight-line segments - as shown in the three figures below.

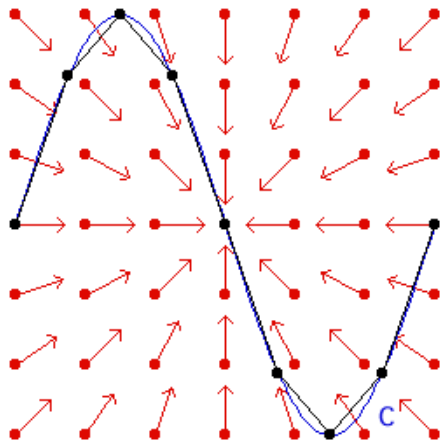


Figure # 71

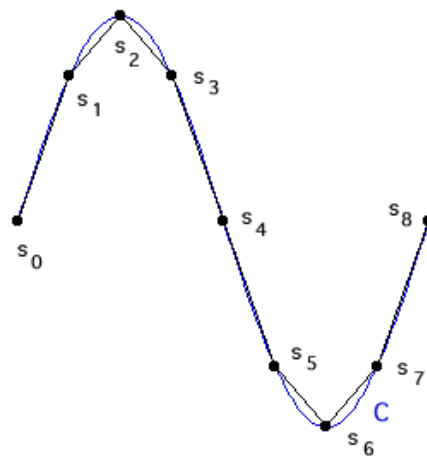


Figure # 72

Path integral may refer to: Line integral, Suppose a force  $F$  acting at each point on a smooth curve  $C$ .

### Line Integral

Any Integral which is evaluated along the curve is called Line Integral, and it is denoted by  $\int_C \vec{F} \cdot d\vec{r}$  where  $\vec{F}$  is a vector point function,  $\vec{r}$  is a position vector and C is the curve

**Theorem:** the work performed by a vector field on a particle moving along a parametric curve C is obtained by integrating the scalar tangential component of force along C.

$$W = \int_C \vec{F} \cdot d\vec{r}$$

Q# 65: Establish the path Integral  $\int_C \vec{F} \cdot d\vec{S} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{F}(\vec{S}_{i-1}) \cdot (\vec{S}_i - \vec{S}_{i-1})$

Answer:

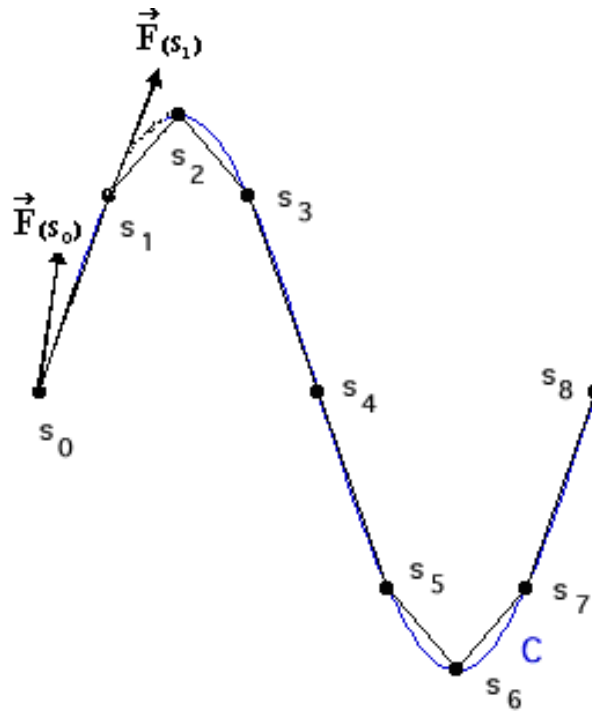


Figure # 73

We choose points  $S_0, S_1, S_2, \dots, S_n$  along the path C and then connect these points as shown in the figure above.

and the amount of total work done on the whole path by :

$$W_1 = \vec{F}(S_0) \cdot (\vec{S}_1 - \vec{S}_0) = \vec{F}(S_{i-1}) \cdot (\vec{S}_i - \vec{S}_{i-1})$$

$$\mathbf{W}_2 = \vec{F}_{(S_1)} \cdot (\vec{S}_2 - \vec{S}_1) = \vec{F}_{(S_{2-1})} \cdot (\vec{S}_2 - \vec{S}_{2-1})$$

$$\mathbf{W}_3 = \vec{F}_{(S_2)} \cdot (\vec{S}_3 - \vec{S}_2) = \vec{F}_{(S_{3-1})} \cdot (\vec{S}_3 - \vec{S}_{3-1})$$

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$$\mathbf{W}_n = \vec{F}_{(S_{n-1})} \cdot (\vec{S}_n - \vec{S}_{n-1})$$

$$\text{Total Work: } \mathbf{W}_1 + \mathbf{W}_2 + \mathbf{W}_3 + \text{-----} + \mathbf{W}_n = \sum_{i=1}^n \mathbf{W}_i = \sum_{i=1}^n \vec{F}_{(\vec{S}_{i-1})} \cdot (\vec{S}_i - \vec{S}_{i-1})$$

Then we estimate the Total work done on the i-th segment of the path by

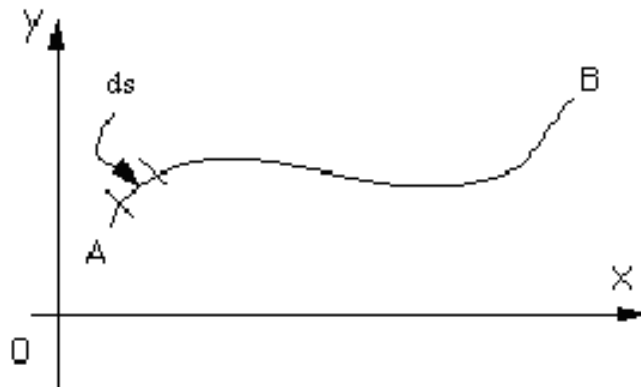
$$\sum_{i=1}^n \mathbf{W}_i = \vec{F}_{(S_{i-1})} \cdot (\vec{S}_i - \vec{S}_{i-1})$$

By using a large number of small segments we can obtain a very good estimate for the amount of work done. The exact amount of work done is obtained by taking the limit of these estimates. This limit is called the **line integral** of the vector field **F** over the path **C** and the amount of work done on the whole path by:

$$\text{Total Work done} = \int_C \vec{F} \cdot d\vec{S} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{F}_{(\vec{S}_{i-1})} \cdot (\vec{S}_i - \vec{S}_{i-1}) \quad \text{Answer}$$

[Where  $ds = s_1 - s_0 = s_2 - s_1 = s_3 - s_2 = \dots = s_n - s_{n-1}$ ]

**Mathematical Expression to find out work done along a curve**



**Figure # 74**

Then the work done by that force in moving the particle small distance  $ds$  is given by:  
Work Done =  $\mathbf{F} \times d\mathbf{s}$

Now if we want to know the total work done in moving the particle all the way from A to B, we need to add up all the small contributions, each of the form  $\mathbf{F} \times d\mathbf{s}$ .

However  $\mathbf{F}$  may have different strengths at different positions, i.e.  $\mathbf{F}$  is a function of position, so what we need to add up are lots of contributions like  $\mathbf{F}(\mathbf{s}) \times d\mathbf{s}$ .

It should be familiar to you that when we add up lots of small things like that we do it by **integration**. The integral in this case is then:  $\int_{AB} \mathbf{F} d\mathbf{s}$

Notice that instead of upper and lower limits we just have AB written at the bottom of the integral sign; to show we're integrating **along the line from A to B**.

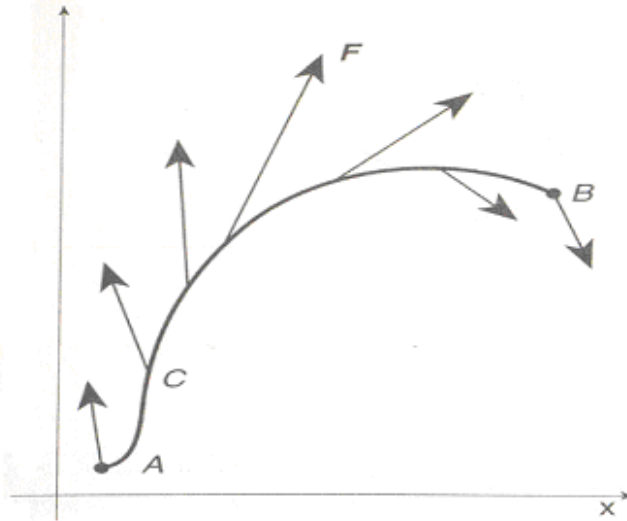


Figure # 75

A force  $\vec{\mathbf{F}}$  acting at each point on a smooth curve C.

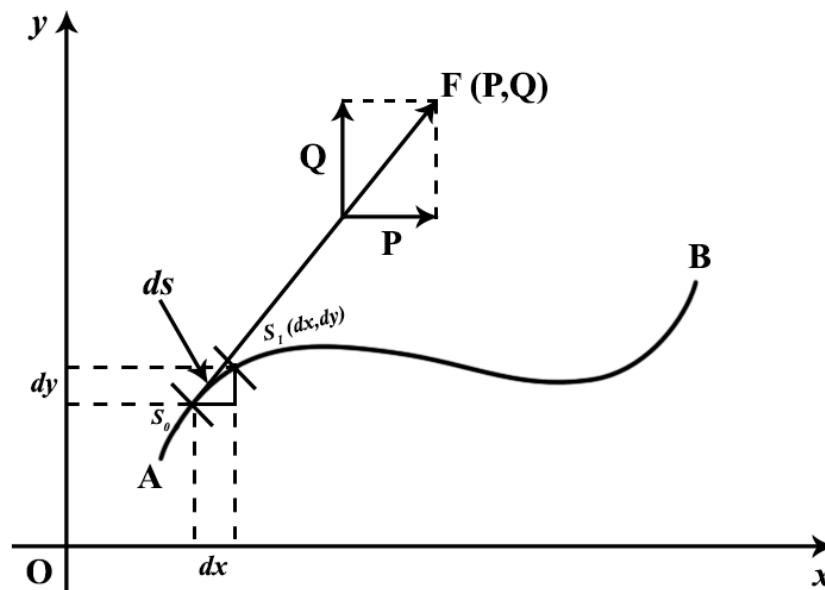


Figure # 76

Here I have divided the force  $\vec{F}$  into a component in the x-direction, called P, and a component in the y-direction, called Q. The work done,  $\vec{F} \cdot d\vec{s}$ , can therefore be written as  $Pdx + Qdy$ .

So we can rewrite the integral above:  $\int_{AB} \vec{F} \cdot d\vec{s} = \int_{AB} Pdx + Qdy$

Because,

The work done by  $\vec{F}$  in moving the particle from the tail to the head  $d\vec{s}$  is approximately:

$$\vec{F} \cdot d\vec{s} \text{ -----(i)}$$

If  $\vec{F}(x, y) = P(x, y)\hat{i} + Q(x, y)\hat{j}$  (Where P and Q are functions of x and y)

and  $d\vec{s} = dx\hat{i} + dy\hat{j}$

Then,  $\vec{F} \cdot d\vec{s} = (P\hat{i} + Q\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$

Then,  $\vec{F} \cdot d\vec{s} = Pdx + Qdy$  [ $\hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1$ ] -----(ii)

Since  $Pdx + Qdy$  is a local estimate (from the tail to the head  $d\vec{s}$ ) of the work, the total work is represented by a line integral:

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{s} = \int_{AB} (P\hat{i} + Q\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) = \int_{AB} (Pdx + Qdy) \text{ -----(iii)}$$

In three dimension,

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{s} = \int_{AB} (P\hat{i} + Q\hat{j} + R\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \int_{AB} (Pdx + Qdy + Rdz) \text{ ----(iv)}$$

**Q # 66:** How much work is accomplished by the force  $\vec{F}(x,y) = xy\hat{i} + y\hat{j}$  in pushing a particle from  $(0,0)$  to  $(3,9)$  along the parabola  $y = x^2$  ?

$$f(x) = x^2$$

### Figure # 77

**Answer:**

Figure shows the path of the particle, Call this path OP. Then

Here,  $\vec{F}(x,y) = xy\hat{i} + y\hat{j}$ ,-----(i)

We know,

$\vec{F}(x,y) = P(x,y)\hat{i} + Q(x,y)\hat{j}$ -----(ii)

Comparing (i) and (ii),

$P = xy, Q = y$

We know,

Work =  $\int_{OP} \vec{F} \cdot d\vec{S} = \int_{OP} (Pdx + Qdy)$

Work =  $\int_{OP} (xydx + ydy)$  [  $\because P = xy, Q = y$  ]

To evaluate this integral, let us use x as the parameter,

Then, Given  $y = x^2$

$$\frac{dy}{dx} = \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 2x$$

$$\therefore dy = 2xdx$$

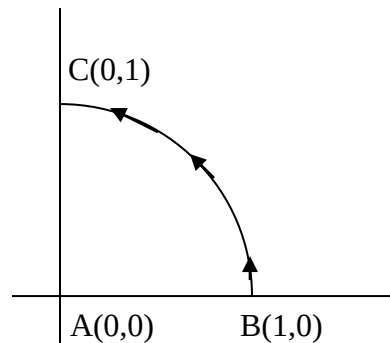
$$\text{Work} = \int_0^3 (xydx + ydy)$$

$$\text{Work} = \int_0^3 \left[ (x \cdot x^2 dx + x^2 \cdot 2x dx) \right]$$

$$\text{Work} = \int_0^3 (x^3 dx + 2x^3 dx)$$

$$\text{Work} = \int_0^3 3x^3 dx = 3 \left[ \frac{x^{3+1}}{3+1} \right]_0^3 = \frac{3}{4} [x^4]_0^3 = \frac{3}{4} \times 3^4 = \frac{243}{4} \text{ Answer.}$$

**Q # 67:** Evaluate  $\int_C xy dx$  from **B(1,0)** to **C(0,1)** along the curve C that is the portion of  $x^2 + y^2 = 1$  in the first quadrant.



**Figure # 78**

Answer:

Given  $x^2 + y^2 = 1$

$$\Rightarrow y = \sqrt{1 - x^2}$$

The curve BC is the first quadrant of the unit circle as shown in figure. On the

curve  $y = \sqrt{1 - x^2}$ , so that,

$$\int_C xy dx = \int_1^0 x \sqrt{1 - x^2} dx$$

$$\text{Let } 1 - x^2 = z$$

$$-2x dx = dz$$

$$x dx = -\frac{dz}{2}$$

| x                        | 1             | 0             |
|--------------------------|---------------|---------------|
| $1 - x^2 = z$            | $z = 1 - x^2$ | $z = 1 - x^2$ |
| $\therefore z = 1 - x^2$ | $z = 1 - 1$   | $z = 1 - 0$   |
|                          | $z = 0$       | $z = 1$       |

$$\int_C xy dx = \int_1^0 x \sqrt{1 - x^2} dx =$$

$$-\int_0^1 \sqrt{z} \frac{dz}{2} = -\frac{1}{2} \int_0^1 \sqrt{z} dz = -\frac{1}{2} \left[ \frac{z^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1 = -\frac{1}{2} \left[ \frac{1^{\frac{3}{2}}}{\frac{3}{2}} - \frac{0^{\frac{3}{2}}}{\frac{3}{2}} \right] = -\frac{1}{2} \left[ \frac{1}{\frac{3}{2}} - \frac{0}{\frac{3}{2}} \right]$$

$$= -\frac{1}{2} \times \frac{2}{3} (1 - 0) = -\frac{1}{3} \text{ Answer}$$

**Q # 68:** Find the value of the line integral when  $\vec{F}(\mathbf{r}) = -y\hat{i} - xy\hat{j}$ , where  $\vec{r}$  is a function of  $t$  and  $C$  is the circular arc in Figure from A to B.

Or

Find the work done in moving a particle once around a circle  $C$  in the  $xy$  plane, if the circle has center at the origin and radius 1 and if the force field is given by

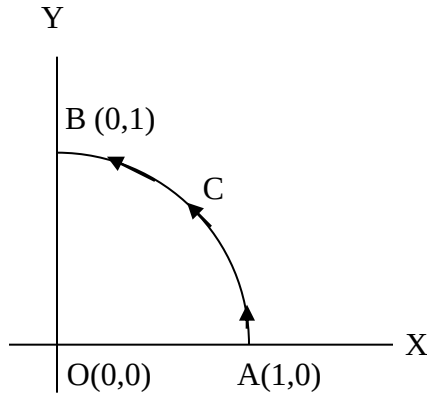


Figure # 79

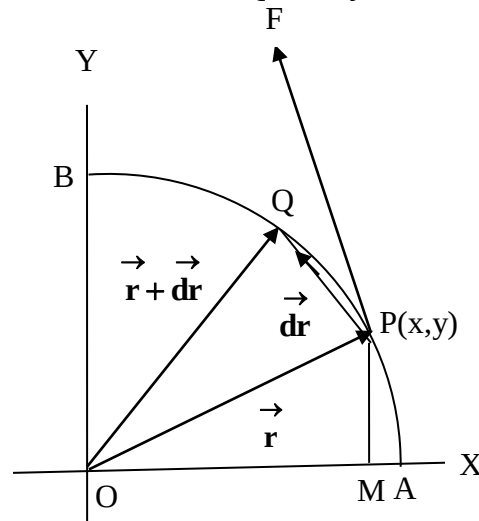


Figure # 80

$$\vec{F}(\mathbf{r}) = -y\hat{i} - xy\hat{j}$$

Since  $\vec{r}$  is a function of  $t$ . i.e.

$\Delta OPM$ ,

Let,  $\angle POM = t$

$$\therefore \frac{OM}{OP} = \cos t \Rightarrow OM = OP \cos t \Rightarrow x = 1 \cdot \cos t \text{ [Since radius is 1, i.e. } |\vec{OP}| = 1]$$

$$x = \cos t \text{ -----(i)}$$

$\Delta OPM$ ,

$$\frac{PM}{OP} = \sin t \Rightarrow PM = OP \sin t \Rightarrow y = 1 \cdot \sin t \text{ [Since radius is 1, i.e. } |\vec{OP}| = 1]$$

$$y = \sin t \text{ -----(ii)}$$

$$\therefore \vec{OP} = \mathbf{r}(t) = \begin{pmatrix} x \\ y \end{pmatrix} = x\hat{i} + y\hat{j} = \cos t\hat{i} + \sin t\hat{j}$$

$$\frac{d\mathbf{r}}{dt} = -\sin t\hat{i} + \cos t\hat{j}$$

$$d\mathbf{r} = (-\sin t\hat{i} + \cos t\hat{j})dt \text{ -----(iii)}$$

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{r} = \int_{AB} (-y \hat{i} - xy \hat{j}) \cdot (-\sin t \hat{i} + \cos t \hat{j}) dt$$

$$\text{Here } 0 \leq t \leq \frac{\pi}{2}$$

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} (-y \hat{i} - xy \hat{j}) \cdot (-\sin t \hat{i} + \cos t \hat{j}) dt$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} (y \sin t - xy \cos t) dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} (\sin t \cdot \sin t - \cos t \cdot \sin t \cos t) dt$$

[From i.  $x = \cos t$  and from ii.  $y = \sin t$ ]

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} (\sin^2 t - \cos^2 t \cdot \sin t) dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} \sin^2 t dt - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \int_0^{\pi/2} 2 \sin^2 t dt - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2t) dt - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt \quad [2 \sin^2 t = 1 - \cos 2t]$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \left[ t - \frac{\sin 2t}{2} \right]_0^{\pi/2} - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \left[ \left( \frac{\pi}{2} - \frac{\sin 2 \times \frac{\pi}{2}}{2} \right) - \left( 0 - \frac{\sin 2 \times 0}{2} \right) \right] - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \left[ \left( \frac{\pi}{2} - \frac{\sin \pi}{2} \right) - \left( 0 - \frac{\sin 0}{2} \right) \right] - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \left[ \left( \frac{\pi}{2} - \frac{0}{2} \right) - (0 - 0) \right] - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{1}{2} \left[ \left( \frac{\pi}{2} \right) \right] - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{\pi}{4} - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt \text{ -----(iv)}$$

Let,  $z = \cos t$

$$\Rightarrow \frac{dz}{dt} = \frac{d}{dt}(\cos t) = -\sin t$$

$$\Rightarrow dz = -\sin t dt$$

| t            | $\frac{\pi}{2}$                     | 0                       |
|--------------|-------------------------------------|-------------------------|
| $z = \cos t$ | $z = \cos \frac{\pi}{2}$<br>$z = 0$ | $z = \cos 0$<br>$z = 1$ |

From equation (iv),

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{\pi}{4} - \int_0^{\pi/2} \cos^2 t \cdot \sin t dt$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{\pi}{4} + \int_1^0 z^2 dz$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{\pi}{4} + \left[ \frac{z^3}{3} \right]_1^0$$

$$\text{Work} = \int_0^{\pi/2} \vec{F} \cdot d\vec{r} = \frac{\pi}{4} + \left[ \frac{0}{3} - \frac{1}{3} \right] = \frac{\pi}{4} - \frac{1}{3} \approx 0.4521 \text{ Answer}$$

Q # 69: Find the work done by a)  $\vec{F} = x\hat{i} + y\hat{j}$  and b)  $\vec{F} = \frac{3}{4}\hat{i} + \frac{1}{2}\hat{j}$  along the curve C

traced by  $\vec{r}(t) = \cos t\hat{i} + \sin t\hat{j}$  from  $t = 0$  to  $t = \pi$

Answer:

a) The vector function  $\vec{r}(t)$  gives the parametric equations  $x = \cos t, y = \sin t$ ,

$0 \leq t \leq \pi$  which recognize as a half circle. As seen in Figure 78, the force field  $\vec{F}$  is

perpendicular to C at every point. Because the tangential components of  $\vec{F}$  are zero, the work done along C is zero.

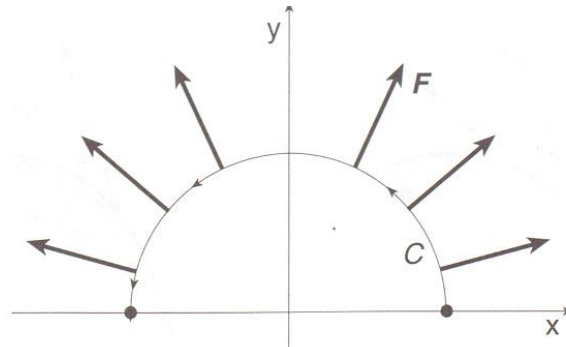


Figure # 81

Half-circle C, with force  $\vec{F}$  perpendicular to C

Given,  $\vec{r}(t) = \cos t \hat{i} + \sin t \hat{j}$

$$\therefore \frac{d}{dt}\{\vec{r}(t)\} = \frac{d}{dt}\{\cos t \hat{i} + \sin t \hat{j}\}$$

$$\therefore \frac{d}{dt}\{\vec{r}(t)\} = -\sin t \hat{i} + \cos t \hat{j}$$

$$\therefore d\vec{r}(t) = (-\sin t \hat{i} + \cos t \hat{j})dt$$

$$\therefore d\vec{r} = (-\sin t \hat{i} + \cos t \hat{j})dt$$

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_C (x \hat{i} + y \hat{j}) \cdot d\vec{r}$$

$$= \int_0^\pi (\cos t \hat{i} + \sin t \hat{j}) \cdot (-\sin t \hat{i} + \cos t \hat{j}) dt$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^\pi (-\cos t \sin t + \sin t \cos t) dt = 0$$

b) In Figure 79 the vectors tangent to the semi-circle are the projections of  $\vec{F}$  on the unit tangent vectors.

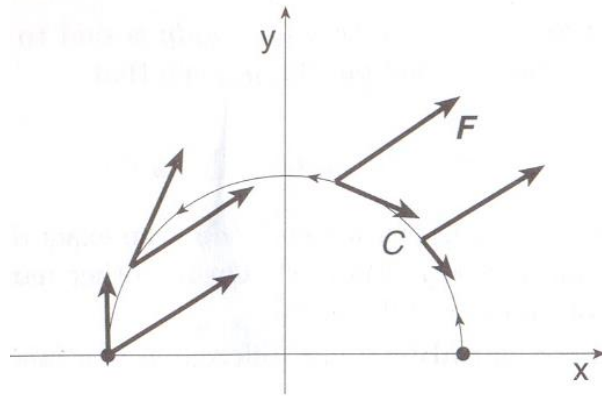


Figure # 82

The work done by  $\vec{F}$  is:

$$W = \int_C \vec{F} \cdot d\vec{r} = \int_C \left(\frac{3}{4} \hat{i} + \frac{1}{2} \hat{j}\right) \cdot d\vec{r}$$

$$= \int_0^\pi \left(\frac{3}{4} \hat{i} + \frac{1}{2} \hat{j}\right) \cdot (-\sin t \hat{i} + \cos t \hat{j}) dt$$

$$\begin{aligned}
&= \int_0^{\pi} \left(-\frac{3}{4} \sin t + \frac{1}{2} \cos t\right) dt \quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1] \\
&= \left[ \frac{3}{4} \cos t + \frac{1}{2} \sin t \right]_0^{\pi} = \left[ \frac{3}{4} \cos \pi + \frac{1}{2} \sin \pi \right] - \left[ \frac{3}{4} \cos 0 + \frac{1}{2} \sin 0 \right] \\
&= \left[ \frac{3}{4}(-1) + \frac{1}{2}(0) \right] - \left[ \frac{3}{4}(1) + \frac{1}{2}(0) \right] = \left[ -\frac{3}{4} + 0 - \frac{3}{4} + 0 \right] = -\frac{3}{2} \text{ Answer}
\end{aligned}$$

**Q# 70:** Find the work done by the force field  $\vec{F}(x, y) = x^3 y \hat{i} + (x - y) \hat{j}$  on a particle that moves along the parabola  $y = x^2$  from  $(-2, 4)$  to  $(1, 1)$

Answer:

$$\boxed{f(x) = x^2/2}$$

**Figure # 83**

The work  $W$  performed by the field is:

$$\mathbf{W} = \oint_C \vec{F} \cdot d\vec{r} \quad \text{-----(i)}$$

We have the position vector

$$\therefore \vec{r} = x \hat{i} + y \hat{j} + z \hat{k} +$$

$$\therefore d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\vec{F} \cdot d\vec{r} = [x^3 y \hat{i} + (x - y) \hat{j}] \cdot [dx \hat{i} + dy \hat{j} + dz \hat{k}]$$

$$\vec{F} \cdot d\vec{r} = [x^3 y \hat{i} + (x - y) \hat{j} + 0 \hat{k}] \cdot [dx \hat{i} + dy \hat{j} + dz \hat{k}]$$

$$\vec{F} \cdot d\vec{r} = [x^3 y dx + (x - y) dy] \quad \text{-----(ii)}$$

Let  $x = t$  as the parameter, As  $x = t$ ,

|          |            |          |
|----------|------------|----------|
| <b>x</b> | <b>- 2</b> | <b>1</b> |
| <b>t</b> | <b>- 2</b> | <b>1</b> |

Then the path C of the particle can be expressed parametrically as

$$x = t, y = t^2; -2 \leq t \leq 1$$

Now,

$$x = t$$

$$\therefore \frac{dx}{dt} = \frac{d}{dt}(t)$$

$$\therefore \frac{dx}{dt} = 1$$

$$\therefore dx = dt \quad \text{-----(iii)}$$

Again,  $y = t^2$

$$\therefore \frac{dy}{dt} = \frac{d}{dt}(t^2)$$

$$\therefore \frac{dy}{dt} = 2t$$

$$\therefore dy = 2t dt \quad \text{-----(iv)}$$

From (i) & (ii)

$$W = \oint_C \vec{F} \cdot d\vec{r}$$

$$W = \oint_C [x^3 y dx + (x - y) dy]$$

$$W = \int_{-2}^1 [t^3 \cdot t^2 dt + (t - t^2) 2t dt]$$

$$W = \int_{-2}^1 [t^5 dt + (2t^2 - 2t^3) dt]$$

$$W = \int_{-2}^1 t^5 dt + \int_{-2}^1 2t^2 dt - \int_{-2}^1 2t^3 dt$$

$$W = \left[ \frac{t^6}{6} \right]_{-2}^1 + \left[ \frac{2t^3}{3} \right]_{-2}^1 - \left[ \frac{2t^4}{4} \right]_{-2}^1$$

$$W = \left[ \frac{1^6}{6} - \frac{(-2)^6}{6} \right] + \left[ \frac{2 \cdot 1^3}{3} - \frac{2(-2)^3}{3} \right] - \left[ \frac{2 \cdot (1)^4}{4} - \frac{2(-2)^4}{4} \right]$$

$$W = \left[ \frac{1}{6} - \frac{64}{6} \right] + \left[ \frac{2}{3} + \frac{16}{3} \right] - \left[ \frac{2}{4} - \frac{32}{4} \right]$$

$$W = \left[ \frac{1-64}{6} \right] + \left[ \frac{2+16}{3} \right] - \left[ \frac{2-32}{4} \right]$$

$$W = \left[ \frac{-63}{6} \right] + \left[ \frac{18}{3} \right] + \left[ \frac{30}{4} \right]$$

$$W = \frac{-126+72+90}{12}$$

$$W = \frac{-126+162}{12}$$

$$W = \frac{36}{12} = 3; \text{ where the units for } W \text{ depend on the units chosen for force and distance}$$

**Or**

Figure shows the path of the particle, Call this path PO. Then

$$\text{Here, } \vec{F}(x, y) = x^3 y \hat{i} + (x - y) \hat{j}, \text{-----(i)}$$

We know,

$$\vec{F}(x, y) = P(x, y) \hat{i} + Q(x, y) \hat{j} \text{-----(ii)}$$

Comparing (i) and (ii),

$$P = x^3 y, Q = (x - y)$$

We know,

$$\text{Work} = \int_C \vec{F} \cdot d\vec{s} = \int_C (Pdx + Qdy)$$

$$\text{Work} = W = \oint_C [x^3 y dx + (x - y) dy] \quad [P = x^3 y, Q = x - y]$$

To evaluate this integral, let us use x as the parameter,

Then, Given  $y = x^2$

$$\frac{dy}{dx} = \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 2x$$

$$\therefore dy = 2x dx$$

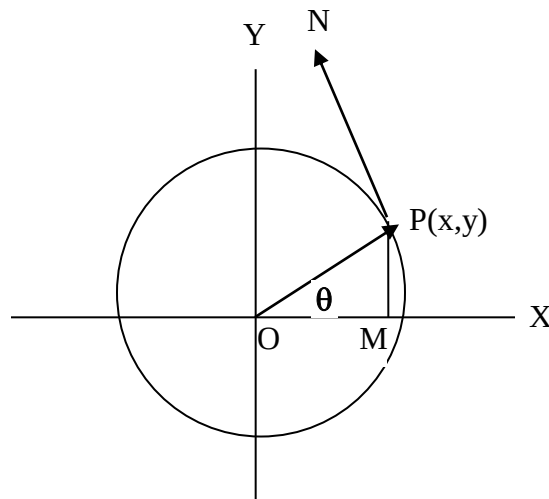
$$W = \oint_C \vec{F} \cdot d\vec{s}$$

$$W = \oint_C [x^3 y dx + (x - y) dy]$$

$$W = \int_{-2}^1 [x^3 \cdot x^2 dx + (x - x^2) 2x dx]$$

$$\begin{aligned}
 W &= \int_{-2}^1 [x^5 dx + (2x^2 - 2x^3) dx] \\
 W &= \int_{-2}^1 x^5 dx + \int_{-2}^1 2x^2 dx - \int_{-2}^1 2x^3 dx \\
 W &= \left[ \frac{x^6}{6} \right]_{-2}^1 + \left[ \frac{2x^3}{3} \right]_{-2}^1 - \left[ \frac{2x^4}{4} \right]_{-2}^1 \\
 W &= \left[ \frac{1^6}{6} - \frac{(-2)^6}{6} \right] + \left[ \frac{2 \cdot 1^3}{3} - \frac{2(-2)^3}{3} \right] - \left[ \frac{2 \cdot (1)^4}{4} - \frac{2(-2)^4}{4} \right] \\
 W &= \left[ \frac{1}{6} - \frac{64}{6} \right] + \left[ \frac{2}{3} + \frac{16}{3} \right] - \left[ \frac{2}{4} - \frac{32}{4} \right] \\
 W &= \left[ \frac{1-64}{6} \right] + \left[ \frac{2+16}{3} \right] - \left[ \frac{2-32}{4} \right] \\
 W &= \left[ \frac{-63}{6} \right] + \left[ \frac{18}{3} \right] + \left[ \frac{30}{4} \right] \\
 W &= \frac{-126+72+90}{12} \\
 W &= \frac{-126+162}{12} \\
 W &= \frac{36}{12} = 3; \text{ where the units for } W \text{ depend on the units chosen for force and distance}
 \end{aligned}$$

**Q # 71:** Find the value of  $\int_C \vec{F} \cdot d\vec{r}$  where  $\vec{F} = (y - 2x)\hat{i} + (3x + 2y)\hat{j}$  and C is a circle in the xy- plane with center the origin and radius 2.



**Figure # 84**

The position Vector is

$$\vec{r} = x\hat{i} + y\hat{j} \text{-----(i)}$$

Let,

$$\angle POM = \theta, OP = 2$$

$$\frac{PM}{OP} = \sin\theta$$

$$\frac{y}{2} = \sin\theta \Rightarrow y = 2\sin\theta$$

Similarly,

$$\frac{OM}{OP} = \cos\theta \Rightarrow \frac{x}{2} = \cos\theta \Rightarrow x = 2\cos\theta$$

From (i),

$$\vec{r}(\theta) = 2\cos\theta\hat{i} + 2\sin\theta\hat{j}$$

$$\frac{d\vec{r}}{d\theta} = -2\sin\theta\hat{i} + 2\cos\theta\hat{j}$$

$$\Rightarrow d\vec{r} = (-2\sin\theta\hat{i} + 2\cos\theta\hat{j})d\theta$$

Given,

$$\vec{F} = (y - 2x)\hat{i} + (3x + 2y)\hat{j}$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(y - 2x)\hat{i} + (3x + 2y)\hat{j}\} \cdot (-2\sin\theta\hat{i} + 2\cos\theta\hat{j})d\theta$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(y - 2x)(-2\sin\theta)d\theta + (3x + 2y)(2\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(2\sin\theta - 2 \times 2\cos\theta)(-2\sin\theta)d\theta + (3 \times 2\cos\theta + 2 \times 2\sin\theta)(2\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(2\sin\theta - 4\cos\theta)(-2\sin\theta)d\theta + (6\cos\theta + 4\sin\theta)(2\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int \{(-4\sin^2\theta + 8\sin\theta\cos\theta)d\theta + (12\cos^2\theta + 8\sin\theta\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int (-4\sin^2\theta + 16\sin\theta\cos\theta + 12\cos^2\theta)d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (-4\sin^2\theta + 16\sin\theta\cos\theta + 12\cos^2\theta)d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (-2 \times 2\sin^2\theta + 8 \times 2\sin\theta\cos\theta + 6 \times 2\cos^2\theta)d\theta.$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \{-2(1 - \cos 2\theta) + 8 \times \sin 2\theta + 6(1 + \cos 2\theta)\}d\theta.$$

$$\begin{aligned}
\therefore \int_C \vec{F} \cdot d\vec{r} &= \left[ -2\left(\theta - \frac{\sin 2\theta}{2}\right) - 8 \times \frac{\cos 2\theta}{2} + 6\left(\theta + \frac{\sin 2\theta}{2}\right) \right]_0^{2\pi} \\
\therefore \int_C \vec{F} \cdot d\vec{r} &= -2 \left[ \left(2\pi - \frac{\sin 2 \times 2\pi}{2}\right) - \left(0 - \frac{\sin 2 \times 0}{2}\right) \right] - 8 \left[ \left(\frac{\cos 2 \times 2\pi}{2} - \frac{\cos 2 \times 0}{2}\right) \right] + 6 \left[ \left(2\pi + \frac{\sin 2 \times 2\pi}{2}\right) - \left(0 + \frac{\sin 2 \times 0}{2}\right) \right] \\
&= -2 \left[ \left(2\pi - \frac{\sin 4\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right) \right] - 8 \left[ \left(\frac{\cos 4\pi}{2} - \frac{\cos 0}{2}\right) \right] + 6 \left[ \left(2\pi + \frac{\sin 4\pi}{2}\right) - \left(0 + \frac{\sin 0}{2}\right) \right] \\
&= -2 \left[ \left(2\pi - \frac{0}{2}\right) - \left(0 - \frac{0}{2}\right) \right] - 8 \left[ \left(\frac{1}{2} - \frac{1}{2}\right) \right] + 6 \left[ \left(2\pi + \frac{0}{2}\right) - \left(0 + \frac{0}{2}\right) \right] \\
&= -2 \left[ 2\pi - \left(-\frac{0}{2}\right) \right] - 8[0] + 6[2\pi - 0] \\
&= -2[2\pi + 0] - 0 + 6[2\pi] \\
&= -2[2\pi] + 6[2\pi] \\
\therefore \int_C \vec{F} \cdot d\vec{r} &= -4\pi + 12\pi = 8\pi \text{ Answer}
\end{aligned}$$

**Q # 72:** Find the value of  $\int_C \vec{F} \cdot d\vec{r}$  where  $\vec{F} = x^2 \hat{i} + 3xy \hat{j}$  if

- C is the straight line path from **(0,0)** to **(1,2)**
- C is the parabolic path  $y = 2x^2$  from **(0,0)** to **(1,2)**
- C is composed of two straight-line paths the x axis from **(0,0)** to **(1,0)** and then a line parallel to the y-axis from **(1,0)** to **(1,2)**

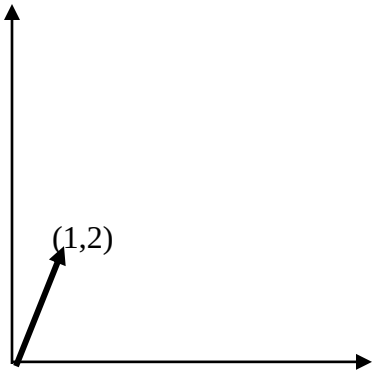


Figure # 85

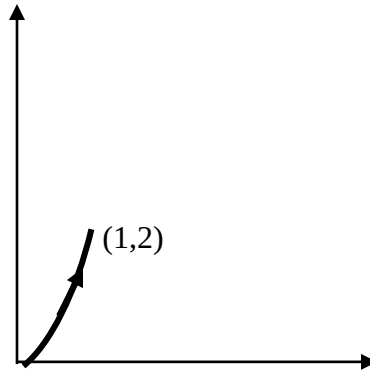


Figure # 86

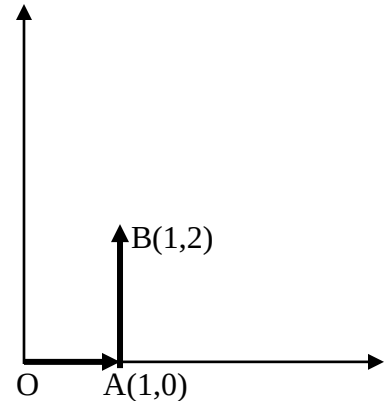


Figure # 87

Let, the position vector is:  $\vec{r} = x \hat{i} + y \hat{j}$

Since the line element  $d\vec{r}$  lies in the xy-plane, we can express it as  $d\vec{r} = dx \hat{i} + dy \hat{j}$ , so that

$$\begin{aligned}\int_C \vec{F} \cdot d\vec{r} &= \int_C (x^2 \hat{i} + 3xy \hat{j}) \cdot (dx \hat{i} + dy \hat{j}) \\ &= \int_C (x^2 dx + 3xy dy) \text{-----(i)}\end{aligned}$$

- a) On this path  $y = 2x$ , so we can convert (i) to a definite integral with respect to  $x$ , this is effectively a simple parameterization,  $x = t, y = 2t, 0 \leq t \leq 1$

Hence,

$$\begin{aligned}\int_C (x^2 dx + 3xy dy) &= \int_0^1 [(x^2 dx + 3x(2x)2 dx)] \text{ [since } y = 2x ; \therefore dy = 2x dx \text{]} \\ &= \int_0^1 (x^2 dx + 12x^2 dx) = \int_0^1 13x^2 dx = 13 \left[ \frac{x^3}{3} \right]_0^1 = 13 \left[ \frac{1^3}{3} - \frac{0^3}{3} \right] = \frac{13}{3}\end{aligned}$$

Alternatively, of course, we could choose  $y$  as the integration variable or parameter and put  $x = \frac{y}{2}$  instead of  $y = 2x$ , in which case (i) becomes,

$$\begin{aligned}\int_C (x^2 dx + 3xy dy) &= \int_0^2 (x^2 dx + 3xy dy) = \int_0^2 \left( \frac{y^2}{4} \frac{dy}{2} + 3 \frac{y}{2} y dy \right) \\ &\quad \text{[Since } y = 2x ; \therefore dy = 2dx \therefore dx = \frac{dy}{2}; x = \frac{y}{2}] \\ &= \int_0^2 \left( \frac{y^2}{8} dy + \frac{3}{2} y^2 dy \right) = \int_0^2 \frac{13y^2}{8} dy = \frac{13}{8} \left[ \frac{y^3}{3} \right]_0^2 = \frac{13}{8} \left[ \frac{2^3}{3} - \frac{0^3}{3} \right] = \frac{13}{8} \times \frac{8}{3} = \frac{13}{3}\end{aligned}$$

- b) Here we put  $y = 2x^2$  and  $\therefore dy = 4x dx$  in (i), giving the definite integral:

$$\begin{aligned}\int_C (x^2 dx + 3xy dy) &= \int_0^1 (x^2 dx + 3x(2x^2)4x dx) = \int_0^1 (x^2 dx + 24x^4 dx) \\ &= \left[ \frac{x^3}{3} + 24 \frac{x^5}{5} \right]_0^1 = \left[ \frac{1^3}{3} + 24 \frac{1^5}{5} - \frac{0^3}{3} - 24 \frac{0^5}{5} \right] = \left[ \frac{1}{3} + \frac{24}{5} - \frac{0}{3} - \frac{0}{5} \right] = \frac{77}{5}\end{aligned}$$

- c) Referring to figure (c), we must integrate on the horizontal and vertical portions separately and add the two contributions. On the horizontal section,  $y$  is a constant i.e the equation of horizontal line is  $y = 0$  so  $dy = 0$  and (i) is simply:

$$\int_C (x^2 dx + 3xy dy) = \int_0^1 (x^2 dx + 3x(0).0) = \int_0^1 x^2 dx = \frac{1}{3}$$

On the vertical section,  $x$  is constant at 1, i.e the equation of the line parallel to  $y$  axis from  $(1,0)$  to  $(1,2)$  is  $x = 1$  so  $dx = 0$  and (i) yield:

$$\int_C (x^2 dx + 3xy dy) = \int_0^2 (1^2 .0 + 3.1y dy) = \int_0^2 3y dy = 3 \left[ \frac{y^2}{2} \right]_0^2 = 3 \left[ \frac{2^2}{2} - \frac{0^2}{2} \right] = 6$$

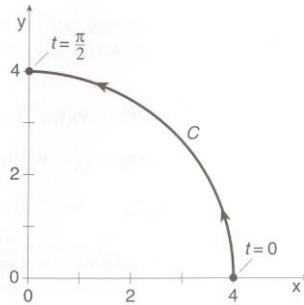
Adding the two values, we conclude that the value of the line integral along OA and then AB is  $= \frac{1}{3} + 6 = \frac{19}{3}$

**Q # 73:** Evaluate the following integrals:

1.  $\int_C xy^2 dx$

2.  $\int_C xy^2 dy$

On the quarter circle C defined by  $x = 4\cos t, y = 4\sin t, 0 \leq t \leq \frac{\pi}{2}$



**Figure # 88**

Given,

$$x = 4\cos t, y = 4\sin t, 0 \leq t \leq \frac{\pi}{2}$$

$$\therefore dx = -4\sin t dt, dy = 4\cos t dt$$

$$1. \int_C xy^2 dx = \int_0^{\pi/2} (4\cos t)(16\sin^2 t)(-4\sin t dt) = -256 \int_0^{\pi/2} \sin^3 t \cos t dt$$

Let,  $z = \sin t$   
 $dz = \cos t dt$

|             |                                      |
|-------------|--------------------------------------|
| $t = 0$     | $z = \sin t$<br>$z = \sin 0 = 0$     |
| $t = \pi/2$ | $z = \sin t$<br>$z = \sin \pi/2 = 1$ |

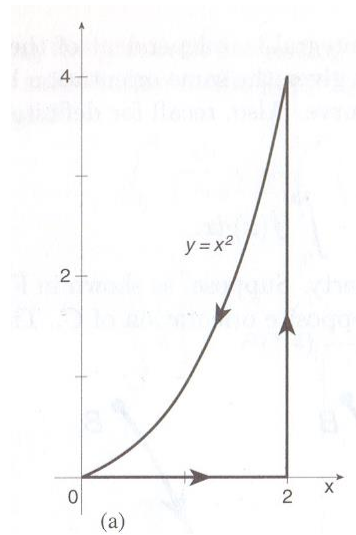
$$= -256 \int_0^1 z^3 dz = -256 \left[ \frac{z^4}{4} \right]_0^1 = -256 \left[ \frac{1^4}{4} - 0 \right] = -64 \text{ Answer}$$

$$2. \int_C xy^2 dy = \int_0^{\pi/2} (4\cos t)(16\sin^2 t)(4\cos t dt) = 256 \int_0^{\pi/2} \sin^2 t \cos^2 t dt$$

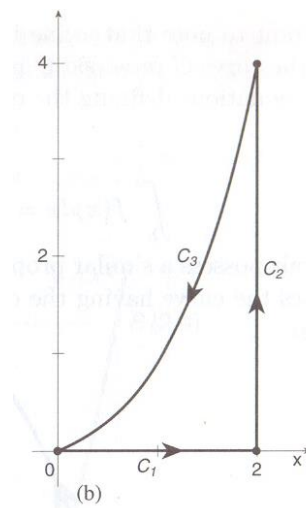
$$= 256 \int_0^{\pi/2} \frac{1}{4} (4\sin^2 t \cos^2 t) dt = 256 \int_0^{\pi/2} \frac{1}{4} (2\sin t \cos t)(2\sin t \cos t) dt$$

$$\begin{aligned}
&= 256 \int_0^{\pi/2} \frac{1}{4} (\sin 2t)(\sin 2t) dt = 256 \int_0^{\pi/2} \frac{1}{4} \sin^2 2t dt = 64 \int_0^{\pi/2} \sin^2 2t dt \\
&= 64 \int_0^{\pi/2} \frac{1}{2} (2 \sin^2 2t) dt = 64 \int_0^{\pi/2} \frac{1}{2} (1 - \cos 4t) dt = 32 \left[ t - \frac{\sin 4t}{4} \right]_0^{\pi/2} \\
&= 32 \left[ \frac{\pi}{2} - \frac{\sin 4 \times \frac{\pi}{2}}{4} - 0 + 0 \right] = 32 \left[ \frac{\pi}{2} - \frac{\sin 2\pi}{4} \right] = 32 \left[ \frac{\pi}{2} - 0 \right] = 16\pi \text{ Answer}
\end{aligned}$$

**Q # 74:** Evaluate  $\oint_C y^2 dx - x^2 dy$  on the closed curve C that is shown in the figure



**Figure # 89**



**Figure # 90**

**Answer:** Because C is piecewise smooth, we can express the integral as a sum of integrals. Symbolically, we write:  $\oint_C = \int_{C_1} + \int_{C_2} + \int_{C_3}$  where  $C_1, C_2$  and  $C_3$  are the curves shown in figure.

- i. On  $C_1$ , we use  $x$  as a parameter. Because  $y = 0, dy = 0$ ; therefore

$$\int_{C_1} [y^2 dx - x^2 dy] = \int_0^2 [(0)dx - x^2(0)] = 0$$

- ii. On  $C_2$ , we use  $y$  as a parameter. From  $x = 2, dx = 0$ ; therefore

$$\int_{C_2} [y^2 dx - x^2 dy] = \int_0^4 [y^2(0) - 4dy] = -\int_0^4 4dy = -[4y]_0^4 = -16$$

- iii. Finally, on  $C_3$ , we again use  $x$  as a parameter. From

$$y = x^2; dy = 2x dx; \text{ therefore}$$

$$\oint_{C_3} [y^2 dx - x^2 dy] = \int_2^0 [x^4 dx - x^2 (2x dx)] = \int_2^0 (x^4 dx - 2x^3) dx$$

$$= \left[ \frac{1}{5} x^5 - \frac{1}{2} x^4 \right]_2^0 = \frac{8}{5}$$

Therefore,  $\oint_C y^2 dx - x^2 dy = 0 - 16 + \frac{8}{5} = -\frac{72}{5}$

## Surface Integrals

### Surface Integral

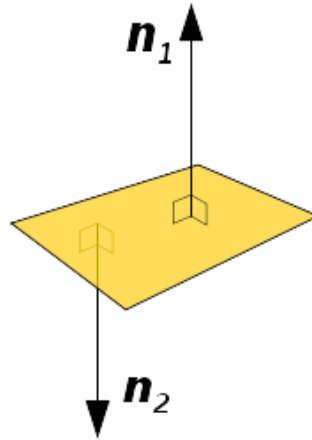
The Integral which is evaluated over a surface is called Surface Integral.

If  $S$  is any surface and  $\hat{n}$  is the outward drawn unit normal vector to the surface  $S$  then

$\int_S \vec{F} \cdot \hat{n} dS$  is called the Surface Integral.

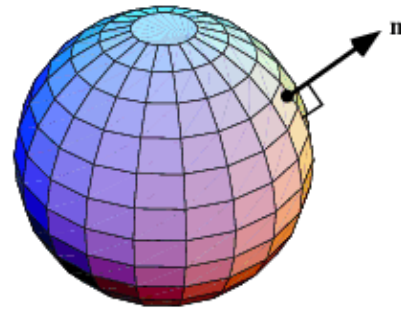
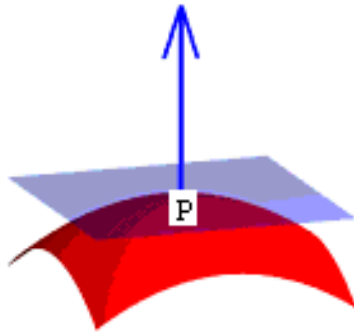
### Normal Vector:

A surface normal, or simply normal, to a flat surface is a vector that is perpendicular to that surface.

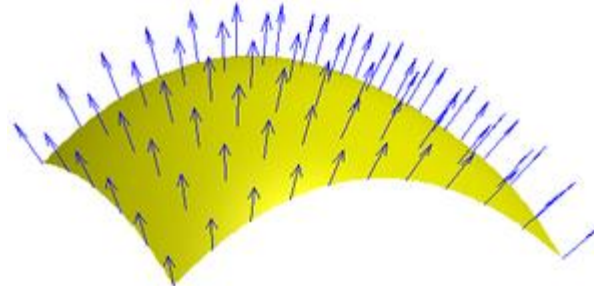


**Figure # 91**

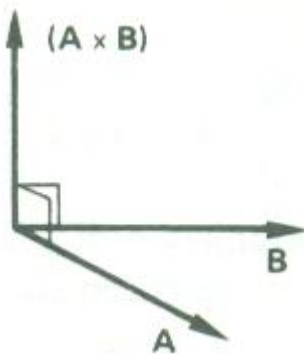
A normal to a non-flat surface at a point  $P$  on the surface is a vector perpendicular to the tangent plane to that surface at  $P$ .



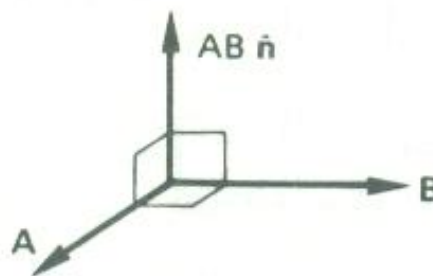
**Figure # 92:** A normal to a surface at a point is the same as a normal to the tangent plane to that surface at that point P



**Figure # 93:** A vector field of normals to a surface



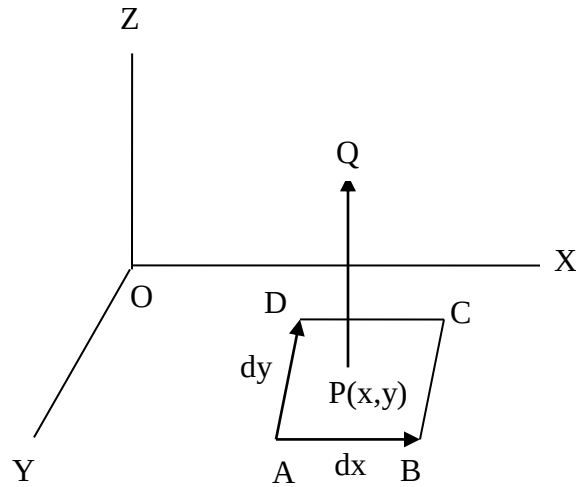
**Figure # 94**



**Figure # 95**

We know,  $\vec{A} \times \vec{B} = \|\vec{A}\| \|\vec{B}\| \sin \theta \hat{n}$

If  $\theta = \frac{\pi}{2}$ , then  $\vec{A} \times \vec{B} = \|\vec{A}\| \|\vec{B}\| \sin \theta \hat{n} = \|\vec{A}\| \|\vec{B}\| \sin \frac{\pi}{2} \hat{n} = \|\vec{A}\| \|\vec{B}\| .1. \hat{n} = \|\vec{A}\| \|\vec{B}\| \hat{n}$ , where  $\hat{n}$  is a unit normal to the plane A and B.



**Figure # 96**

If  $P(x, y)$  is a point in the  $xy$  plane, the element of area  $dx dy$  has a vector area

$$d\vec{S} = \left| \vec{dS} \right|$$

From figure

$$\vec{PQ} = \left| \vec{PQ} \right| \hat{\eta} \quad [\because \text{Any Vector} = \text{Length of this Vector} \times \text{Unit Vector}]$$

$$\vec{PQ} = \vec{AB} \times \vec{AD} = \left| \vec{AB} \right| \left| \vec{AD} \right| \sin \theta \hat{\eta} \text{-----(i)}$$

$$\text{Again, } \vec{AB} = \left| \vec{AB} \right| \hat{\eta} \quad [\because \text{Any Vector} = \text{Length of this Vector} \times \text{Unit Vector}]$$

$$\vec{AB} = dx \hat{\eta} \quad [\because \text{Any Vector} = \text{Length of this Vector} \times \text{Unit Vector}]$$

$$\vec{AB} = dx \hat{i} \quad [\hat{\eta} = \hat{i}] \text{-----(ii)}$$

Again,

$$\vec{AD} = \left| \vec{AD} \right| \hat{\eta}$$

$$\vec{AD} = dy \hat{\eta}$$

$$\vec{AD} = dy \hat{j} \quad [\hat{\eta} = \hat{j}] \text{-----(iii)}$$

From (i)

$$\vec{PQ} = \vec{AB} \times \vec{AD} = \left| \vec{AB} \right| \left| \vec{AD} \right| \sin \theta \hat{\eta}$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = \left| dx \hat{i} \right| \left| dy \hat{j} \right| \sin \theta \hat{\eta} \quad [\text{From (ii) \& (iii)}]$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = dx dy \sin \theta \hat{\eta}$$

$$[\because |dx \hat{i}| = dx \text{ \& } |dy \hat{j}| = dy]$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = dx dy \sin\left(\frac{\pi}{2}\right) \hat{\eta} \quad [\theta = \frac{\pi}{2}]$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = dx dy \cdot 1 \cdot \hat{\eta}$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = dx dy \hat{\eta} \text{-----(iv)}$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = dS \hat{\eta} \text{-----(v)} \quad [\text{Let } dS = dx dy]$$

$$\Rightarrow \vec{PQ} = dx \hat{i} \times dy \hat{j} = dS \hat{k} \text{-----(vi)}$$

$$[\text{Here } \hat{\eta} = \hat{k} \text{ since } \hat{i} \times \hat{j} = \hat{k} \text{ and } dS = dx dy]$$

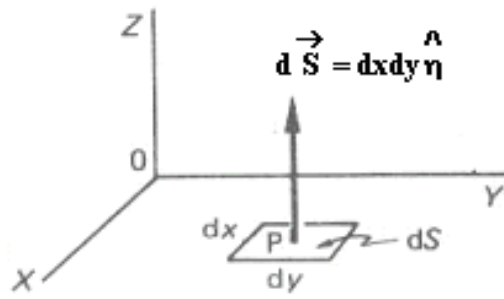


Figure # 97

$\vec{dS}$  is the vector which is perpendicular to the plane  $dx dy$  and  $dS$  is the length of the perpendicular vector  $\vec{dS}$  as well as  $dS$  is the area of parallelogram  $dx dy$  and  $\hat{\eta}$  is the unit normal vector of  $\vec{dS}$  to the plane  $dx dy$

$$\Rightarrow \vec{dS} = dx \hat{i} \times dy \hat{j} = dS \hat{\eta} \text{-----(vii)} \quad [\text{Here } \vec{PQ} = \vec{dS}]$$

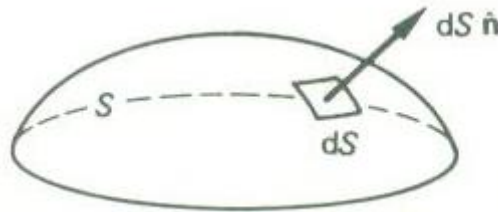


Figure # 98

So we can write, From Figure # 95

$$01. \vec{PQ} = dx \hat{i} \times dy \hat{j} = dx dy \hat{k} \quad [\text{Anti-Clockwise; xy-plane; } \hat{i} \times \hat{j} = \hat{k}]; dS = dx dy]$$

$$02. \vec{PQ} = dy \hat{j} \times dx \hat{i} = -dx dy \hat{k} \quad [\text{Clockwise; xy-plane; } \hat{j} \times \hat{i} = -\hat{k}; dS = dx dy]$$

$$03. \vec{PQ} = dz \hat{k} \times dx \hat{i} = dx dz \hat{j}$$

[Anti-Clockwise; xz-plane;  $\hat{k} \times \hat{i} = \hat{j}$ ;  $dS = dx dz$ ]

$$04. \vec{PQ} = dx \hat{i} \times dz \hat{k} = -dx dz \hat{j}$$

[Clockwise; xz-plane;  $\hat{i} \times \hat{k} = -\hat{j}$ ;  $dS = dx dz$ ]

$$05. \vec{PQ} = dy \hat{j} \times dz \hat{k} = dy dz \hat{i}$$

[Anti-Clockwise; yz-plane;  $\hat{j} \times \hat{k} = \hat{i}$ ;  $dS = dy dz$ ]

$$06. \vec{PQ} = dz \hat{k} \times dy \hat{j} = -dy dz \hat{i}$$

[Clockwise; yz-plane;  $\hat{k} \times \hat{j} = -\hat{i}$ ;  $dS = dy dz$ ]

Example:

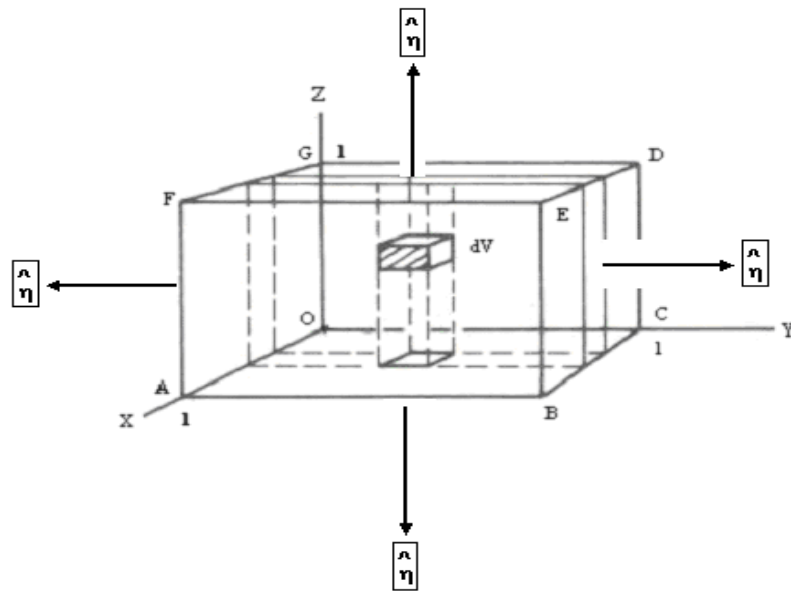


Figure # 99

From Figure # 96

| Serial no | Surface | $\hat{n} = ?$        | $dS$    | Plane   |
|-----------|---------|----------------------|---------|---------|
| 1         | OABC    | $\hat{n} = -\hat{k}$ | $dx dy$ | $z = 0$ |
| 2         | DEFG    | $\hat{n} = \hat{k}$  | $dx dy$ | $z = 1$ |
| 3         | OAFG    | $\hat{n} = -\hat{j}$ | $dx dz$ | $y = 0$ |
| 4         | BCDE    | $\hat{n} = \hat{j}$  | $dx dz$ | $y = 1$ |
| 5         | OCDG    | $\hat{n} = -\hat{i}$ | $dy dz$ | $x = 0$ |
| 6         | ABEF    | $\hat{n} = \hat{i}$  | $dy dz$ | $x = 1$ |

## Volume Integrals

### Volume Integral

If  $\vec{F}$  is a vector point function bounded by the region R with volume V, then  $\int_V \vec{F} dV$  is called as Volume Integral

If V is a closed region bounded by a surface S and  $\vec{F}$  is a vector field at each point of V and on its boundary surface S, then  $\int_V \vec{F} dV$  is the volume integral of  $\vec{F}$  throughout the region.

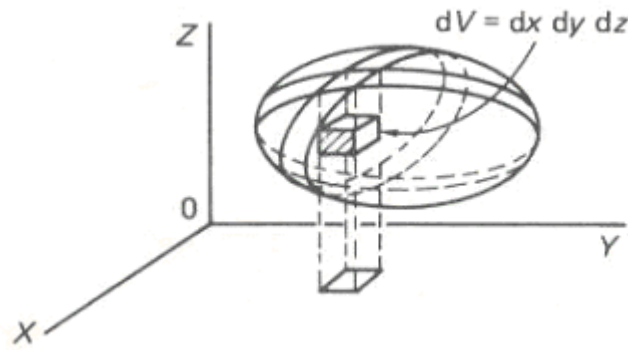


Figure # 100

$$dV = dx dy dz$$

$$\text{Then, } \int_V \vec{F} dV = \int_{x_1}^{x_2} \int_{y_1}^{y_2} \int_{z_1}^{z_2} \vec{F} dz dy dx$$

### Double Integral

$$\begin{aligned} \text{Example: } & \int_0^1 \int_0^1 x^2 dx dy \\ &= \int_0^1 \left[ \int_0^1 x^2 dx \right] dy \\ &= \int_0^1 \left[ \frac{x^3}{3} \right]_0^1 dy \\ &= \int_0^1 \left[ \frac{1^3}{3} - \frac{0^3}{3} \right] dy \\ &= \int_0^1 \left[ \frac{1}{3} - 0 \right] dy \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \left[ \frac{1}{3} \right] dy \\
&= \frac{1}{3} \int_0^1 dy \\
&= \frac{1}{3} [y]_0^1 \\
&= \frac{1}{3} [1 - 0] \\
&= \frac{1}{3} [1] \\
&= \frac{1}{3}
\end{aligned}$$

Or

$$\begin{aligned}
\text{Example: } &\int_0^1 \int_0^1 x^2 dy dx \\
&= \int_0^1 \left[ \int_0^1 x^2 dy \right] dx \\
&= \int_0^1 \left[ x^2 \int_0^1 dy \right] dx \\
&= \int_0^1 x^2 [y]_0^1 dx \\
&= \int_0^1 x^2 [1 - 0] dx \\
&= \int_0^1 x^2 dx \\
&= \left[ \frac{x^3}{3} \right]_0^1 \\
&= \left[ \frac{1^3}{3} - \frac{0^3}{3} \right] \\
&= \left[ \frac{1}{3} - 0 \right] \\
&= \frac{1}{3}
\end{aligned}$$

**Q # 75:** If  $\vec{F} = 2z\hat{i} - x\hat{j} + y\hat{k}$ , Evaluate  $\int_V \vec{F} dV$  where v is the bounded by the surfaces.

$$x = 0, x = 2, y = 0, y = 4, z = x^2, z = 2.$$

Answer:  $\int_V \vec{F} dV = \iiint_V (2z\hat{i} - x\hat{j} + y\hat{k}) dz dy dx$

$$\begin{aligned} &= \int_0^2 \int_0^4 \int_{x^2}^2 (2z\hat{i} - x\hat{j} + y\hat{k}) dz dy dx \\ &= \int_0^2 \int_0^4 \left[ 2z^2/2 \hat{i} - xz\hat{j} + yz\hat{k} \right]_{x^2}^2 dy dx \\ &= \int_0^2 \int_0^4 \left[ z^2 \hat{i} - xz\hat{j} + yz\hat{k} \right]_{x^2}^2 dy dx \\ &= \int_0^2 \int_0^4 \left[ 4\hat{i} - 2x\hat{j} + 2y\hat{k} - x^4\hat{i} + x^3\hat{j} - x^2y\hat{k} \right] dy dx \\ &= \int_0^2 \left[ 4y\hat{i} - 2xy\hat{j} + y^2\hat{k} - x^4y\hat{i} + x^3y\hat{j} - x^2 \frac{y^2}{2} \hat{k} \right]_0^4 dx \\ &= \int_0^2 \left[ 16\hat{i} - 8x\hat{j} + 16\hat{k} - 4x^4\hat{i} + 4x^3\hat{j} - 8x^2\hat{k} \right] dx \\ &= \left[ 16x\hat{i} - 4x^2\hat{j} + 16x\hat{k} - \frac{4}{5}x^5\hat{i} + x^4\hat{j} - \frac{8}{3}x^3\hat{k} \right]_0^2 \\ &= 32\hat{i} - 16\hat{j} + 32\hat{k} - \frac{128}{5}\hat{i} + 16\hat{j} - \frac{8}{3} \times 8\hat{k} \\ &= \frac{160-128}{5}\hat{i} + \frac{96-64}{3}\hat{k} \\ &= \frac{32}{5}\hat{i} + \frac{32}{3}\hat{k} \text{ Answer} \end{aligned}$$

**Q # 76:** Evaluate  $\int_V \vec{F} dV$  where V is the region bounded by the planes:

$$x = 0, x = 2, y = 0, y = 3, z = 0, z = 4 \text{ and } \vec{F} = xy\hat{i} + z\hat{j} - x^2\hat{k}.$$

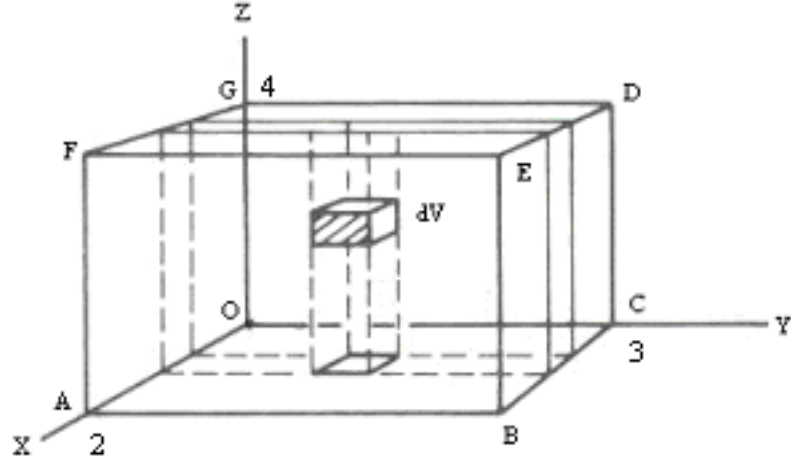


Figure # 101

Answer:  $\int_V \vec{F} dV = \iiint_V (xy\hat{i} + z\hat{j} - x^2\hat{k}) dx dy dz$

$$= \int_0^4 \int_0^3 \left[ \int_0^2 (xy\hat{i} + z\hat{j} - x^2\hat{k}) dx \right] dy dz$$

$$= \int_0^4 \int_0^3 \left[ y \frac{x^2}{2} \hat{i} + xz\hat{j} - \frac{x^3}{3} \hat{k} \right]_0^2 dy dz \quad \left[ \int x^n dx = \frac{x^{n+1}}{n+1}; \int dx = x \right]$$

$$= \int_0^4 \int_0^3 \left[ \left( y \frac{2^2}{2} \hat{i} + 2z\hat{j} - \frac{2^3}{3} \hat{k} \right) - \left( y \frac{0^2}{2} \hat{i} + 0z\hat{j} - \frac{0^3}{3} \hat{k} \right) \right] dy dz$$

$$= \int_0^4 \int_0^3 \left[ \left( 4y \hat{i} + 2z\hat{j} - \frac{8}{3} \hat{k} \right) \right] dy dz$$

$$= \int_0^4 \left[ \int_0^3 \left( 2y\hat{i} + 2z\hat{j} - \frac{8}{3} \hat{k} \right) dy \right] dz$$

$$= \int_0^4 \left[ 2 \frac{y^2}{2} \hat{i} + 2yz\hat{j} - \frac{8}{3} y\hat{k} \right]_0^3 dz \quad \left[ \because \int x^n dx = \frac{x^{n+1}}{n+1}; \int dy = y \right]$$

$$= \int_0^4 \left[ y^2 \hat{i} + 2yz\hat{j} - \frac{8}{3} y\hat{k} \right]_0^3 dz$$

$$= \int_0^4 \left[ \left( 3^2 \hat{i} + 2 \times 3.z.\hat{j} - \frac{8}{3} \times 3\hat{k} \right) - \left( 0^2 \hat{i} + 2 \times 0.z\hat{j} - \frac{8}{3} \times 0.\hat{k} \right) \right] dz$$

$$= \int_0^4 \left[ 9\hat{i} + 6z\hat{j} - 8\hat{k} \right] dz$$

$$= \int_0^4 9\hat{i} dz + \int_0^4 6z\hat{j} dz + \int_0^4 -8\hat{k} dz$$

$$\begin{aligned}
&= \left[ 9z\hat{i} + 6\frac{z^2}{2}\hat{j} - 8z\hat{k} \right]_0^4 \\
&= \left[ 9z\hat{i} + 3z^2\hat{j} - 8z\hat{k} \right]_0^4 \\
&= \left[ (9 \times 4\hat{i} + 3 \times 4^2\hat{j} - 8 \times 4\hat{k}) - (9 \times 0\hat{i} + 3 \times 0^2\hat{j} - 8 \times 0\hat{k}) \right] \\
&= \left[ 36\hat{i} + 3 \times 16\hat{j} - 32\hat{k} \right] \\
&= 4[9\hat{i} + 12\hat{j} - 8\hat{k}]
\end{aligned}
\quad \left[ \because \int x^n dx = \frac{x^{n+1}}{n+1}; \int dz = z \right]$$

### Divergence theorem (Gauss' theorem)

For a closed surface  $S$ , enclosing a region  $V$  in a vector field  $\vec{F}$ .

$$\text{Then, } \int_V \text{div} \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$$

**Q # 77:** Verify the divergence theorem for the vector field  $\vec{F} = x^2\hat{i} + z\hat{j} + y\hat{k}$  taken over the region bounded by the planes  $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$

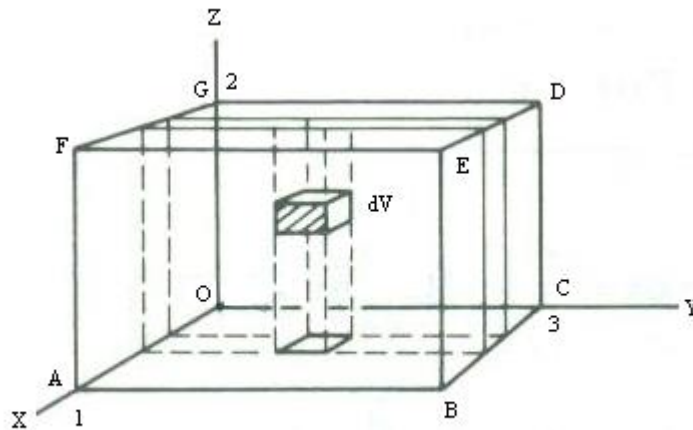


Figure # 102

Answer:

$$dV = dx dy dz$$

$$\text{We have to show that } \int_V \text{div} \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$$

**L.H.S.**

$$\text{Given, } \vec{F} = x^2\hat{i} + z\hat{j} + y\hat{k}$$

$$\therefore \text{div} \vec{F} = \nabla \cdot \vec{F} = \left( \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \cdot (x^2\hat{i} + z\hat{j} + y\hat{k})$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \left( \frac{\delta}{\delta x}(x^2) + \frac{\delta}{\delta y}(z) + \frac{\delta}{\delta z}(y) \right)$$

$$= 2x + 0 + 0 = 2x$$

L.H.S.

$$\therefore \int_V \text{div} \vec{F} dV = \int_V 2x dV = \iiint_V 2x dz dy dx = \int_0^1 \int_0^3 \int_0^2 2x dz dy dx$$

$$= \int_0^1 \int_0^3 [2xz]_0^2 dy dx \quad [\int dz = z]$$

$$= \int_0^1 \int_0^3 [(2x \times 2) - (2x \times 0)] dy dx$$

$$= \int_0^1 \int_0^3 [4x] dy dx$$

$$= \int_0^1 [4xy]_0^3 dx \quad [\int dy = y]$$

$$= \int_0^1 [4x \times 3 - 4x \times 0] dx$$

$$= \int_0^1 [12x] dx$$

$$= \left[ \frac{12x^2}{2} \right]_0^1 \quad [\because \int x^n dx = \frac{x^{n+1}}{n+1}]$$

$$= [6x^2]_0^1 = [6 \times 1^2 - 6 \times 0^2] = 6$$

$$\therefore \int_V \text{div} \vec{F} dV = 6$$

R.H.S.

We have,

**Any Vector = Length of this Vector  $\times$  Unit Vector**

$$\therefore d\vec{S} = |ds| \hat{\eta}$$

or

$$\therefore d\vec{S} = dS \hat{\eta}$$

So, we can write

$$\int_S \vec{F} \cdot d\vec{S} = \int_S \vec{F} \cdot dS \hat{\eta}$$

The enclosing surface S consists of six separate plane faces denoted as

$S_1, S_2, S_3, S_4, S_5, S_6$  as shown. We consider each face in turn  $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$

(i)  $S_1$  (OABC Base):  $z = 0$ ;  $\hat{\eta} = -\hat{k}$  (clockwise)

$\therefore \vec{F} = x^2 \hat{i} + 0 \hat{j} + y \hat{k} = x^2 \hat{i} + y \hat{k}$  and  $dS_1 = dx dy$  [  $dS = dx dy$  ]

$$\int_{S_1} \vec{F} \cdot \hat{\eta} dS_1 = \iint_{S_1} (x^2 \hat{i} + y \hat{k}) \cdot (-\hat{k}) dy dx = \int_0^1 \int_0^3 (-y) dy dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[ -\frac{y^2}{2} \right]_0^3 dx = \int_0^1 \left[ -\frac{3^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[ -\frac{9}{2} \right] dx = \left[ -\frac{9}{2} x \right]_0^1 = -\frac{9}{2} (1 - 0) = -\frac{9}{2}$$

$$[\because \int dx = x]$$

ii)  $S_2$  (DEFG Top):  $z = 2$ ;  $\hat{\eta} = \hat{k}$

$\therefore \vec{F} = x^2 \hat{i} + 2 \hat{j} + y \hat{k}$  and  $dS_2 = dx dy$

$$\int_{S_2} \vec{F} \cdot \hat{\eta} dS_2 = \iint_{S_2} (x^2 \hat{i} + 2 \hat{j} + y \hat{k}) \cdot (\hat{k}) dy dx = \int_0^1 \int_0^3 (y) dy dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[ \frac{y^2}{2} \right]_0^3 dx = \int_0^1 \left[ \frac{3^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[ \frac{9}{2} \right] dx = \left[ \frac{9}{2} x \right]_0^1 = \frac{9}{2} (1 - 0) = \frac{9}{2} \quad [\because \int dx = x]$$

iii)  $S_3$  (BCDE, Right-hand end):  $y = 3$ ;  $\hat{\eta} = \hat{j}$

$\therefore \vec{F} = x^2 \hat{i} + z \hat{j} + 3 \hat{k}$  and  $dS_3 = dx dz$

$$\int_{S_3} \vec{F} \cdot \hat{\eta} dS_3 = \iint_{S_3} (x^2 \hat{i} + z \hat{j} + 3 \hat{k}) \cdot (\hat{j}) dz dx = \int_0^1 \int_0^2 (z) dz dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[ \frac{z^2}{2} \right]_0^2 dx = \int_0^1 \left[ \frac{2^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[ \frac{4}{2} \right] dx = [2x]_0^1 = 2(1 - 0) = 2 \quad [\because \int dx = x]$$

iv)  $S_4$  (OAFG, Left-hand end):  $y = 0$ ;  $\hat{\eta} = -\hat{j}$

$\therefore \vec{F} = x^2 \hat{i} + z \hat{j} + 0 \hat{k}$  and  $dS_4 = dx dz$

$$\int_{S_4} \vec{F} \cdot \hat{\eta} dS_4 = \iint_{S_4} (x^2 \hat{i} + z \hat{j} + 0 \hat{k}) \cdot (-\hat{j}) dz dx = \int_0^1 \int_0^2 (-z) dz dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[ -\frac{z^2}{2} \right]_0^2 dx = \int_0^1 \left[ -\frac{2^2}{2} + -\frac{0^2}{2} \right] dx = \int_0^1 \left[ -\frac{4}{2} \right] dx = [-2x]_0^1 = -2(1-0) = -2 \quad [\because \int dx = x]$$

v)  $S_5$  (ABEF, Front):  $x = 1$ ;  $\hat{n} = \hat{i}$

$$\therefore \vec{F} = 1^2 \hat{i} + z \hat{j} + y \hat{k} \text{ and } dS_5 = dydz$$

$$\int_{S_5} \vec{F} \cdot \hat{n} dS_5 = \iint_{S_5} (1^2 \hat{i} + z \hat{j} + y \hat{k}) \cdot (\hat{i}) dydz = \int_0^2 \int_0^3 dydz$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^2 [y]_0^3 dz = \int_0^2 [3-0] dz = \int_0^2 [3] dz = [3z]_0^2 = 3(2-0) = 6 \quad [\because \int dy = y]$$

vi)  $S_6$  (OCDG, Back):  $x = 0$ ;  $\hat{n} = -\hat{i}$

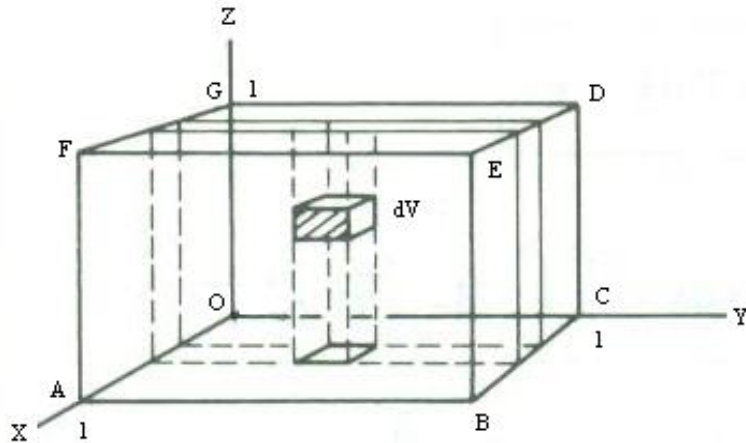
$$\therefore \vec{F} = 0 \cdot \hat{i} + z \hat{j} + y \hat{k} \text{ and } dS_6 = dydz$$

$$\int_{S_6} \vec{F} \cdot \hat{n} dS_6 = \iint_{S_6} (0 \cdot \hat{i} + z \hat{j} + y \hat{k}) \cdot (-\hat{i}) dydz = 0$$

For the whole surface  $S$  we therefore have

$$\int_S \vec{F} \cdot d\vec{S} = -\frac{9}{2} + \frac{9}{2} + 2 - 2 + 6 + 0 = 6$$

**Q # 78:** Show that  $\iint_S \vec{F} \cdot \hat{n} ds = \frac{3}{2}$ ; Where  $\vec{F} = 4xz \hat{i} - y^2 \hat{j} + yz \hat{k}$  and  $S$  is the surface of the cube bounded by the planes  $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$



**Figure # 103**

| Serial no | Surface | $\hat{\eta}=?$          | $ds$    | Plane   |
|-----------|---------|-------------------------|---------|---------|
| 1         | OABC    | $\hat{\eta} = -\hat{k}$ | $dx dy$ | $z = 0$ |
| 2         | DEFG    | $\hat{\eta} = \hat{k}$  | $dx dy$ | $z = 1$ |
| 3         | OAFG    | $\hat{\eta} = -\hat{j}$ | $dx dz$ | $y = 0$ |
| 4         | BCDE    | $\hat{\eta} = \hat{j}$  | $dx dz$ | $y = 1$ |
| 5         | OCDG    | $\hat{\eta} = -\hat{i}$ | $dy dz$ | $x = 0$ |
| 6         | ABEF    | $\hat{\eta} = \hat{i}$  | $dy dz$ | $x = 1$ |

Now,

$$\iint_S \vec{F} \cdot \hat{\eta} ds = \iint_{OABC} \vec{F} \cdot \hat{\eta} ds + \iint_{DEFG} \vec{F} \cdot \hat{\eta} ds + \iint_{OAFG} \vec{F} \cdot \hat{\eta} ds + \iint_{BCDE} \vec{F} \cdot \hat{\eta} ds + \iint_{OCDG} \vec{F} \cdot \hat{\eta} ds + \iint_{ABEF} \vec{F} \cdot \hat{\eta} ds \quad (1)$$

$$1. \iint_{OABC} \vec{F} \cdot \hat{\eta} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{\eta} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{k}) dx dy$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 -yz dx dy$$

$$= - \int_0^1 \int_0^1 yz dx dy = - \int_0^1 [yzx]_0^1 dy \quad [\because \int dx = x]$$

$$= - \int_0^1 [yz \times 1 - yz \times 0] dy = - \int_0^1 [yz(1-0)] dy$$

$$= - \left[ z \frac{y^2}{2} \right]_0^1 = - \left[ z \frac{1^2}{2} - z \frac{0^2}{2} \right] = - \frac{z}{2} (1-0) = - \frac{z}{2} = 0 [\because z = 0]$$

2.

$$\iint_{DEFG} \vec{F} \cdot \hat{\eta} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{\eta} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{k}) dx dy = \int_0^1 \int_0^1 yz dx dy$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 yz dx dy = \int_0^1 [yzx]_0^1 dy = \int_0^1 [yz \times 1 - yz \times 0] dy \quad [\because \int dx = x]$$

$$= \int_0^1 [yz] dy = \left[ z \frac{y^2}{2} \right]_0^1 = \left[ z \frac{1^2}{2} - z \frac{0^2}{2} \right] = \frac{z}{2} (1-0) = \frac{z}{2} = \frac{1}{2} [\because z = 1]$$

3.

$$\begin{aligned}
 \iint_{\text{OAFG}} \vec{F} \cdot \hat{n} \, ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} \, ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{j}) \, dx \, dz = \int_0^1 \int_0^1 y^2 \, dx \, dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= \int_0^1 \int_0^1 y^2 \, dx \, dz = \int_0^1 [xy^2]_0^1 \, dz \quad [\because \int dx = x] \\
 &= \int_0^1 [1 \times y^2 - 0 \times y^2] \, dz \\
 &= \int_0^1 [y^2 - 0] \, dz = [y^2 z]_0^1 = [y^2 \times 1 - y^2 \times 0] \quad [\because \int dz = z] \\
 &= y^2(1 - 0) = y^2 = 0 \quad [\because y = 0]
 \end{aligned}$$

4.

$$\begin{aligned}
 \iint_{\text{BCDE}} \vec{F} \cdot \hat{n} \, ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} \, ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{j}) \, dx \, dz = \int_0^1 \int_0^1 -y^2 \, dx \, dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= -\int_0^1 \int_0^1 y^2 \, dx \, dz = -\int_0^1 [xy^2]_0^1 \, dz = -\int_0^1 [1 \times y^2 - 0 \times y^2] \, dz \quad [\because \int dx = x] \\
 &= -\int_0^1 [y^2 - 0] \, dz = -[y^2 z]_0^1 \quad [\because \int dz = z] \\
 &= -[y^2 \times 1 - y^2 \times 0] \\
 &= -y^2(1 - 0) = -y^2 = -1 \quad [\because y = 1]
 \end{aligned}$$

$$\begin{aligned}
 5. \quad \iint_{\text{OC DG}} \vec{F} \cdot \hat{n} \, ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} \, ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{i}) \, dy \, dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= \int_0^1 \int_0^1 -4xz \, dy \, dz \\
 &= -\int_0^1 \int_0^1 4xz \, dy \, dz = -\int_0^1 [4xyz]_0^1 \, dz = -\int_0^1 [4x \times 1 \times z - 4x \times 0 \times z] \, dz \quad [\because \int dy = y] \\
 &= -\int_0^1 [4xz(1 - 0)] \, dz = -\left[4x \frac{z^2}{2}\right]_0^1 = -\left[4x \frac{1^2}{2} - 4x \frac{0^2}{2}\right] \\
 &= -4x\left(\frac{1}{2} - 0\right) = -2x = 0 \quad [\because x = 0]
 \end{aligned}$$

$$6. \quad \iint_{\text{ABEF}} \vec{F} \cdot \hat{n} \, ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} \, ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{i}) \, dy \, dz$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 4xyz dy dz$$

$$= \int_0^1 \int_0^1 4xyz dy dz = \int_0^1 [4xyz]_0^1 dz = \int_0^1 [4x \times 1 \times z - 4x \times 0 \times z] dz \quad [\because \int dy = y]$$

$$= \int_0^1 [4xz(1-0)] dz = \left[ 4x \frac{z^2}{2} \right]_0^1 = \left[ 4x \frac{1^2}{2} - 4x \frac{0^2}{2} \right]$$

$$= 4x \left( \frac{1}{2} - 0 \right) = 2x = 2 \quad [\because x = 1]$$

Putting the values in (1),

Now,

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} ds &= \iint_{OABC} \vec{F} \cdot \hat{n} ds + \iint_{DEFG} \vec{F} \cdot \hat{n} ds + \iint_{OAFG} \vec{F} \cdot \hat{n} ds + \iint_{BCDE} \vec{F} \cdot \hat{n} ds + \iint_{OCDG} \vec{F} \cdot \hat{n} ds + \iint_{ABEF} \vec{F} \cdot \hat{n} ds \\ \iint_S \vec{F} \cdot \hat{n} ds &= 0 + \frac{1}{2} + 0 - 1 + 0 + 2 = \frac{1}{2} + 1 = \frac{3}{2} \text{ (Proved)} \end{aligned}$$

### Area Enclosed by a closed Curve:

One of the earliest applications of integration is finding the area of a plane figure bounded by the x-axis, the curve  $y = f(x)$  and ordinates at  $x = x_1$  and  $x = x_2$ .

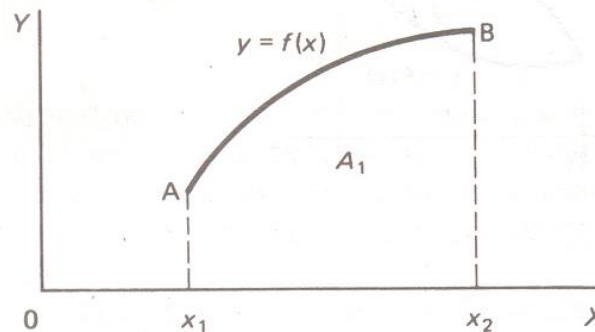


Figure # 104

$$A_1 = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} f(x) dx \text{ -----(i)}$$

If points A and B are joined by another curve  $y = F(x)$

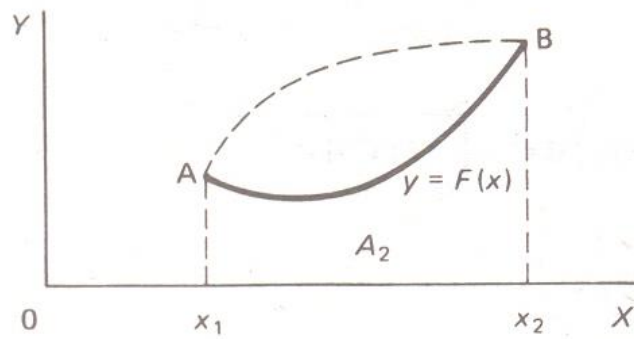


Figure # 105

$$A_2 = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} F(x) dx \text{ -----(ii)}$$

Combining the two figures, we have

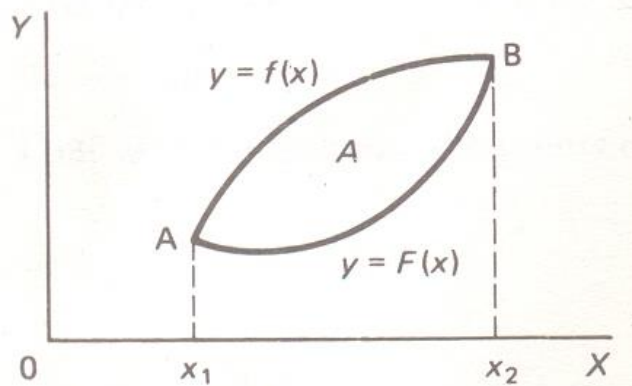


Figure # 106

$$A = A_1 - A_2$$

$$\therefore A = \int_{x_1}^{x_2} f(x) dx - \int_{x_1}^{x_2} F(x) dx \text{ -----(iii)}$$

It is convenient on occasions to arrange the limits so that the integration follows the path round the enclosed area in a regular order.

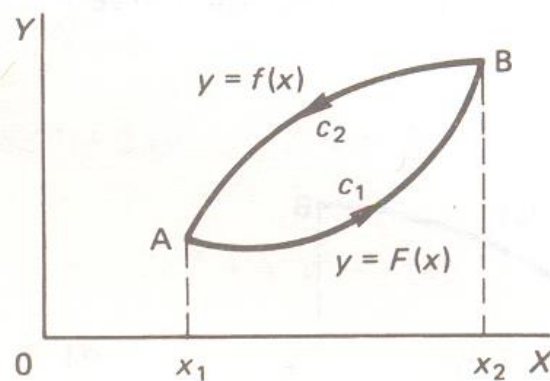


Figure # 107

For example

$\int_{x_1}^{x_2} F(x)dx$  gives  $A_2$  as before, but integrating from B to A along  $c_2$  with  $y = f(x)$ , i.e.

$\int_{x_1}^{x_2} f(x)dx$ , is the integral for  $A_1$  with the sign changed, i.e.

We can write, For Upper Curve  $y = f(x)$ ;

$$\int_{x_2}^{x_1} f(x)dx = - \int_{x_1}^{x_2} f(x)dx \quad \text{-----(iv)}$$

$$\Rightarrow - \int_{x_2}^{x_1} f(x)dx = - \left[ - \int_{x_1}^{x_2} f(x)dx \right] \quad \text{[Multiplying by negative sign on both sides]}$$

$$\Rightarrow - \int_{x_2}^{x_1} f(x)dx = \int_{x_1}^{x_2} f(x)dx$$

$$\therefore \int_{x_1}^{x_2} f(x)dx = - \int_{x_2}^{x_1} f(x)dx \quad \text{[Side change]----- (v)}$$

From (iii),

$$\therefore \text{The result } A = A_1 - A_2 = \int_{x_1}^{x_2} f(x)dx - \int_{x_1}^{x_2} F(x)dx \quad \text{[From (iii)]}$$

$$= - \int_{x_2}^{x_1} f(x)dx - \int_{x_1}^{x_2} F(x)dx \quad \text{[From (v)]}$$

$$= - \left\{ \int_{x_2}^{x_1} f(x)dx + \int_{x_1}^{x_2} F(x)dx \right\}$$

$$= - \left\{ \int_{x_1}^{x_2} F(x)dx + \int_{x_2}^{x_1} f(x)dx \right\} \quad \text{----- (vi)}$$

If we proceed round the boundary in an anticlockwise manner, the enclosed area is kept on the left-hand side and the resulting area is considered positive. If we proceed round the boundary in a clockwise manner, the enclosed area remains on the right-hand side and the resulting area is negative.

The final result above can be written in the form

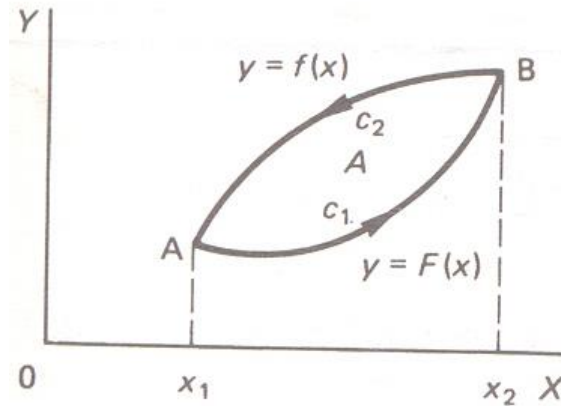


Figure # 108

$A = -\oint y dx$  ; Where the symbol  $\oint$  indicates that the integral is to be evaluated round the closed boundary in the positive (i.e. anticlockwise) direction.

$$\therefore A = -\oint y dx = -\left\{ \int_{x_1}^{x_2} F(x) dx + \int_{x_2}^{x_1} f(x) dx \right\} \text{-----(vii)}$$

**Q # 79:** Determine the area enclosed by the graphs of  $y = x^3$  and  $y = 4x$  for  $x \geq 0$

Answer: First we need to know the points of intersection.

Given,  $y = x^3$  and  $y = 4x$

Then, we can write,  $x^3 = 4x$

$$\Rightarrow x^3 - 4x = 0$$

$$\Rightarrow x(x^2 - 4) = 0$$

$$\Rightarrow x(x + 2)(x - 2) = 0$$

$$\Rightarrow x = 0 \text{ and } x = -2, x = 2$$

$$\text{But } x \geq 0$$

$$\therefore x = 0 \text{ and } x = 2$$

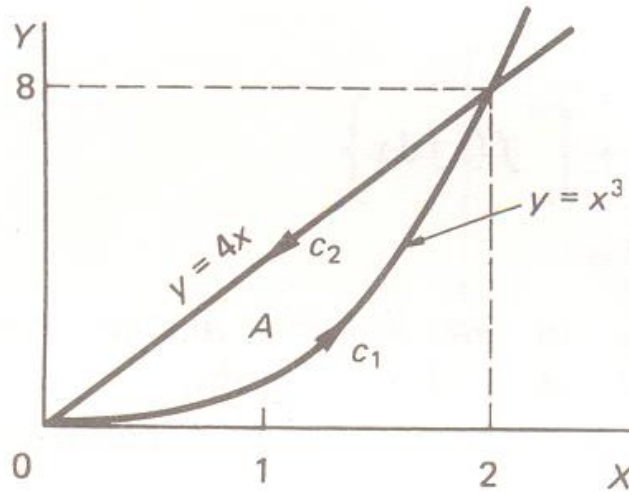


Figure # 109

We integrate in an anticlockwise manner

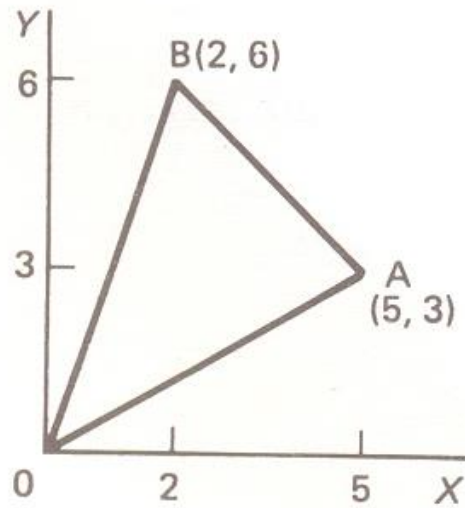
$c_1 : y = x^3$ , Limits  $x = 0$  to  $x = 2$

$c_2 : y = 4x$ , Limits  $x = 2$  to  $x = 0$

We have,

$$\begin{aligned}
 \therefore A &= -\oint y dx = -\left\{ \int_{x_1}^{x_2} F(x) dx + \int_{x_2}^{x_1} f(x) dx \right\} \\
 &= -\left\{ \int_0^2 F(x) dx + \int_2^0 f(x) dx \right\} \\
 &= -\left\{ \int_0^2 x^3 dx + \int_2^0 4x dx \right\} \\
 &= -\left\{ \left[ \frac{x^4}{4} \right]_0^2 + 4 \left[ \frac{x^2}{2} \right]_2^0 \right\} \\
 &= -\left\{ \left[ \frac{2^4}{4} - 0 \right] + 4 \left[ \frac{0^2}{2} - \frac{2^2}{2} \right] \right\} \\
 &= -\left\{ \left[ \frac{16}{4} \right] + 4 \left[ -\frac{4}{2} \right] \right\} \\
 &= 4 \text{ Answer}
 \end{aligned}$$

**Q # 80:** Find the area of the triangle with vertices **(0,0)** ,**(5,3)** and **(2,6)**



**Figure # 110**

Answer: We have,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

∴ The equation of **OA** or **AO** is

$$\frac{y - 0}{0 - 3} = \frac{x - 0}{0 - 5}$$

$$\Rightarrow \frac{y}{-3} = \frac{x}{-5}$$

$$\Rightarrow \frac{y}{3} = \frac{x}{5}$$

$$\Rightarrow y = \frac{3x}{5}$$

The equation of **BA** or **AB** is

$$\frac{y - 6}{6 - 3} = \frac{x - 2}{2 - 5}$$

$$\Rightarrow \frac{y - 6}{3} = \frac{x - 2}{-3}$$

$$\Rightarrow \frac{y - 6}{1} = \frac{x - 2}{-1}$$

$$\Rightarrow \frac{y - 6}{1} = -x + 2$$

$$\Rightarrow y = 8 - x$$

The equation of **OB** or **BO** is

$$\frac{y - 0}{0 - 6} = \frac{x - 0}{0 - 2}$$

$$\Rightarrow \frac{y - 0}{-6} = \frac{x - 0}{-2}$$

$$\Rightarrow \frac{y}{6} = \frac{x}{2}$$

$$\Rightarrow \frac{y}{3} = \frac{x}{1}$$

$$\Rightarrow y = 3x$$

Then,

$$\therefore A = -\oint y dx = -\left[ \int_{OA} f(x) dx + \int_{AB} f(x) dx + \int_{BO} f(x) dx \right]$$

$$\therefore A = -\oint y dx = -\left\{ \int_0^5 \frac{3}{5} x dx + \int_5^2 (8-x) dx + \int_2^0 3x dx \right\}$$

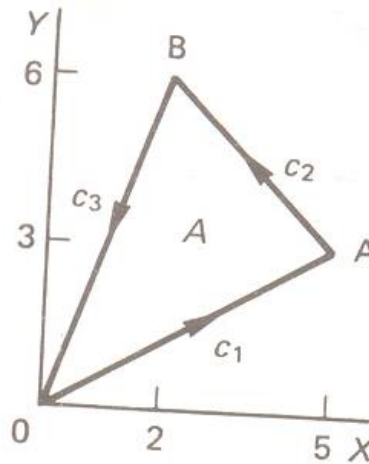


Figure # 111

The limits chosen must progress the integration round the boundary of the figure in an anticlockwise manner.

$$\begin{aligned} \text{We get, } \therefore A &= -\oint y dx = -\left\{ \int_0^5 \frac{3}{5} x dx + \int_5^2 (8-x) dx + \int_2^0 3x dx \right\} \\ &= -\left\{ \frac{3}{5} \left[ \frac{x^2}{2} \right]_0^5 + \left[ 8x - \frac{x^2}{2} \right]_5^2 + 3 \left[ \frac{x^2}{2} \right]_2^0 \right\} \\ &= -\left\{ \frac{3}{5} \left[ \frac{5^2}{2} - 0 \right] + \left[ (8 \times 2 - \frac{2^2}{2}) - (8 \times 5 - \frac{5^2}{2}) \right] + 3 \left[ \frac{0^2}{2} - \frac{2^2}{2} \right] \right\} \\ &= -\left\{ \frac{3}{5} \left[ \frac{25}{2} \right] + \left[ (16 - \frac{4}{2}) - (40 - \frac{25}{2}) \right] + 3 \left[ 0 - \frac{4}{2} \right] \right\} \\ &= -\left\{ \frac{3}{5} \left[ \frac{25}{2} \right] + \left[ (16 - 2) - (40 - \frac{25}{2}) \right] + 3 \left[ -\frac{4}{2} \right] \right\} \\ &= -\left\{ \frac{3}{5} \left[ \frac{25}{2} \right] + \left[ 14 - (40 - \frac{25}{2}) \right] - 3 \left[ \frac{4}{2} \right] \right\} \end{aligned}$$

$$\begin{aligned}
&= -\left\{3\left[\frac{5}{2}\right] + \left[14 - \left(40 - \frac{25}{2}\right)\right] - 6\right\} \\
&= -\left\{\frac{15}{2} + \left[14 - \left(40 - \frac{25}{2}\right)\right] - 6\right\} \\
&= -\left\{\frac{15 + 28 - 80 + 25 - 12}{2}\right\} \\
&= -\left\{\frac{-24}{2}\right\} \\
&= 12 \text{ Square units}
\end{aligned}$$

### Another method of a line Integral

It is often more convenient to integrate with respect to x or y than to take arc length as the variable.

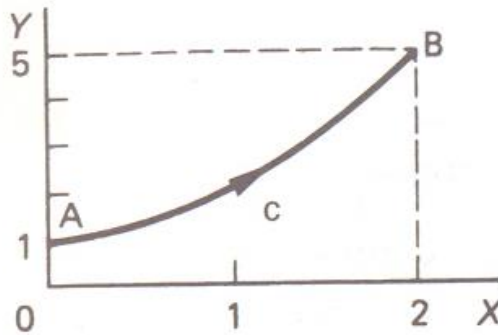
**We have,**

$$\text{Work done} = \int_{AB} \vec{F} \cdot d\vec{S} = \int_{AB} (Pdx + Qdy) \text{ ----- (iii)}$$

In three dimension,

$$\text{Work done} = \int_{AB} \vec{F} \cdot d\vec{S} = \int_{AB} (Pdx + Qdy + Rdz) \text{ ----- (iv)}$$

**Q # 81:** Evaluate  $\int_c (x + 3y)dx$  from **A(0,1)** to **B(2,5)** along the curve  $y = 1 + x^2$



**Figure # 112**

Answer:

The line integral is of the form:  $\int_c (Pdx + Qdy)$  where, in this case,  $P = x + 3y$  and

$Q = 0$  and c is the curve  $y = 1 + x^2$

It can be converted at once into an ordinary integral by substituting for y and applying the appropriate limits of x.

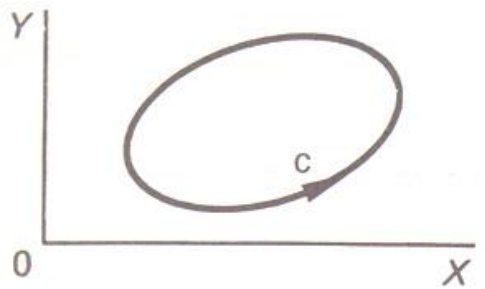
$$I = \int_c (Pdx + Qdy) = \int_c (x + 3y)dx = \int_0^2 \{x + 3(1 + x^2)\} = \int_0^2 (x + 3 + 3x^2)dx$$

$$= \left[ \frac{x^2}{2} + 3x + 3 \frac{x^3}{3} \right]_0^2 = \left[ \left( \frac{2^2}{2} + 3 \times 2 + 3 \times \frac{2^3}{3} \right) - 0 \right] = 16$$

### Line Integrals round a closed curve

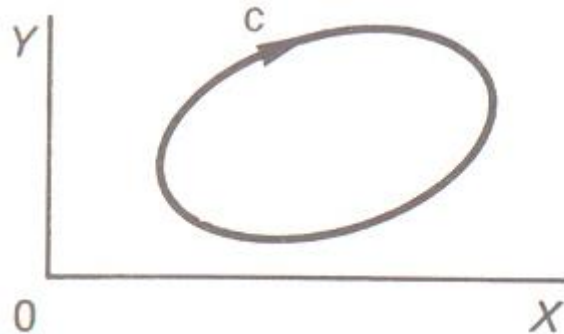
We have already introduced the symbol  $\oint$  to indicate that an integral is to be evaluated round a closed curve in the positive (anticlockwise) direction.

Positive (anticlockwise) direction line integral denoted by  $\oint$



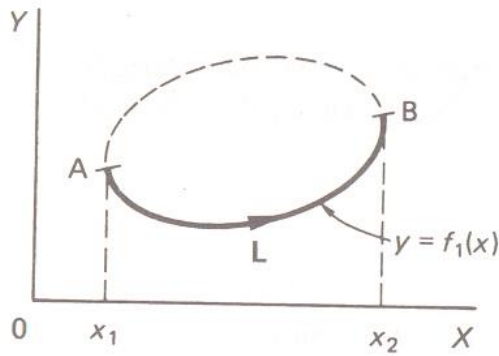
**Figure # 113**

Negative (clockwise) direction line integral denoted by  $-\oint$

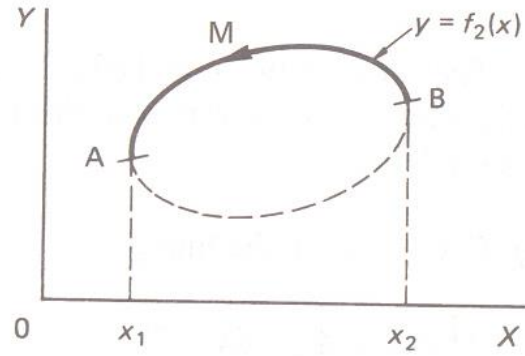


**Figure # 114**

With a closed curve, the path  $c$  cannot be single-valued. Therefore, we divide the path into two or more parts and treat each separately as a single-valued curve.



(i) Use  $y = f_1(x)$  for ALB



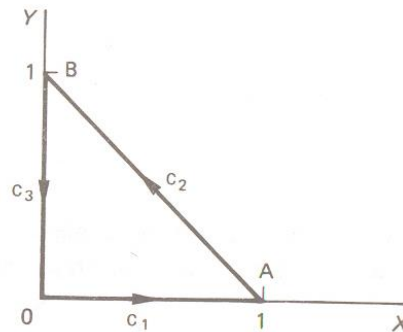
(ii) Use  $y = f_2(x)$  for BMA.

**Figure # 115**

Unless specially required otherwise, we always proceed round the closed curve in an anticlockwise direction

**Q # 82:** Evaluate the line integral  $I = \oint_C (x^2 dx - 2xy dy)$  where  $c$  comprises the three sides of the triangle joining  $O(0,0)$ ,  $A(1,0)$  and  $B(0,1)$

**Answer:** First draw the diagram and mark in  $c_1, c_2$  and  $c_3$  the proposed directions of integration. Do just that.



**Figure # 116**

The three sections of the path of integration must be arranged in an anticlockwise manner round the figure. Now we deal with each part separately.

a) The equation of OA is:

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2} \Rightarrow \frac{y - 0}{0 - 0} = \frac{x - 0}{0 - 1} \Rightarrow \frac{y}{0} = \frac{x}{-1} \Rightarrow -y = x.0 \Rightarrow y = 0$$

**OA :**  $c_1$  is the line  $y = 0 \therefore dy = 0$ , Then

$I_1 = \oint (x^2 dx - 2xy dy)$  for this part becomes

$$I_1 = \int_0^1 (x^2 dx - 2x \cdot 0 \cdot 0) = \int_0^1 x^2 dx = \left[ \frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

b) **AB** :  $c_2$  is the line:  $\frac{y-0}{0-1} = \frac{x-1}{1-0} \Rightarrow \frac{y}{-1} = \frac{x-1}{1} \Rightarrow y = -x+1$

$$\therefore \frac{dy}{dx} = -1+0 \Rightarrow dy = -dx$$

Then

$I_2 = \oint (x^2 dx - 2xy dy)$  for this part becomes

$$I_2 = \int_1^0 \{x^2 dx - 2x(-x+1)(-dx)\} = \int_1^0 (x^2 - 2x^2 + 2x)(dx) = \int_1^0 (2x - x^2) dx = \left[ \frac{2x^2}{2} - \frac{x^3}{3} \right]_1^0 = -\frac{2}{3}$$

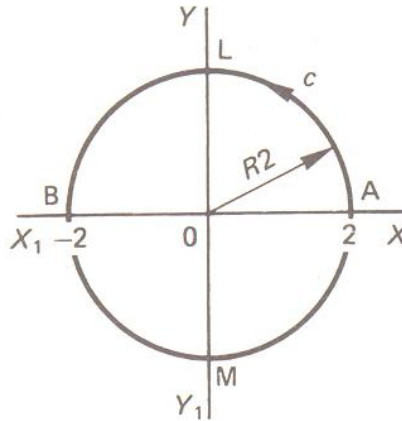
c) **BO** :  $c_3$  is the line  $x = 0 \therefore dx = 0$ , Then

$I_1 = \oint (x^2 dx - 2xy dy)$  for this part becomes

$$I_1 = \oint (0^2 \cdot 0 - 2 \cdot 0 \cdot y dy) = 0$$

Finally  $I = I_1 + I_2 + I_3 = \frac{1}{3} - \frac{2}{3} + 0 = -\frac{1}{3}$

**Q # 83:** Evaluate the area of a circle  $x^2 + y^2 = 4$ .



**Figure # 117**

Answer: Given  $x^2 + y^2 = 4$

$$\Rightarrow y^2 = 4 - x^2$$

$$\Rightarrow y = \pm \sqrt{4 - x^2}$$

The Equation of the upper curve ALB:  $y = +\sqrt{4 - x^2}$  between  $x = 2$  and  $x = -2$

And The equation of the lower curve BMA :  $y = -\sqrt{4 - x^2}$  between  $x = -2$  and  $x = 2$

We have, Area:  $A = -\oint y dx = -\left\{ \int_{x_1}^{x_2} F(x) dx + \int_{x_2}^{x_1} f(x) dx \right\}$

$$\therefore A = -\left[ \int_2^{-2} \sqrt{4-x^2} dx + \int_{-2}^2 \{-(\sqrt{4-x^2}) dx\} \right]$$

$$\therefore A = -\left[ \int_2^{-2} \sqrt{4-x^2} dx - \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[ \int_2^{-2} \sqrt{4-x^2} dx + \int_2^{-2} \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[ 2 \int_2^{-2} \sqrt{4-x^2} dx \right]$$

$$\therefore A = (-2)(-1) \int_{-2}^2 \sqrt{4-x^2} dx$$

$$\therefore A = 2 \int_{-2}^2 \sqrt{4-x^2} dx \text{ -----(i)}$$

$$\therefore A = 2 \times 2 \int_0^2 \sqrt{4-x^2} dx \quad \left[ \because \int_{-a}^{+a} f(x) dx = 2 \int_0^a f(x) dx \right]$$

$$\therefore A = 4 \int_0^2 \sqrt{4-x^2} dx \text{ -----(ii)}$$

Let,  $x = 2 \sin \theta$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} (2 \sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \cos \theta$$

$$\Rightarrow dx = 2 \cos \theta d\theta$$

$$\text{Now, } \sqrt{4-x^2} = \sqrt{4-(2 \sin \theta)^2} = \sqrt{4-4 \sin^2 \theta} = \sqrt{4(1-\sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

Given,  $x = 2 \sin \theta$

$$\Rightarrow \theta = \sin^{-1} \left( \frac{x}{2} \right)$$

| x                                               | 0                                                                                                                                                                                                                             | 2                                                                                                                                                                                                                            |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\theta = \sin^{-1} \left( \frac{x}{2} \right)$ | $\theta = \sin^{-1} \left( \frac{0}{2} \right)$<br>$\Rightarrow \theta = \sin^{-1} \left( \frac{0}{2} \right)$<br>$\Rightarrow \theta = \sin^{-1} (0)$<br>$\Rightarrow \theta = \sin^{-1} \sin 0$<br>$\Rightarrow \theta = 0$ | $\theta = \sin^{-1} \left( \frac{x}{2} \right)$<br>$\Rightarrow \theta = \sin^{-1} \left( \frac{2}{2} \right)$<br>$\Rightarrow \theta = \sin^{-1} (1)$<br>$\Rightarrow \theta = \sin^{-1} \left( \sin \frac{\pi}{2} \right)$ |

|  |  |                                      |
|--|--|--------------------------------------|
|  |  | $\Rightarrow \theta = \frac{\pi}{2}$ |
|--|--|--------------------------------------|

From (ii),

$$\therefore A = 4 \int_0^2 \sqrt{4 - x^2} dx$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 - (2\sin\theta)^2} 2\cos\theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 - 4\sin^2\theta} 2\cos\theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4(1 - \sin^2\theta)} 2\cos\theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4\cos^2\theta} 2\cos\theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} 2\cos\theta \times 2\cos\theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} 4\cos^2\theta d\theta$$

$$\therefore A = 4 \times 2 \int_0^{\pi/2} 2\cos^2\theta d\theta$$

$$\therefore A = 8 \int_0^{\pi/2} 2\cos^2\theta d\theta$$

$$\therefore A = 8 \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$$

$$[[2\cos^2\theta = 1 + \cos 2\theta]]$$

$$\therefore A = 8 \left[ \theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$\therefore A = 8 \left[ \left( \frac{\pi}{2} - 0 \right) + \left( \frac{\sin 2 \times \frac{\pi}{2}}{2} - 0 \right) \right]$$

$$\therefore A = 8 \left[ \frac{\pi}{2} + \frac{\sin \pi}{2} \right]$$

$$\therefore A = 8 \left[ \frac{\pi}{2} + \frac{0}{2} \right]$$

$$\therefore A = 8 \left[ \frac{\pi}{2} \right]$$

$$\therefore A = 4\pi \quad \text{Answer}$$

Or from (i),

$$\therefore A = 2 \int_{-2}^2 \sqrt{4 - x^2} dx \text{-----(i)}$$

$$\text{Let, } x = 2\sin\theta$$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta}(2\sin\theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2\cos\theta$$

$$\Rightarrow dx = 2\cos\theta d\theta$$

$$\text{Now, } \sqrt{4 - x^2} = \sqrt{4 - (2\sin\theta)^2} = \sqrt{4 - 4\sin^2\theta} = \sqrt{4(1 - \sin^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta$$

$$\text{Given, } x = 2\sin\theta$$

$$\frac{x}{2} = \sin\theta$$

$$\sin\theta = \frac{x}{2}$$

$$\Rightarrow \theta = \sin^{-1}(x/2)$$

| x                         | - 2                                                                                                                                                                                                                                                   | 2                                                                                                                                                                                                         |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\theta = \sin^{-1}(x/2)$ | $\theta = \sin^{-1}(x/2)$<br>$\Rightarrow \theta = \sin^{-1}(-2/2)$<br>$\Rightarrow \theta = \sin^{-1}(-1)$<br>$\Rightarrow \theta = -\sin^{-1}(1)$<br>$\Rightarrow \theta = -\sin^{-1}(\sin \frac{\pi}{2})$<br>$\Rightarrow \theta = -\frac{\pi}{2}$ | $\theta = \sin^{-1}(x/2)$<br>$\Rightarrow \theta = \sin^{-1}(2/2)$<br>$\Rightarrow \theta = \sin^{-1}(1)$<br>$\Rightarrow \theta = \sin^{-1}(\sin \frac{\pi}{2})$<br>$\Rightarrow \theta = \frac{\pi}{2}$ |

From (i),

$$\therefore A = 2 \int_{-2}^2 \sqrt{4 - x^2} dx$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 - (2\sin\theta)^2} 2\cos\theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 - 4\sin^2 \theta} \cdot 2\cos\theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4(1 - \sin^2 \theta)} \cdot 2\cos\theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4\cos^2 \theta} \cdot 2\cos\theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} 2\cos\theta \times 2\cos\theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} 4\cos^2 \theta d\theta$$

$$\therefore A = 2 \times 2 \int_{-\pi/2}^{\pi/2} 2\cos^2 \theta d\theta$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} 2\cos^2 \theta d\theta$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta \quad [2\cos^2 \theta = 1 + \cos 2\theta]$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} 1 d\theta + 4 \int_{-\pi/2}^{\pi/2} \cos 2\theta d\theta$$

$$\therefore A = 4[\theta]_{-\pi/2}^{\pi/2} + 4\left[\frac{\sin 2\theta}{2}\right]_{-\pi/2}^{\pi/2}$$

$$\therefore A = 4\left[\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right)\right] + 4\left[\left(\frac{\sin 2 \times \frac{\pi}{2}}{2} - \frac{\sin 2 \times \left(-\frac{\pi}{2}\right)}{2}\right)\right]$$

$$\therefore A = 4\left[\left(\frac{\pi}{2} + \frac{\pi}{2}\right)\right] + 4\left[\left(\frac{\sin \pi}{2} - \frac{\sin(-\pi)}{2}\right)\right]$$

$$\therefore A = 4\left[\left(\frac{2\pi}{2}\right)\right] + 4\left[\left(\frac{\sin \pi}{2} + \frac{\sin \pi}{2}\right)\right] \quad [\because \sin(-\theta) = -\sin\theta]$$

$$\therefore A = 4[\pi] + 4\left[\left(\frac{0}{2} + \frac{0}{2}\right)\right] \quad [\because \sin \pi = 0]$$

$$\therefore A = 4[\pi] + 4.0$$

$$\therefore A = 4\pi \quad \text{Answer}$$

**Justification**

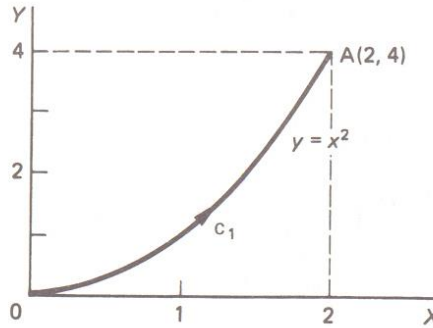
We know Area of a circle is  $\pi r^2$

**Here radius =  $r = 2$**

$$\therefore \text{Area of a circle is } \pi r^2 = \pi \times 2^2 = \pi \times 4 = 4\pi$$

**Q # 84:** Evaluate  $I = \oint_C (3x^2y^2dx + 2x^3ydy)$  between  $O(0,0)$  and  $A(2,4)$

- a) along  $c_1$  i.e.  $y = x^2$
- b) along  $c_2$  i.e.  $y = 2x$
- c) along  $c_3$  i.e.  $x = 0$  from  $(0,0)$  to  $(0,4)$  and  $y = 4$  from  $(0,4)$  to  $(2,4)$



**Figure # 118**

$$I = \oint_C (3x^2y^2dx + 2x^3ydy)$$

The path  $c_1$  is  $y = x^2$   
 $\Rightarrow dy = 2xdx$

$$I_1 = \int_0^2 (3x^2y^2dx + 2x^3ydy)$$

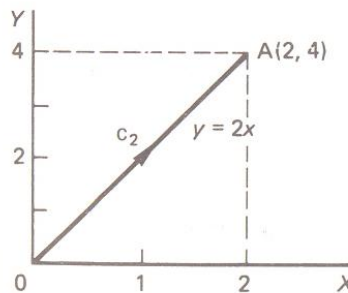
$$I_1 = \int_0^2 (3x^2(x^2)^2dx + 2x^3 \times x^2 \times 2xdx)$$

$$I_1 = \int_0^2 (3x^6 + 4x^6)dx$$

$$I_1 = \int_0^2 7x^6dx = \left[ 7\frac{x^7}{7} \right]_0^2$$

$$I_1 = 128$$

- a) In (b), the path of integration changes to  $c_2$  i.e.  $y = 2x$



**Figure # 119**

$$I = \oint_C (3x^2y^2dx + 2x^3ydy)$$

The path  $c_2$  is  $y = 2x$

$$\Rightarrow dy = 2dx$$

$$I_2 = \int_0^2 (3x^2y^2dx + 2x^3ydy)$$

$$I_2 = \int_0^2 (3x^2(2x)^2dx + 2x^3 \times 2x \times 2dx)$$

$$I_2 = \int_0^2 (12x^4 + 8x^4)dx$$

$$I_2 = \int_0^2 20x^4dx$$

$$I_2 = \left[ 20 \frac{x^5}{5} \right]_0^2$$

$$I_2 = 128$$

c) In the third case, the path  $c_3$  is split

i.  $x = 0$  from  $(0,0)$  to  $(0,4)$

ii.  $y = 4$  from  $(0,4)$  to  $(2,4)$

Sketch the diagram and determine  $I_3$

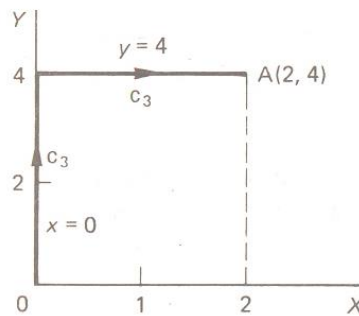


Figure # 120

i. From  $(0,0)$  to  $(0,4)$

$$x = 0 \Rightarrow dx = 0$$

$$I = \oint_C (3x^2y^2dx + 2x^3ydy)$$

The path  $c_3$  is  $x = 0$

$$\Rightarrow dx = 0$$

$$I_3 = \int_0^4 (3x^2y^2dx + 2x^3ydy)$$

$$I_3 = \int_0^2 (3(0)^2 y^2(0) + 2(0)^3 \times y dy)$$

$$I_3 = 0$$

ii. From (0,4) to (2,4)  $y = 4 \Rightarrow dy = 0$

$$I = \oint_C (3x^2 y^2 dx + 2x^3 y dy)$$

The path  $c_3$  is  $y = 4 \Rightarrow dy = 0$

$$I_3 = \int_0^2 (3x^2 \cdot 4^2 dx + 2x^3 \cdot 4 \cdot 0)$$

$$I_3 = \int_0^2 48x^2 dx$$

$$I_3 = 128$$

In the above example, we have just worked through, we took three different paths and in each case, the line integral produced the same result. It appears, therefore, that in this case, the value of the integral is independent of the path of integration taken.

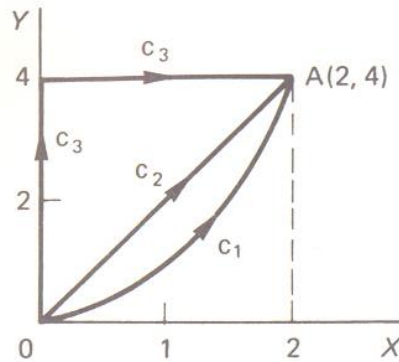


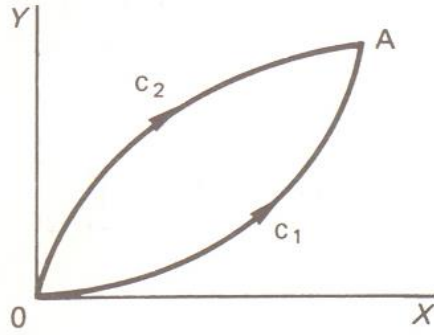
Figure # 121

We have been dealing with  $I = \oint_C (3x^2 y^2 dx + 2x^3 y dy)$

On reflection, we see that the integrand  $3x^2 y^2 dx + 2x^3 y dy$  is of the form  $Pdx + Qdy$  which we have met before and that it is, in fact, an exact differential of the function  $z = x^3 y^2$ , for

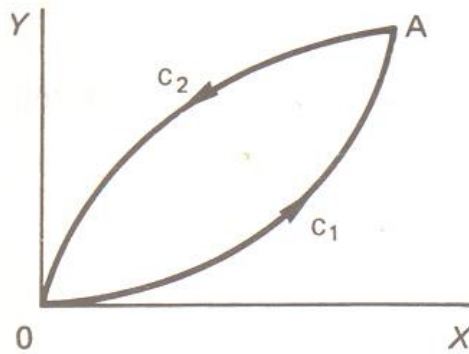
$$\frac{\partial z}{\partial x} = 3x^2 y^2 \text{ and } \frac{\partial z}{\partial y} = 2x^3 y$$

This always happens. If the integrand of the given integral is seen to be an exact differential, then the value of the line integral is independent of the path taken and depends only on the coordinates of the two end points.



**Figure # 122**

If  $I = \int_C (Pdx + Qdy)$  and  $(Pdx + Qdy)$  is an exact differential, then  $I_{C1} = I_{C2}$



**Figure # 123**

If we reverse the direction of  $C_2$ , then  $I_{C1} = -I_{C2} \Rightarrow I_{C1} + I_{C2} = 0$

Hence the integration taken round a closed curve is zero, provided  $(Pdx + Qdy)$  is an exact differential.

$\therefore$  If  $(Pdx + Qdy)$  is an exact differential,  $I = \oint_C (Pdx + Qdy) = 0$

**Q# 85:** Evaluate  $I = \int_C \{3ydx + (3x + 2y)dy\}$  from  $A(1,2)$  to  $B(3,5)$ .

Now path is given, so the integrand is doubtless an exact differential of some function  $z = f(x, y)$ .

Here,  $(Pdx + Qdy) = 3ydx + (3x + 2y)dy$

$\therefore P = 3y$  and  $Q = 3x + 2y$

In fact  $\frac{\delta P}{\delta y} = 3$  and  $\frac{\delta Q}{\delta x} = 3$

We have already dealt with the integration of exact differentials, so there is no difficulty.

Compare with  $I = \int_C \{Pdx + Qdy\}$ .

$$P = \frac{\delta z}{\delta x} = 3y \quad \therefore z = \int 3ydx = 3xy + f(y) \text{-----(i)}$$

$$Q = \frac{\delta z}{\delta y} = 3x + 2y \quad \therefore z = \int (3x + 2y)dy = 3xy + y^2 + F(x) \text{-----(ii)}$$

For (i) and (ii) to agree

$$f(y) = y^2 \text{ and } F(x) = 0$$

$$\text{Hence } z = 3xy + y^2$$

$$I = \int_C \{3ydx + (3x + 2y)dy\}$$

$$I = \int_C \{3ydx + 3xdy + 2ydy\}$$

$$I = \int_C \{3xdy + 3ydx + 2ydy\}$$

$$I = \int_C \{d(3xy) + d(y^2)\}$$

$$I = \int_{(1,2)}^{(3,5)} d(3xy + y^2)$$

$$I = [3xy + y^2]_{(1,2)}^{(3,5)}$$

$$I = [3 \times 3 \times 5 + 5^2 - 3 \times 1 \times 2 - 2^2]$$

$$I = [45 + 25 - 6 - 4] = [70 - 10] = 60$$

### Exact differentials in three independent variables

A line integral in space naturally involves three independent variables, but the method is very much like that for two independent variables.

$dz = Pdx + Qdy + Rdw$  is an exact differential of  $z = f(x, y, w)$

$$\text{if } \frac{\delta P}{\delta y} = \frac{\delta Q}{\delta x}; \frac{\delta P}{\delta w} = \frac{\delta R}{\delta x}; \frac{\delta R}{\delta y} = \frac{\delta Q}{\delta w};$$

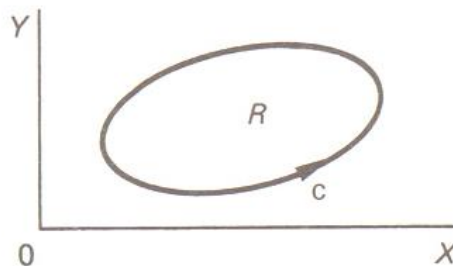
If the test is successful, then

$$\text{a) } I = \oint_C (Pdx + Qdy + Rdw) \text{ is independent of the path of integration}$$

$$\text{b) } I = \oint_C (Pdx + Qdy + Rdw) \text{ is zero}$$

### **Green's Theorem:**

Let  $P$  and  $Q$  be two functions of  $x$  and  $y$  that are finite and continuous inside and on the boundary  $c$  of a region  $R$  in the  $xy$ -plane.



**Figure # 124**

If the first partial derivatives are continuous within the region and on the boundary, then Green's theorem states that

$$\iint_R \left( \frac{\delta P}{\delta y} - \frac{\delta Q}{\delta x} \right) dx dy = - \oint (P dx + Q dy)$$

That is, a double integral over the plane region R can be transformed into a line integral over the boundary c of a region and the action is reversible.

### Green's Theorem

Green's Theorem enables an integral over a plane area to be expressed in terms of a line integral round its boundary curve.

If P and Q are two single-valued functions of x and y, continuous over a plane surface S, and c is its boundary curve, then

$$\therefore \oint_c (P dx + Q dy) = - \iint_R \left( \frac{\delta P}{\delta y} - \frac{\delta Q}{\delta x} \right) dx dy \text{ -----(i)}$$

$$\therefore \oint_c (P dx + Q dy) = \iint_R \left( \frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy \text{ -----(ii)}$$

Where the line integral is taken round c in an anticlockwise manner. In vector terms, this becomes:

S is a two-dimensional space enclosed by a simple closed curve c.

$$\left| \vec{dS} \right| = dS = dx dy$$

$$\vec{dS} = \hat{n} dS = \hat{k} dx dy$$

If  $\vec{F} = P \hat{i} + Q \hat{j}$  Where  $P = P(x, y)$  and  $Q = Q(x, y)$  then

$$\begin{aligned} \text{Curl } \vec{F} = \nabla \times \vec{F} &= \left( \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \times (P \hat{i} + Q \hat{j}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ P & Q & 0 \end{vmatrix} \\ &= \hat{i} \left[ \frac{\delta}{\delta y} (0) - \frac{\delta}{\delta z} (Q) \right] - \hat{j} \left[ \frac{\delta}{\delta x} (0) - \frac{\delta}{\delta z} (P) \right] + \hat{k} \left[ \frac{\delta}{\delta x} (Q) - \frac{\delta}{\delta y} (P) \right] \\ &= \hat{i} [0] - \hat{j} [0 - 0] + \hat{k} \left[ \frac{\delta}{\delta x} (Q) - \frac{\delta}{\delta y} (P) \right] \\ &= \hat{k} \left[ \frac{\delta}{\delta x} (Q) - \frac{\delta}{\delta y} (P) \right] \end{aligned}$$

Since P and Q are functions of x and y, that is in the xy-plane  $\frac{\delta Q}{\delta z} = \frac{\delta P}{\delta z} = 0$

So  $\int \text{curl } \vec{F} \cdot \vec{dS} = \int \text{curl } \vec{F} \cdot \hat{n} dS$  and in the xy plane  $\hat{n} = \hat{k}$

$$\int_S \text{curl} \vec{F} \cdot d\vec{S} = \int_S \hat{k} \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \cdot \hat{n} dS = \int_S \hat{k} \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \cdot \hat{k} dS = \iint_S \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \quad [\because \hat{k} \cdot \hat{k} = 1]$$

$$\therefore \int_S \text{curl} \vec{F} \cdot d\vec{S} = \iint_S \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \quad \text{-----(i)}$$

We have, from Stoke's theorem:

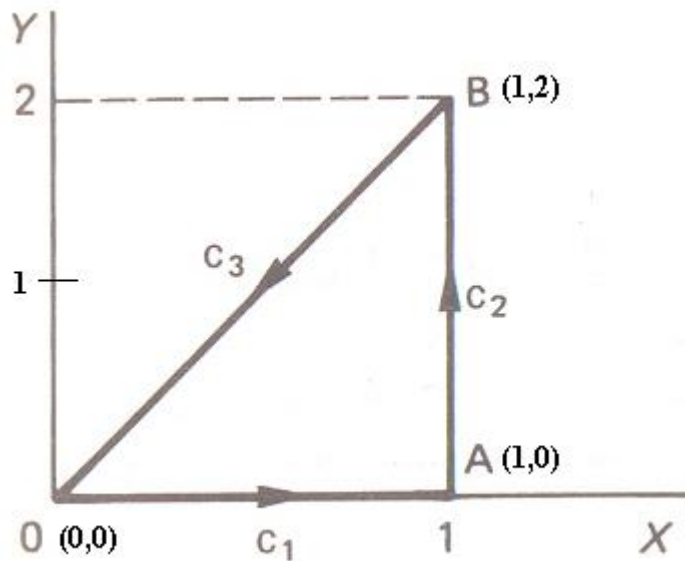
$$\begin{aligned} \therefore \int_S \text{curl} \vec{F} \cdot d\vec{S} &= \oint_C \vec{F} \cdot d\vec{r} \\ &= \oint_C (P \hat{i} + Q \hat{j}) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k}) \quad [\because \vec{r} = x \hat{i} + y \hat{j} + z \hat{k}] \\ &= \oint_C (P dx + Q dy) \quad \text{-----(ii)} \end{aligned}$$

From (i) and (ii)

$$\therefore \int_S \text{curl} \vec{F} \cdot d\vec{S} = \iint_S \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_C (P dx + Q dy)$$

$$\therefore \iint_S \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_C (P dx + Q dy)$$

**Q# 86:** Evaluate  $I = \oint_C \{(2x + y)dx + (3x - 2y)dy\}$  taken in anticlockwise manner round the triangle with vertices at **O(0,0), A(1,0), B(1,2)**



**Figure # 125**

$$I = \oint_C \{(2x + y)dx + (3x - 2y)dy\}$$

a) There are clearly three stages with  $c_1$ ,  $c_2$ ,  $c_3$ . Work through the complete evaluation to determine the value of  $I$ . It will be good revision.

i) The equation of OA is:

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2} \Rightarrow \frac{y - 0}{0 - 0} = \frac{x - 0}{0 - 1} \Rightarrow \frac{y}{0} = \frac{x}{-1} \Rightarrow -y = x \cdot 0 \Rightarrow y = 0$$

$c_1$  is  $y = 0 \therefore dy = 0$

$$I = \oint_C \{(2x + y)dx + (3x - 2y)dy\}$$

$$\Rightarrow I = \oint_C \{(2x + 0)dx + (3x - 2 \cdot 0) \cdot 0\}$$

$$\therefore I_1 = \int_0^1 2x dx = \left[ \frac{2x^2}{2} \right]_0^1 = 1$$

ii) The equation of AB is:

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2} \Rightarrow \frac{y - 0}{0 - 2} = \frac{x - 1}{1 - 1} \Rightarrow \frac{y}{-2} = \frac{x - 1}{0} \Rightarrow -2x + 2 = 0 \Rightarrow -2x = -2 \Rightarrow x = 1$$

$c_2$  is  $x = 1 \therefore dx = 0$

$$I = \oint_C \{(2x + y)dx + (3x - 2y)dy\}$$

$$\Rightarrow I = \oint_C \{(2 \cdot 1 + y) \cdot 0 + (3 \cdot 1 - 2y)dy\}$$

$$\therefore I_2 = \int_0^2 (3 - 2y)dy = \left[ 3y - \frac{2y^2}{2} \right]_0^2 = 2$$

iii) The equation of OB is:

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2} \Rightarrow \frac{y - 0}{0 - 2} = \frac{x - 0}{0 - 1} \Rightarrow \frac{y}{-2} = \frac{x}{-1} \Rightarrow -y = -2x \Rightarrow y = 2x$$

$c_3$  is  $y = 2x$

Then, Given  $y = 2x$

$$\frac{dy}{dx} = \frac{d}{dx}(2x)$$

$$\frac{dy}{dx} = 2$$

$$\therefore dy = 2dx$$

$$I = \oint_C \{(2x + y)dx + (3x - 2y)dy\}$$

$$I = \oint_C \{(2x + 2x)dx + (3x - 2 \times 2x)2dx\}$$

$$\Rightarrow I = \oint_C \{4x dx + (3x - 4x)2dx\}$$

$$\Rightarrow I = \oint_C \{4x + 6x - 8x\}dx$$

$$\therefore I_1 = \int_1^0 2x dx = \left[ \frac{2x^2}{2} \right]_1^0 = -1$$

$$\text{Finally } I = I_1 + I_2 + I_3 = 1 + 2 + (-1) = 2$$

b) By Green's theorem

$$I = \oint_C \{(2x + y)dx + (3x - 2y)dy\}$$

$$\text{Here, } P = 2x + y$$

$$\therefore \frac{\delta P}{\delta y} = 1$$

and

$$Q = 3x - 2y$$

$$\therefore \frac{\delta Q}{\delta x} = 3$$

We have, Green's theorem

$$\oint_C (Pdx + Qdy) = - \iint_R \left( \frac{\delta P}{\delta y} - \frac{\delta Q}{\delta x} \right) dx dy$$

Now,

$$I = - \iint_R \left( \frac{\delta P}{\delta y} - \frac{\delta Q}{\delta x} \right) dx dy$$

$$I = - \iint_R (1 - 3) dx dy$$

$$I = - \iint_R -2 dx dy$$

$$I = 2 \iint_R dx dy = 2A \quad \text{-----(i)}$$

$$[ \iint_R dx dy = A ]$$

$$\text{Now, The area of the triangle: } A = \frac{1}{2} \times \text{base} \times \text{height}$$

$$A = \frac{1}{2} \times OA \times AB \quad \text{[Figure 122]}$$

$$A = \frac{1}{2} \times 1 \times 2 = 1$$

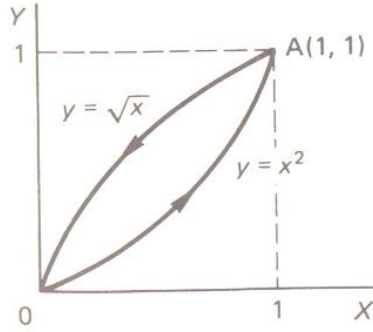
From (i),

$$I = 2 \times A = 2 \times 1 = 2 \quad [A = 1]$$

**L.H.S = R.H.S** (Proved)

**Q# 87:** Evaluate the line Integral  $I = \oint_C \{xydx + (2x - y)dy\}$  round the region bounded by

the curves  $y = x^2$  and  $x = y^2$  by the use of Green's theorem.



**Figure # 126**

Answer: Points of intersection are  $O(0,0)$  and  $A(1,1)$ . P and Q are known, so there is no difficulty.

Now, By Green's theorem

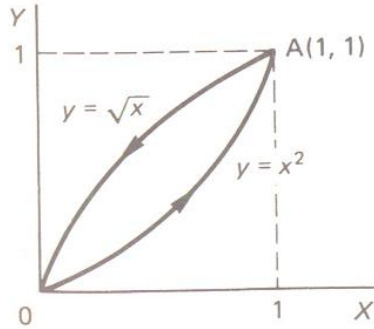
$$I = \oint_C \{xydx + (2x - y)dy\}$$

Here,  $P = xy$

$$\therefore \frac{\partial P}{\partial y} = x$$

and  $Q = 2x - y$

$$\therefore \frac{\partial Q}{\partial x} = 2$$



**Figure # 127**

$$\text{We have, } \iint_R \left( \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy = - \oint_C (Pdx + Qdy)$$

$$\therefore - \iint_R \left( \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy = \oint_C (Pdx + Qdy)$$

$$\therefore \oint_C (Pdx + Qdy) = - \iint_R \left( \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy$$

$$\text{Now, } I = - \iint_R \left( \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy$$

$$I = - \iint_R (x - 2) dx dy$$

$$I = - \int_0^1 \int_{y=x^2}^{y=\sqrt{x}} (x-2) dy dx \quad [\text{Here upper limit of } y \text{ is } y = \sqrt{x} \text{ and the lower limit is } y = x^2]$$

$$I = - \int_0^1 (x-2) \left[ y \right]_{x^2}^{\sqrt{x}} dx$$

$$\therefore I = - \int_0^1 (x-2) \left[ \sqrt{x} - x^2 \right] dx$$

$$\therefore I = - \int_0^1 (x^{3/2} - x^3 - 2x^{1/2} + 2x^2) dx$$

$$\therefore I = - \left[ \frac{2}{5} x^{5/2} - \frac{1}{4} x^4 - \frac{4}{3} x^{3/2} + \frac{2}{3} x^3 \right]_0^1 = \frac{31}{60}$$

Now, L.H.S.  $\oint P dx + Q dy$

$$= \oint xy dx + (2x - y) dy$$

$$I_1 = \int_0^1 x \cdot x^2 dx + (2x - x^2) 2x dx$$

$$= \int_0^1 x^3 dx + 4x^2 - 2x^3 dx$$

$$= \int_0^1 (-x^3 dx + 4x^2) dx$$

$$= \left[ -\frac{x^4}{4} + 4 \frac{x^3}{3} \right]_0^1$$

$$= \left[ -\frac{1}{4} + \frac{4}{3} \right] = \frac{13}{12}$$

and  $\oint P dx + Q dy$

$$= \oint xy dx + (2x - y) dy$$

$$I_2 = \int_1^0 xy dx + (2x - y) dy$$

$$I_2 = \int_1^0 x \cdot \sqrt{x} dx + (2x - \sqrt{x}) \frac{1}{2} x^{-1/2} dx$$

$$I_2 = \int_1^0 (x^{3/2} + x^{1/2} - \frac{1}{2}) dx$$

$$\mathbf{I}_2 = \left[ \frac{\mathbf{x}^{5/2}}{\frac{5}{2}} + \frac{\mathbf{x}^{3/2}}{\frac{3}{2}} - \frac{1}{2}\mathbf{x} \right]_1^0$$

$$\mathbf{I}_2 = \left[ 0 - \frac{2}{5} - \frac{2}{3} + \frac{1}{2} \right] = -\frac{17}{30}$$

Finally  $\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 = \frac{13}{12} - \frac{17}{30} = \frac{31}{60}$

**L.H.S = R.H.S** (Proved)

## Changing Coordinates

### 01. A Cartesian coordinate system:

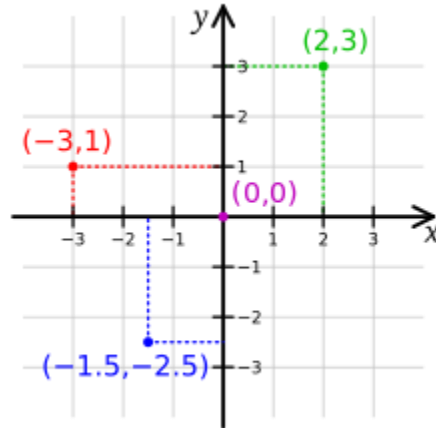


Figure # 128

### Rectangular coordinates in 3-D:

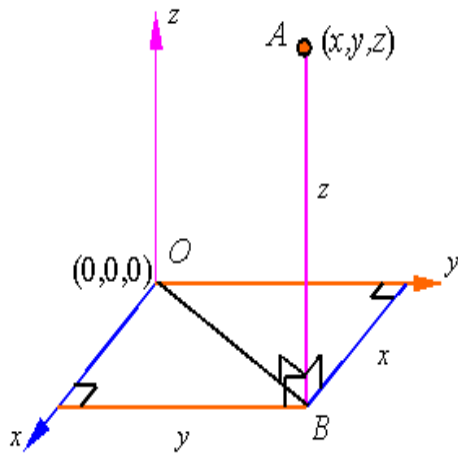


Figure # 129

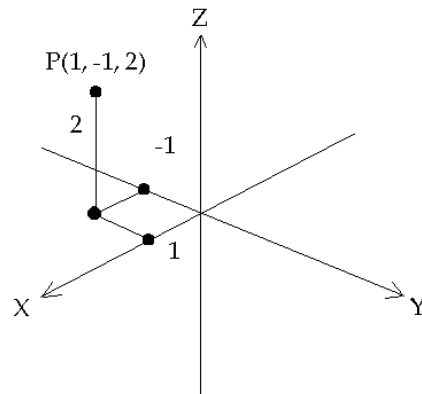


Figure # 130

**The polar coordinate:** the polar coordinate system in which each point on a plane is determined by a distance from a fixed point and an angle from a fixed direction.

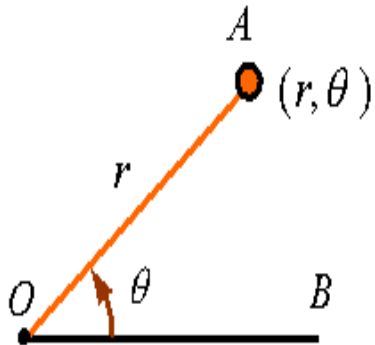


Figure # 131

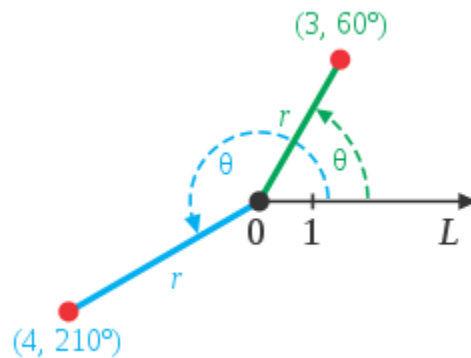
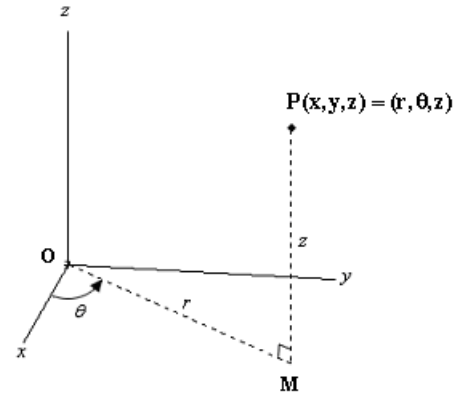
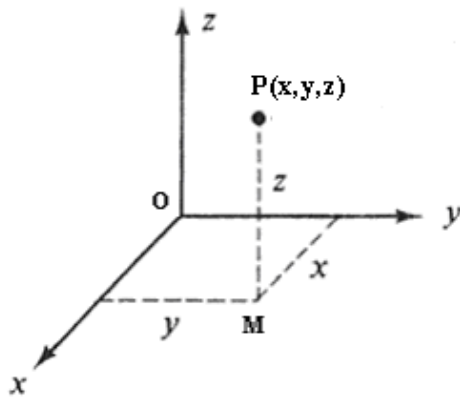
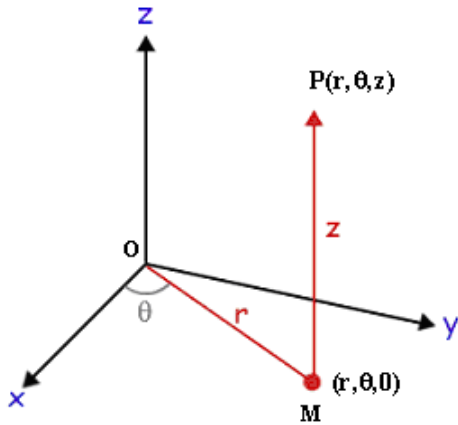


Figure # 132

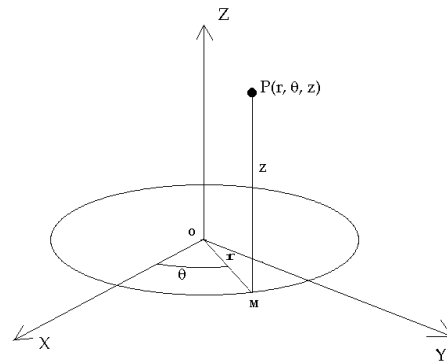
## 02. Changing Coordinate Systems: Rectangular and Cylindrical



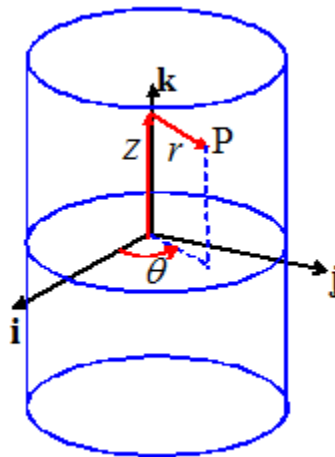
**Figure # 133:** Rectangular Coordinates (Cartesian Coordinates) **Figure # 134**



**Figure # 135**



**Figure # 136**



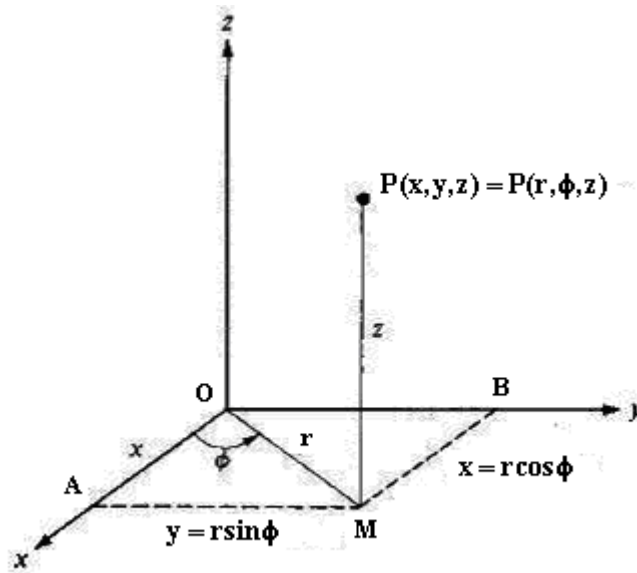
**Figure # 137:** Cylindrical Polar Coordinates

$r$  is the radial distance of  $P$  from the axis of the cylinder

$\theta$  is the angle between the  $\mathbf{i}$  direction and the projection of  $OP$  onto the  $xy$ - plane

$z$  is the length of the projection of  $OP$  on the axis of the cylinder.

*If the relationships between the variables  $(x, y, z)$  of the Cartesian coordinate system and those of the cylindrical system  $(r, \phi, z)$  are easily obtained from Figure below:*



**Figure # 138: Cylindrical Polar Coordinates**

$$\frac{OA}{OM} = \cos \phi$$

$$\frac{x}{r} = \cos \phi$$

$$x = r \cos \phi \text{ -----(i)}$$

$$\frac{AM}{OM} = \sin \phi$$

$$\frac{y}{r} = \sin \phi$$

$$y = r \sin \phi \text{ -----(ii)}$$

**Finally we get, the Cylindrical Polar Coordinates are**

$$x = r \cos \phi \text{ -----(iii)}$$

$$y = r \sin \phi \text{ -----(iv)}$$

$$z = z \text{ -----(v)}$$

### 03. Changing Coordinate Systems: Rectangular and Spherical

#### A SPHERE

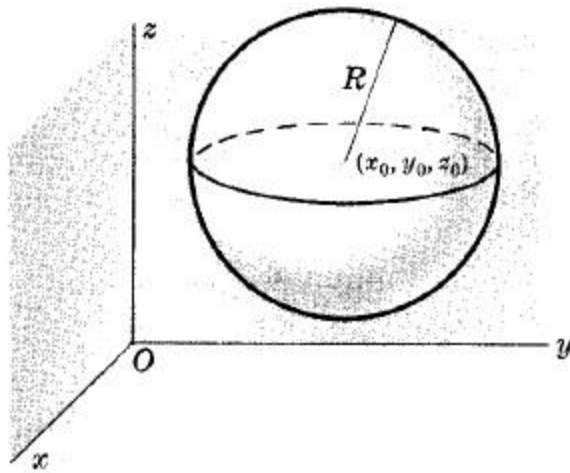


Figure # 139

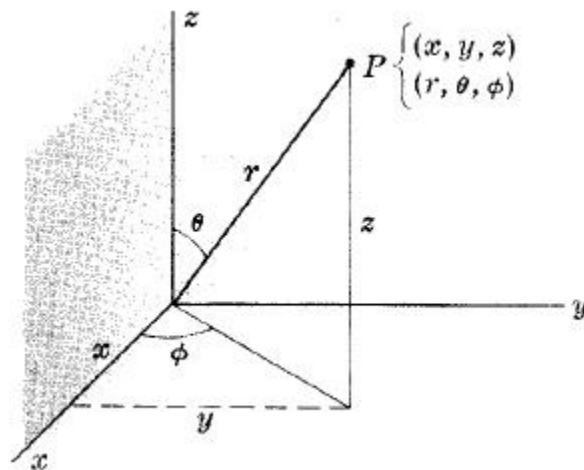


Figure # 140

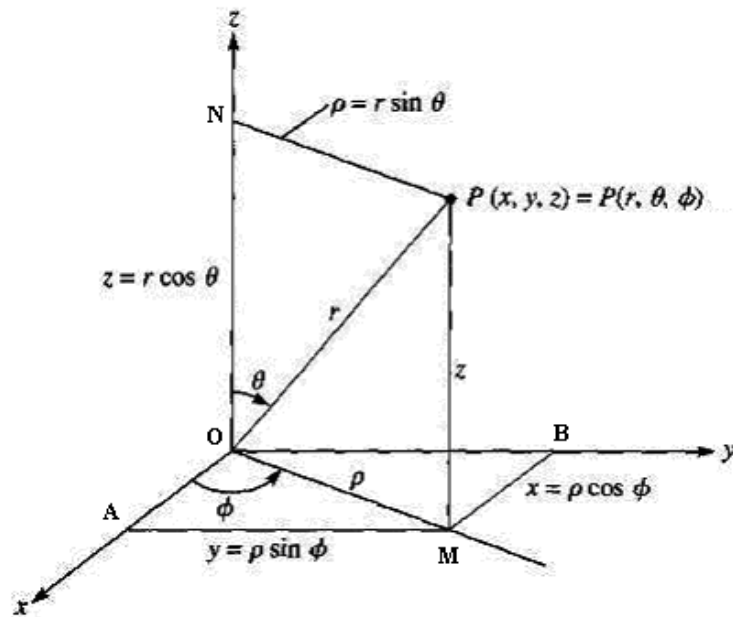


Figure # 141: Spherical Polar Coordinates

$$\frac{OA}{OM} = \cos \phi$$

$$\frac{x}{\rho} = \cos \phi$$

$$x = \rho \cos \phi \text{-----(i)}$$

&

$$\frac{AM}{OM} = \sin \phi$$

$$\frac{y}{\rho} = \sin \phi$$

$$y = \rho \sin \phi \text{-----(ii)}$$

Again,

$$\frac{ON}{OP} = \cos \theta$$

$$\frac{z}{r} = \cos \theta$$

$$z = r \cos \theta \text{-----(iii)}$$

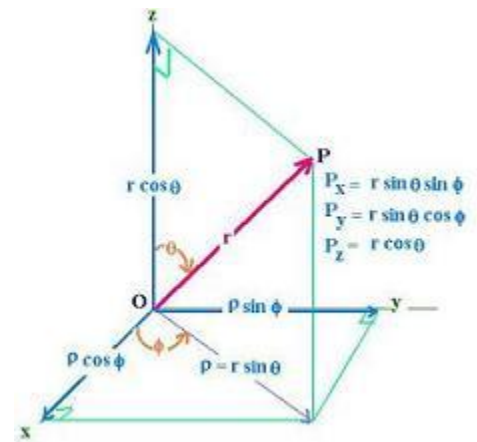


Figure # 142

&

$$\frac{PN}{OP} = \sin\theta$$

$$\frac{PN}{r} = \sin\theta$$

$$\Rightarrow PN = r \sin\theta$$

$$\Rightarrow PN = OM = r \sin\theta \quad [\because PN = OM]$$

$$\Rightarrow PN = \rho = r \sin\theta \quad [\because OM = \rho]$$

$$\therefore PN = \rho = r \sin\theta \text{-----(iv)}$$

Again,

$\Delta OAM$ ,

$$OM^2 = OA^2 + AM^2$$

$$\rho^2 = x^2 + y^2 \text{-----(v)}$$

Again,

$\Delta OPN$ ,

$$OP^2 = ON^2 + PN^2$$

$$r^2 = z^2 + \rho^2 \quad [\because PN = \rho]$$

$$r^2 = z^2 + \rho^2$$

$$r^2 = z^2 + x^2 + y^2 \quad [\because \rho^2 = x^2 + y^2]$$

Again, putting the value of  $\rho$  in (i) and (ii),

$$x = \rho \cos\phi$$

$$x = r \sin\theta \cos\phi \quad [\because PN = \rho = r \sin\theta]$$

&

$$y = \rho \sin\phi$$

$$y = r \sin\theta \sin\phi \quad [\because PN = \rho = r \sin\theta]$$

Finally we get, the Spherical Polar Coordinates are:

$$\therefore x = r \sin\theta \cos\phi \text{-----(vi)}$$

$$y = r \sin\theta \sin\phi \text{-----(vii)}$$

$$z = r \cos\theta \text{-----(viii)}$$

$R$  is the distance of  $P$  from the origin

$\theta$  is the angle between the  $\mathbf{k}$  direction and  $OP$

$\phi$  is the angle between the  $\mathbf{i}$  direction and the projection of  $OP$  onto a plane through  $O$  normal to  $\mathbf{k}$

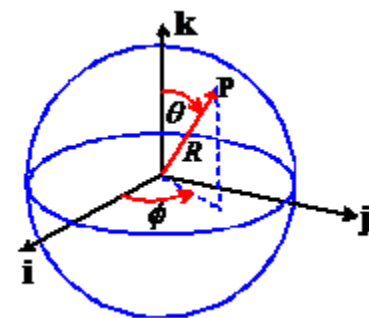


Figure # 143

04. **Jacobian Determinant:** In order to change variables in a double integral or triple integral we will need the **Jacobian** of the transformation. The **Jacobian** of the transformation  $\mathbf{x} = \mathbf{x}(\mathbf{u}, \mathbf{v})$  and  $\mathbf{y} = \mathbf{y}(\mathbf{u}, \mathbf{v})$  is denoted by:

$$|\mathbf{J}(\mathbf{u}, \mathbf{v})| = \frac{\partial(\mathbf{x}, \mathbf{y})}{\partial(\mathbf{u}, \mathbf{v})} = \begin{vmatrix} \frac{\partial \mathbf{x}}{\partial \mathbf{u}} & \frac{\partial \mathbf{x}}{\partial \mathbf{v}} \\ \frac{\partial \mathbf{y}}{\partial \mathbf{u}} & \frac{\partial \mathbf{y}}{\partial \mathbf{v}} \end{vmatrix}$$

**Q # 88 :** To see how this works we can start with one dimension. If we have an integral in rectangular coordinates such as  $\int_{x_1}^{x_2} f(\mathbf{x}) d\mathbf{x}$

We can change coordinate systems if we define  $\mathbf{x} = \mathbf{x}(\mathbf{u})$ . Then we have  $d\mathbf{x} = \frac{d\mathbf{x}}{d\mathbf{u}} d\mathbf{u}$ .

To transform the limits of the integral, we need to invert the definition to get  $\mathbf{u} = \mathbf{u}(\mathbf{x})$ .

Then the integral becomes

$$\int_{x_1}^{x_2} f(\mathbf{x}) d\mathbf{x} = \int_{u(x_1)}^{u(x_2)} f(\mathbf{x}(\mathbf{u})) \frac{d\mathbf{x}}{d\mathbf{u}} d\mathbf{u} = \int_{u(x_1)}^{u(x_2)} f(\mathbf{x}(\mathbf{u})) |\mathbf{J}(\mathbf{u})| d\mathbf{u} \quad [\text{Here } |\mathbf{J}(\mathbf{u})| = \frac{d\mathbf{x}}{d\mathbf{u}}]$$

That is  $\int f(\mathbf{x}) d\mathbf{x} = \int f(\mathbf{u}) |\mathbf{J}(\mathbf{u})| d\mathbf{u}$  -----(i)

**Q # 89:**

Find  $\int \frac{d\mathbf{x}}{(3\mathbf{x} + 2)^5}$

Solution: Let

$$\mathbf{u} = 3\mathbf{x} + 2.$$

Now directly,

$$\mathbf{u} = 3\mathbf{x} + 2.$$

$$\frac{d\mathbf{u}}{d\mathbf{x}} = 3$$

$$\frac{d\mathbf{x}}{d\mathbf{u}} = \frac{1}{3}$$

$$d\mathbf{x} = \frac{1}{3} d\mathbf{u} \text{ -----(i)}$$

Another way, Using Jacobian **Determinant**,

Here Variable  $\mathbf{x} \rightarrow \mathbf{u}$

$$\text{Then } \int f(\mathbf{x}) d\mathbf{x} = \int f(\mathbf{u}) |\mathbf{J}(\mathbf{u})| d\mathbf{u} \text{ -----(ii)}$$

$$\text{That is, } d\mathbf{x} = |\mathbf{J}(\mathbf{u})| d\mathbf{u} \text{ -----(iii)}$$

$$\text{Where, } |\mathbf{J}(\mathbf{u})| = \left| \frac{\partial \mathbf{x}}{\partial \mathbf{u}} \right| \text{ -----(iv)}$$

Now, we have to find out the value of  $|\mathbf{J}(\mathbf{u})|$

$$\mathbf{u} = 3\mathbf{x} + 2.$$

$$\frac{\delta u}{\delta x} = 3$$

$$\frac{\delta x}{\delta u} = \frac{1}{3}$$

From (iv),

$$|J(u)| = \frac{\delta x}{\delta u} = \frac{1}{3} \text{-----(v)}$$

Putting the value of  $|J(u)|$  in (iii),

$$\therefore dx = |J(u)| du$$

$$\therefore dx = \frac{1}{3} du \text{-----(vi)}$$

We see, (i) & (vi) are same.

That is,

$$\int_{x_1}^{x_2} f(x) dx = \int_{u(x_1)}^{u(x_2)} f(x(u)) \frac{dx}{du} du = \int_{u(x_1)}^{u(x_2)} f(x(u)) |J(u)| du \text{-----(vii)}$$

$$\text{Here, } f(x) = \frac{1}{(3x+2)^5} \text{ and } f(x(u)) = \frac{1}{u^5}$$

From (ii),

$$\int f(x) dx = \int f(u) |J(u)| du$$

$$\begin{aligned} \int \frac{1}{(3x+2)^5} dx &= \int \frac{1}{u^5} |J(u)| du \\ &= \int u^{-5} \frac{1}{3} du = \frac{1}{3} \cdot \frac{u^{-5+1}}{-5+1} = \frac{1}{3} \cdot \frac{u^{-4}}{-4} + c \quad [\because |J(u)| = \frac{1}{3}] \\ &= \frac{-1}{12} (3x+2)^{-4} + c \\ &= \frac{-1}{12(3x+2)^4} + c \end{aligned}$$

**Q # 90:** In two dimensions,  $\iint f(x, y) dx dy$  -----(i)

Now if we want to switch to another coordinate system, we define  $x = x(u, v)$  and  $y = y(u, v)$

That is:  $\iint_R f(x, y) dx dy = \iint_R f[x(u, v), y(u, v)] |J(u, v)| du dv$  -----(ii)

$$\text{Where } |J(u, v)| = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

**Q # 91:** Find area of a circle whose center  $(0,0)$  and radius  $r$  that is find  $\iint f(x, y) dx dy$

Answer:  $f(x, y) = x^2 + y^2 = r^2$

Let  $x = r \cos \theta$      $y = r \sin \theta$

Hence,  $dx dy = |J(r, \theta)| dr d\theta$  -----(i)

When  $x = r$ ,  $y = 0$  and  $r = r$  then

$x = r \cos \theta$

$\Rightarrow r = r \cos \theta$

$\Rightarrow 1 = \cos \theta$

$\Rightarrow \cos 0 = \cos \theta$

$\Rightarrow \theta = 0$

Hence the limit of  $\theta$  for a circle is  $0 \leq \theta \leq 2\pi$

Again

When  $x = r$  and  $\theta = 0$  then

$x = r \cos \theta$

$r = r \cos 0$

$\Rightarrow r = r \times 1$

$\Rightarrow r = r$

$\Rightarrow r = r$

Hence the limit of  $r$  is  $0 \leq r \leq r$

$$\therefore |J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta - (-r \sin^2 \theta)$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta + r \sin^2 \theta$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r(\cos^2 \theta + \sin^2 \theta)$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r.1$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \text{ -----(ii)}$$

Putting the value of  $|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r$  in (i), we get

$$dx dy = |J(r, \theta)| dr d\theta$$

$$dx dy = r dr d\theta \text{ -----(iii)}$$

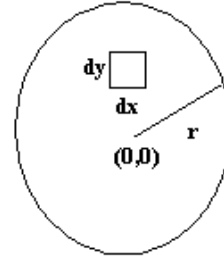


Figure # 144

Now, we can write

$$\iint f(x, y) dx dy = \iint f[x(u, v), y(u, v)] |J(u, v)| du dv$$

$$\iint f(x, y) dx dy = \iint f[x(r, \theta), y(r, \theta)] |J(r, \theta)| dr d\theta$$

$$\iint dx dy = \int_{\theta=0}^{2\pi} \int_{r=0}^r |J(r, \theta)| dr d\theta$$

$$\iint dx dy = \int_{\theta=0}^{2\pi} \int_{r=0}^r r dr d\theta \quad [ |J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r ]$$

$$\iint dx dy = \int_{\theta=0}^{2\pi} \left[ \frac{r^2}{2} \right]_0^r d\theta$$

$$\iint dx dy = \int_{\theta=0}^{2\pi} \left[ \frac{r^2}{2} - \frac{0^2}{2} \right] d\theta$$

$$\iint dx dy = \int_{\theta=0}^{2\pi} \left[ \frac{r^2}{2} - 0 \right] d\theta$$

$$\iint dx dy = \frac{r^2}{2} \int_{\theta=0}^{2\pi} d\theta$$

$$\iint dx dy = \frac{r^2}{2} [\theta]_0^{2\pi}$$

$$\iint dx dy = \frac{r^2}{2} [2\pi - 0]$$

$$\iint dx dy = \frac{r^2}{2} [2\pi]$$

$$\iint dx dy = \pi r^2$$

**Q # 92:** In three dimensions,  $\iiint_V f(x, y, z) dx dy dz$  -----(i)

Now if we want to switch to another coordinate system, we define  $x = x(u, v, w)$ ,  $y = y(u, v, w)$  and  $z = z(u, v, w)$

That is:  $\iiint_V f(x, y, z) dx dy dz = \iiint_V f(u, v, w) |J(u, v, w)| du dv dw$  -----(ii)

$$\text{Where } |J(u, v, w)| = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

**Q# 93:** Verify Green's theorem for the integral

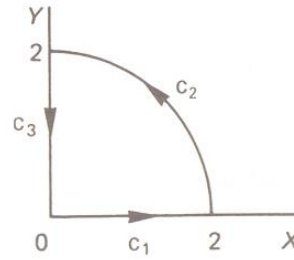
$\oint_c \{(x^2 + y^2)dx + (x + 2y)dy\}$  taken round the

boundary curve  $c$  defined by

$$y = 0; 0 \leq x \leq 2$$

$$x^2 + y^2 = 4; 0 \leq x \leq 2$$

$$x = 0; 0 \leq y \leq 2$$



**Figure # 145**

Answer:

We have Green's theorem  $\iint_s \left( \frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy = \oint_c (P dx + Q dy)$

**L.H.S.**

First we have to evaluate  $\iint_s \left( \frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy$  -----(i)

Given,  $\oint_c \{(x^2 + y^2)dx + (x + 2y)dy\}$  -----(ii)

Since, R.H.S  $\oint_c (P dx + Q dy)$  -----(iii)

Comparing (ii) & (iii),

Here,  $P = x^2 + y^2$

$$\therefore \frac{\delta P}{\delta y} = 0 + 2y = 2y$$

And  $Q = x + 2y$

$$\therefore \frac{\delta Q}{\delta x} = 1 + 0 = 1$$

So, from (i), we get, [Putting the values of  $\frac{\delta P}{\delta y}$  &  $\frac{\delta Q}{\delta x}$  in (i)]

$$\iint_s \left( \frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy = \iint_s (1 - 2y) dx dy$$
 -----(iv)

It will be more convenient to work in polar coordinates, so we make the substitutions

Let  $x = r \cos \theta$      $y = r \sin \theta$

Then  $\iint_s f(x, y) dx dy = \iint_s f(r, \theta) |J(r, \theta)| dr d\theta$  -----(v)

Here  $f(x, y) = 1 - 2y$

From **Jacobian Determinant**,

$$\therefore |J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$
 -----(vi)

We have,  $x = r \cos \theta$

$$\frac{\partial x}{\partial r} = \frac{\partial}{\partial r}(r \cos \theta)$$

$$\frac{\partial x}{\partial r} = \cos \theta \frac{\partial}{\partial r}(r)$$

$$\frac{\partial x}{\partial r} = \cos \theta \cdot 1$$

$$\frac{\partial x}{\partial r} = \cos \theta \quad \text{-----(vii)}$$

Again

$$\frac{\partial x}{\partial \theta} = \frac{\partial}{\partial \theta}(r \cos \theta)$$

$$\frac{\partial x}{\partial \theta} = r \frac{\partial}{\partial \theta}(\cos \theta)$$

$$\frac{\partial x}{\partial \theta} = -r \sin \theta \quad \text{-----(viii)}$$

Again,  $y = r \sin \theta$

$$\frac{\partial y}{\partial r} = \frac{\partial}{\partial r}(r \sin \theta)$$

$$\frac{\partial y}{\partial r} = \sin \theta \frac{\partial}{\partial r}(r)$$

$$\frac{\partial y}{\partial r} = \sin \theta \cdot 1$$

$$\frac{\partial y}{\partial r} = \sin \theta \quad \text{-----(ix)}$$

Again

$$\frac{\partial y}{\partial \theta} = \frac{\partial}{\partial \theta}(r \sin \theta)$$

$$\frac{\partial y}{\partial \theta} = r \frac{\partial}{\partial \theta}(\sin \theta)$$

$$\frac{\partial y}{\partial \theta} = r \cos \theta \quad \text{-----(x)}$$

From (vi),

$$\therefore |J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta - (-r \sin^2 \theta)$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \cos^2 \theta + r \sin^2 \theta$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r(\cos^2 \theta + \sin^2 \theta)$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r.1$$

$$|J(r, \theta)| = \frac{\partial(x, y)}{\partial(r, \theta)} = r \quad \text{-----(xi)}$$

From (v),

$$\iint_s f(x, y) dx dy = \iint_s f(r, \theta) |J(r, \theta)| dr d\theta \quad \text{-----(*)}$$

Hence, From (iv),

$$\iint_s (1 - 2y) dx dy = \iint_s (1 - 2r \sin \theta) r dr d\theta \quad \text{-----(xii)}$$

[Comparing \* & xii,

$$[Here, f(x, y) = 1 - 2y, y = r \sin \theta, f(r, \theta) = 1 - 2r \sin \theta, |J(r, \theta)| = r]$$

Now,

When  $x = 2, y = 0$  and  $r = 2$  then

$$x = r \cos \theta$$

$$\Rightarrow 2 = 2 \cos \theta$$

$$\Rightarrow 1 = \cos \theta$$

$$\Rightarrow \cos 0 = \cos \theta$$

$$\Rightarrow \theta = 0$$

Again

When  $x = 0, y = 2$  and  $r = 2$  then

$$y = r \sin \theta$$

$$\Rightarrow 2 = 2 \sin \theta$$

$$\Rightarrow 1 = \sin \theta$$

$$\Rightarrow \sin \frac{\pi}{2} = \sin \theta$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

Hence the limit of  $\theta$  is  $0 \leq \theta \leq \frac{\pi}{2}$

Again

When  $x = 2$  and  $\theta = 0$  then

$$x = r \cos \theta$$

$$2 = r \cos 0$$

$$\Rightarrow 2 = r \times 1$$

$$\Rightarrow 2 = r$$

$$\Rightarrow r = 2$$

When  $y = 2$  and  $\theta = \pi/2$  then

$$y = r \sin \theta$$

$$2 = r \sin \frac{\pi}{2}$$

$$\Rightarrow 2 = r \times 1$$

$$\Rightarrow 2 = r$$

$$\Rightarrow r = 2$$

Hence the limit of  $r$  is  $0 \leq r \leq 2$

From (xii),

$$\iint_s (1 - 2y) dx dy = \iint_s (1 - 2r \sin \theta) r dr d\theta -$$

$$= \int_0^{\pi/2} \int_0^2 (1 - 2r \sin \theta) r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^2 (1 - 2r \sin \theta) r dr d\theta$$

$$= \int_0^{\pi/2} \int_0^2 (r - 2r^2 \sin \theta) dr d\theta$$

$$= \int_0^{\pi/2} \left[ \frac{r^2}{2} - 2 \frac{r^3}{3} \sin \theta \right]_0^2 d\theta$$

$$= \int_0^{\pi/2} \left[ \frac{2^2}{2} - 2 \frac{2^3}{3} \sin \theta - 0 \right] d\theta$$

$$= \int_0^{\pi/2} \left[ 2 - \frac{16}{3} \sin \theta \right] d\theta$$

$$= \left[ 2\theta + \frac{16}{3} \cos \theta \right]_0^{\pi/2}$$

$$= \left[ 2 \times \frac{\pi}{2} + \frac{16}{3} \cos \frac{\pi}{2} - 0 - \frac{16}{3} \cos 0 \right]$$

$$= \left[ \pi + \frac{16}{3} \times 0 - 0 - \frac{16}{3} \times 1 \right]$$

$$= \left[ \pi - \frac{16}{3} \right]$$

$$\therefore \iint_s \left( \frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y} \right) dx dy = \iint_s (1 - 2y) dx dy = \left[ \pi - \frac{16}{3} \right] \text{-----(xiii)}$$

Now we have to evaluate R.H.S of Green's Theorem  $\oint_C (Pdx + Qdy)$

$$\text{Given } \oint_C \{(x^2 + y^2)dx + (x + 2y)dy\}$$

$$\therefore P = x^2 + y^2 \text{ and } Q = x + 2y$$

We now take  $c_1, c_2, c_3$  in turn

$$\text{i) } c_1 : y = 0; \therefore dy = 0$$

$$\begin{aligned} \therefore \oint_{c_1} (Pdx + Qdy) &= \int_0^2 (x^2 + y^2)dx + (x + 2y)dy = \int_0^2 (x^2 + 0^2)dx + (x + 2.0).0 \\ &= \int_0^2 x^2 dx = \frac{8}{3} \end{aligned}$$

$$\text{ii) } c_2 : x^2 + y^2 = 4 \therefore y^2 = 4 - x^2 \therefore y = \sqrt{4 - x^2} = (4 - x^2)^{1/2}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} (4 - x^2)^{-1/2} \cdot \frac{d}{dx} (4 - x^2)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} (4 - x^2)^{-1/2} \cdot (0 - 2x)$$

$$\therefore dy = -(4 - x^2)^{-1/2} \cdot x dx = \frac{-x dx}{\sqrt{4 - x^2}}$$

$$\therefore \oint_{c_2} (Pdx + Qdy) = \oint_{c_2} (x^2 + y^2)dx + (x + 2y)dy$$

$$= \oint_{c_2} (x^2 + 4 - x^2)dx + (x + 2\sqrt{4 - x^2}) \cdot \frac{-x dx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} 4dx - (x + 2\sqrt{4 - x^2}) \frac{x dx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} 4dx - \frac{x^2 dx}{\sqrt{4 - x^2}} - 2\sqrt{4 - x^2} \frac{x dx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} 4dx - \frac{x^2 dx}{\sqrt{4 - x^2}} - 2x dx$$

$$= \oint_{c_2} (4 - 2x)dx - \frac{x^2 dx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} (4 - 2x - \frac{x^2}{\sqrt{4 - x^2}})dx$$

$$= \int_2^0 (4 - 2x - \frac{x^2}{\sqrt{4 - x^2}})dx \text{-----(xiv)}$$

$$\text{Let } x = 2\sin\theta$$

$$\Rightarrow dx = 2\cos\theta d\theta$$

$$\therefore \sqrt{4 - x^2} = \sqrt{4 - (2\sin\theta)^2} = \sqrt{4 - 4\sin^2\theta} = \sqrt{4(1 - \sin^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta$$

Given,  $x = 2\sin\theta$

$$\Rightarrow \theta = \sin^{-1}(x/2)$$

| $x$      | $0$                                                                                                                                                                              | $2$                                                                                                                                                                                                       |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\theta$ | $\theta = \sin^{-1}(x/2)$<br>$\Rightarrow \theta = \sin^{-1}(0/2)$<br>$\Rightarrow \theta = \sin^{-1}(0)$<br>$\Rightarrow \theta = \sin^{-1} \sin 0$<br>$\Rightarrow \theta = 0$ | $\theta = \sin^{-1}(x/2)$<br>$\Rightarrow \theta = \sin^{-1}(2/2)$<br>$\Rightarrow \theta = \sin^{-1}(1)$<br>$\Rightarrow \theta = \sin^{-1}(\sin \frac{\pi}{2})$<br>$\Rightarrow \theta = \frac{\pi}{2}$ |

$$\begin{aligned} \therefore \oint_{c_2} (Pdx + Qdy) &= \int_2^0 (4 - 2x - \frac{x^2}{\sqrt{4-x^2}}) dx \\ &= \int_{\pi/2}^0 \{ (4 - 2 \times 2\sin\theta - \frac{(2\sin\theta)^2}{2\cos\theta}) \} 2\cos\theta d\theta \\ &= \int_{\pi/2}^0 \{ (8\cos\theta - 4 \times 2\sin\theta\cos\theta - 4\sin^2\theta) \} d\theta \\ \therefore \oint_{c_2} (Pdx + Qdy) &= \int_{\pi/2}^0 8\cos\theta d\theta - \int_{\pi/2}^0 4 \times 2\sin\theta\cos\theta d\theta - \int_{\pi/2}^0 4\sin^2\theta d\theta \text{-----(xv)} \end{aligned}$$

Now,

$$\text{a) } \int_{\pi/2}^0 8\cos\theta d\theta = [8\sin\theta]_{\pi/2}^0 = 8\sin 0 - 8\sin \frac{\pi}{2} = 0 - 8 \times 1 = -8$$

$$\text{b) } \int_{\pi/2}^0 4 \times 2\sin\theta\cos\theta d\theta$$

Let  $z = \sin\theta$

$$\Rightarrow dz = \cos\theta d\theta$$

Since  $z = \sin\theta$

$$\Rightarrow \theta = \sin^{-1}(z/2)$$

| $\theta$ | $0$                                         | $\frac{\pi}{2}$                                         |
|----------|---------------------------------------------|---------------------------------------------------------|
| $z$      | $z = \sin\theta$<br>$z = \sin 0$<br>$z = 0$ | $z = \sin\theta$<br>$z = \sin \frac{\pi}{2}$<br>$z = 1$ |

$$\int_{\pi/2}^0 4 \times 2 \sin \theta \cos \theta d\theta = \int_1^0 4 \times 2z dz = \left[ 8 \times \frac{z^2}{2} \right]_1^0 = 4(0 - 1) = -4$$

$$\begin{aligned} \text{c) } \int_{\pi/2}^0 4 \sin^2 \theta d\theta &= 2 \int_{\pi/2}^0 2 \sin^2 \theta d\theta = 2 \int_{\pi/2}^0 (1 + \cos 2\theta) d\theta \\ &= 2 \left[ \theta - \frac{\sin 2\theta}{2} \right]_{\pi/2}^0 = 2 \left( 0 - \frac{1}{2} \sin 2 \cdot 0 - \frac{\pi}{2} + \frac{1}{2} \sin 2 \cdot \frac{\pi}{2} \right) = 2 \left( 0 - 0 - \frac{\pi}{2} + \frac{1}{2} \cdot \sin \pi \right) = 2 \left( -\frac{\pi}{2} + \frac{1}{2} \cdot 0 \right) \\ &= -\pi \end{aligned}$$

From (xv)

$$\begin{aligned} \therefore \oint_{c_2} (Pdx + Qdy) &= \int_{\pi/2}^0 8 \cos \theta d\theta - \int_{\pi/2}^0 4 \times 2 \sin \theta \cos \theta d\theta - \int_{\pi/2}^0 4 \sin^2 \theta d\theta \\ &= -8 - (-4) - (-\pi) \\ &= -8 + 4 + \pi \\ &= \pi - 4 \end{aligned}$$

iii)  $c_3 : x = 0; \therefore dx = 0$

$$\therefore \oint_{c_3} (Pdx + Qdy) = \oint_{c_3} (0^2 + y^2) \cdot 0 + (0 + 2y) dy = \int_2^0 2y dy = \left[ 2 \frac{y^2}{2} \right]_2^0 = -4$$

Collecting our three partial results

$$\oint_c (Pdx + Qdy) = \frac{8}{3} + \pi - 4 - 4 = \pi - \frac{16}{3} \text{-----(xvi)}$$

From (xiii) and (xvi), we can write

L.H.S. = R.H.S.

$$\therefore \iint_s \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \oint_c (Pdx + Qdy) \text{ (Proved)}$$

**Q# 94:** Use Green's theorem to evaluate  $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$  where C is the square formed by the lines  $y = \pm 1, x = \pm 1$

Answer: From Green's theorem, we have,

$$\int_C \phi dx + \psi dy = \iint_R \left( \frac{\partial \psi}{\partial x} - \frac{\partial \phi}{\partial y} \right) dx dy$$

Here,  $\phi = x^2 + xy$  and  $\psi = x^2 + y^2$

$$\therefore \int_C \phi dx + \psi dy = \iint_R \left( \frac{\partial \psi}{\partial x} - \frac{\partial \phi}{\partial y} \right) dx dy$$

$$\text{Here, } \int_C (x^2 + xy)dx + (x^2 + y^2)dy = \int_{-1}^1 \int_{-1}^1 \left\{ \left( \frac{\partial}{\partial x} (x^2 + y^2) - \frac{\partial}{\partial y} (x^2 + xy) \right) \right\} dx dy$$

$$\begin{aligned}
&= \int_{-1}^1 \int_{-1}^1 (2x - x) dx dy \\
&= \int_{-1}^1 \int_{-1}^1 x dx dy \\
&= \int_{-1}^1 \left[ \frac{x^2}{2} \right]_{-1}^1 dy \\
&= \int_{-1}^1 \left[ \frac{1^2}{2} - \frac{(-1)^2}{2} \right] dy \\
&= \int_{-1}^1 \left[ \frac{1^2}{2} - \frac{1^2}{2} \right] dy \\
&= \mathbf{0} \text{ Answer}
\end{aligned}$$